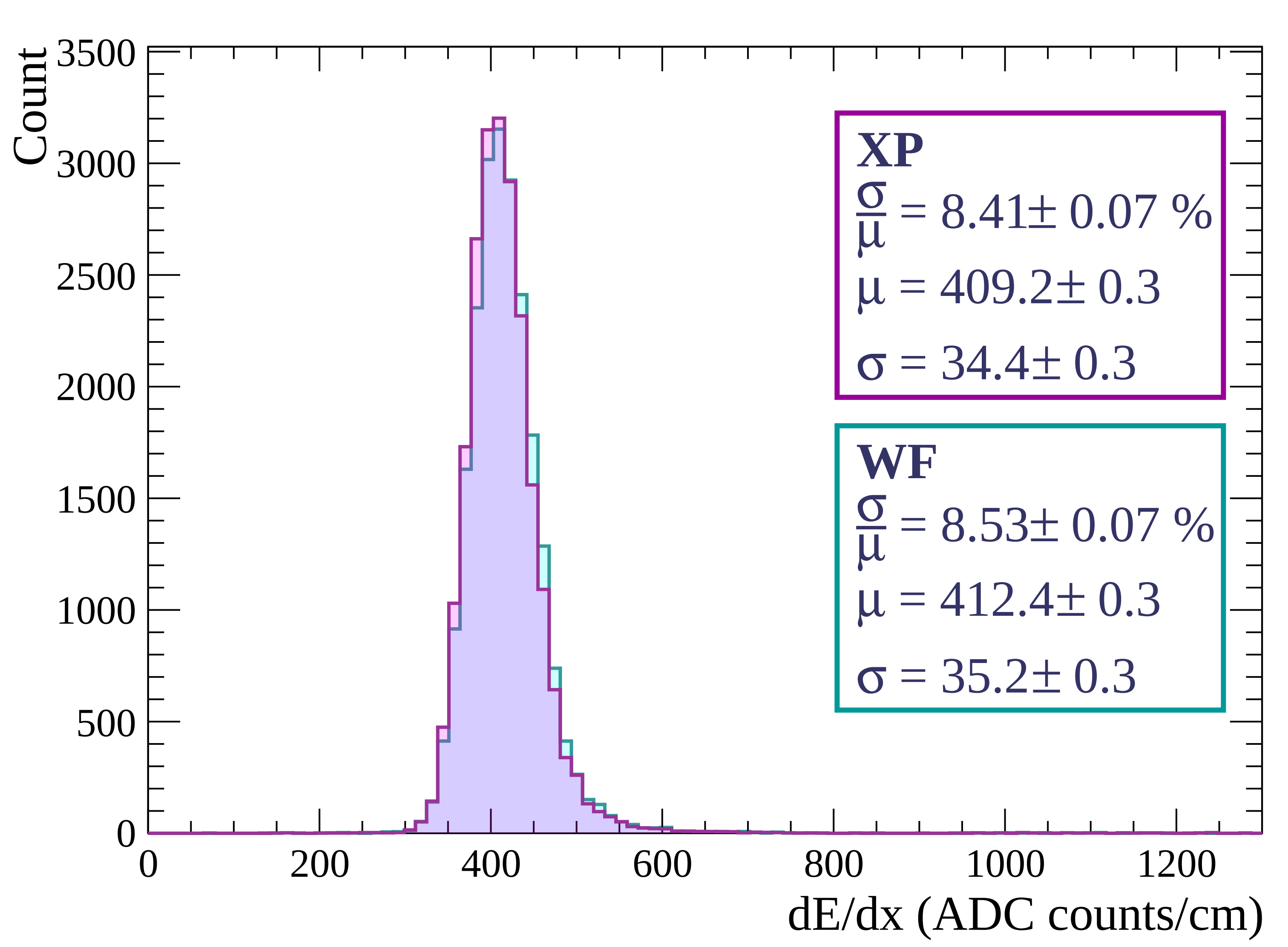
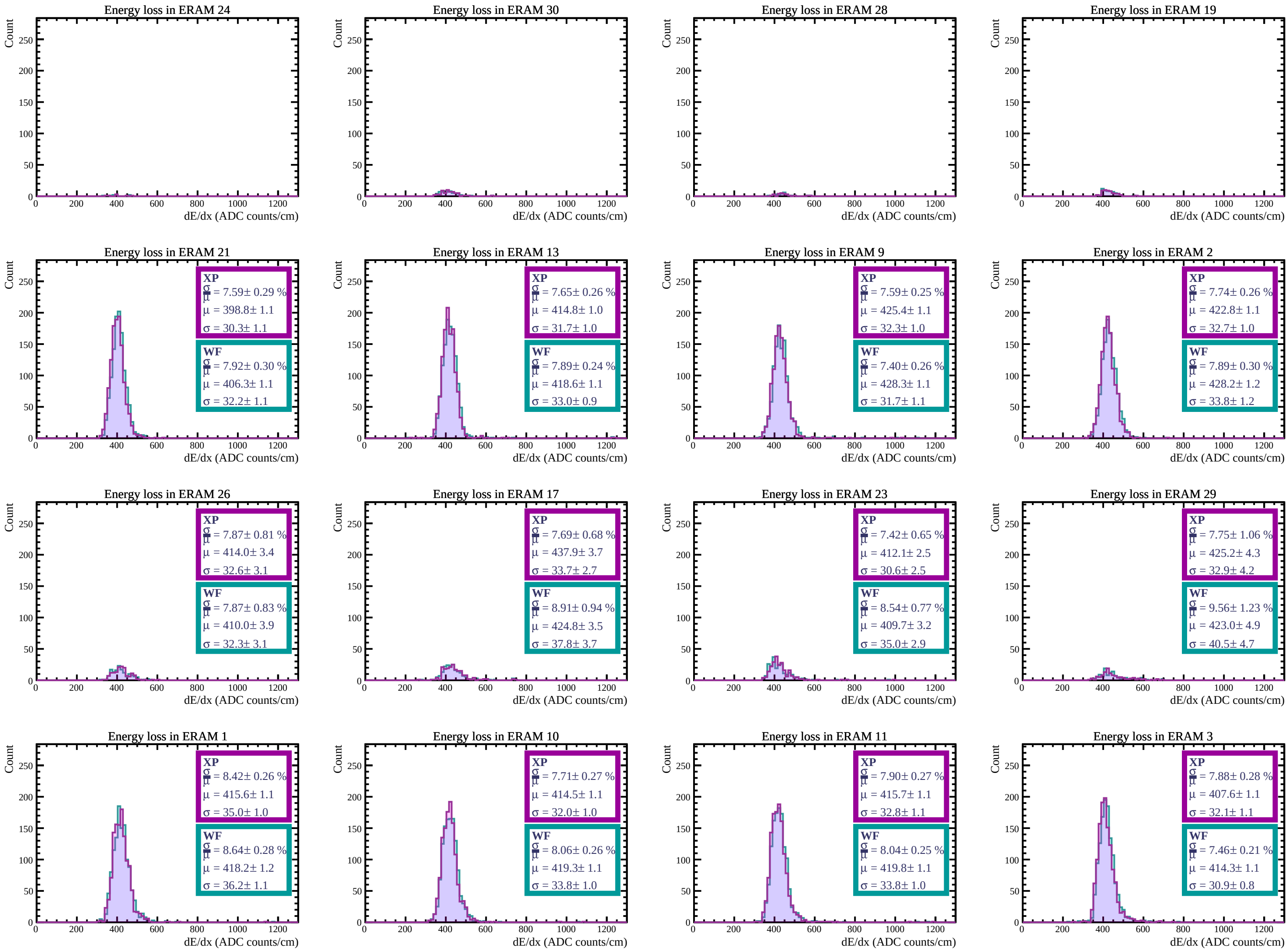
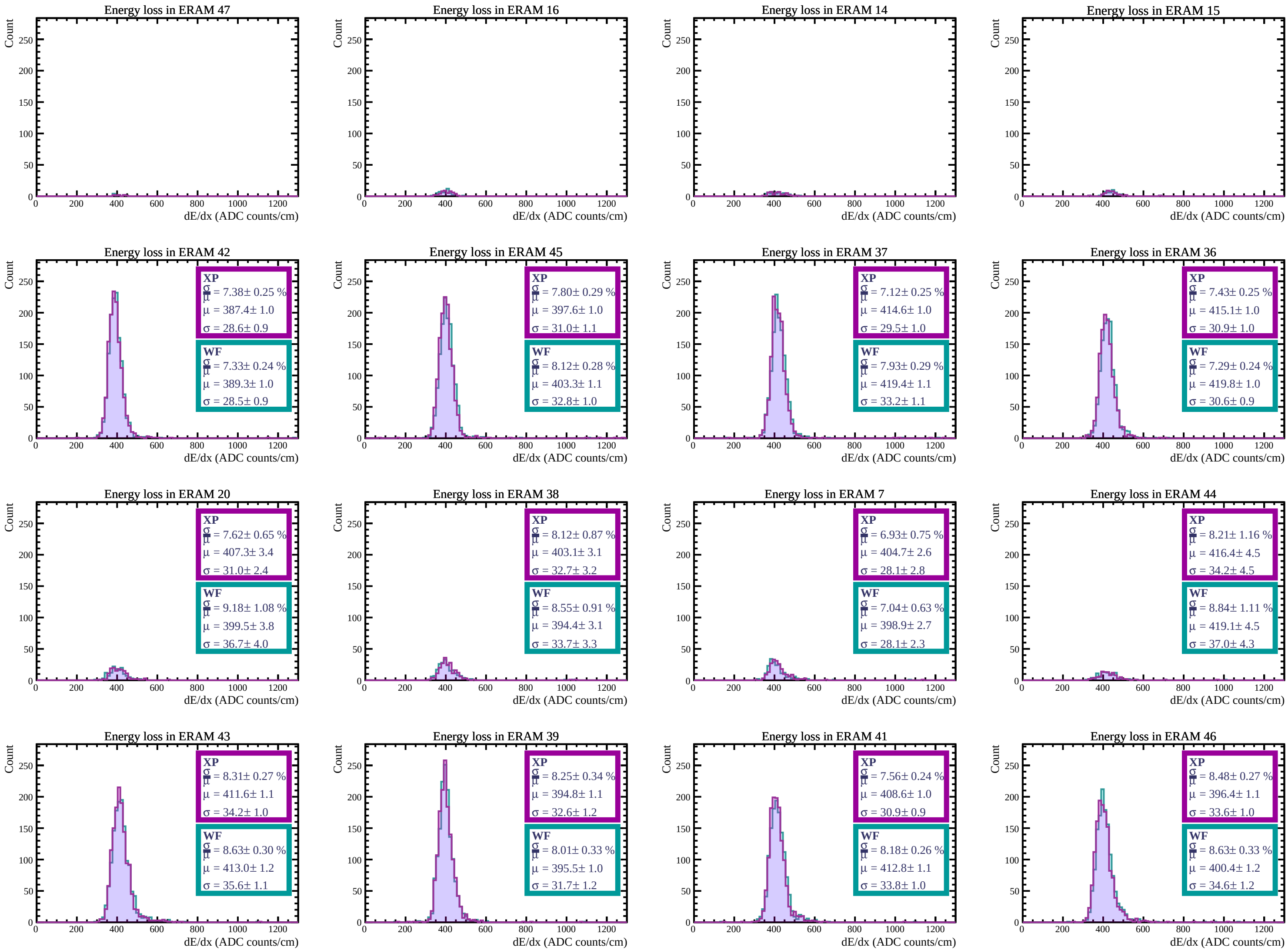


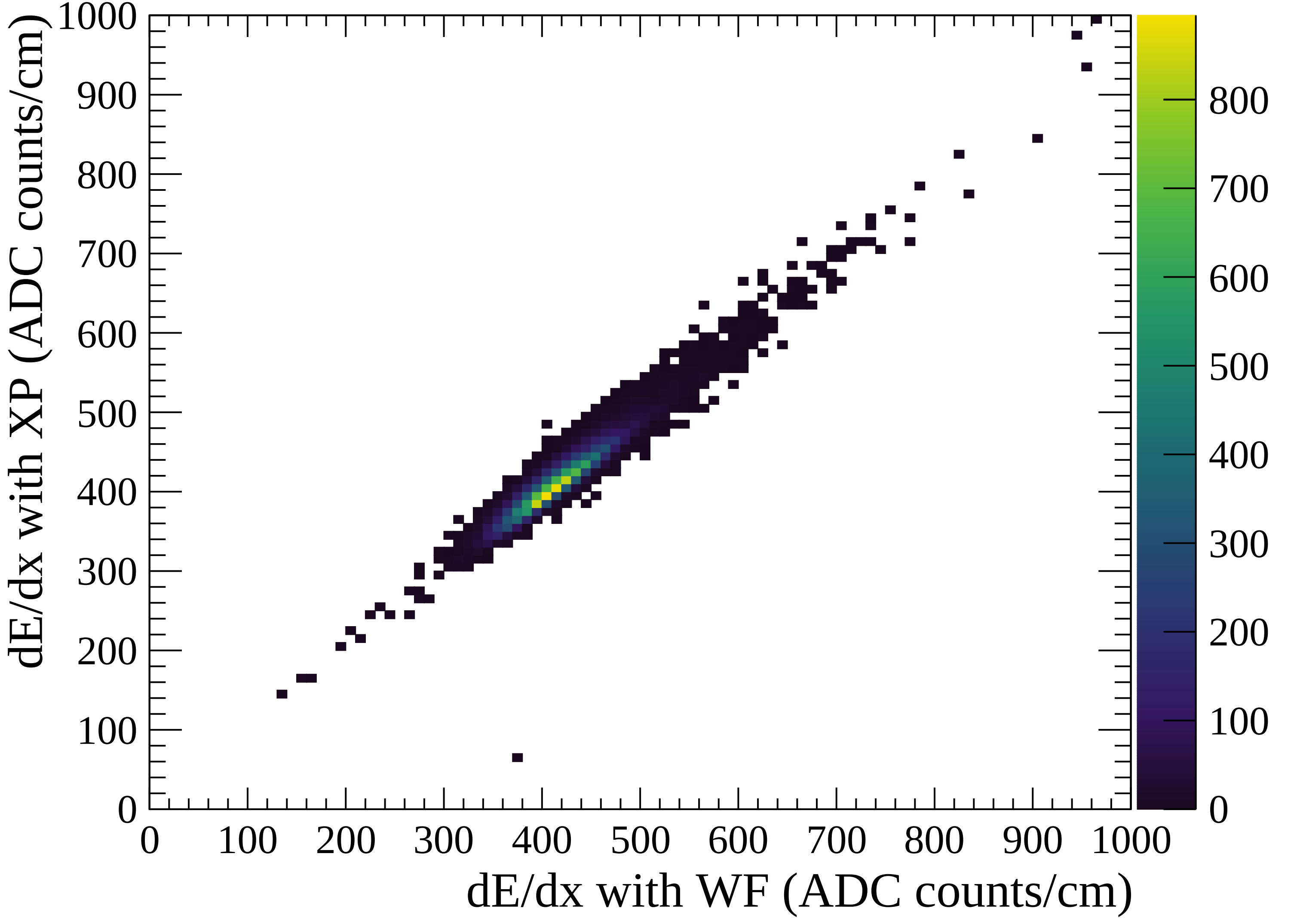
Count
3500
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.41± 0.07 %
μ = 409.2± 0.3
σ = 34.4± 0.3
WF
$\frac{\sigma}{\mu}$ = 8.53± 0.07 %
μ = 412.4± 0.3
σ = 35.2± 0.3

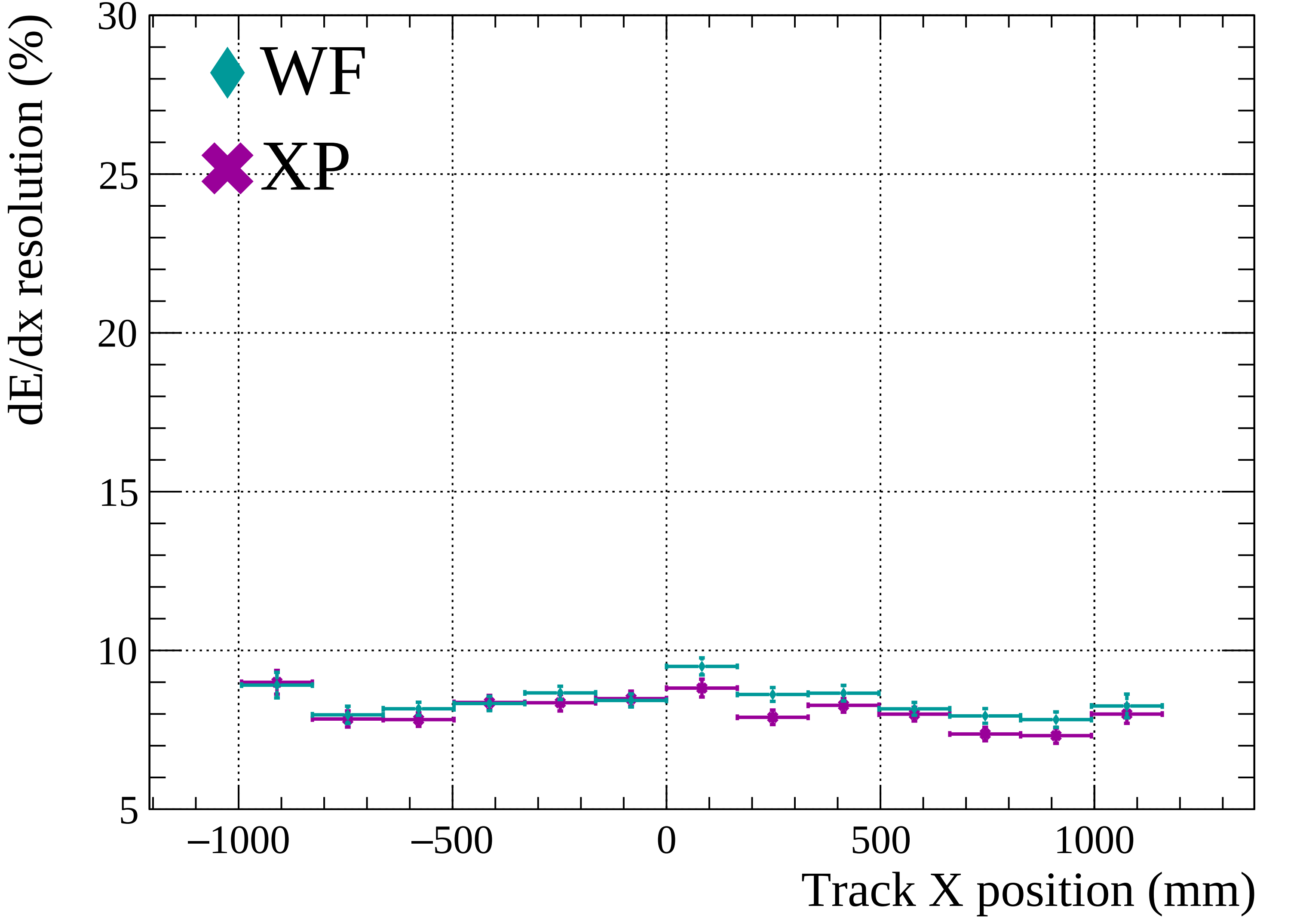


Energy loss in ERAM 24
Energy loss in ERAM 30
Energy loss in ERAM 28
Energy loss in ERAM 19
Energy loss in ERAM 21
XP
σ/μ = 7.59± 0.29 %
μ = 398.8± 1.1
σ = 30.3± 1.1
WF
σ/μ = 7.92± 0.30 %
μ = 406.3± 1.1
σ = 32.2± 1.1
Energy loss in ERAM 13
XP
σ/μ = 7.65± 0.26 %
μ = 414.8± 1.0
σ = 31.7± 1.0
WF
σ/μ = 7.89± 0.24 %
μ = 418.6± 1.1
σ = 33.0± 0.9
Energy loss in ERAM 9
XP
σ/μ = 7.59± 0.25 %
μ = 425.4± 1.1
σ = 32.3± 1.0
WF
σ/μ = 7.40± 0.26 %
μ = 428.3± 1.1
σ = 31.7± 1.1
Energy loss in ERAM 2
XP
σ/μ = 7.74± 0.26 %
μ = 422.8± 1.1
σ = 32.7± 1.0
WF
σ/μ = 7.89± 0.30 %
μ = 428.2± 1.2
σ = 33.8± 1.2
Energy loss in ERAM 26
XP
σ/μ = 7.87± 0.81 %
μ = 414.0± 3.4
σ = 32.6± 3.1
WF
σ/μ = 7.87± 0.83 %
μ = 410.0± 3.9
σ = 32.3± 3.1
Energy loss in ERAM 17
XP
σ/μ = 7.69± 0.68 %
μ = 437.9± 3.7
σ = 33.7± 2.7
WF
σ/μ = 8.91± 0.94 %
μ = 424.8± 3.5
σ = 37.8± 3.7
Energy loss in ERAM 23
XP
σ/μ = 7.42± 0.65 %
μ = 412.1± 2.5
σ = 30.6± 2.5
WF
σ/μ = 8.54± 0.77 %
μ = 409.7± 3.2
σ = 35.0± 2.9
Energy loss in ERAM 29
XP
σ/μ = 7.75± 1.06 %
μ = 425.2± 4.3
σ = 32.9± 4.2
WF
σ/μ = 9.56± 1.23 %
μ = 423.0± 4.9
σ = 40.5± 4.7
Energy loss in ERAM 1
XP
σ/μ = 8.42± 0.26 %
μ = 415.6± 1.1
σ = 35.0± 1.0
WF
σ/μ = 8.64± 0.28 %
μ = 418.2± 1.2
σ = 36.2± 1.1
Energy loss in ERAM 10
XP
σ/μ = 7.71± 0.27 %
μ = 414.5± 1.1
σ = 32.0± 1.0
WF
σ/μ = 8.06± 0.26 %
μ = 419.3± 1.1
σ = 33.8± 1.0
Energy loss in ERAM 11
XP
σ/μ = 7.90± 0.27 %
μ = 415.7± 1.1
σ = 32.8± 1.1
WF
σ/μ = 8.04± 0.25 %
μ = 419.8± 1.1
σ = 33.8± 1.0
Energy loss in ERAM 3
XP
σ/μ = 7.88± 0.28 %
μ = 407.6± 1.1
σ = 32.1± 1.1
WF
σ/μ = 7.46± 0.21 %
μ = 414.3± 1.1
σ = 30.9± 0.8
Count
dE/dx (ADC counts/cm)

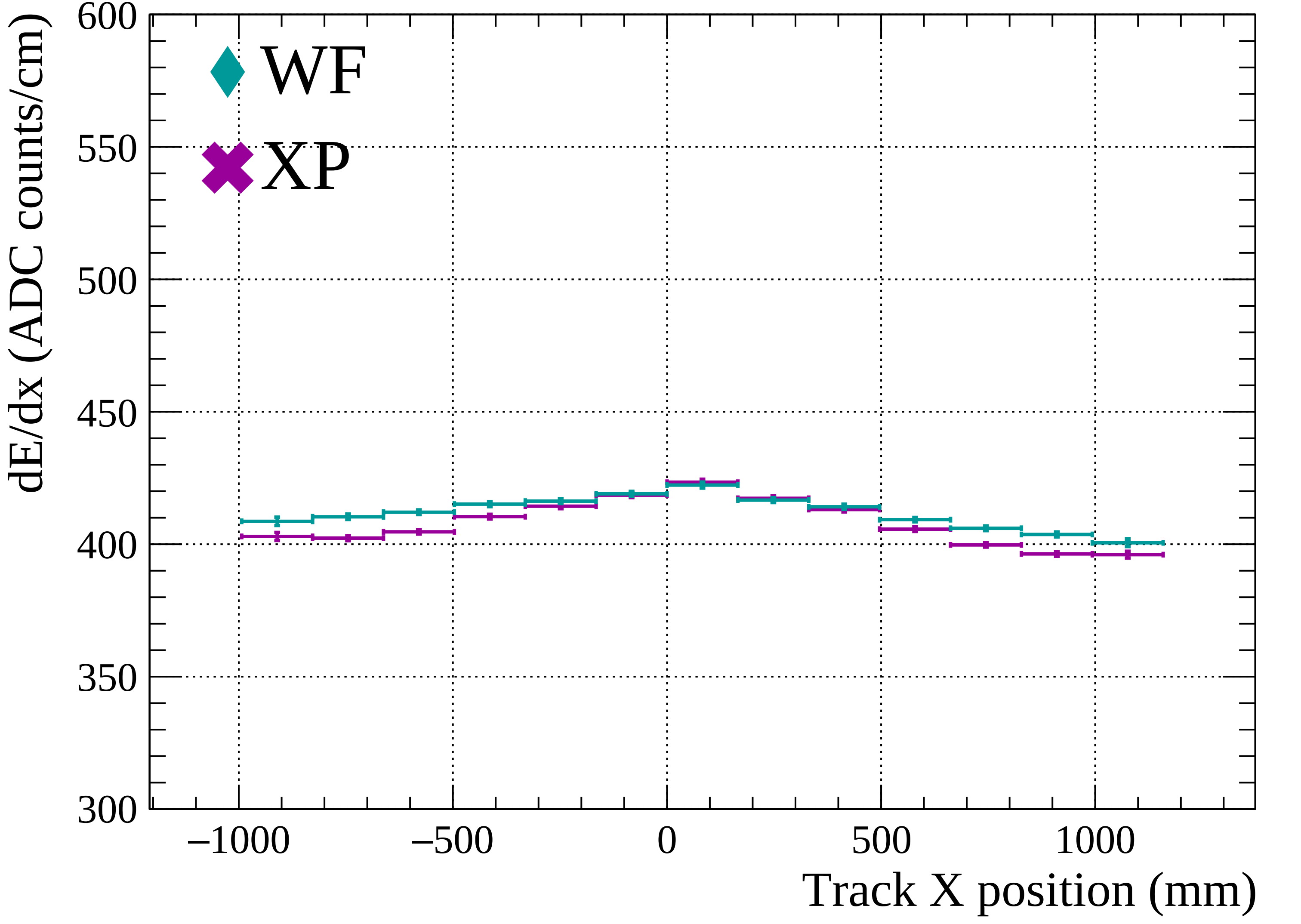


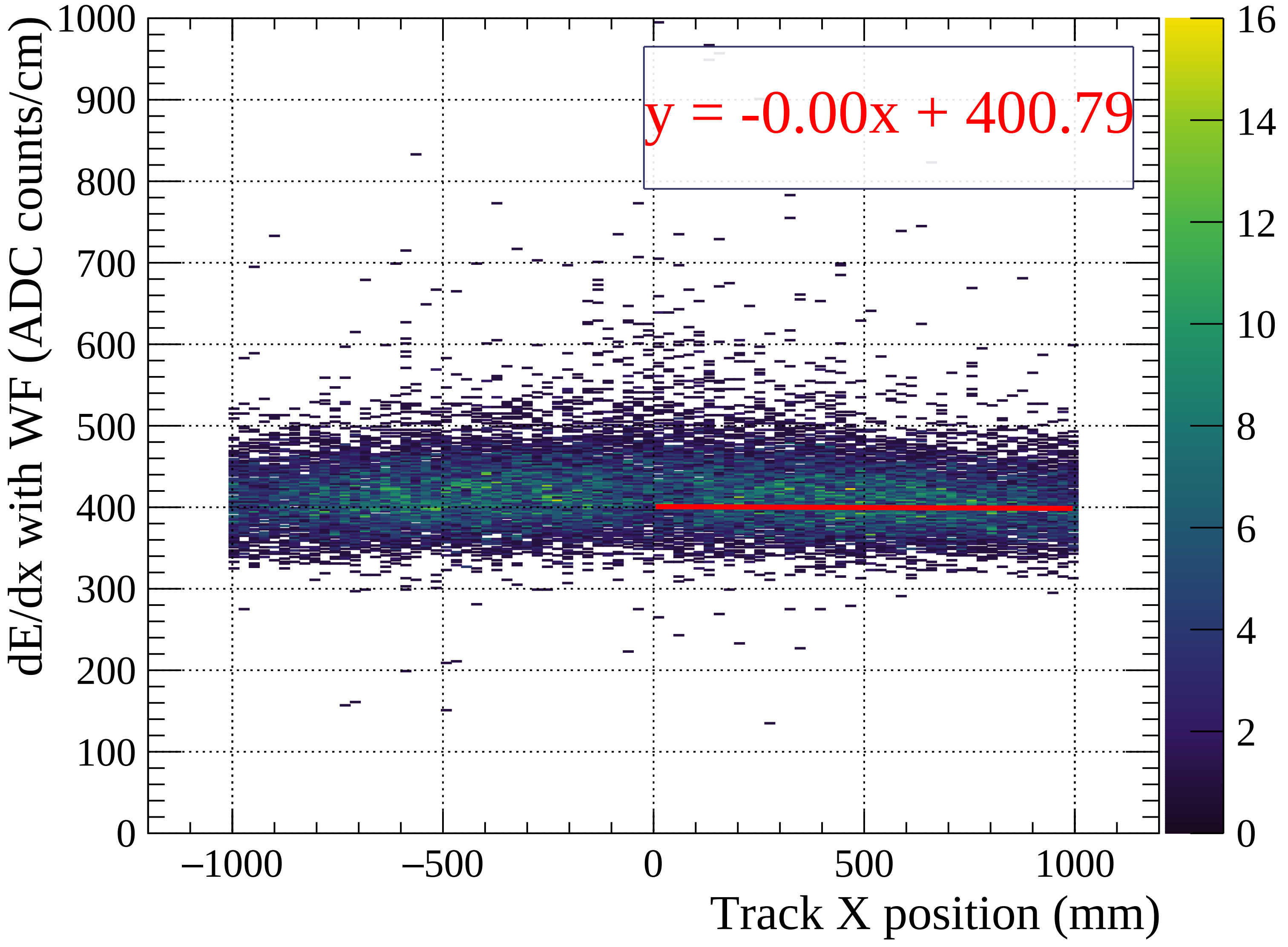
Energy loss in ERAM 47
Energy loss in ERAM 16
Energy loss in ERAM 14
Energy loss in ERAM 15
Energy loss in ERAM 42
XP
σ/μ = 7.38± 0.25 %
μ = 387.4± 1.0
σ = 28.6± 0.9
WF
σ/μ = 7.33± 0.24 %
μ = 389.3± 1.0
σ = 28.5± 0.9
Energy loss in ERAM 45
XP
σ/μ = 7.80± 0.29 %
μ = 397.6± 1.0
σ = 31.0± 1.1
WF
σ/μ = 8.12± 0.28 %
μ = 403.3± 1.1
σ = 32.8± 1.0
Energy loss in ERAM 37
XP
σ/μ = 7.12± 0.25 %
μ = 414.6± 1.0
σ = 29.5± 1.0
WF
σ/μ = 7.93± 0.29 %
μ = 419.4± 1.1
σ = 33.2± 1.1
Energy loss in ERAM 36
XP
σ/μ = 7.43± 0.25 %
μ = 415.1± 1.0
σ = 30.9± 1.0
WF
σ/μ = 7.29± 0.24 %
μ = 419.8± 1.0
σ = 30.6± 0.9
Energy loss in ERAM 20
XP
σ/μ = 7.62± 0.65 %
μ = 407.3± 3.4
σ = 31.0± 2.4
WF
σ/μ = 9.18± 1.08 %
μ = 399.5± 3.8
σ = 36.7± 4.0
Energy loss in ERAM 38
XP
σ/μ = 8.12± 0.87 %
μ = 403.1± 3.1
σ = 32.7± 3.2
WF
σ/μ = 8.55± 0.91 %
μ = 394.4± 3.1
σ = 33.7± 3.3
Energy loss in ERAM 7
XP
σ/μ = 6.93± 0.75 %
μ = 404.7± 2.6
σ = 28.1± 2.8
WF
σ/μ = 7.04± 0.63 %
μ = 398.9± 2.7
σ = 28.1± 2.3
Energy loss in ERAM 44
XP
σ/μ = 8.21± 1.16 %
μ = 416.4± 4.5
σ = 34.2± 4.5
WF
σ/μ = 8.84± 1.11 %
μ = 419.1± 4.5
σ = 37.0± 4.3
Energy loss in ERAM 43
XP
σ/μ = 8.31± 0.27 %
μ = 411.6± 1.1
σ = 34.2± 1.0
WF
σ/μ = 8.63± 0.30 %
μ = 413.0± 1.2
σ = 35.6± 1.1
Energy loss in ERAM 39
XP
σ/μ = 8.25± 0.34 %
μ = 394.8± 1.1
σ = 32.6± 1.2
WF
σ/μ = 8.01± 0.33 %
μ = 395.5± 1.0
σ = 31.7± 1.2
Energy loss in ERAM 41
XP
σ/μ = 7.56± 0.24 %
μ = 408.6± 1.0
σ = 30.9± 0.9
WF
σ/μ = 8.18± 0.26 %
μ = 412.8± 1.1
σ = 33.8± 1.0
Energy loss in ERAM 46
XP
σ/μ = 8.48± 0.27 %
μ = 396.4± 1.1
σ = 33.6± 1.0
WF
σ/μ = 8.63± 0.33 %
μ = 400.4± 1.2
σ = 34.6± 1.2
Count
dE/dx (ADC counts/cm)

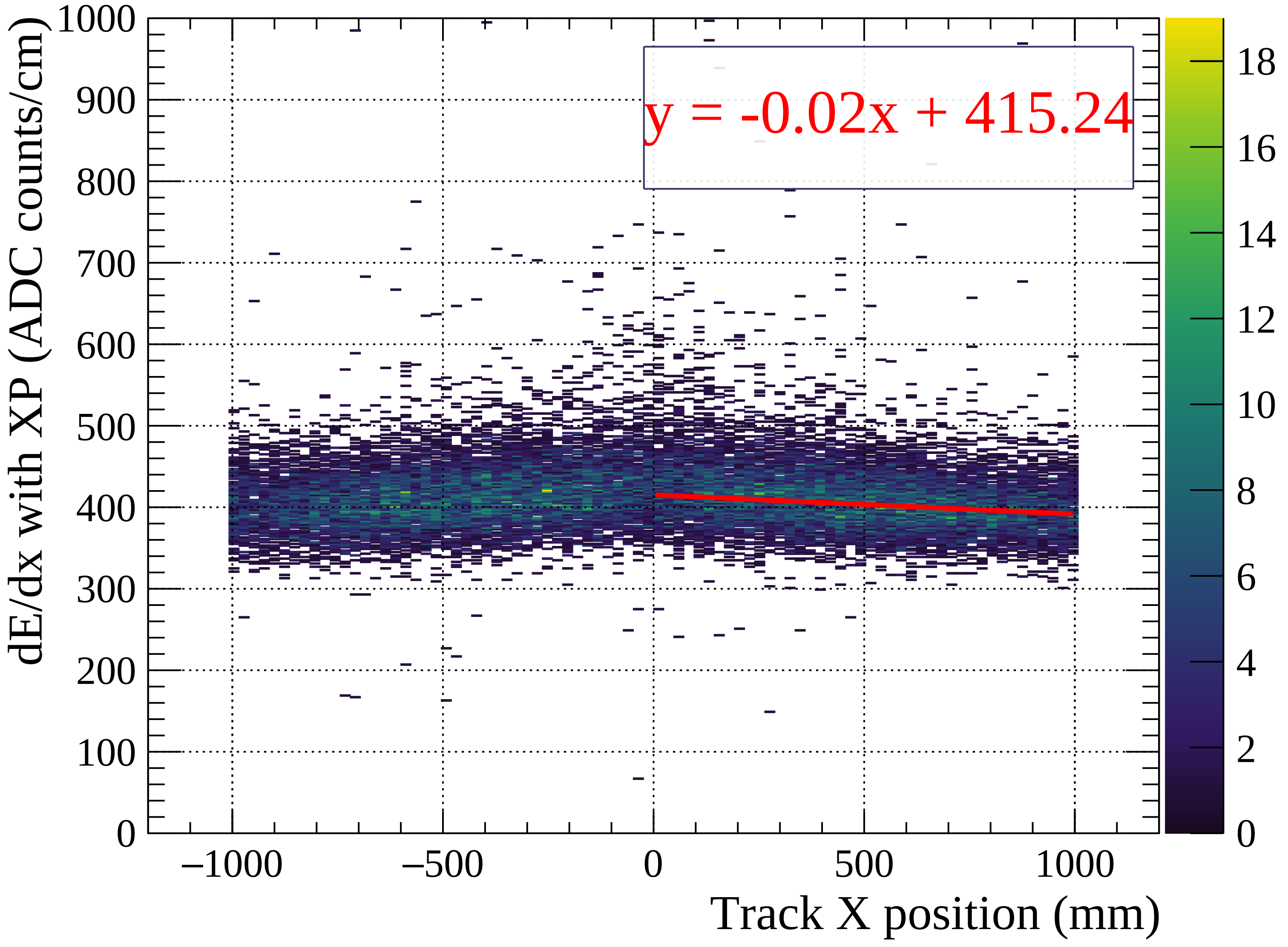

dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000



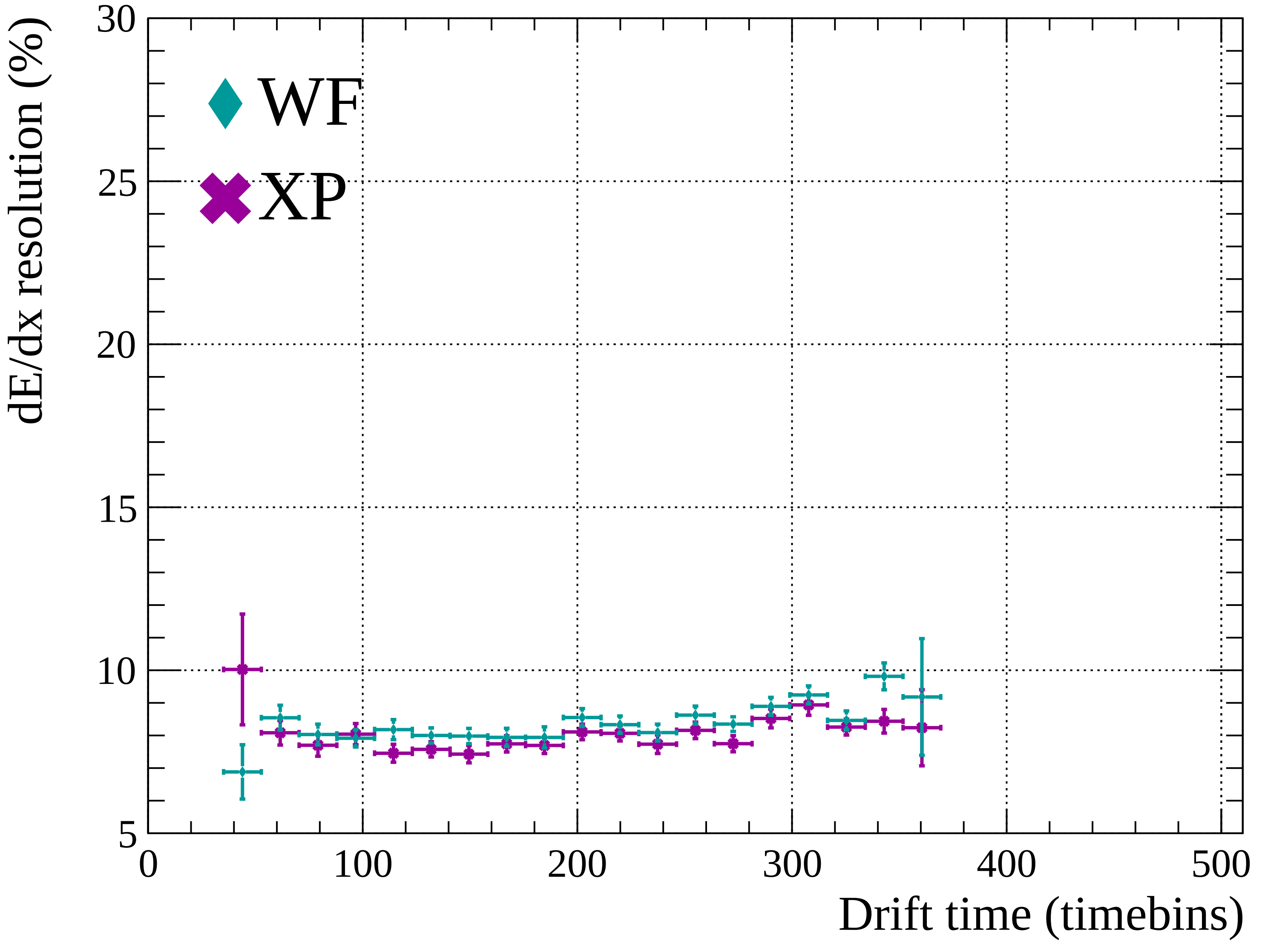
WF
XP
dE/dx resolution (%)
30
25
20
15
10
5
−1000
−500
0
500
1000
Track X position (mm)


WF
XP
600
550
500
450
400
350
300
dE/dx (ADC counts/cm)
−1000
−500
0
500
1000
Track X position (mm)

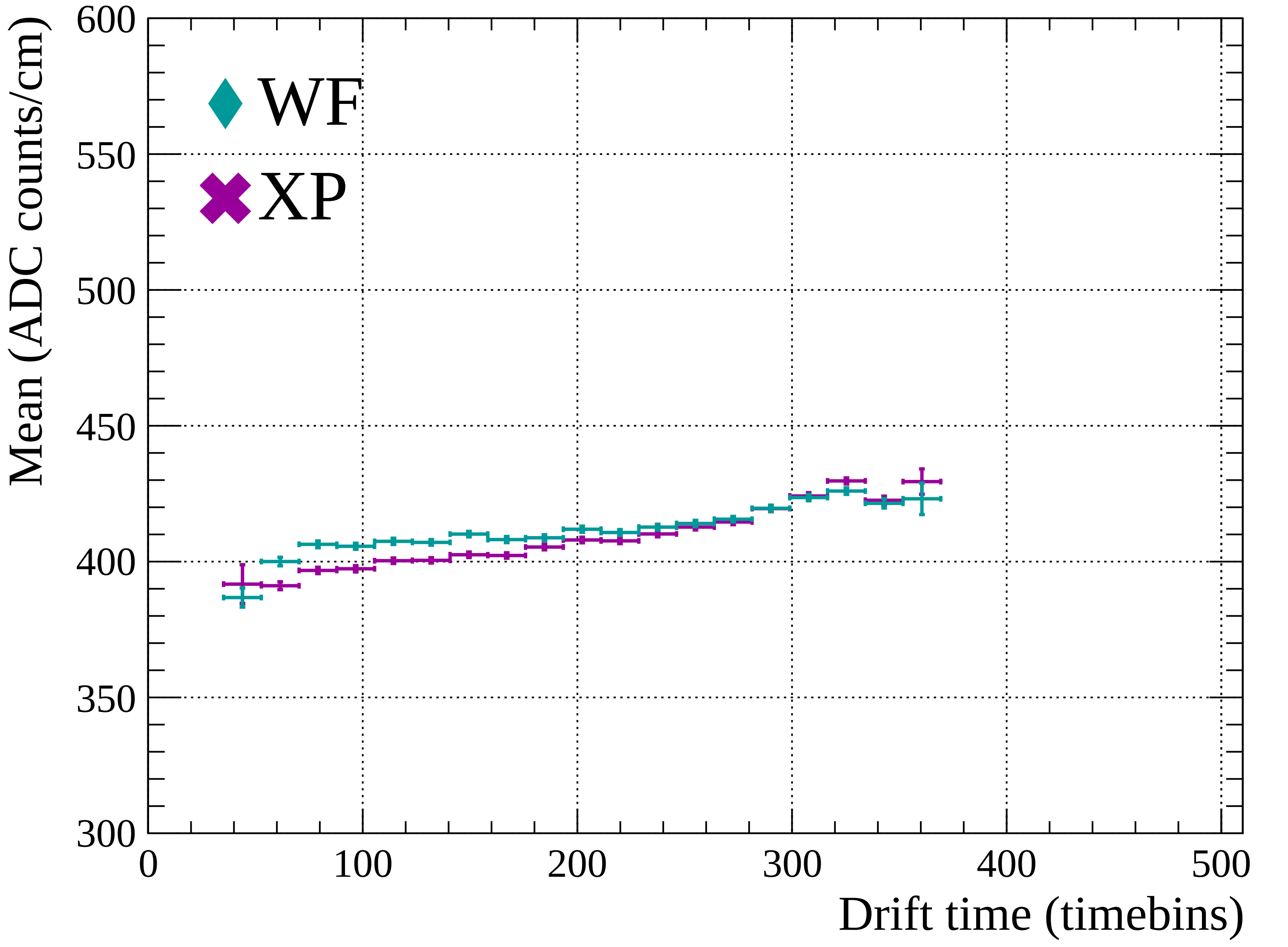

y = -0.00x + 400.79
dE/dx with WF (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
2
4
6
8
10
12
14
16

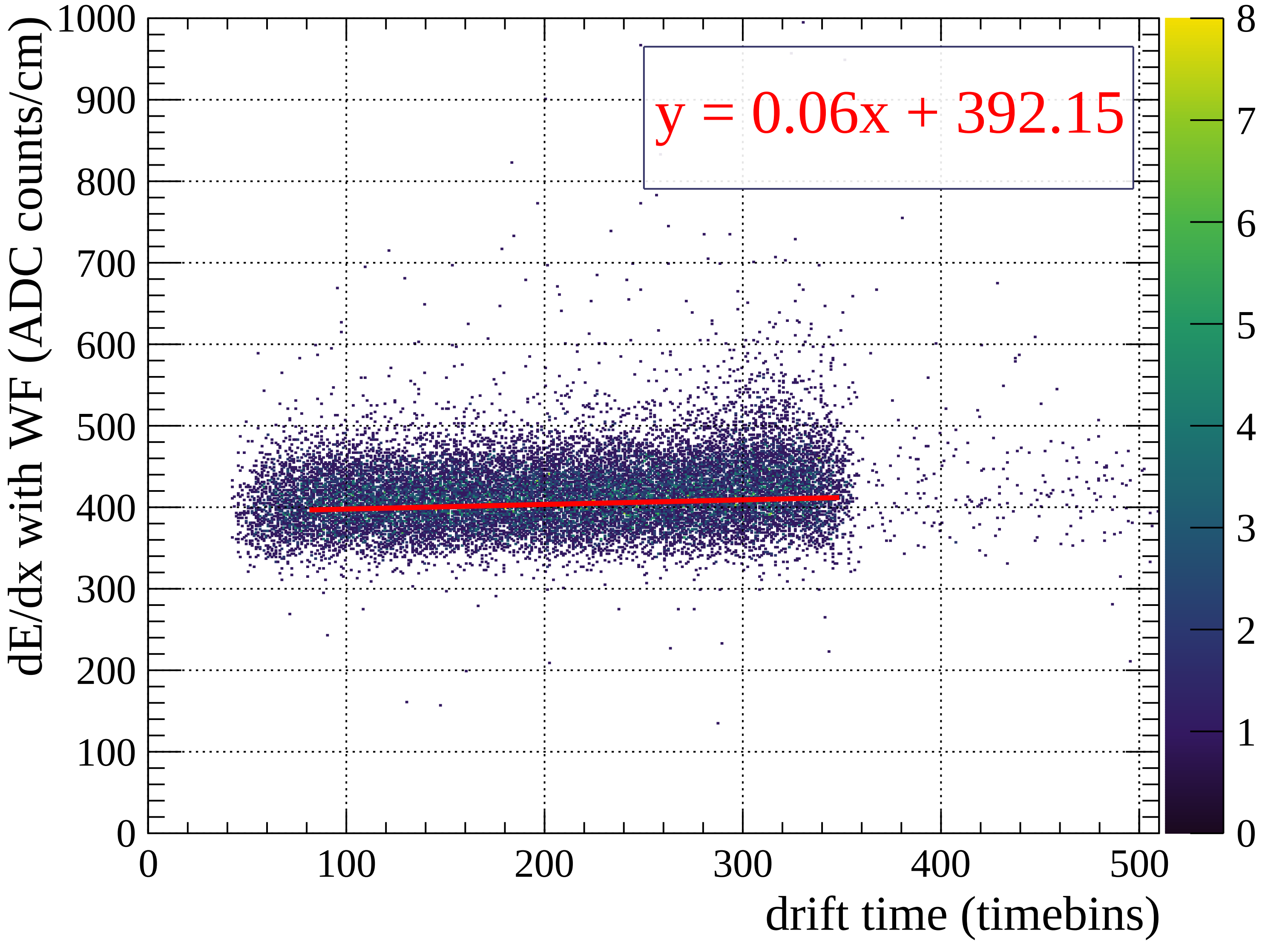


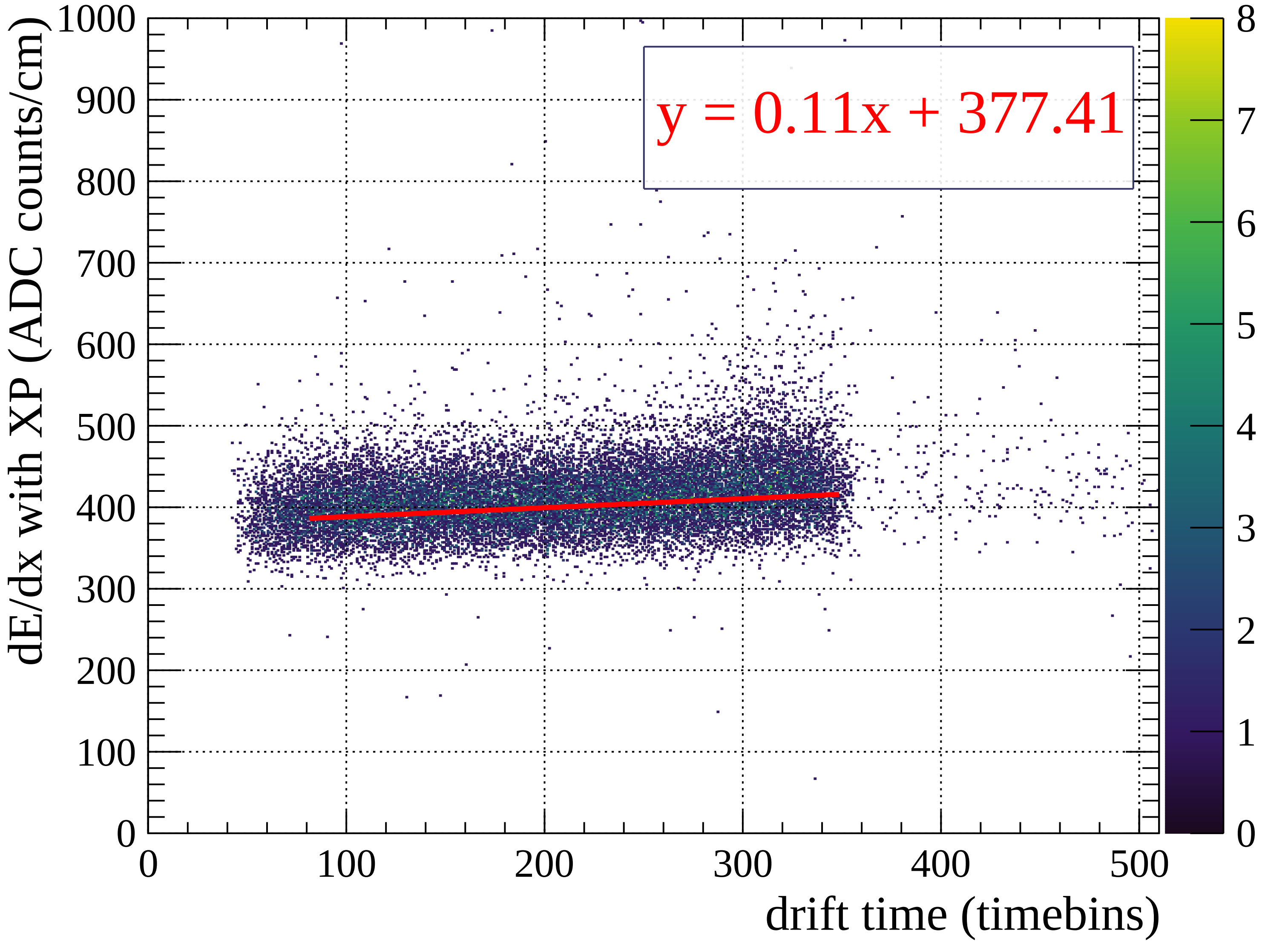
y = -0.02x + 415.24
dE/dx with XP (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
2
4
6
8
10
12
14
16
18



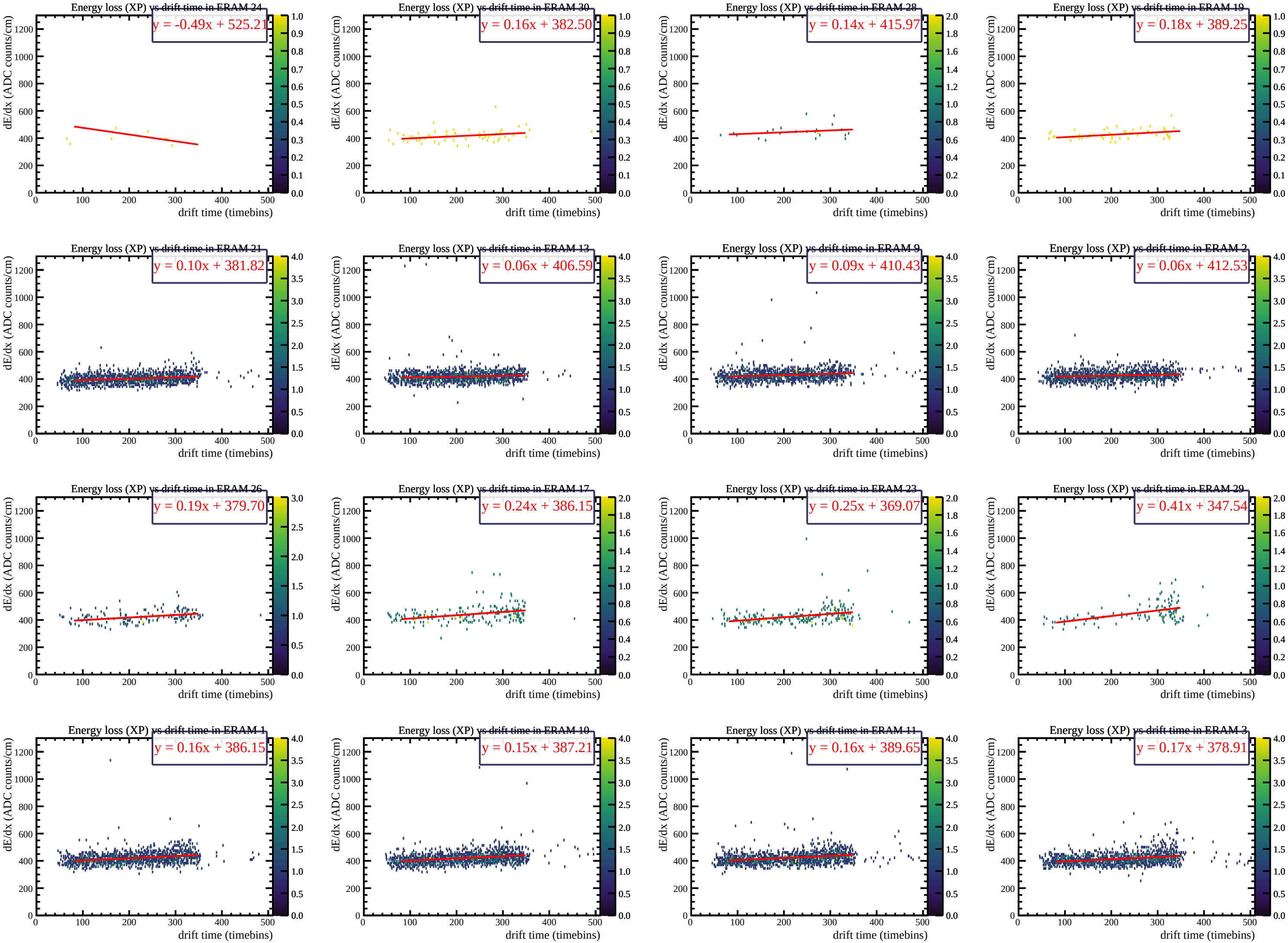
dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
0
100
200
300
400
500
Drift time (timebins)

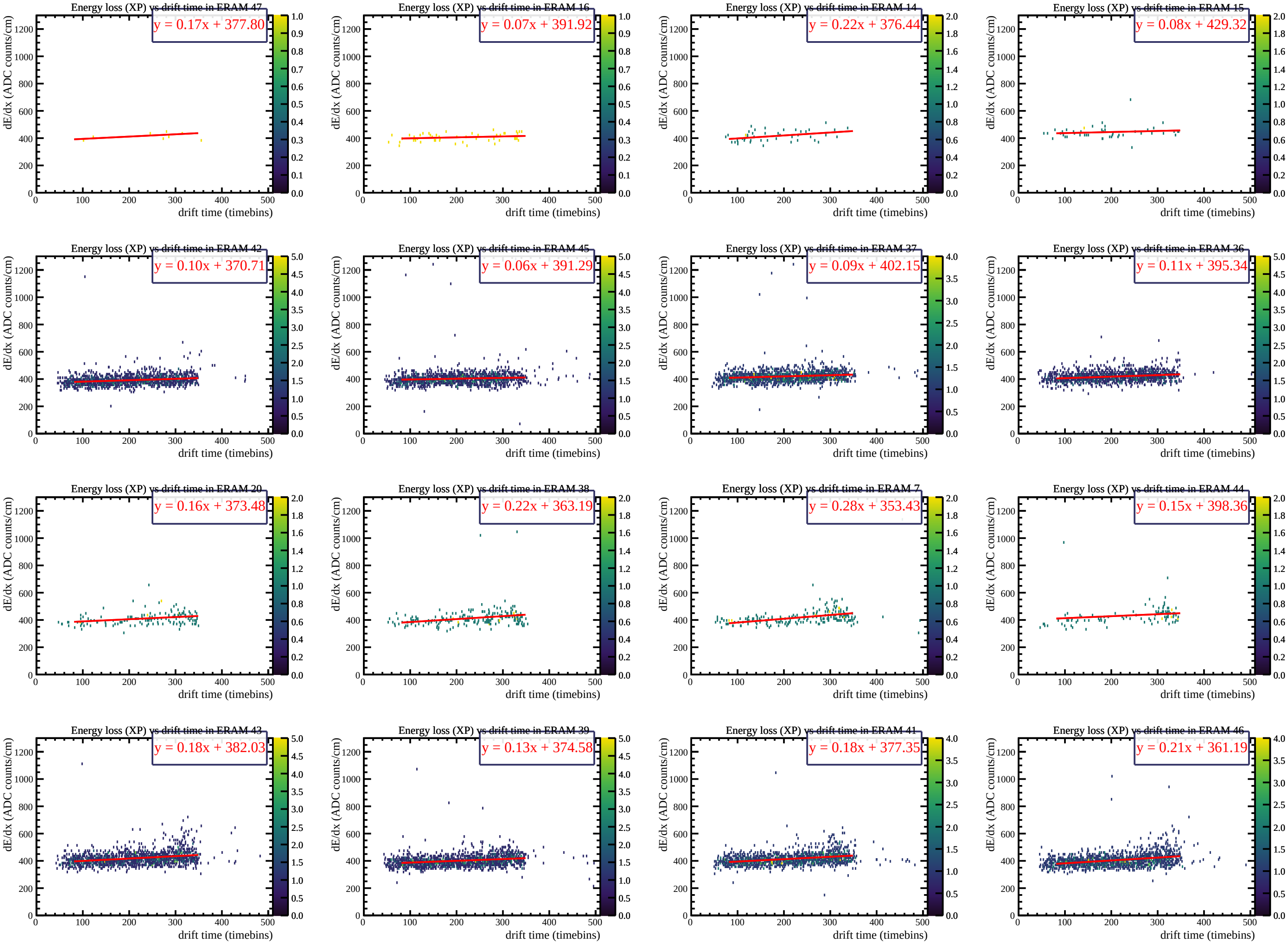

WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
100
200
300
400
500
Drift time (timebins)

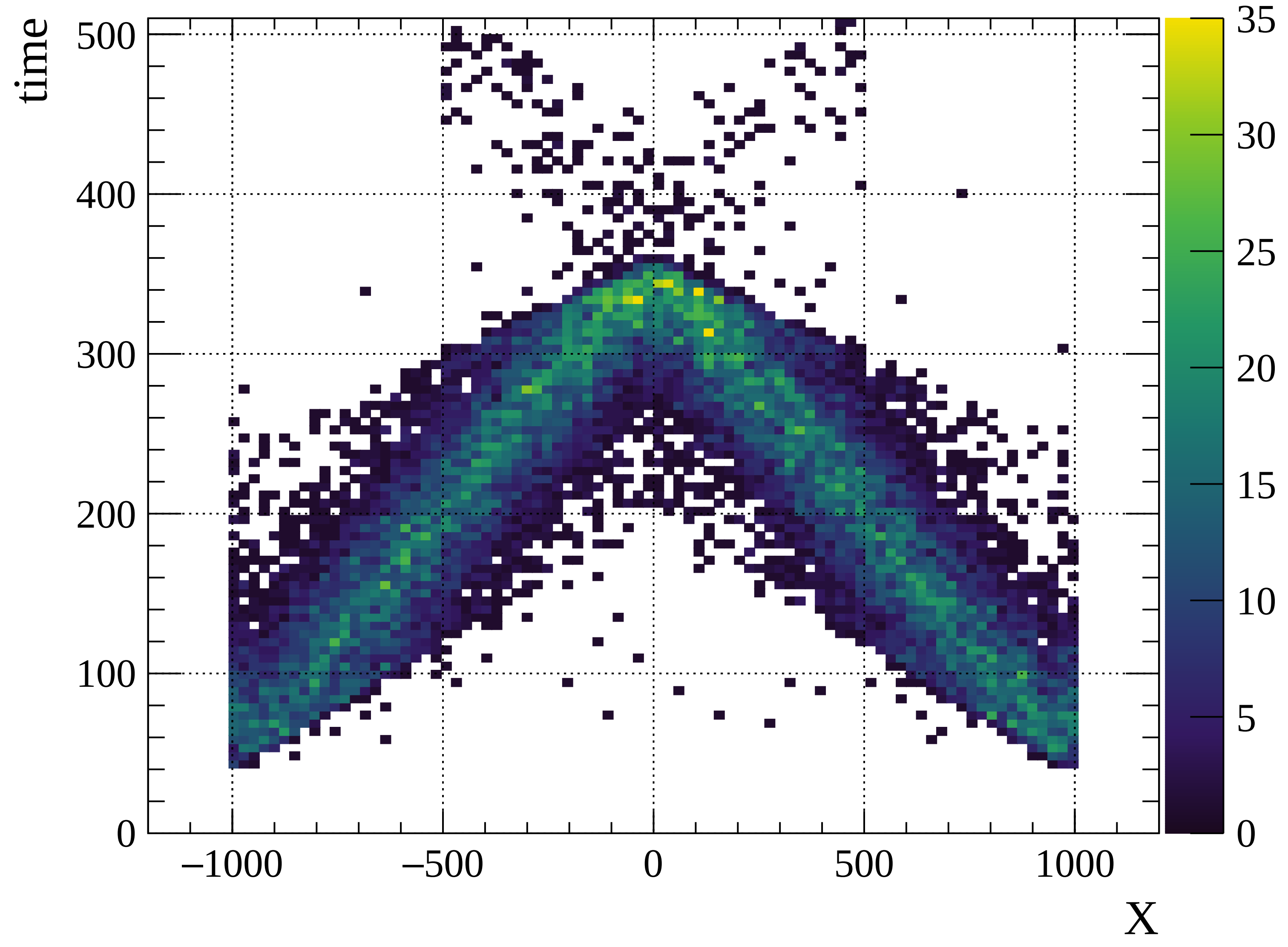

y = 0.06x + 392.15
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
1
2
3
4
5
6
7
8



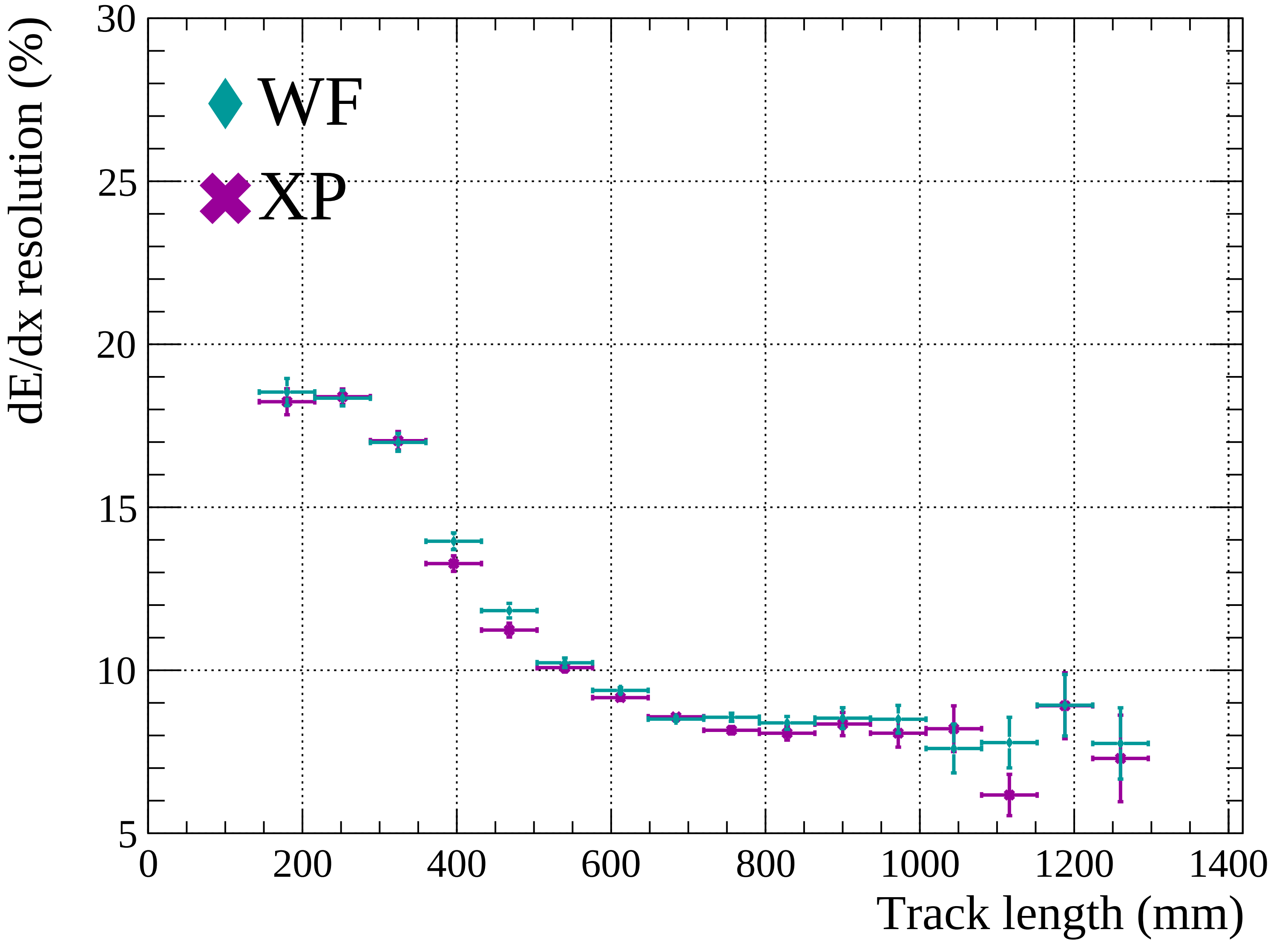
y = 0.11x + 377.41
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
1
2
3
4
5
6
7
8


Energy loss (XP) vs drift time in ERAM 24
y = -0.49x + 525.21
Energy loss (XP) vs drift time in ERAM 30
y = 0.16x + 382.50
Energy loss (XP) vs drift time in ERAM 28
y = 0.14x + 415.97
Energy loss (XP) vs drift time in ERAM 19
y = 0.18x + 389.25
Energy loss (XP) vs drift time in ERAM 21
y = 0.10x + 381.82
Energy loss (XP) vs drift time in ERAM 13
y = 0.06x + 406.59
Energy loss (XP) vs drift time in ERAM 9
y = 0.09x + 410.43
Energy loss (XP) vs drift time in ERAM 2
y = 0.06x + 412.53
Energy loss (XP) vs drift time in ERAM 26
y = 0.19x + 379.70
Energy loss (XP) vs drift time in ERAM 17
y = 0.24x + 386.15
Energy loss (XP) vs drift time in ERAM 23
y = 0.25x + 369.07
Energy loss (XP) vs drift time in ERAM 29
y = 0.41x + 347.54
Energy loss (XP) vs drift time in ERAM 1
y = 0.16x + 386.15
Energy loss (XP) vs drift time in ERAM 10
y = 0.15x + 387.21
Energy loss (XP) vs drift time in ERAM 11
y = 0.16x + 389.65
Energy loss (XP) vs drift time in ERAM 3
y = 0.17x + 378.91
dE/dx (ADC counts/cm)
drift time (timebins)

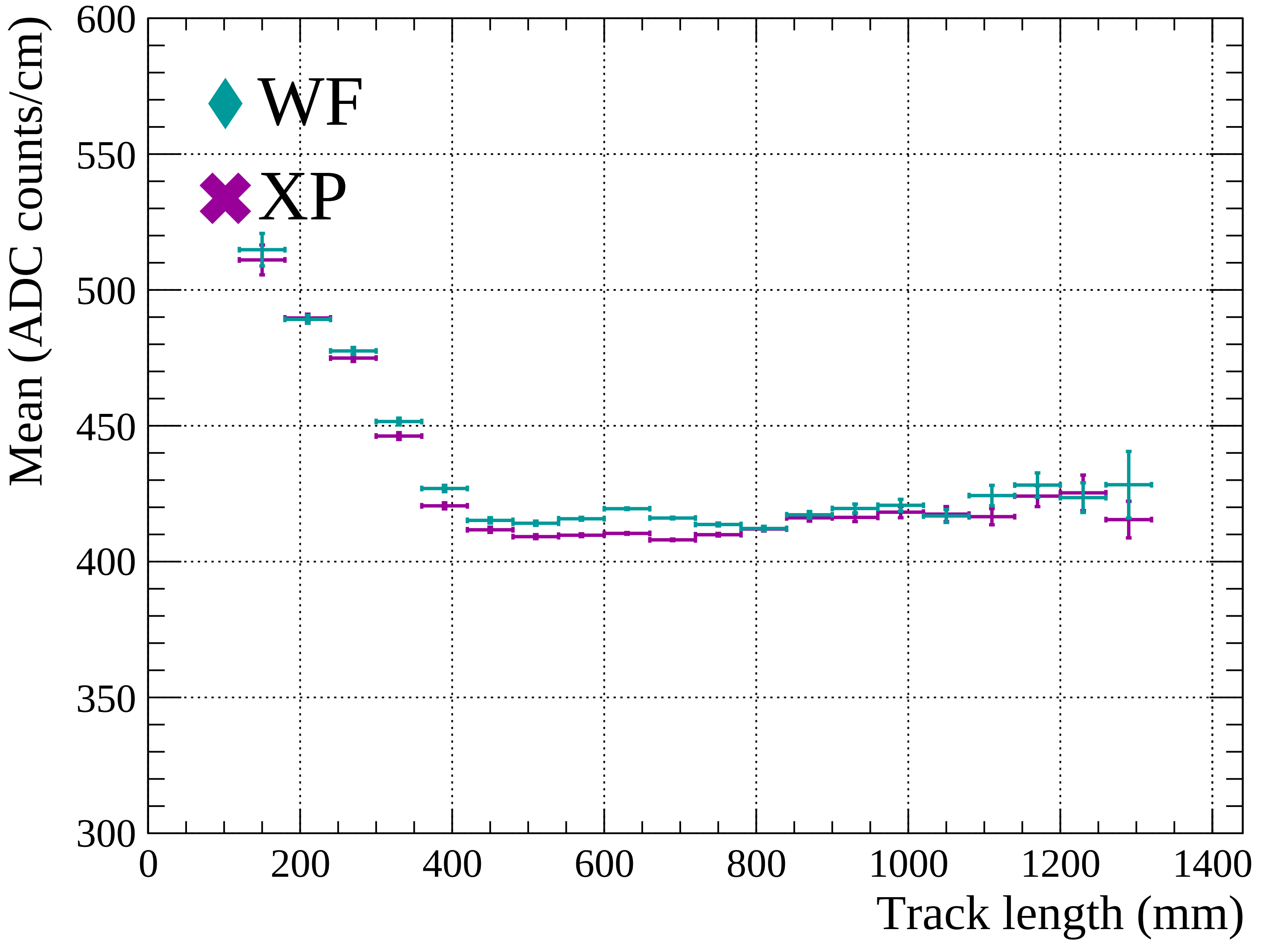

Energy loss (XP) vs drift time in ERAM 47
y = 0.17x + 377.80
Energy loss (XP) vs drift time in ERAM 16
y = 0.07x + 391.92
Energy loss (XP) vs drift time in ERAM 14
y = 0.22x + 376.44
Energy loss (XP) vs drift time in ERAM 15
y = 0.08x + 429.32
Energy loss (XP) vs drift time in ERAM 42
y = 0.10x + 370.71
Energy loss (XP) vs drift time in ERAM 45
y = 0.06x + 391.29
Energy loss (XP) vs drift time in ERAM 37
y = 0.09x + 402.15
Energy loss (XP) vs drift time in ERAM 36
y = 0.11x + 395.34
Energy loss (XP) vs drift time in ERAM 20
y = 0.16x + 373.48
Energy loss (XP) vs drift time in ERAM 38
y = 0.22x + 363.19
Energy loss (XP) vs drift time in ERAM 7
y = 0.28x + 353.43
Energy loss (XP) vs drift time in ERAM 44
y = 0.15x + 398.36
Energy loss (XP) vs drift time in ERAM 43
y = 0.18x + 382.03
Energy loss (XP) vs drift time in ERAM 39
y = 0.13x + 374.58
Energy loss (XP) vs drift time in ERAM 41
y = 0.18x + 377.35
Energy loss (XP) vs drift time in ERAM 46
y = 0.21x + 361.19
dE/dx (ADC counts/cm)
drift time (timebins)



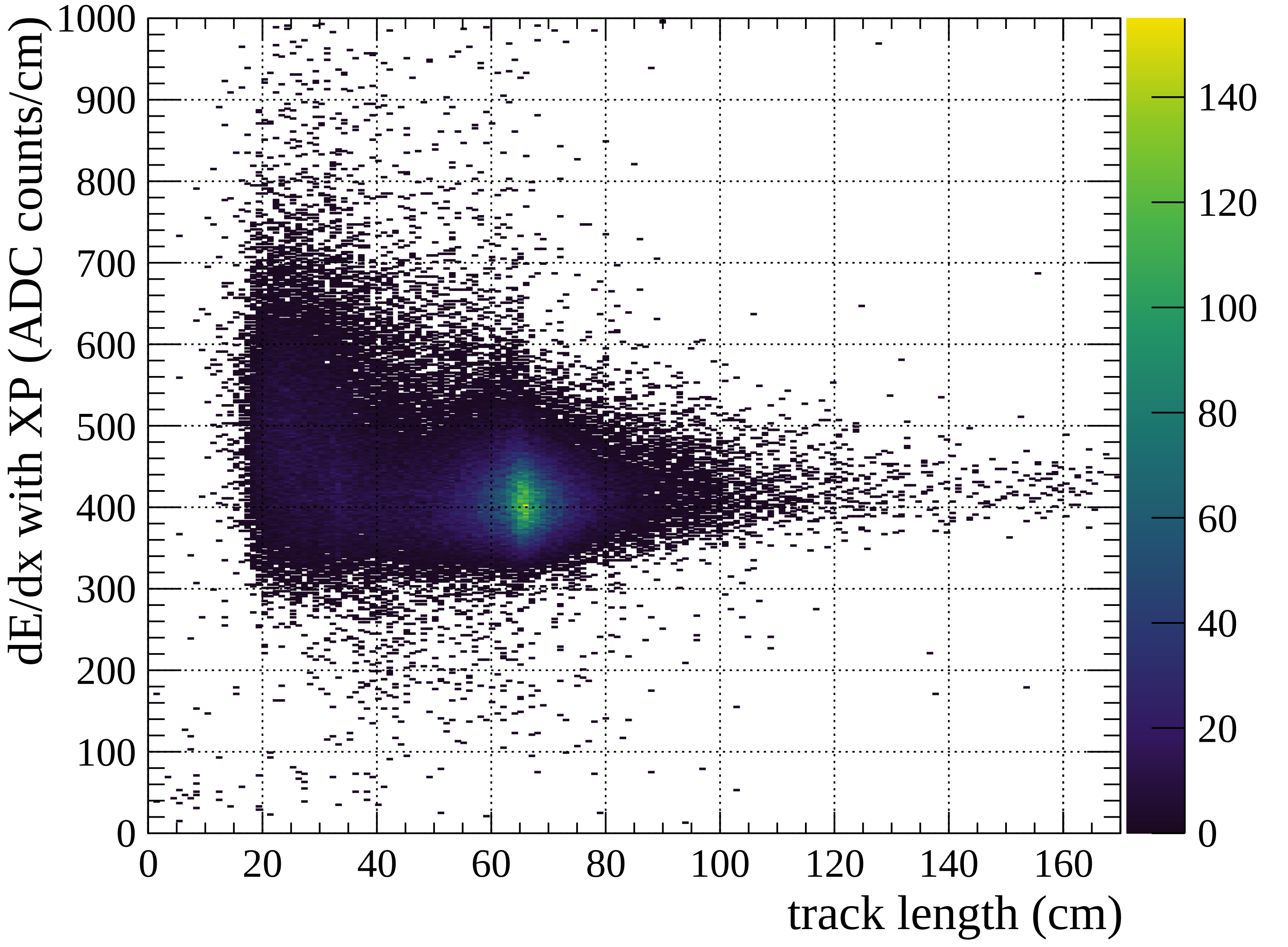
time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
35
30
25
20
15
10
5
0



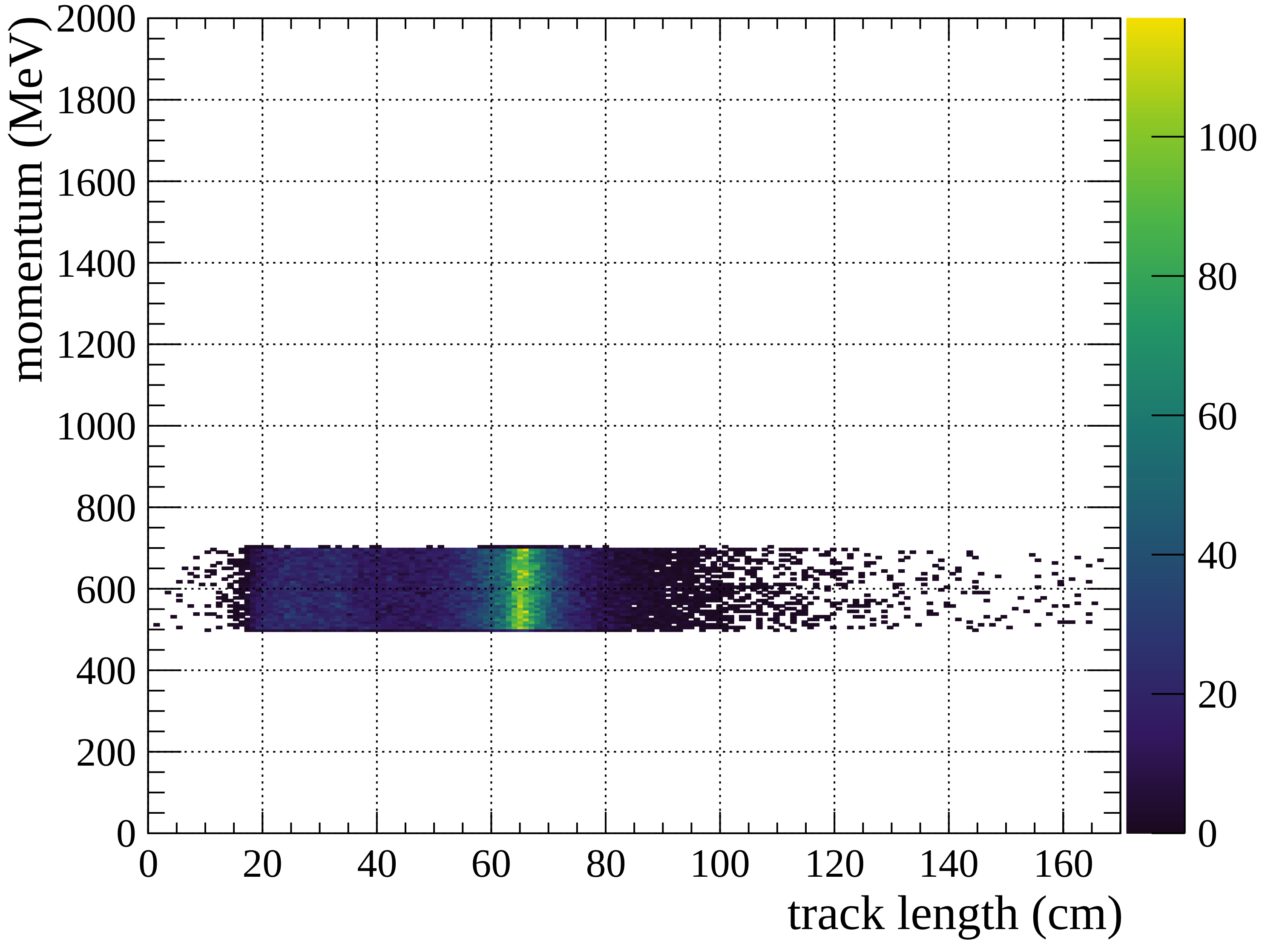
WF
XP
dE/dx resolution (%)
Track length (mm)
30
25
20
15
10
5
0
200
400
600
800
1000
1200
1400



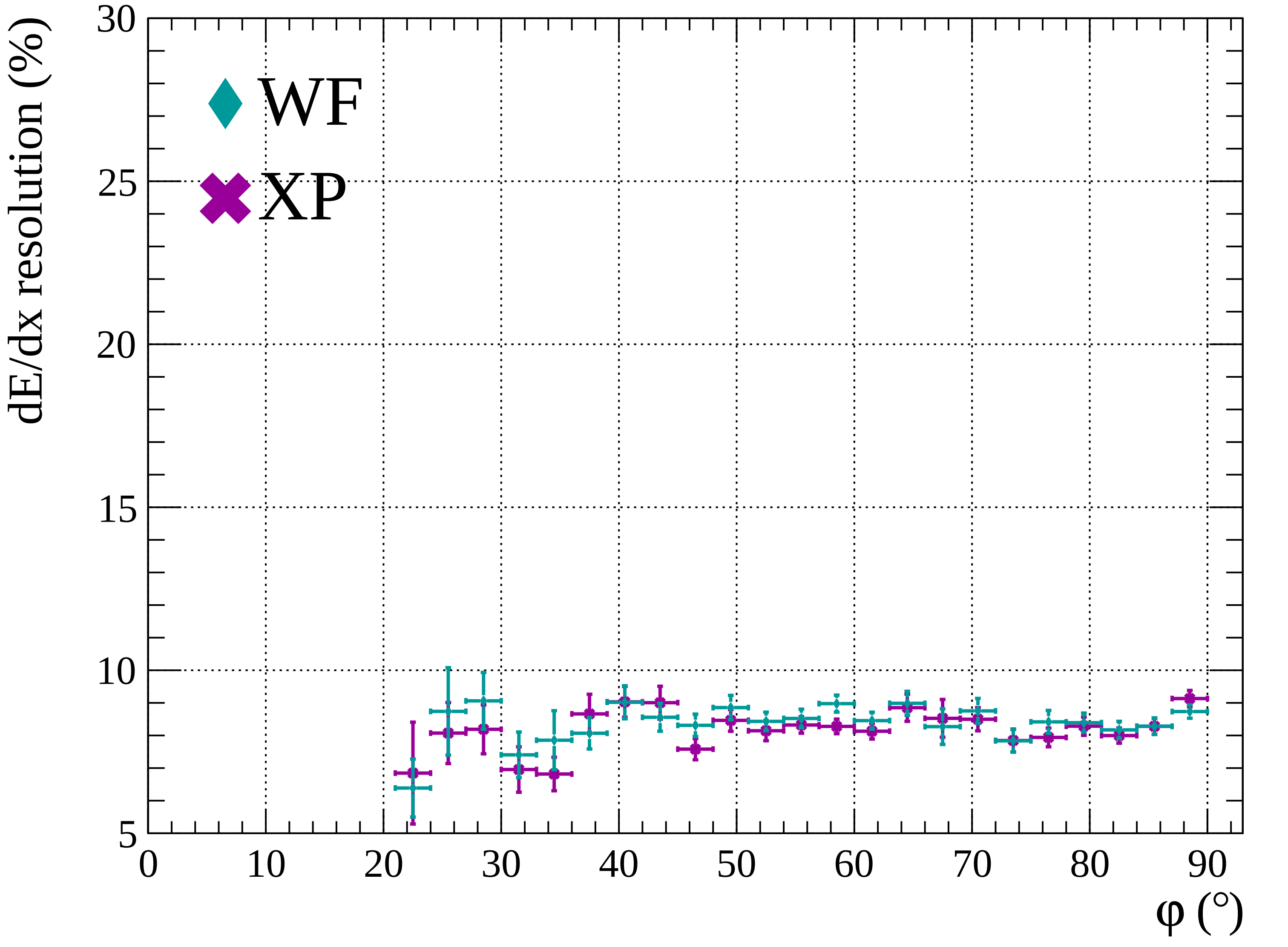
Mean (ADC counts/cm)
Track length (mm)
WF
XP
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400



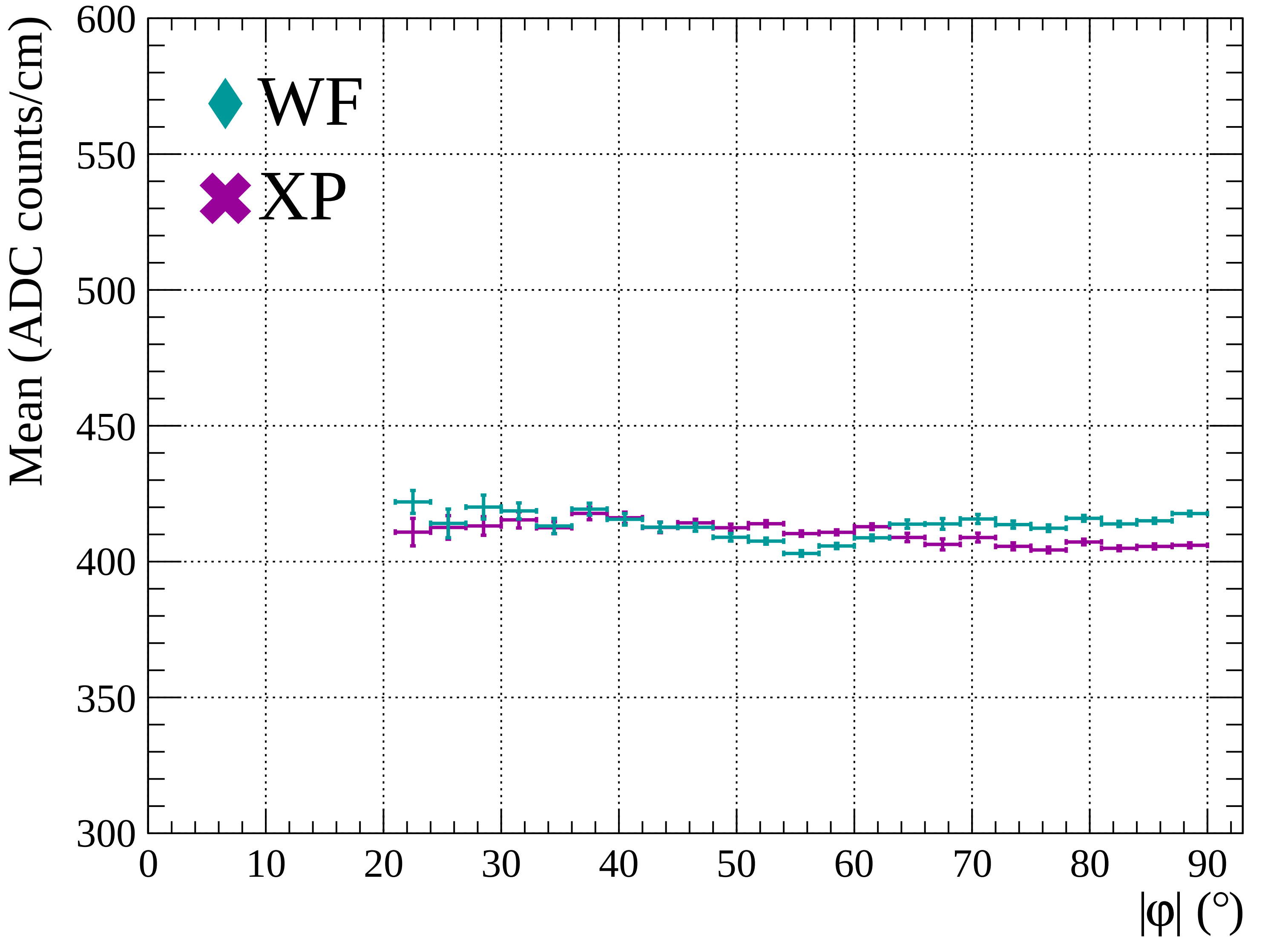
dE/dx with XP (ADC counts/cm)
track length (cm)
0
20
40
60
80
100
120
140
160
0
100
200
300
400
500
600
700
800
900
1000
0
20
40
60
80
100
120
140

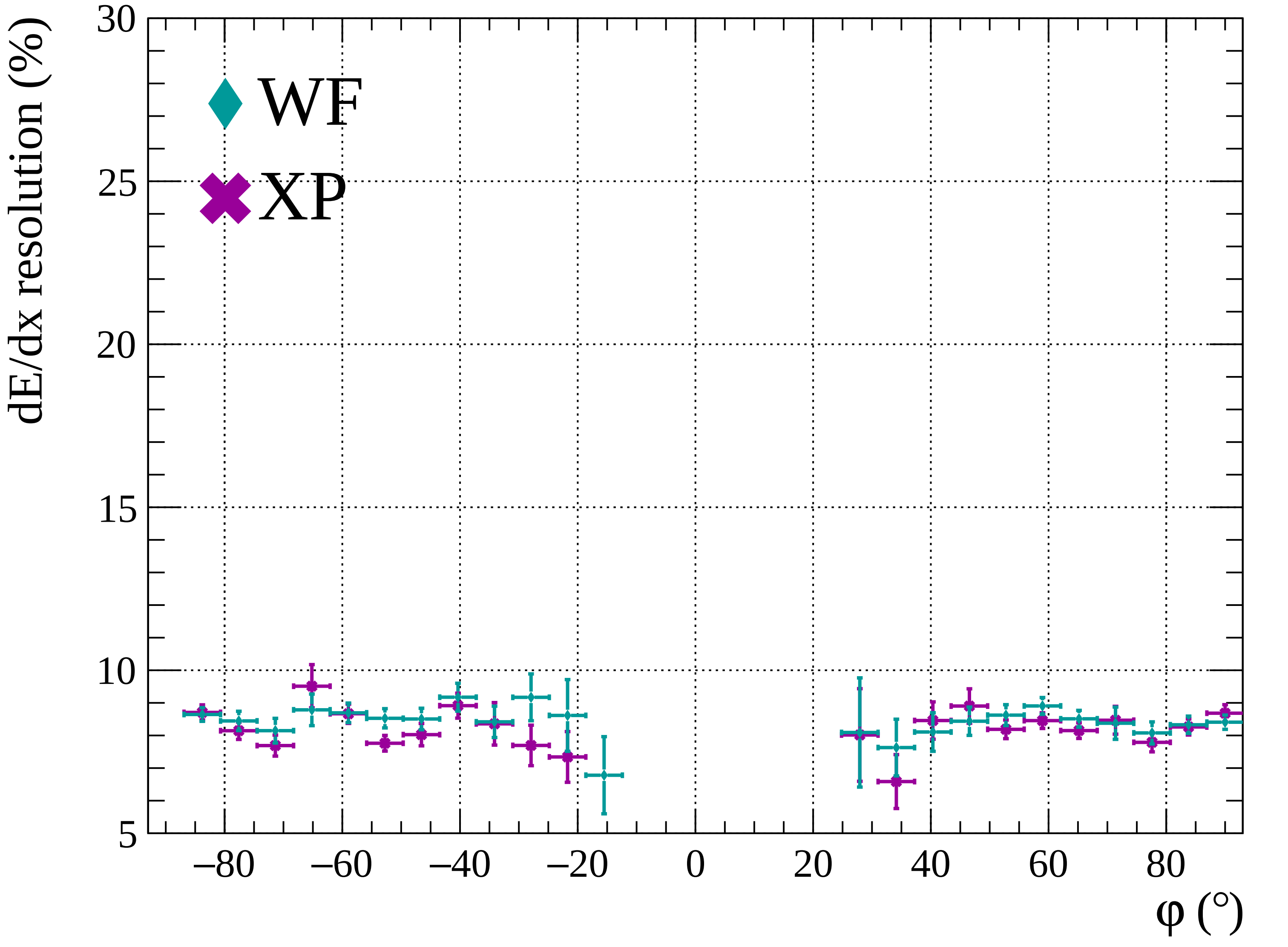


momentum (MeV)
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
20
40
60
80
100
120
140
160
track length (cm)
100
80
60
40
20
0

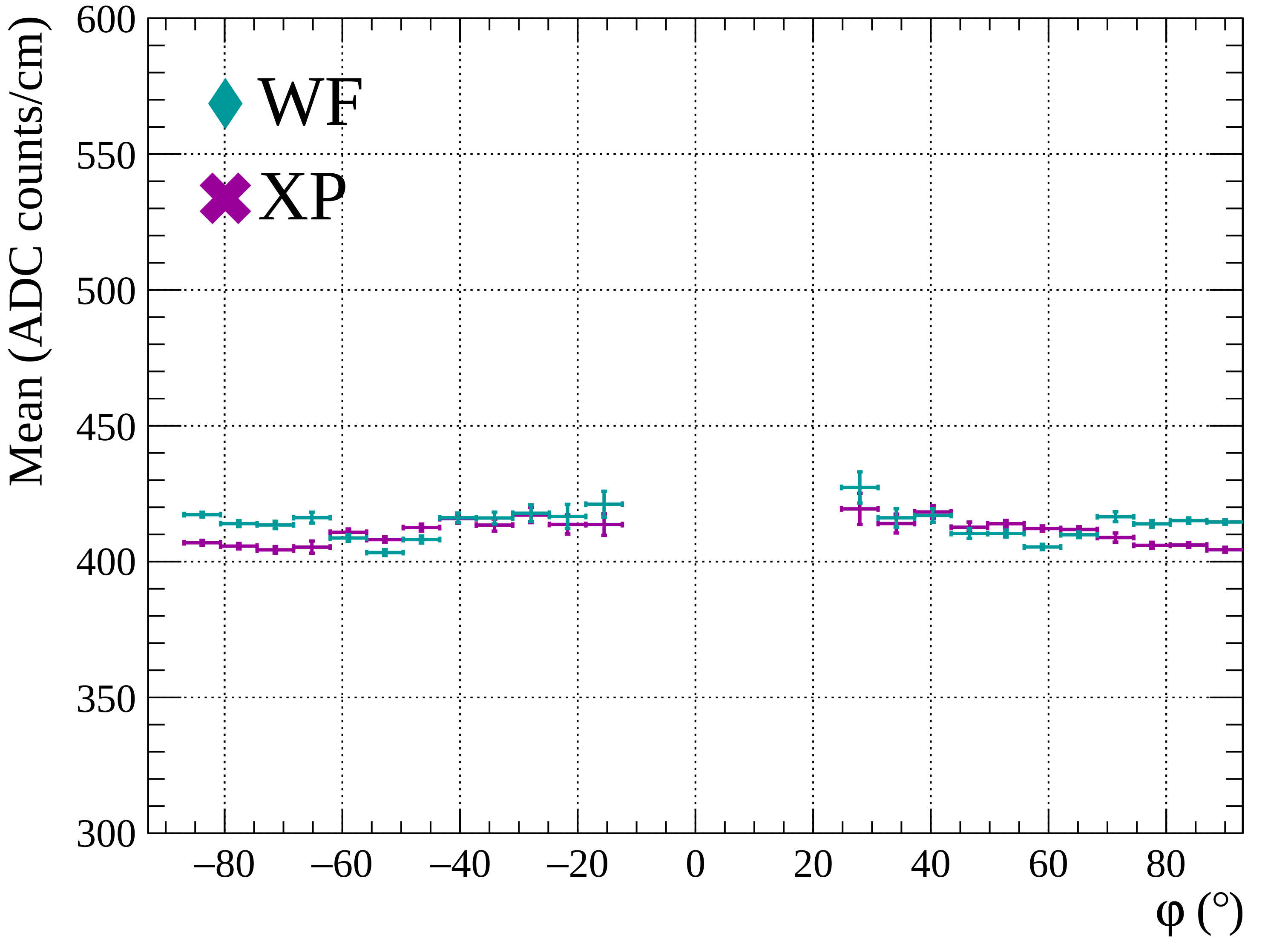


dE/dx resolution (%)
WF
XP
30
25
20
15
10
5
0
10
20
30
40
50
60
70
80
90
φ (°)

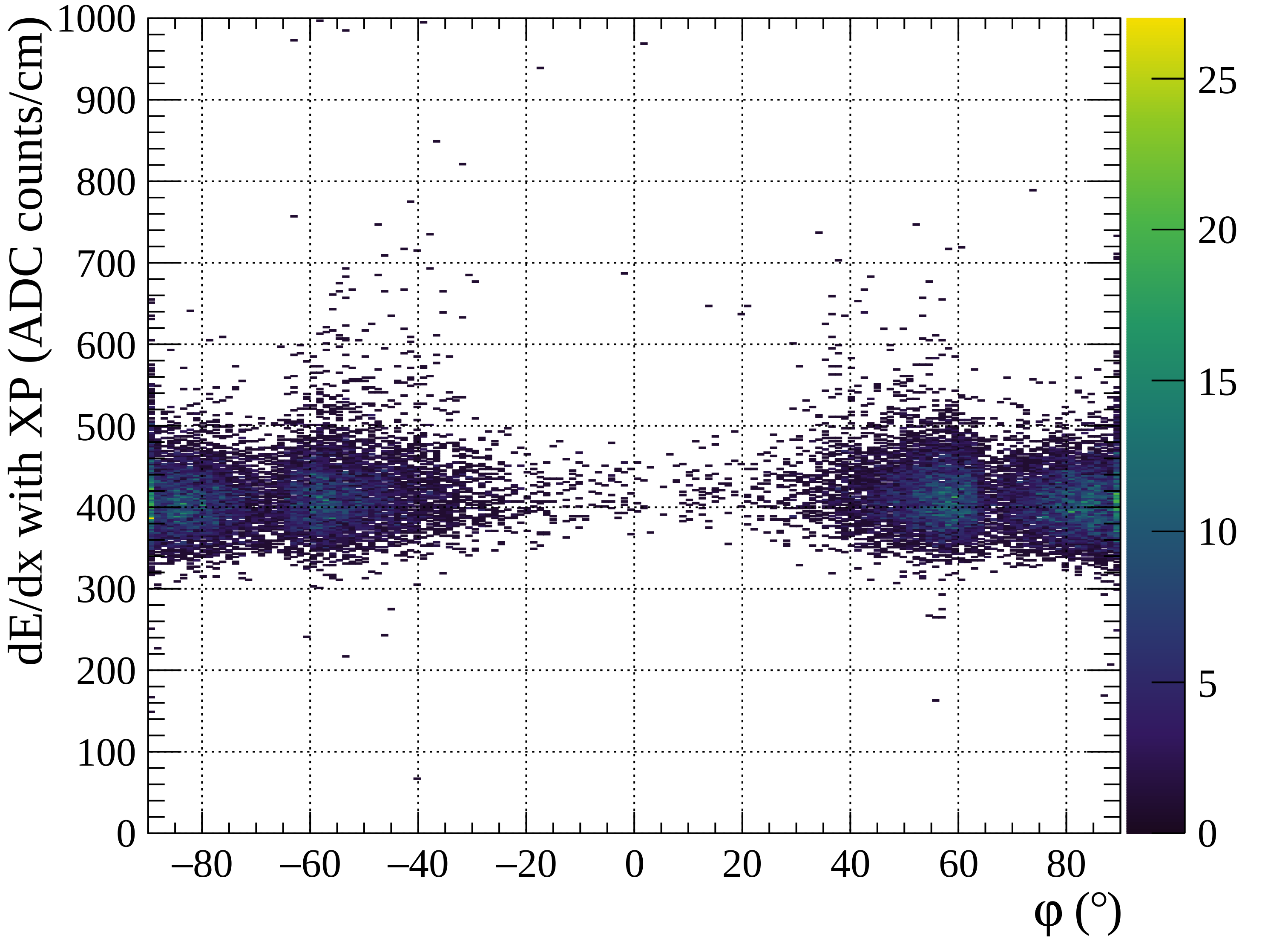

WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90
|φ| (°)



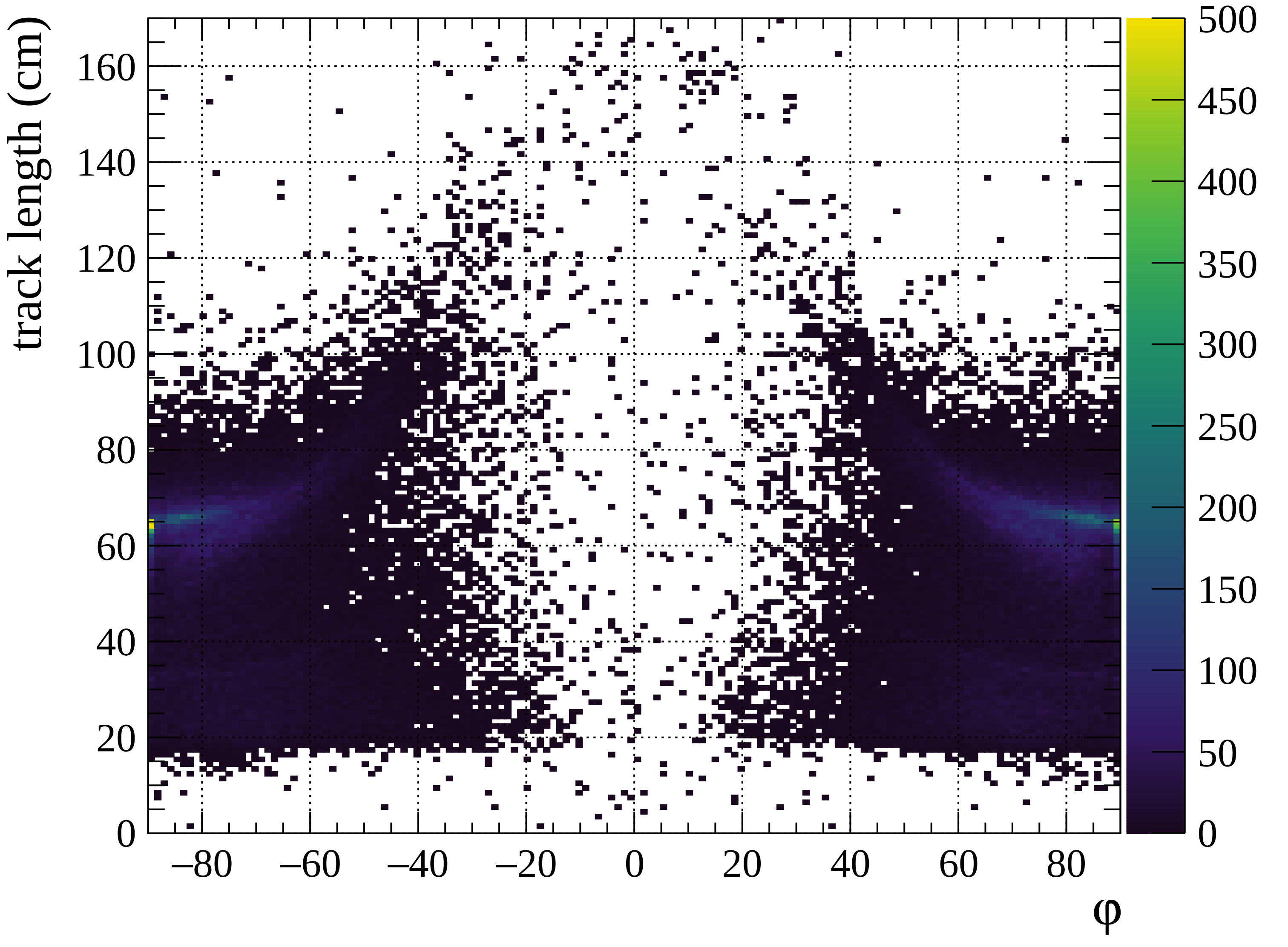
dE/dx resolution (%)
WF
XP
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80
φ (°)

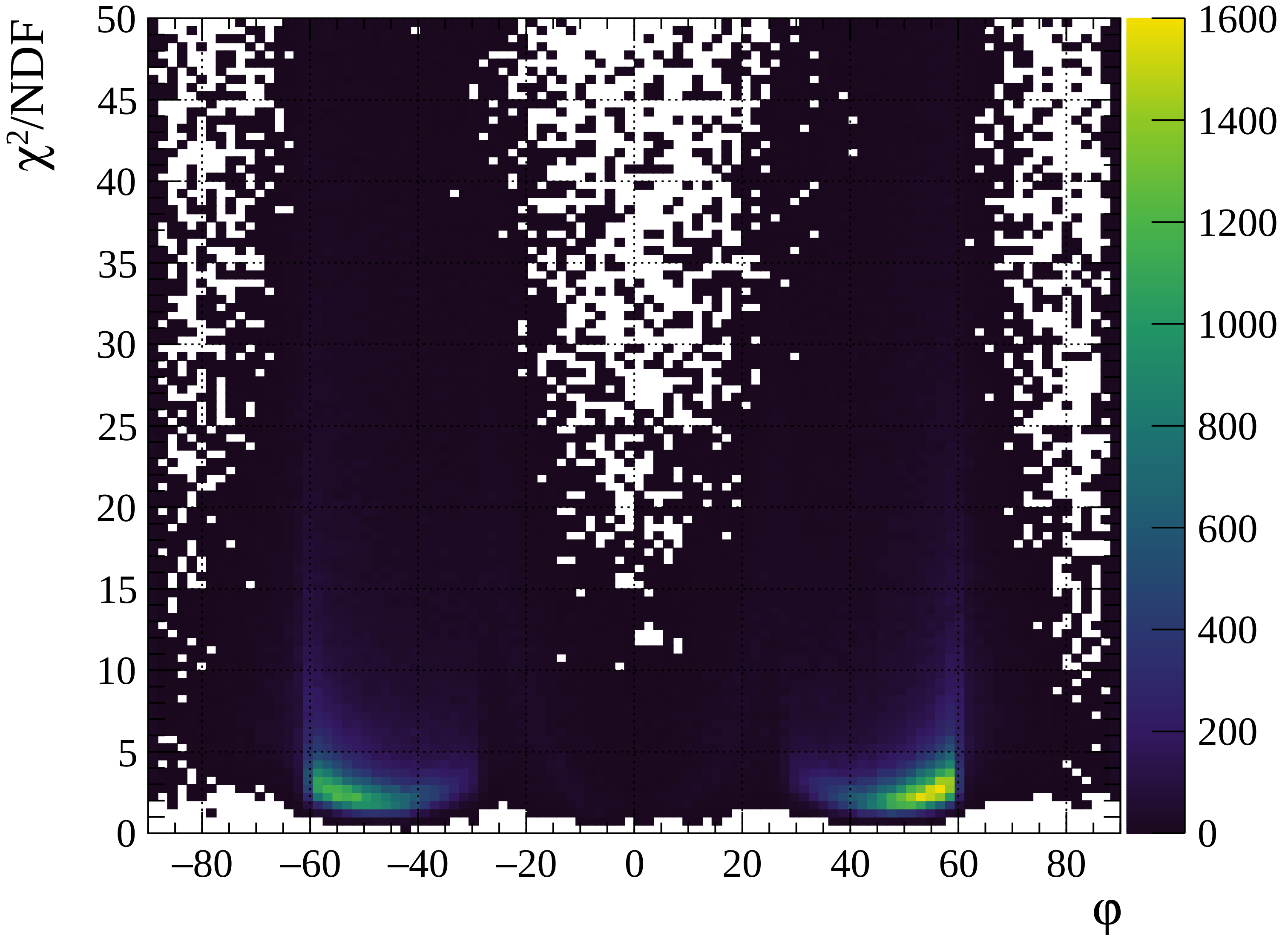


WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80
φ (°)

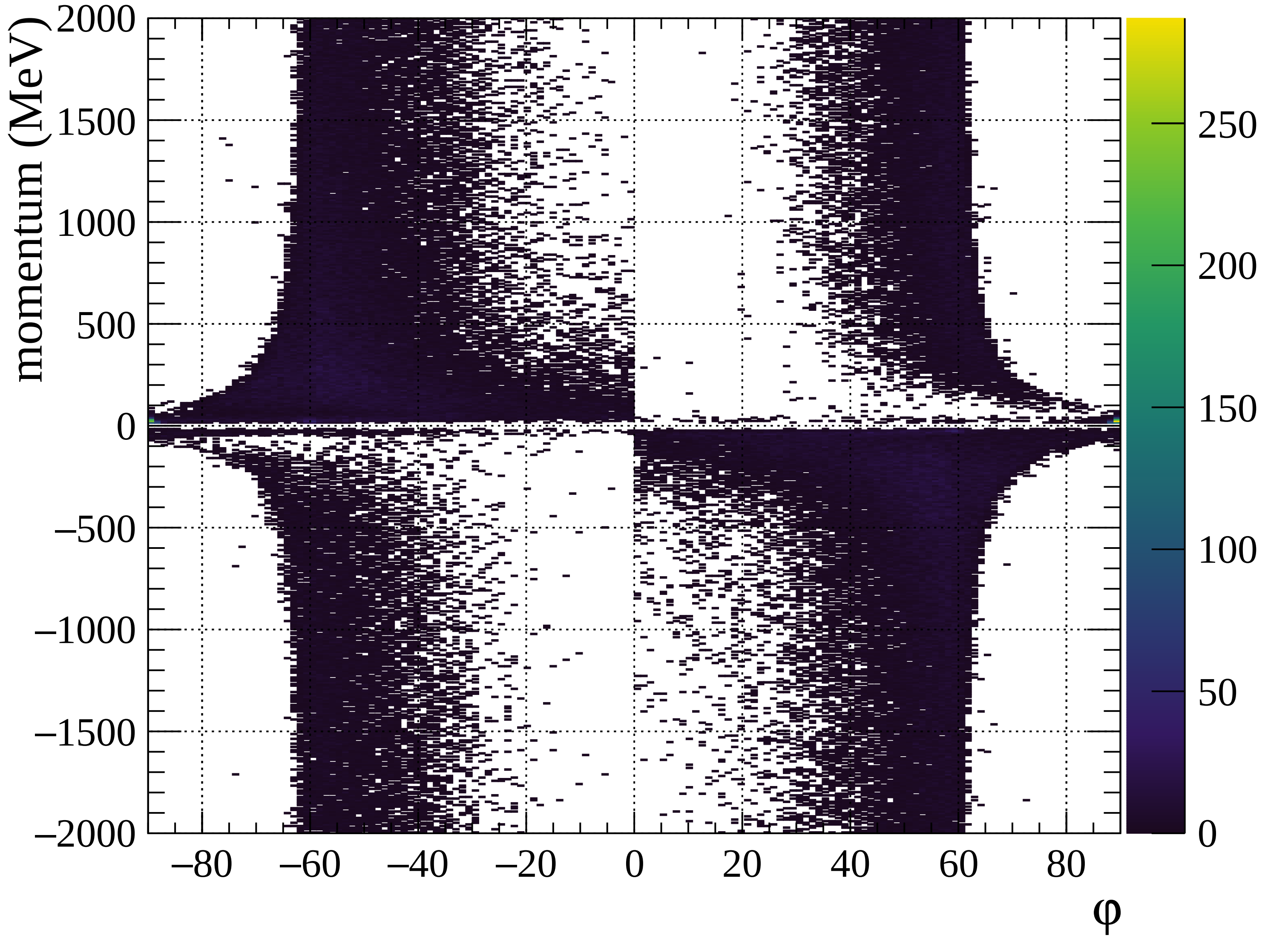


dE/dx with XP (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
−80
−60
−40
−20
0
20
40
60
80
φ (°)
0
5
10
15
20
25

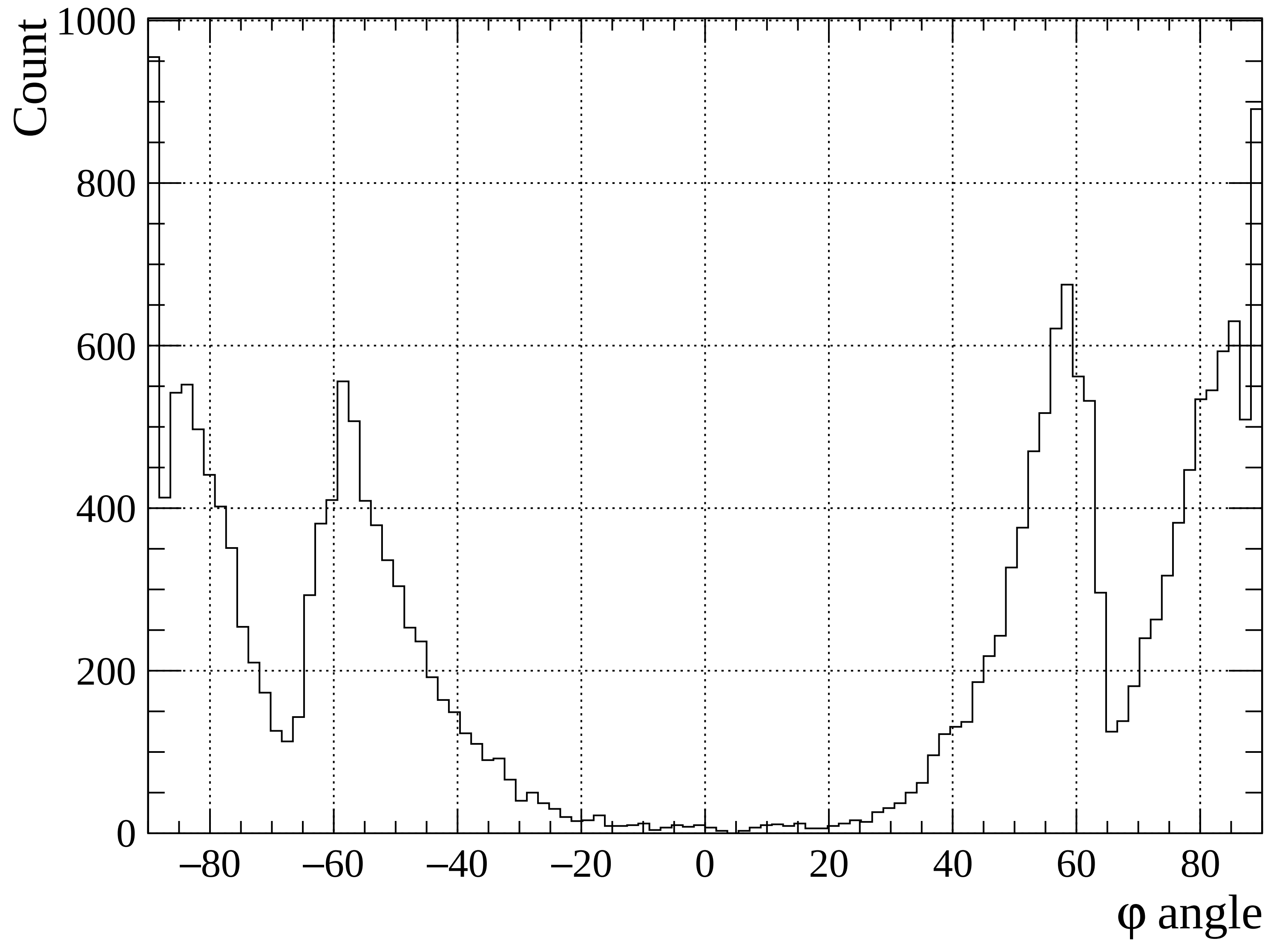

track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
φ
500
450
400
350
300
250
200
150
100
50
0



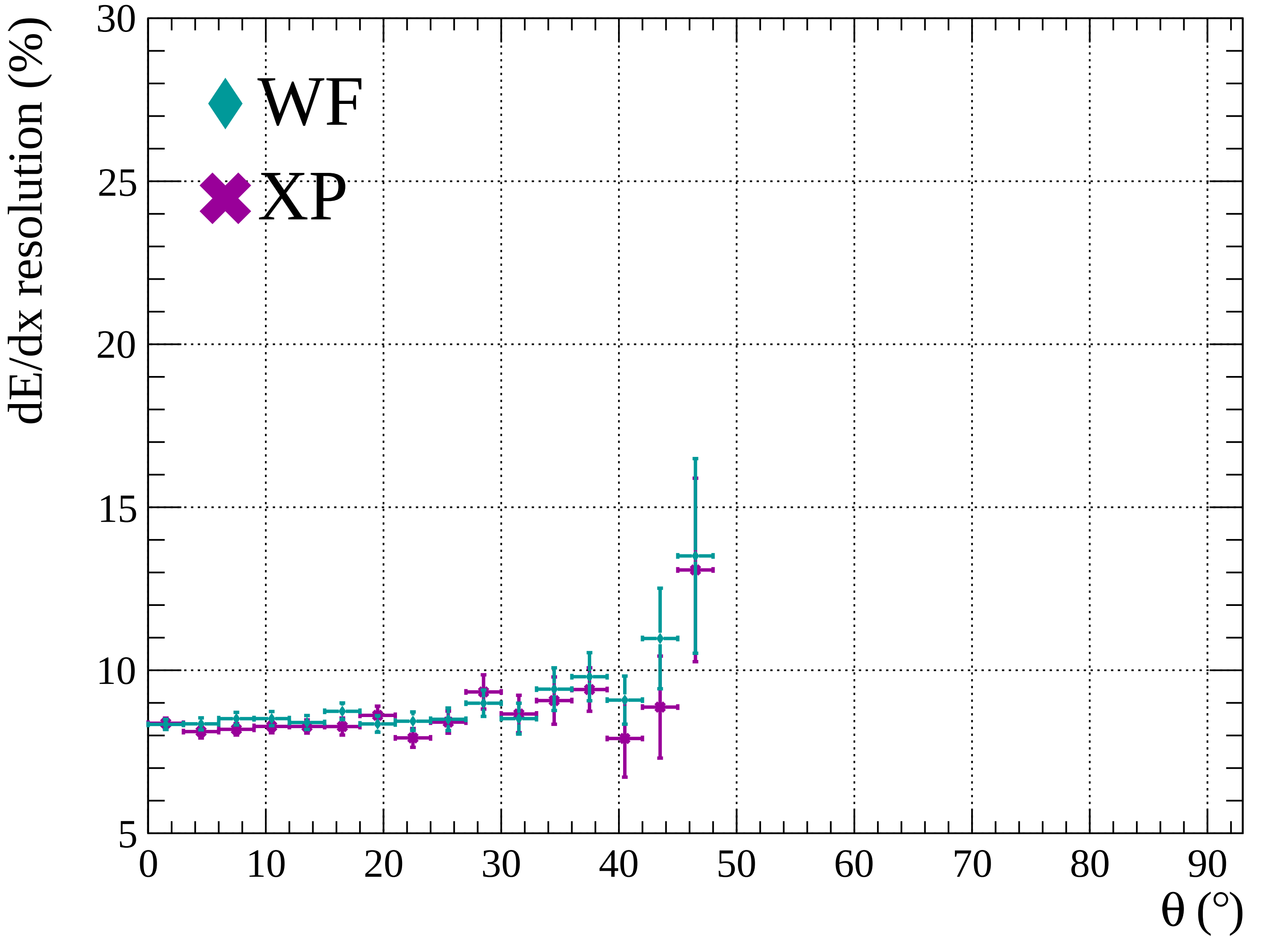
χ^2/NDF
φ



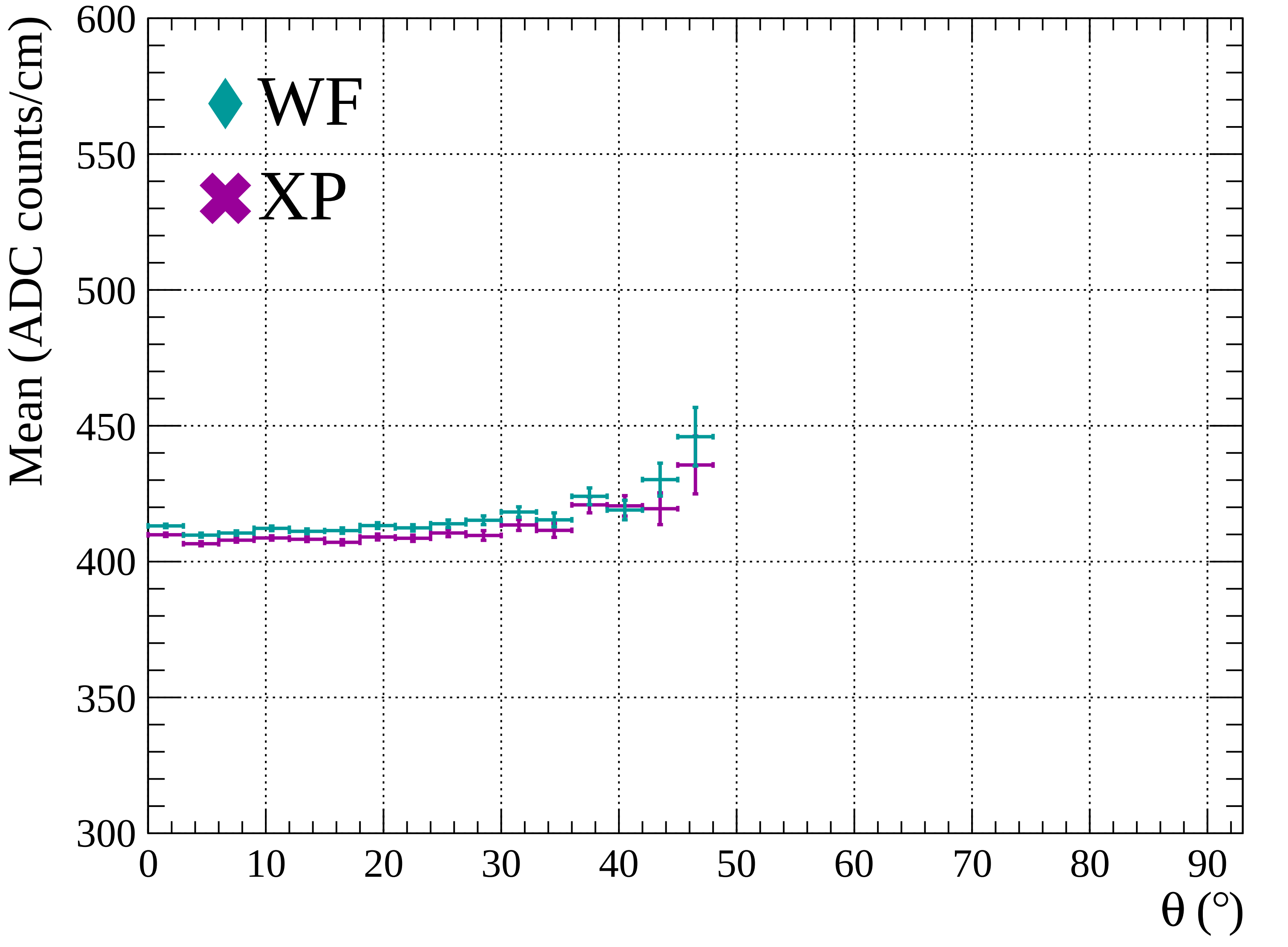
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
250
200
150
100
50
0



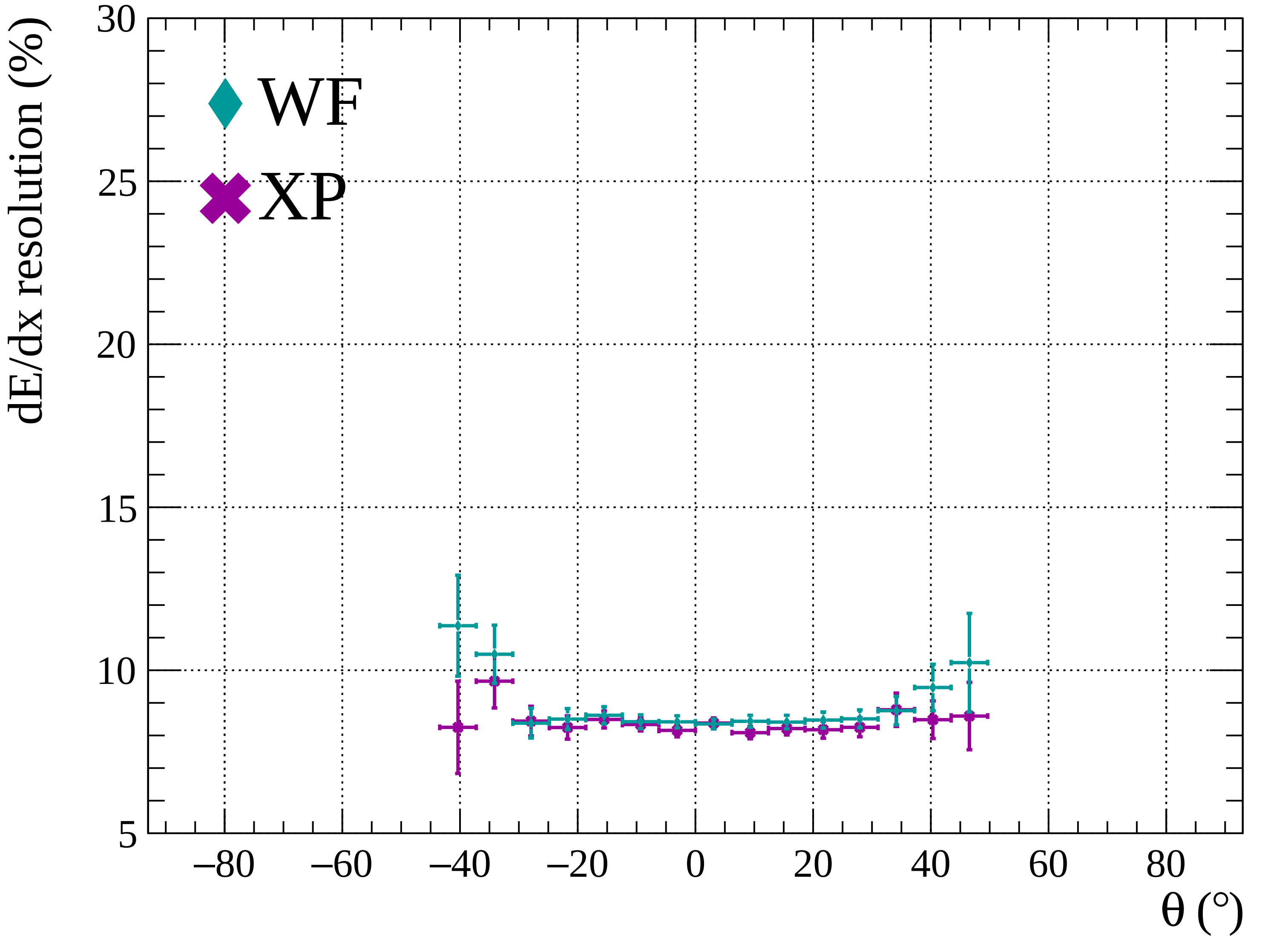
Count
1000
800
600
400
200
0
−80
−60
−40
−20
0
20
40
60
80
φ angle



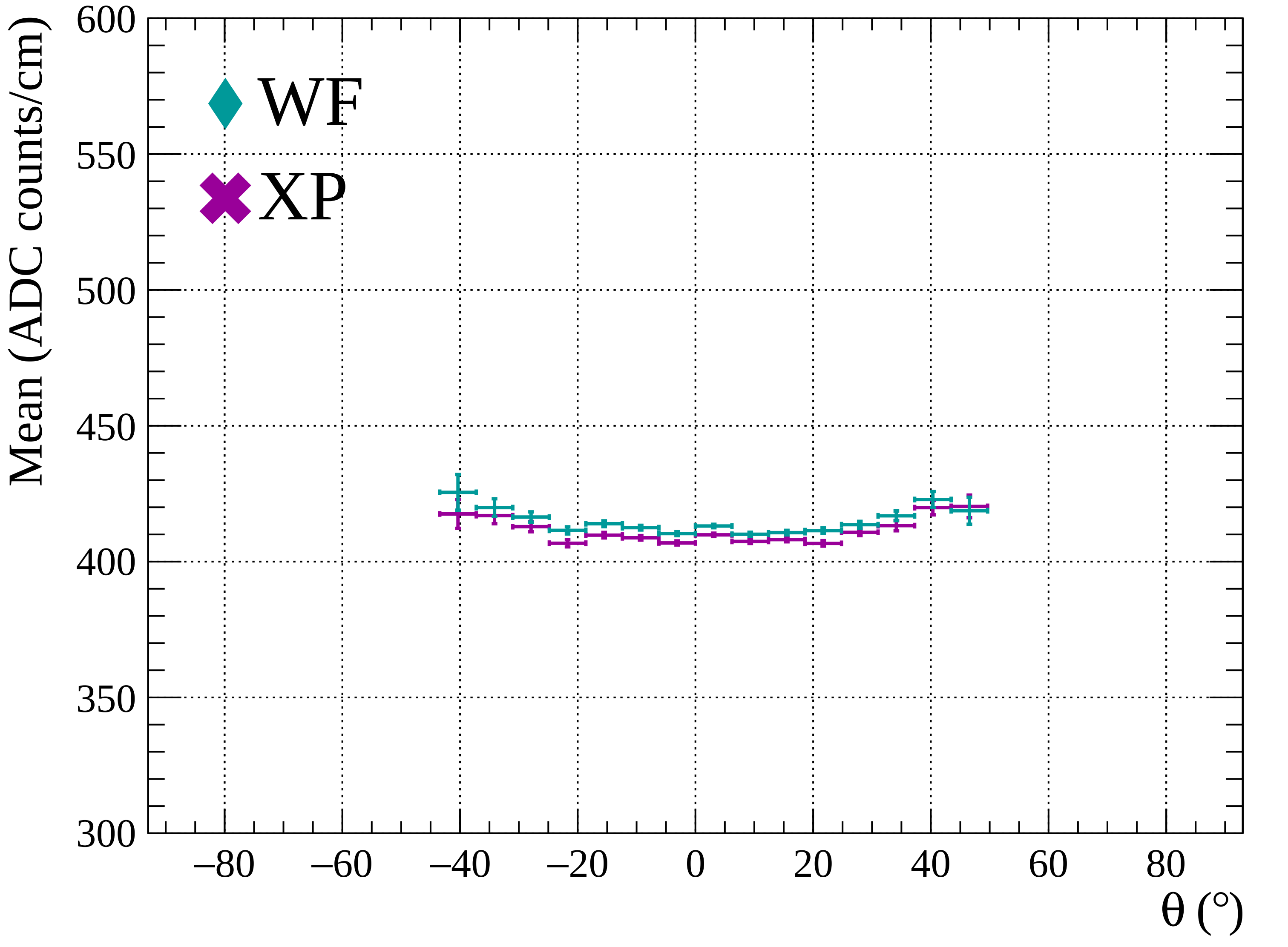
dE/dx resolution (%)
θ (°)
WF
XP
30
25
20
15
10
5
0
10
20
30
40
50
60
70
80
90



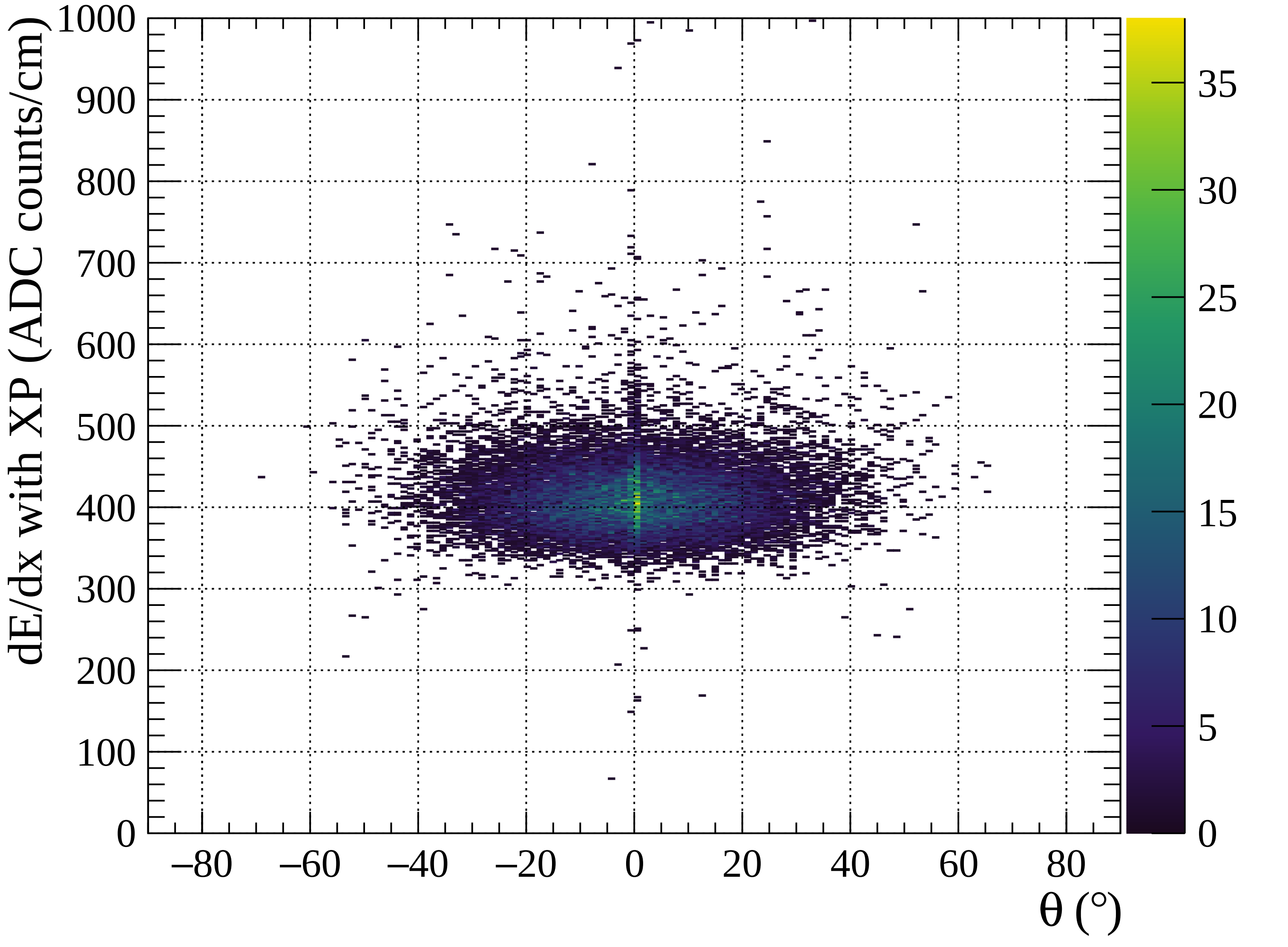
Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
0
10
20
30
40
50
60
70
80
90
θ (°)



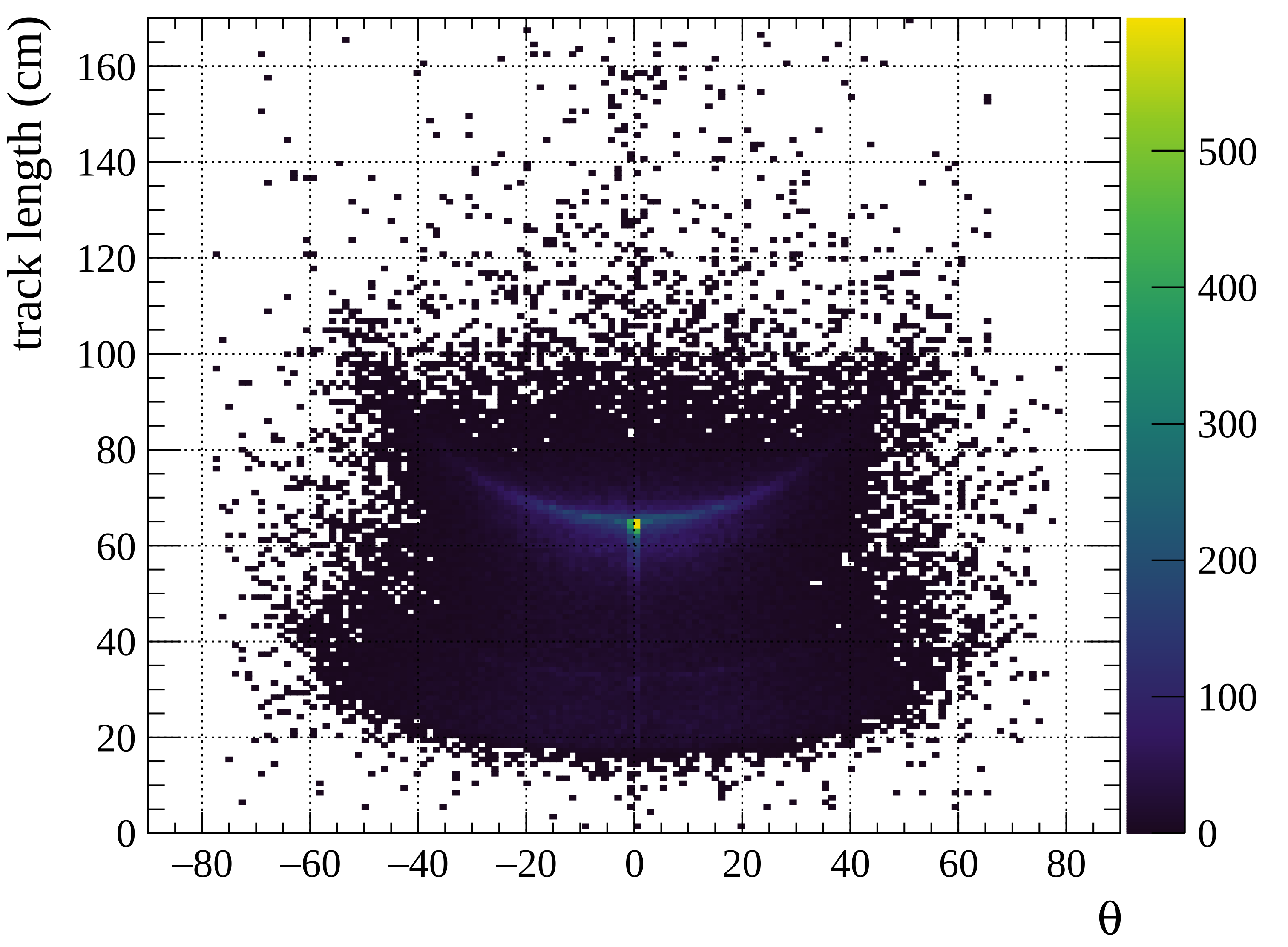
dE/dx resolution (%)
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80
θ (°)
WF
XP



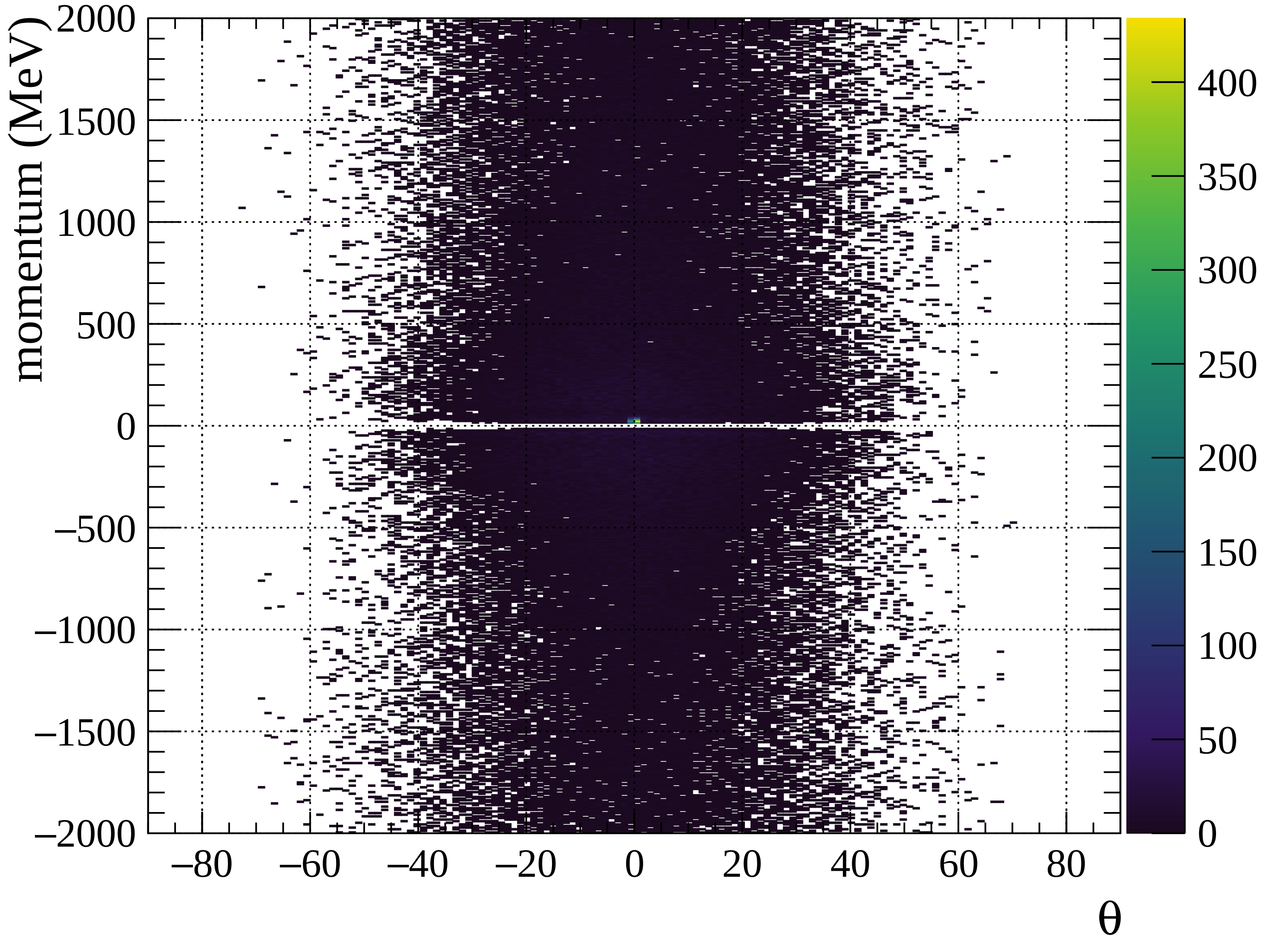
Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
−80
−60
−40
−20
0
20
40
60
80
θ (°)



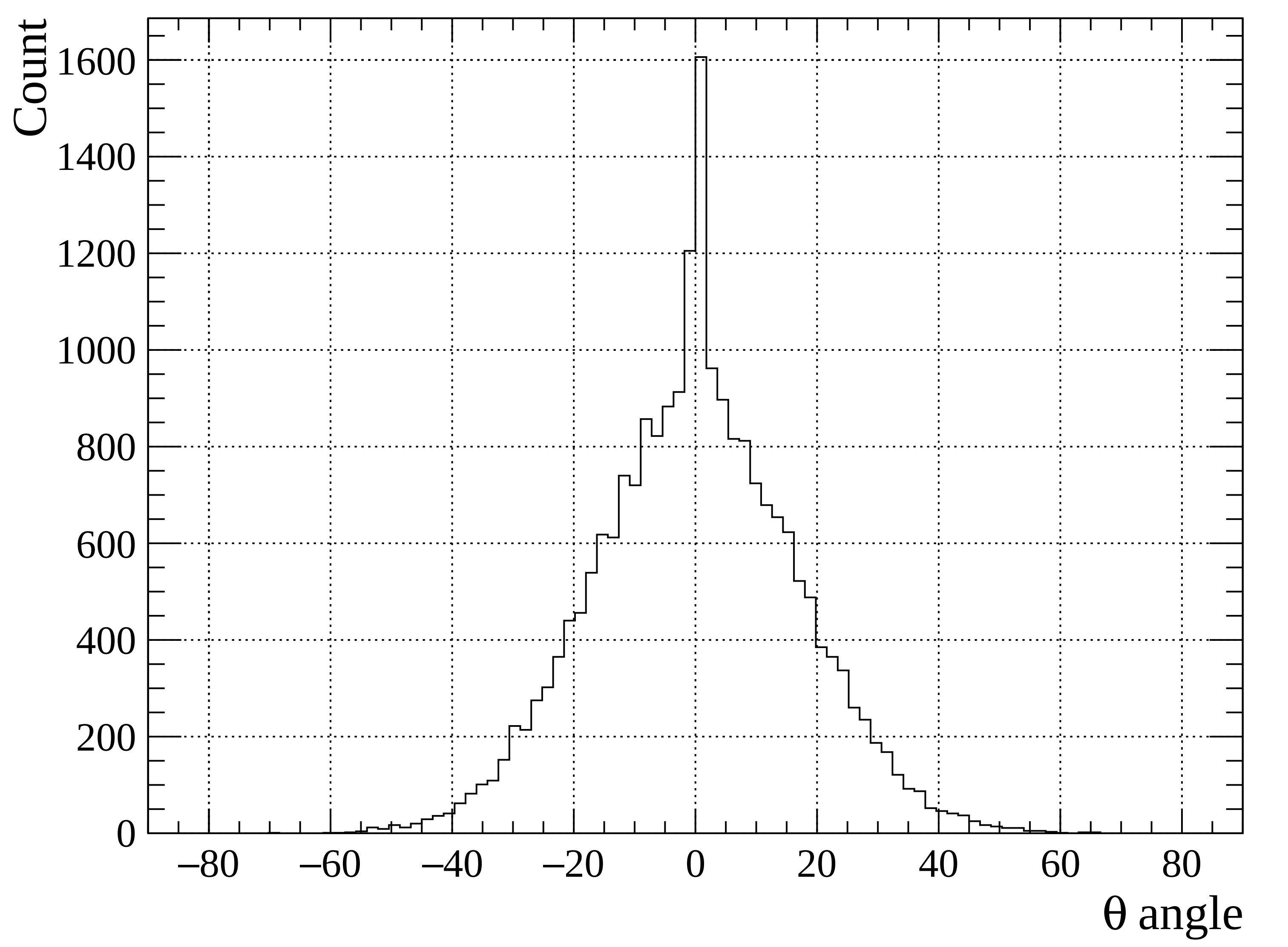
dE/dx with XP (ADC counts/cm)
θ (°)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
35
30
25
20
15
10
5
0

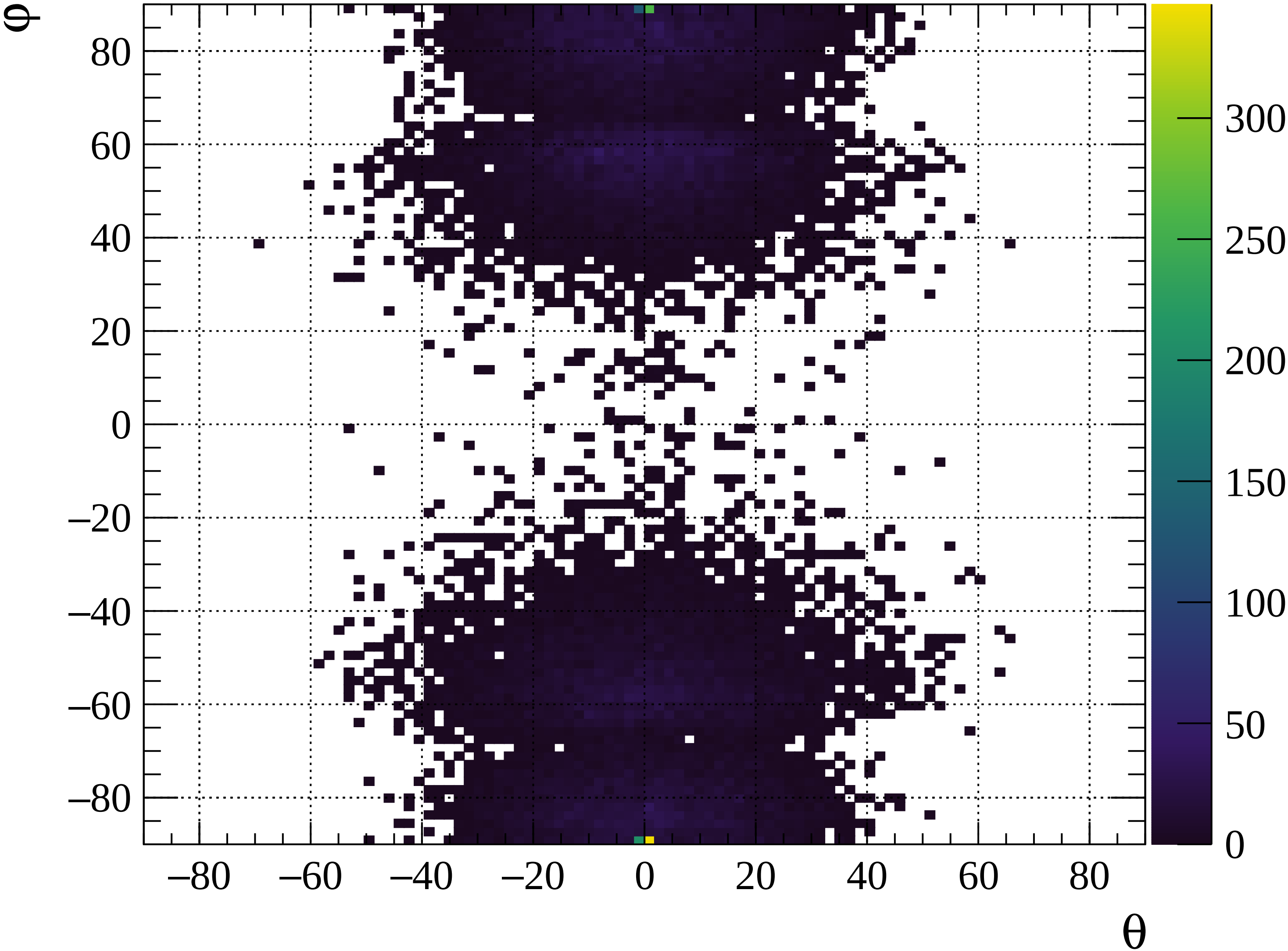


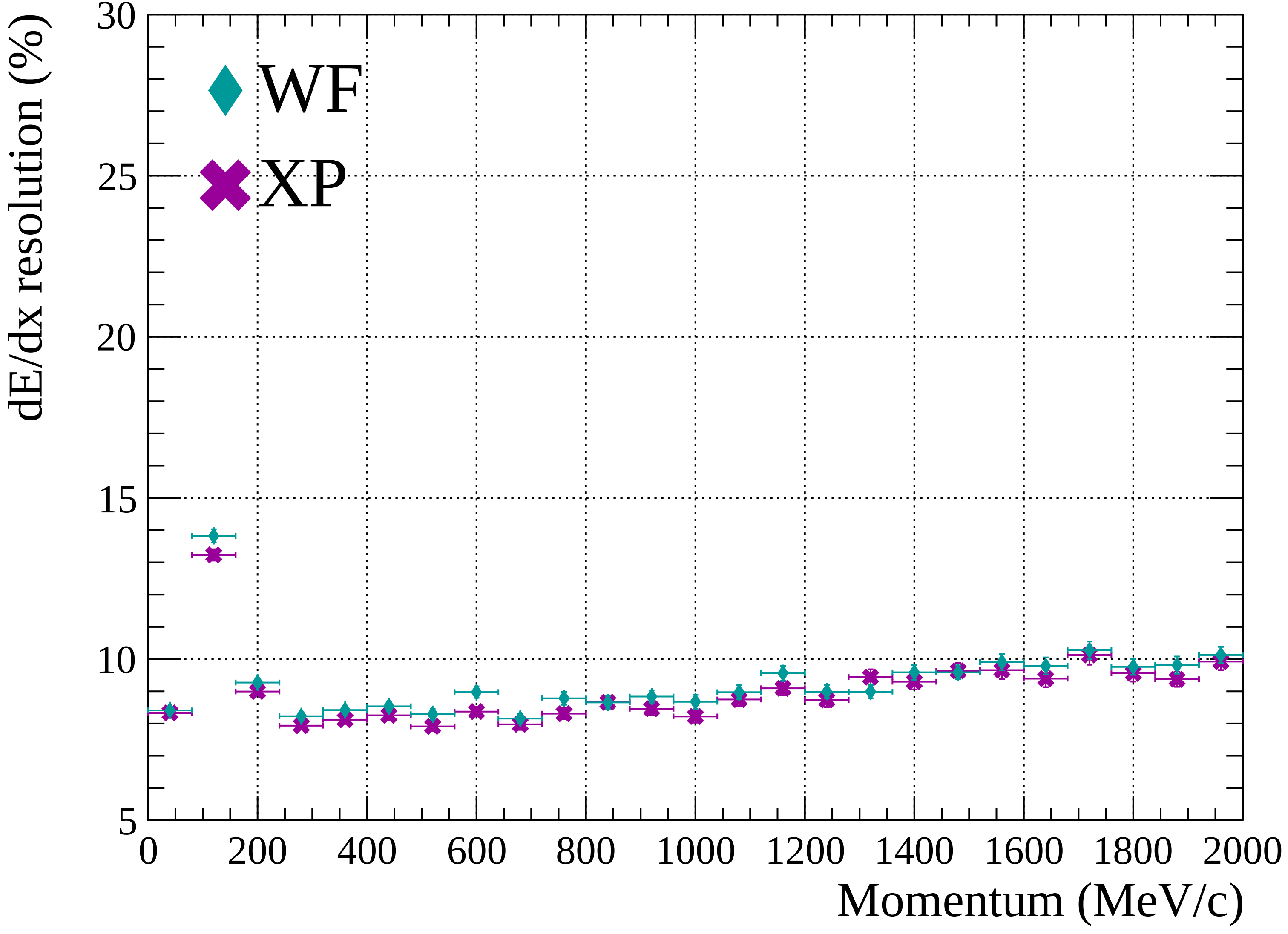
track length (cm)
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
θ
0
100
200
300
400
500

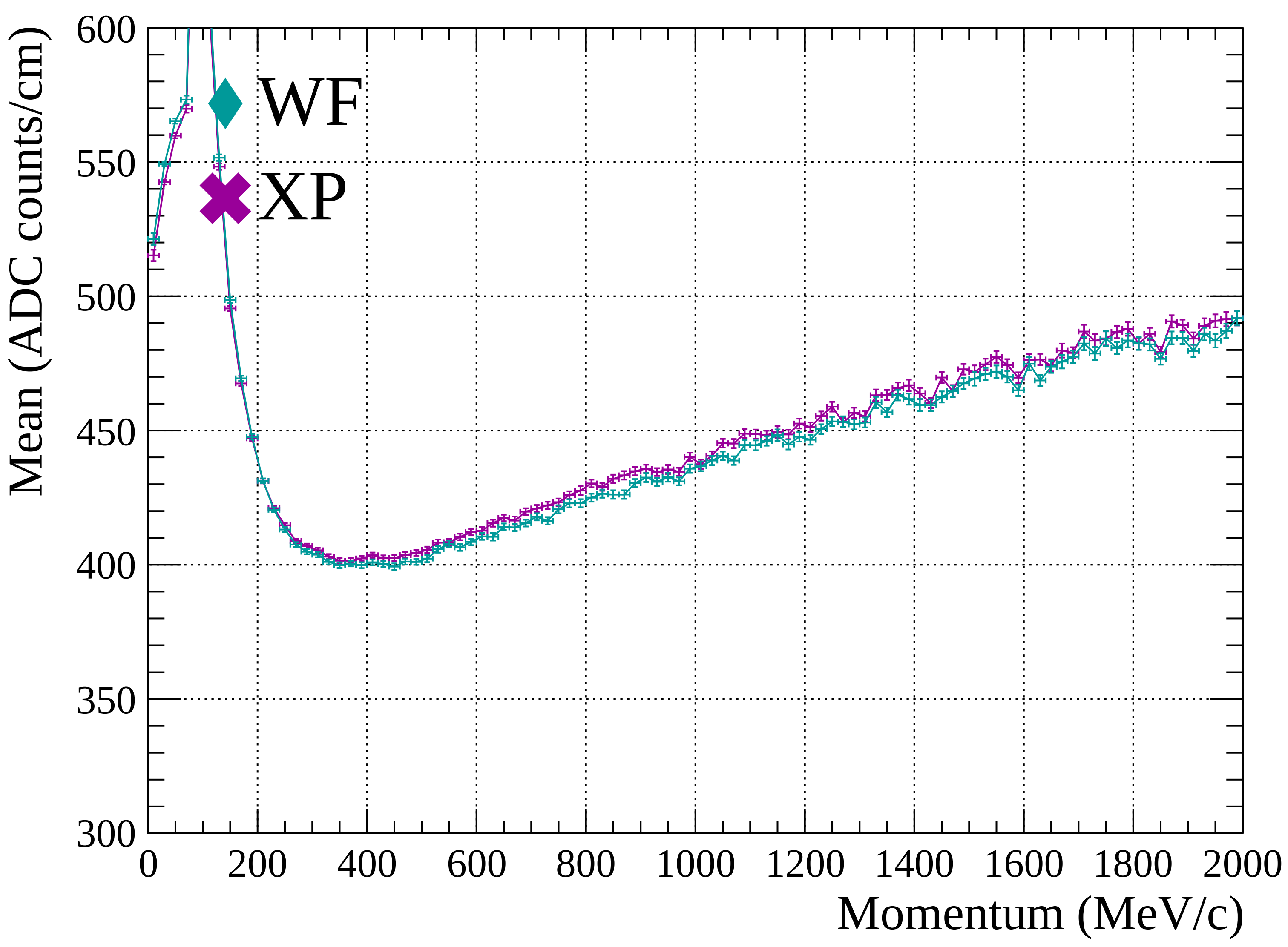


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
400
350
300
250
200
150
100
50
0

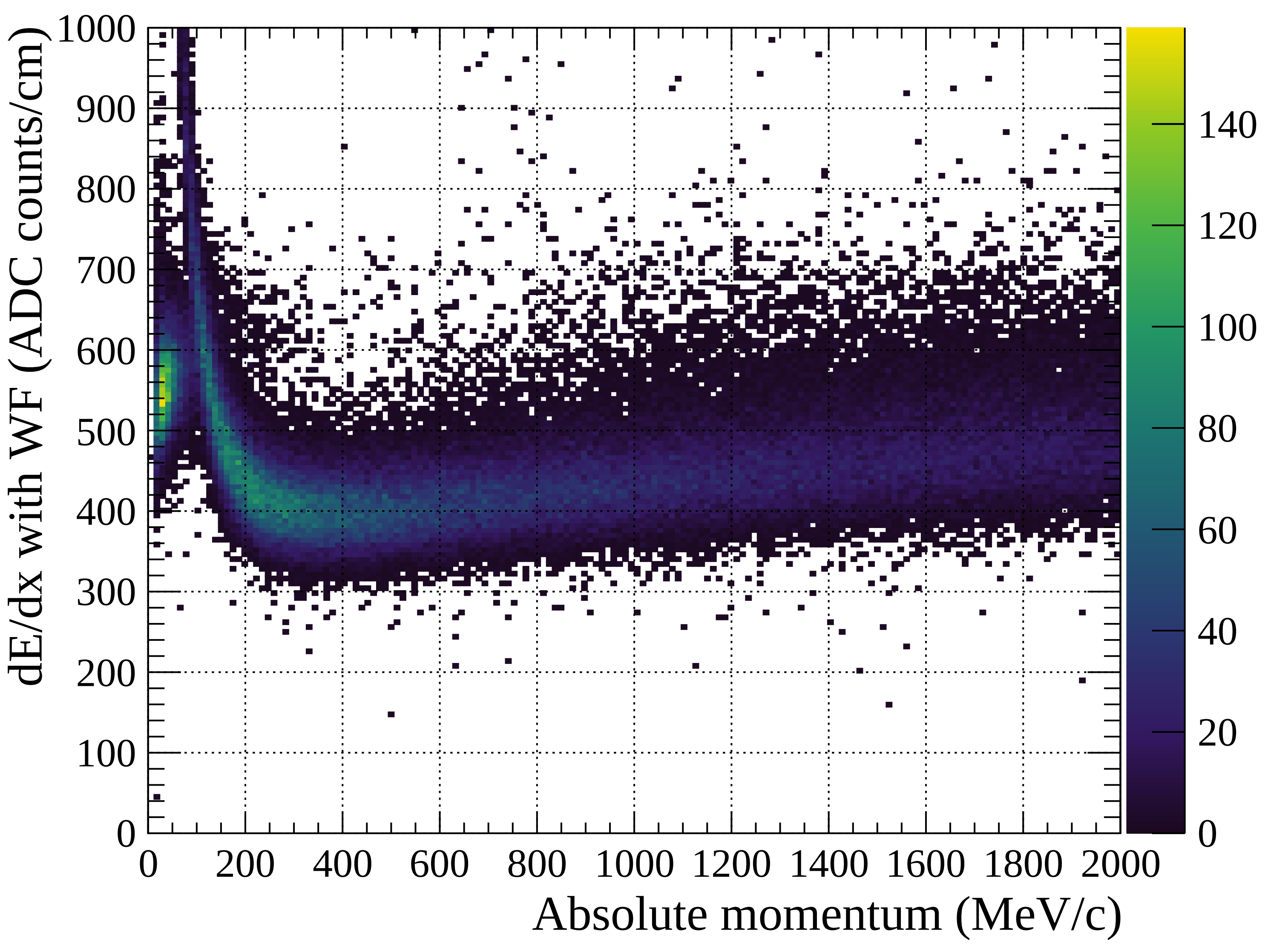

Count
1600
1400
1200
1000
800
600
400
200
0
−80
−60
−40
−20
0
20
40
60
80
θ angle

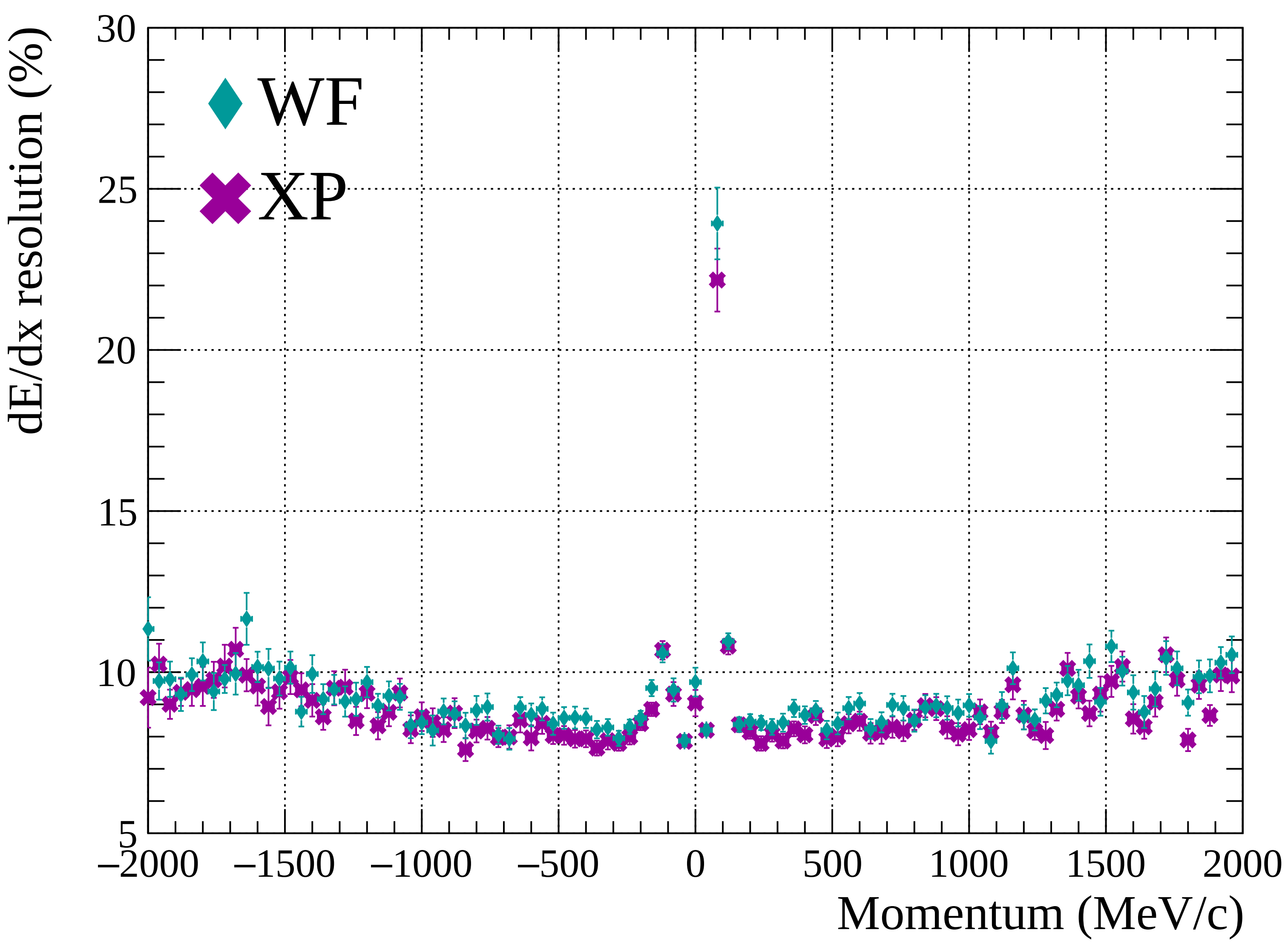

φ
θ
80
60
40
20
0
−20
−40
−60
−80
300
250
200
150
100
50


dE/dx resolution (%)
Momentum (MeV/c)
WF
XP
30
25
20
15
10
5
0
200
400
600
800
1000
1200
1400
1600
1800
2000

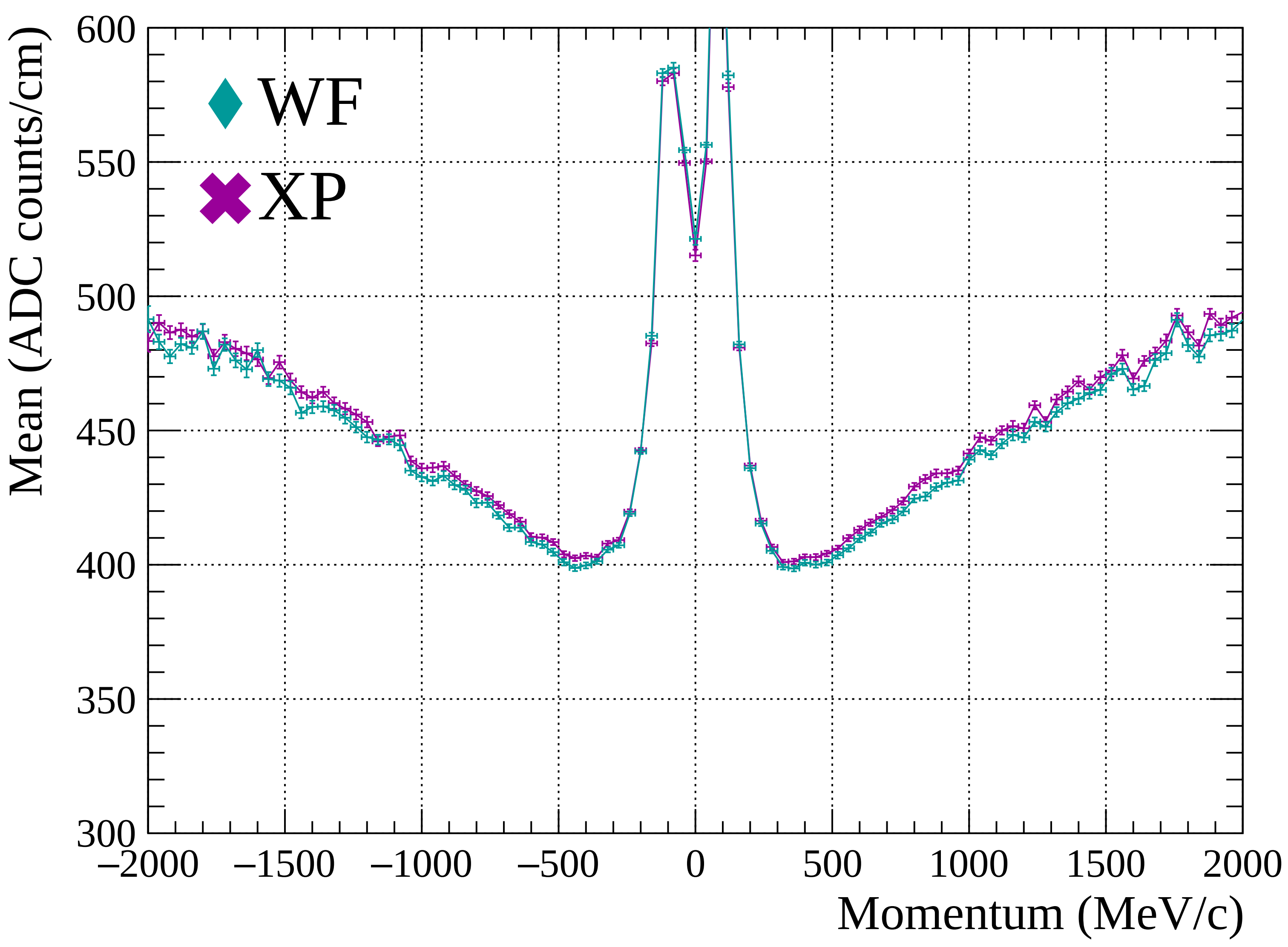


WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800
2000

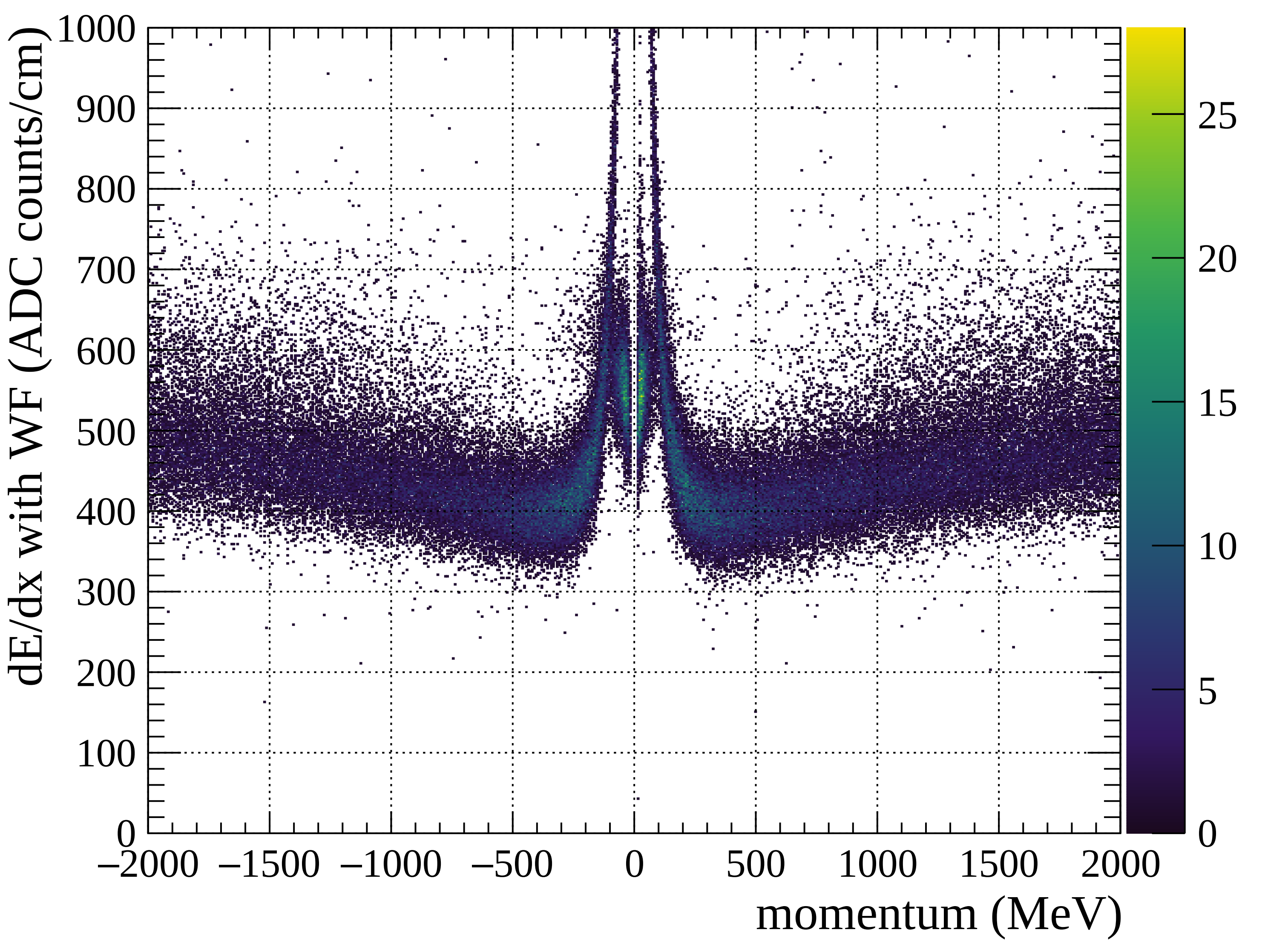

dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
1400
1600
1800
2000
Absolute momentum (MeV/c)
140
120
100
80
60
40
20
0



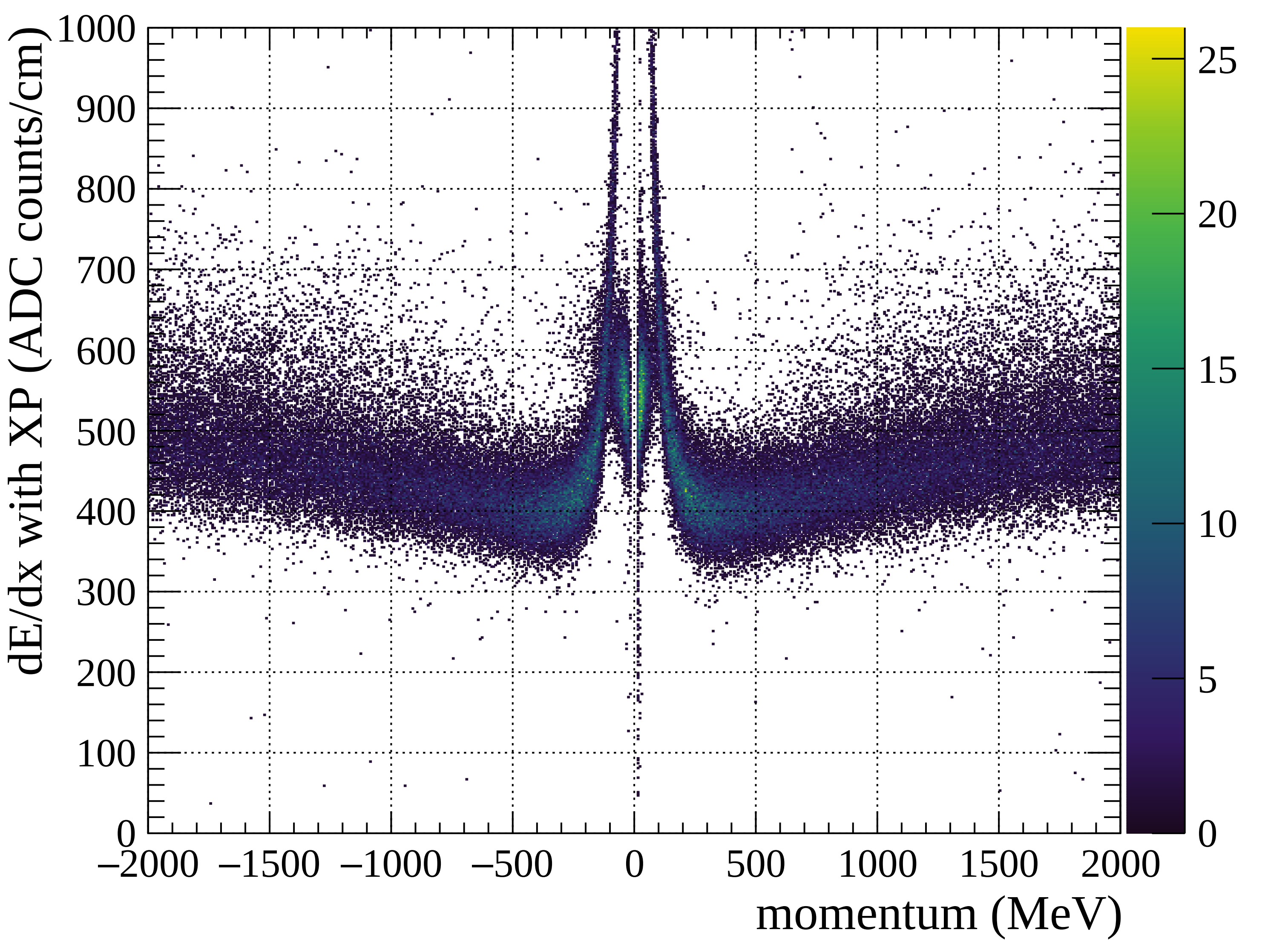
WF
XP
dE/dx resolution (%)
Momentum (MeV/c)
30
25
20
15
10
5
−2000
−1500
−1000
−500
0
500
1000
1500
2000

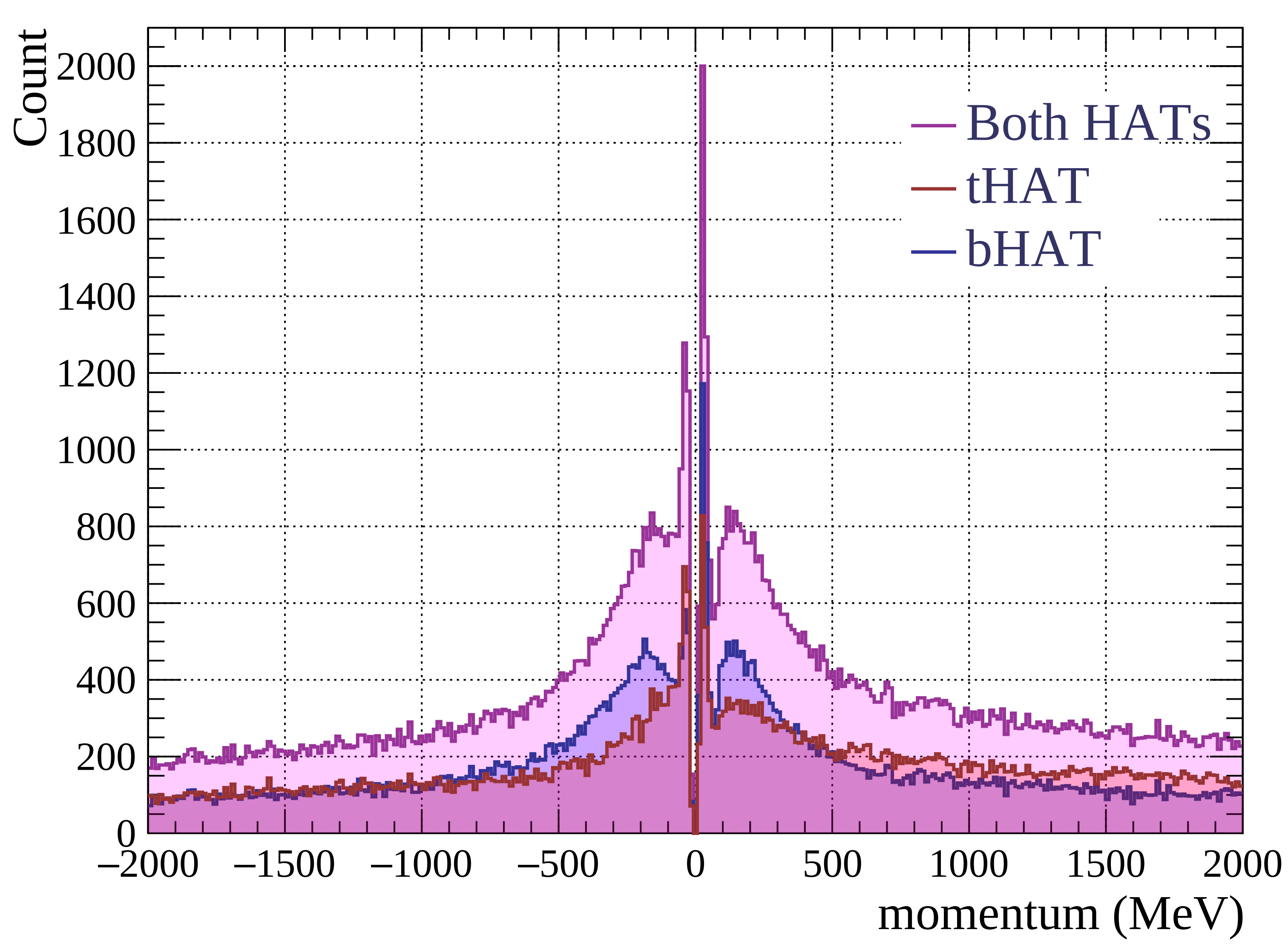


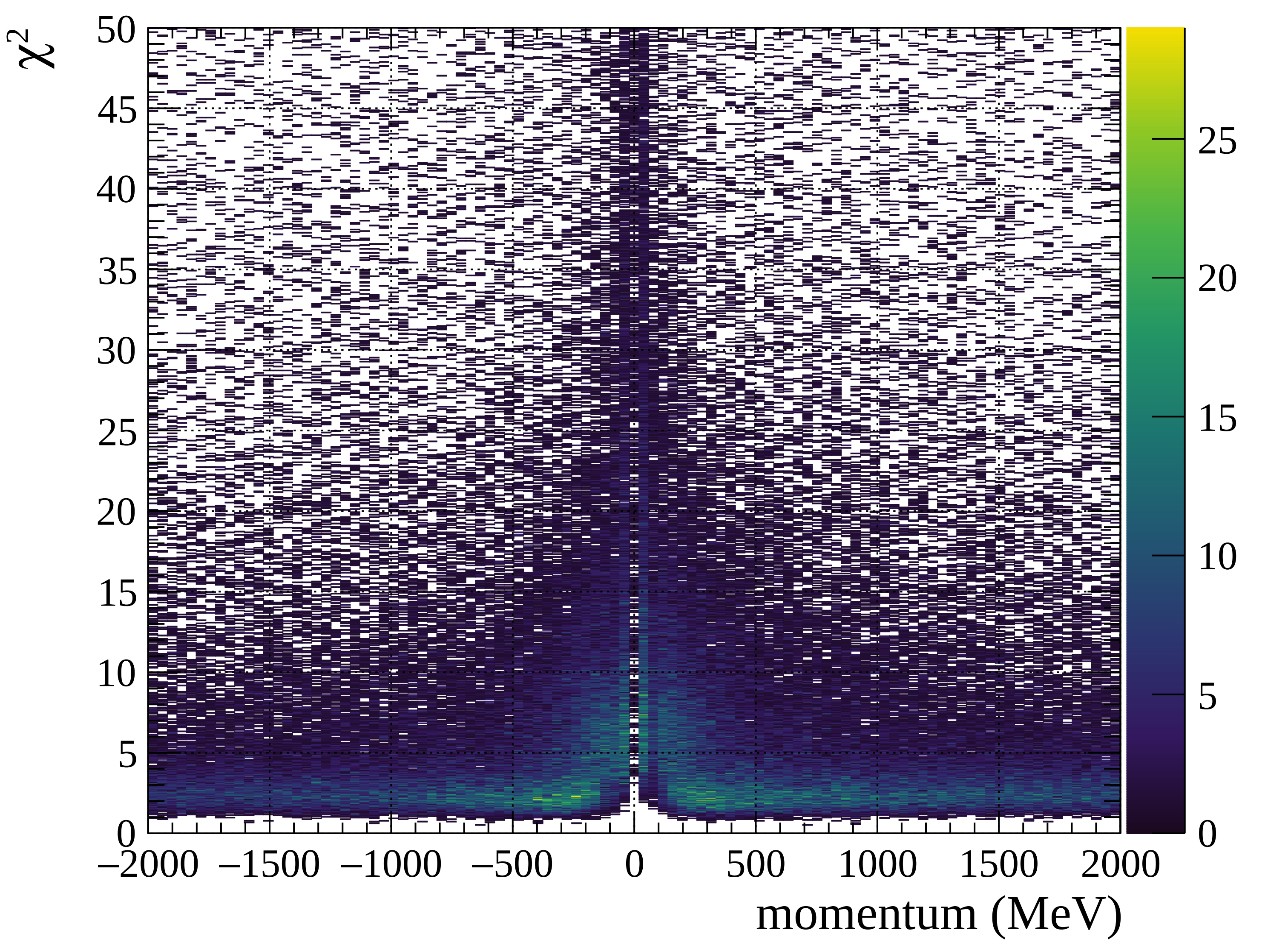
WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
−2000
−1500
−1000
−500
0
500
1000
1500
2000



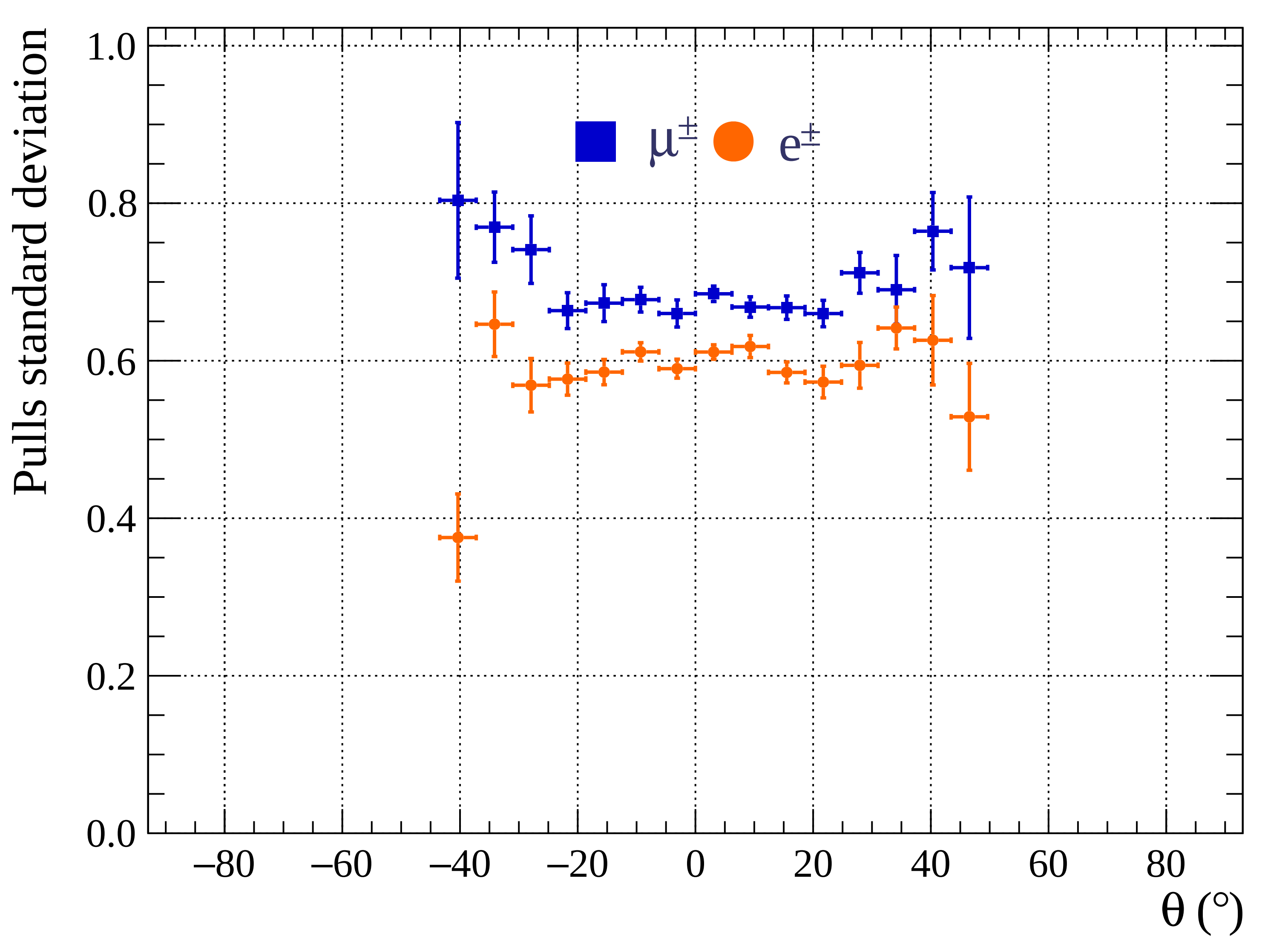
dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
25
20
15
10
5
0


dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
25
20
15
10
5
0

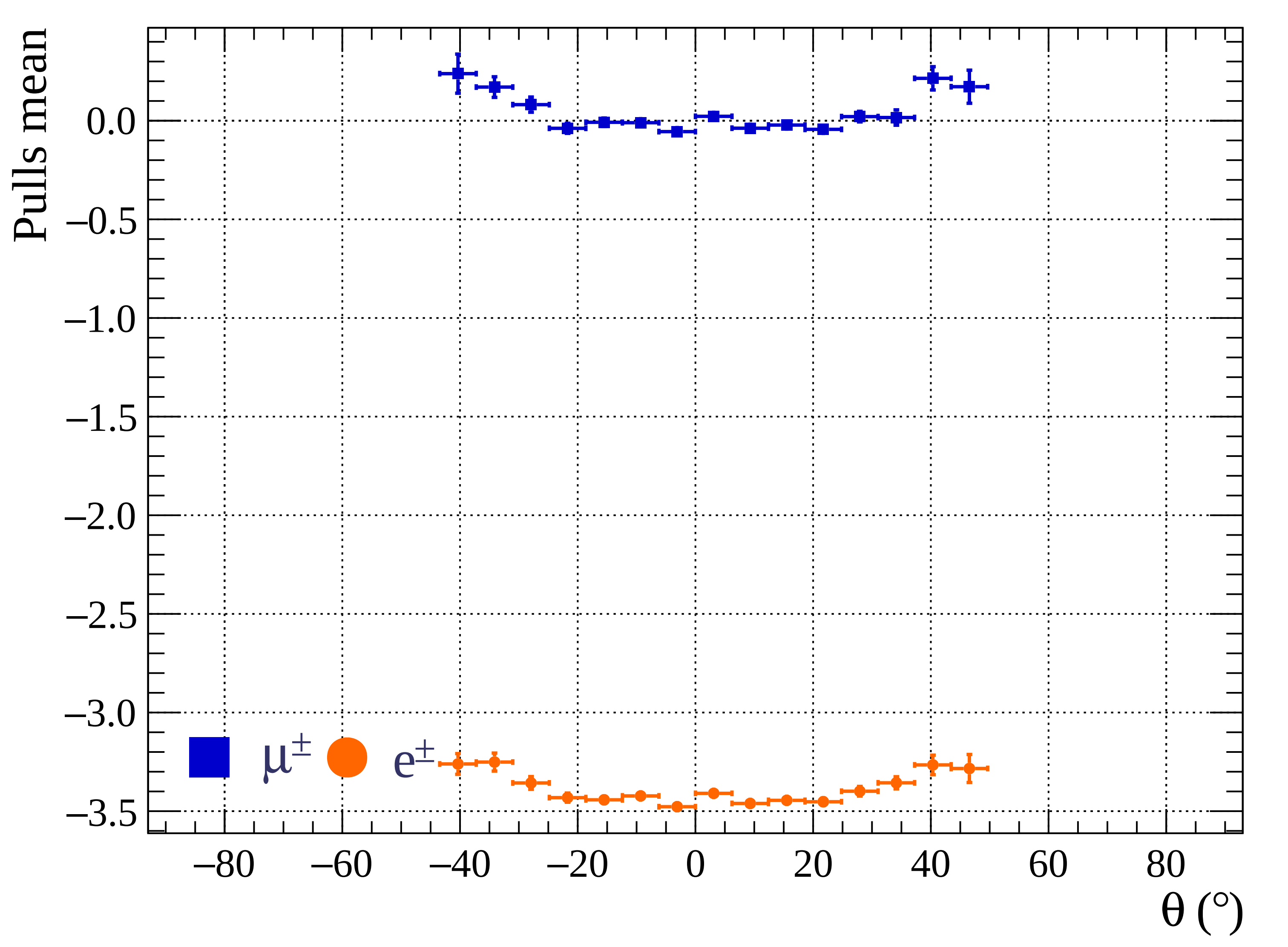

Count
2000
1800
1600
1400
1200
1000
800
600
400
200
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
Both HATs
tHAT
bHAT



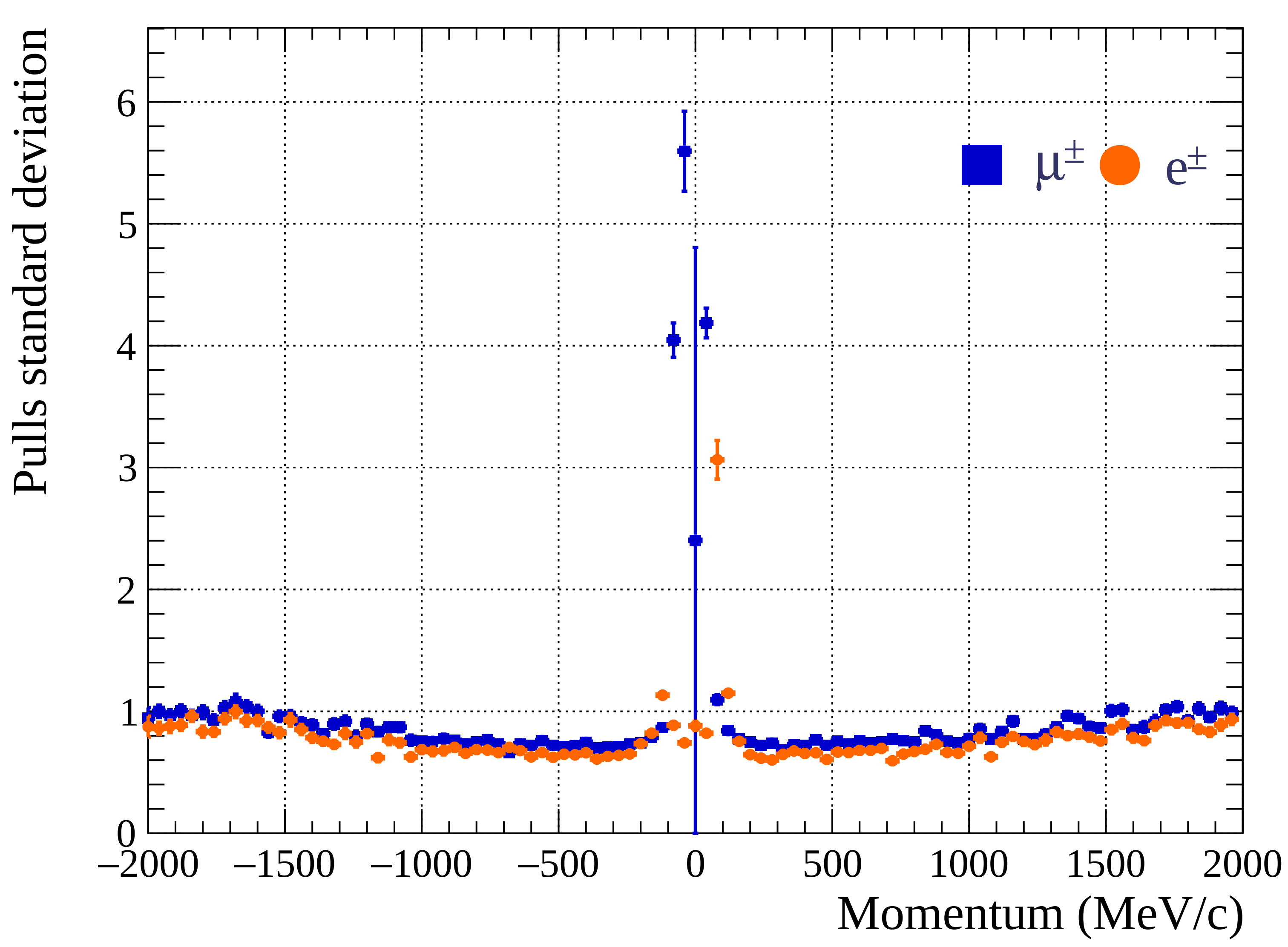
χ^2
50
45
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
25
20
15
10
5
0



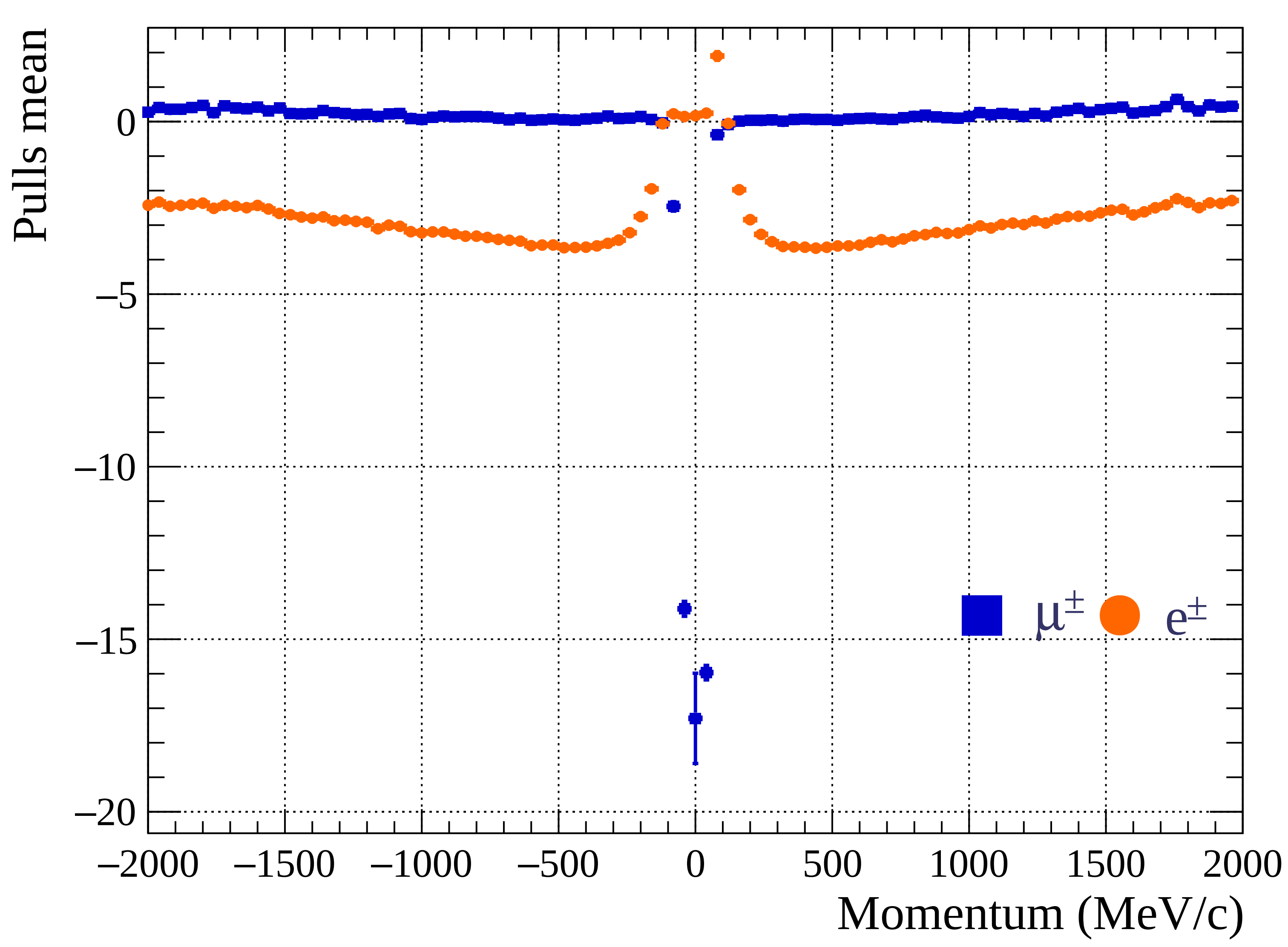
Pulls standard deviation
1.0
0.8
0.6
0.4
0.2
0.0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±

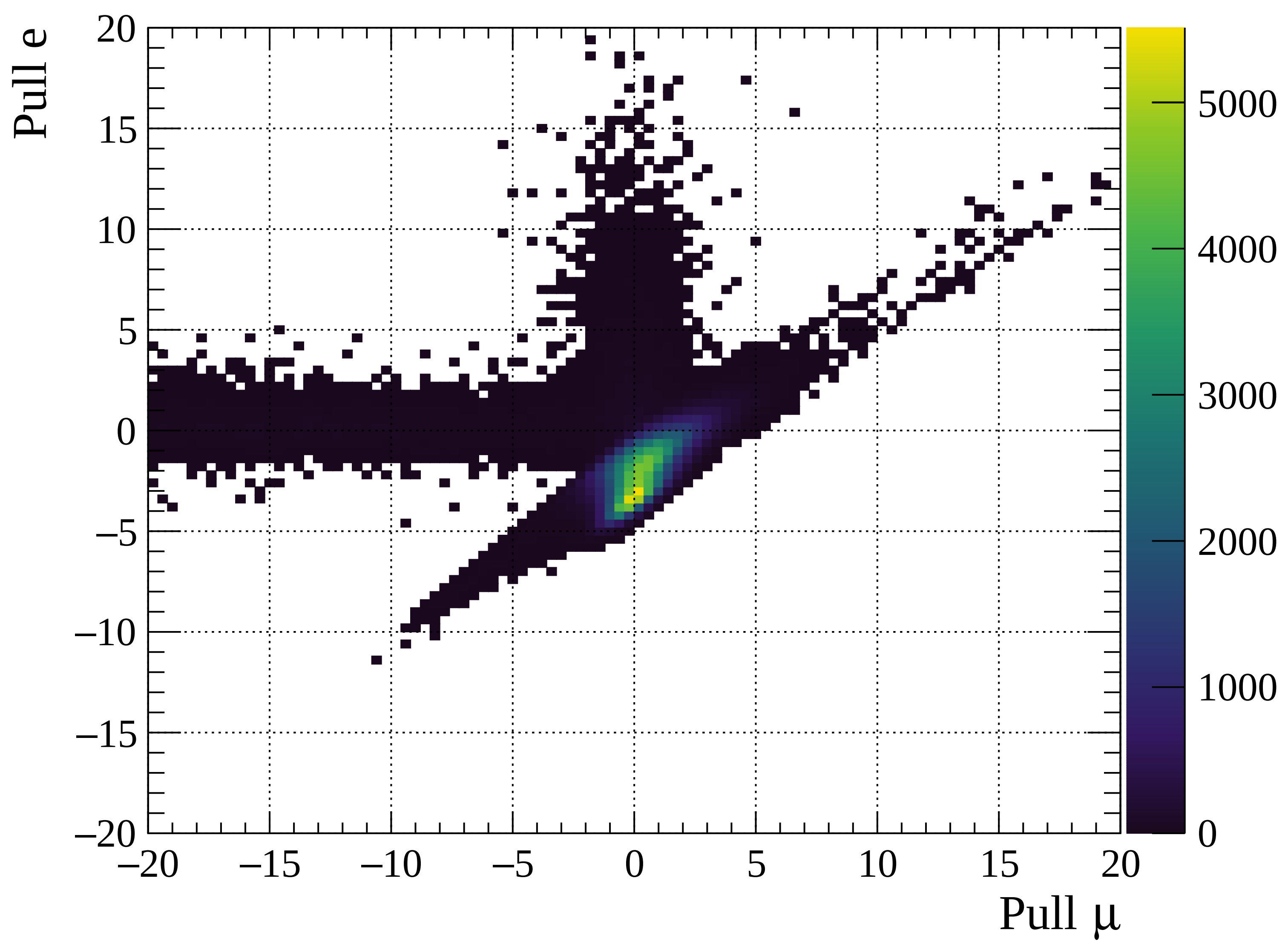


Pulls mean
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
−3.5
$\mu^{\pm}$
$e^{\pm}$
−80
−60
−40
−20
0
20
40
60
80
θ (°)

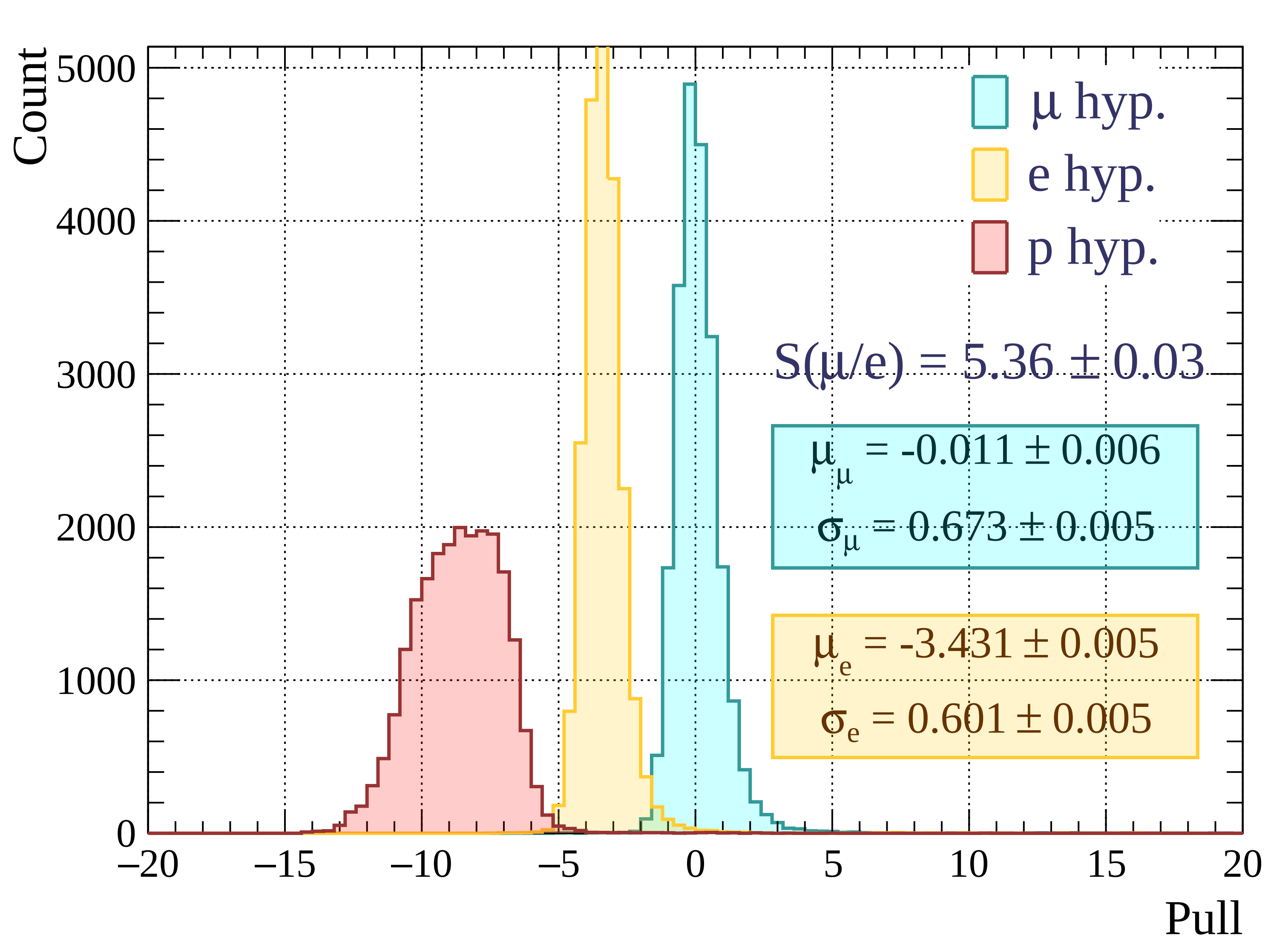


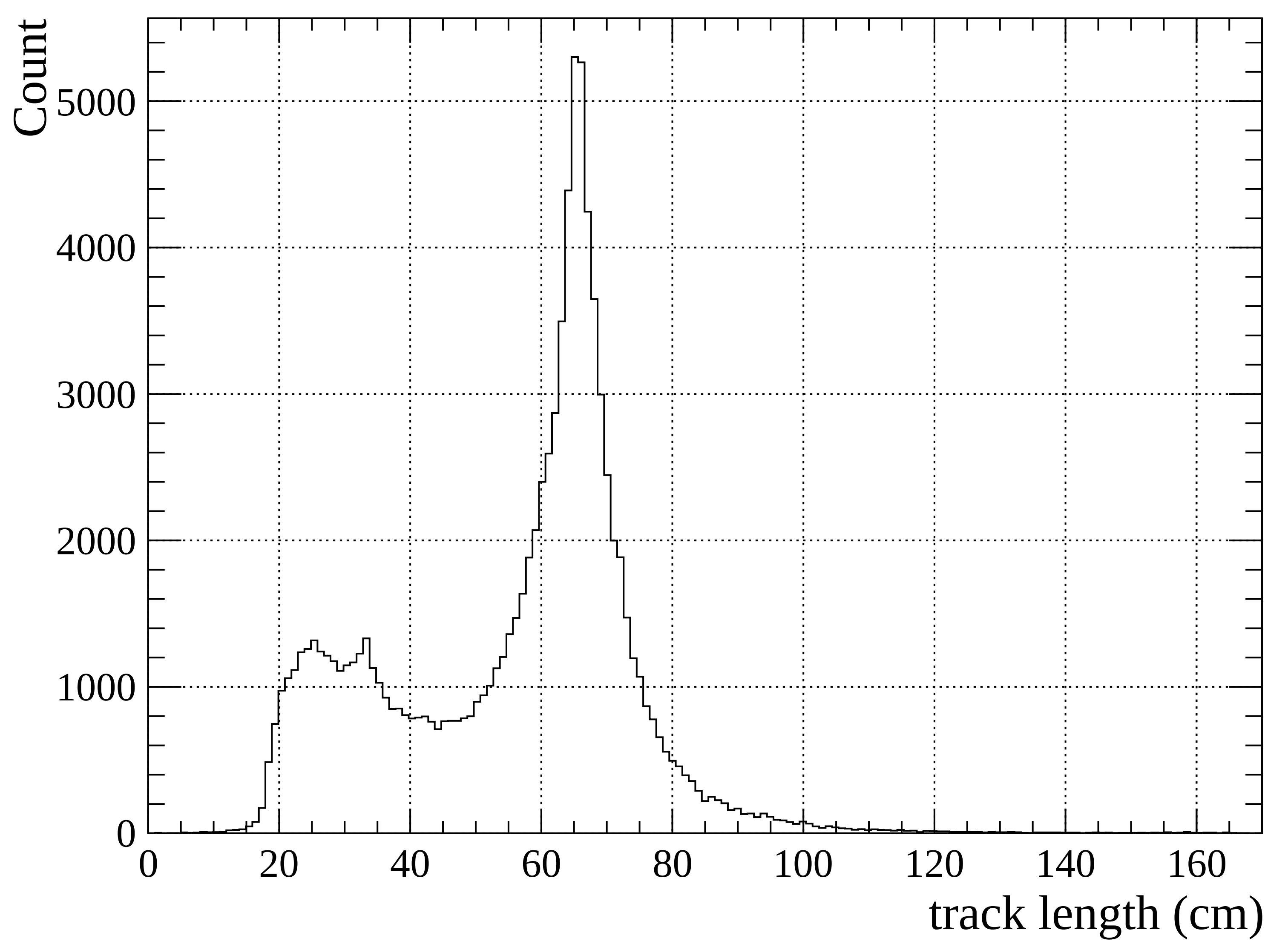
Pulls standard deviation
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$
0
1
2
3
4
5
6
−2000
−1500
−1000
−500
0
500
1000
1500
2000


Pulls mean
0
−5
−10
−15
−20
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
μ±
e±

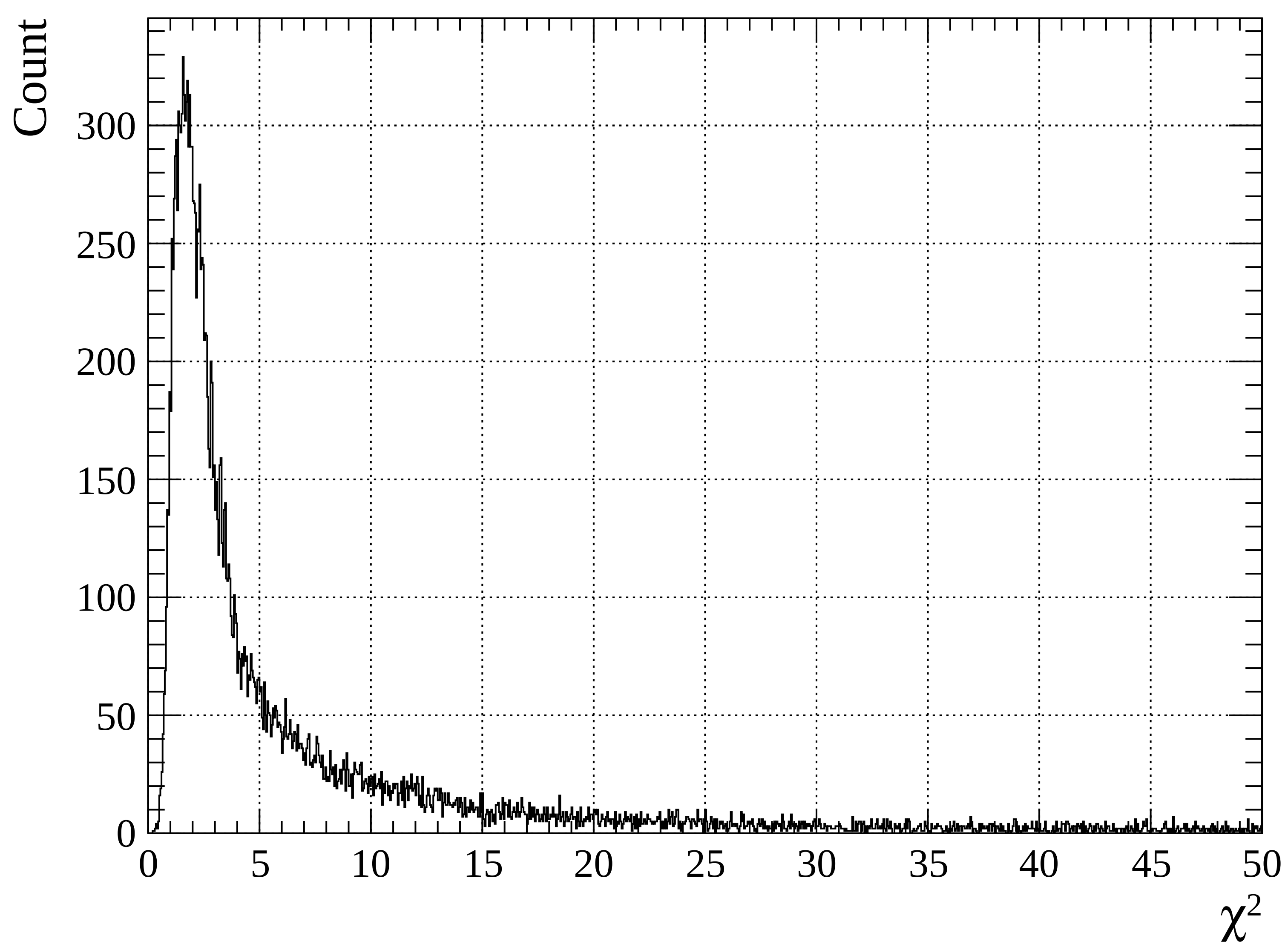


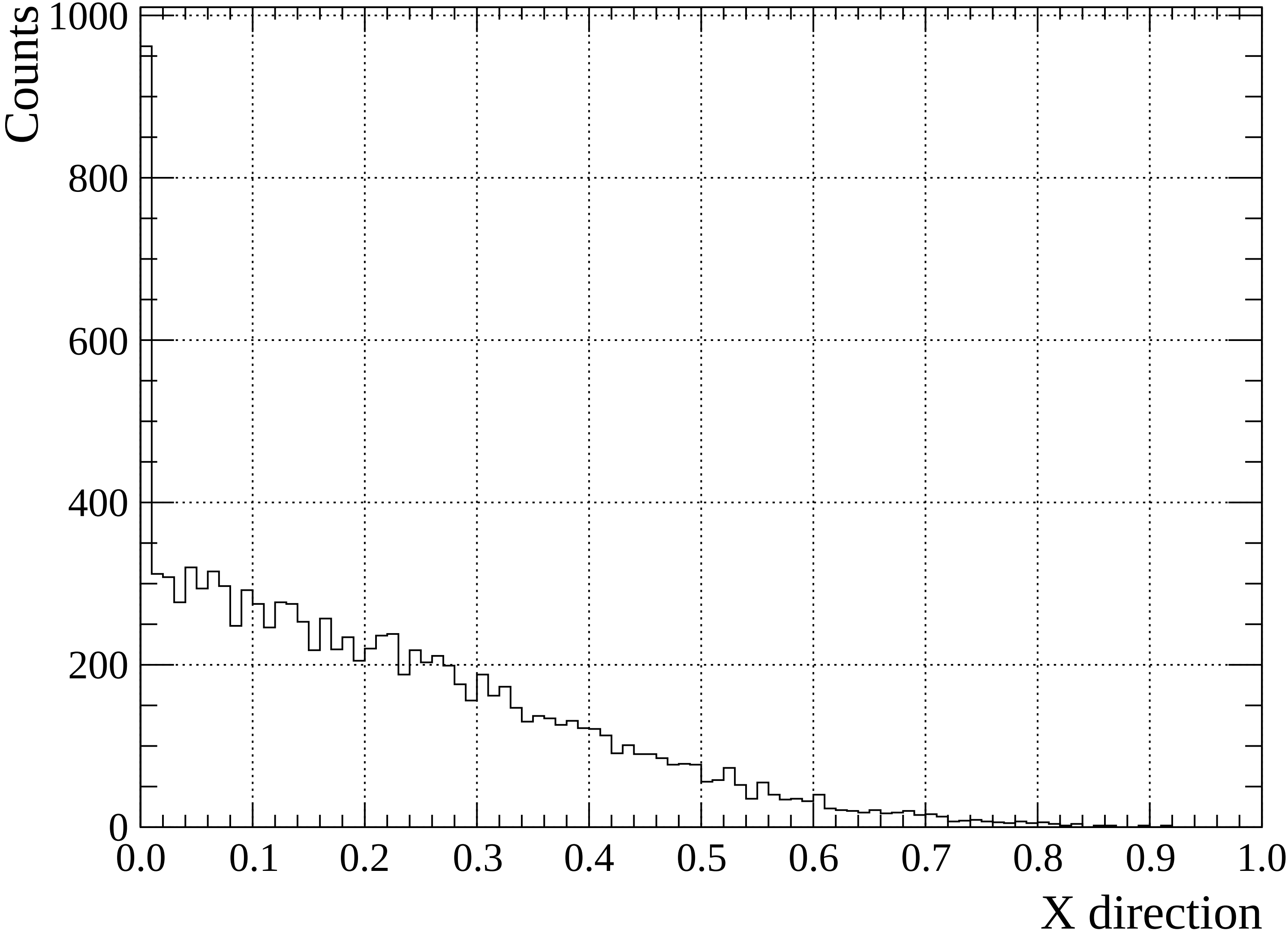
Pull e
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
Pull μ
5000
4000
3000
2000
1000
0


Count
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 5.36 ± 0.03
μ_μ = -0.011 ± 0.006
σ_μ = 0.673 ± 0.005
μ_e = -3.431 ± 0.005
σ_e = 0.601 ± 0.005
5000
4000
3000
2000
1000
0
−20
−15
−10
−5
0
5
10
15
20

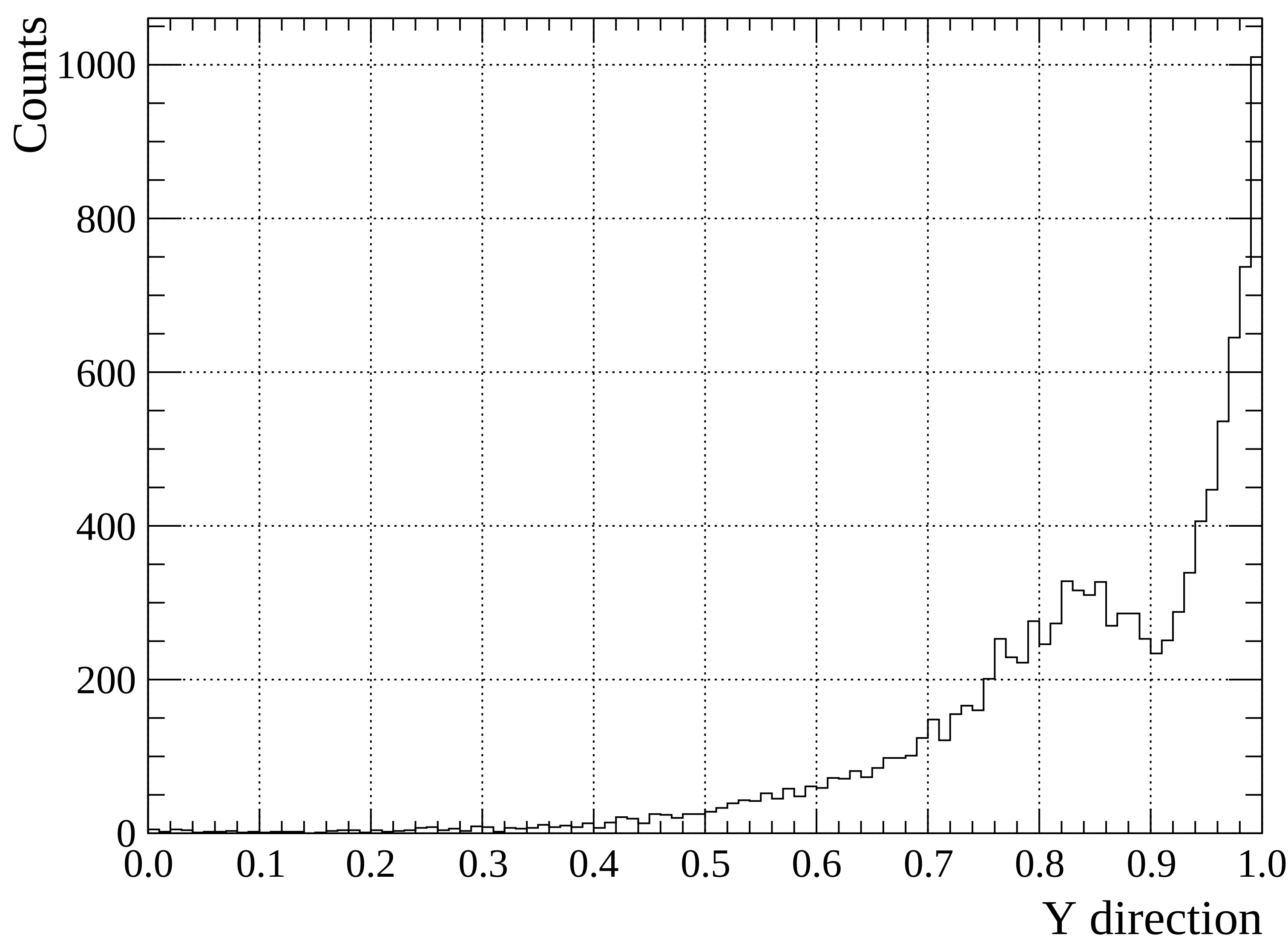


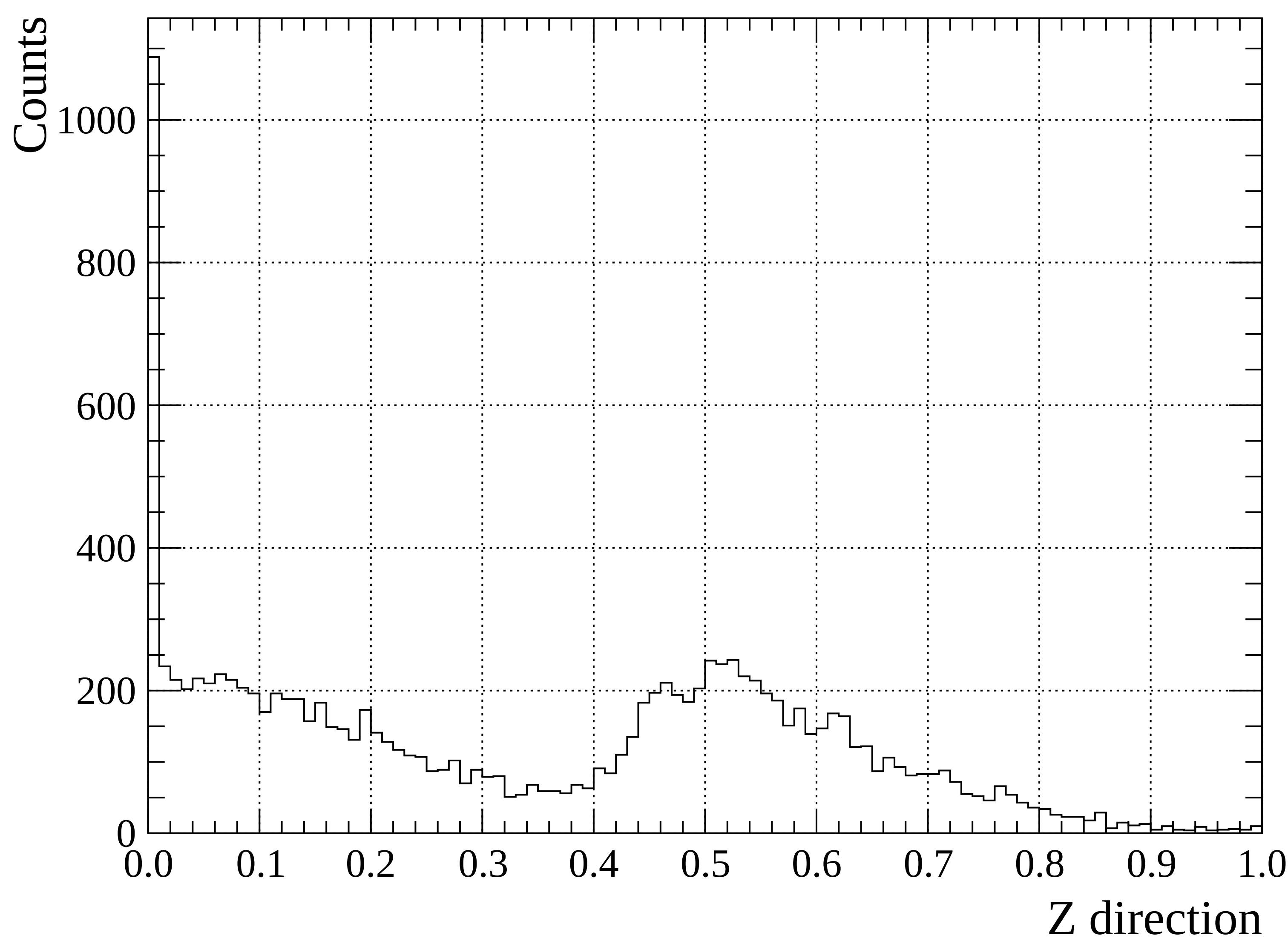
Count
5000
4000
3000
2000
1000
0
0
20
40
60
80
100
120
140
160
track length (cm)

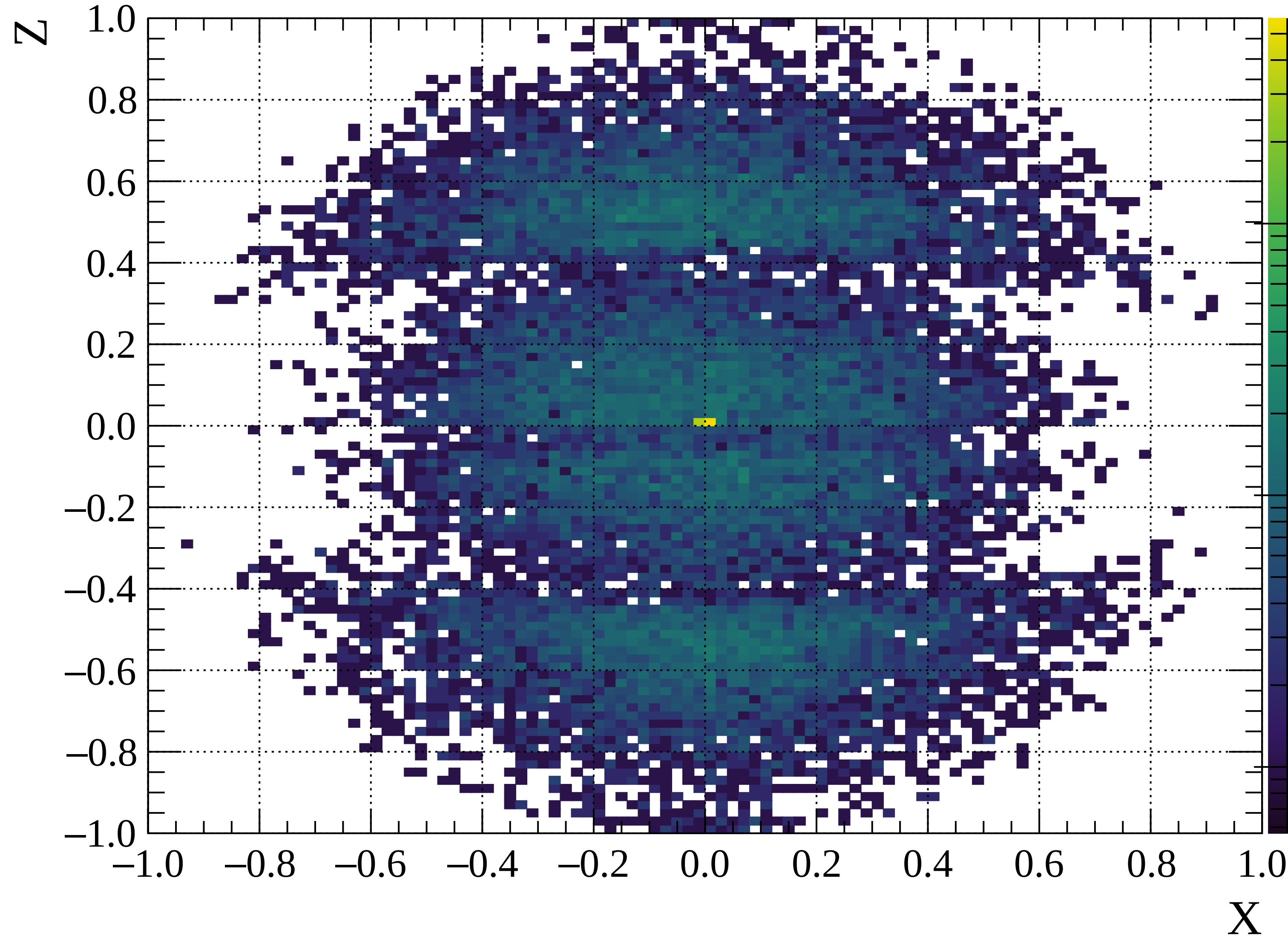

Count
300
250
200
150
100
50
0
0
5
10
15
20
25
30
35
40
45
50
χ^2

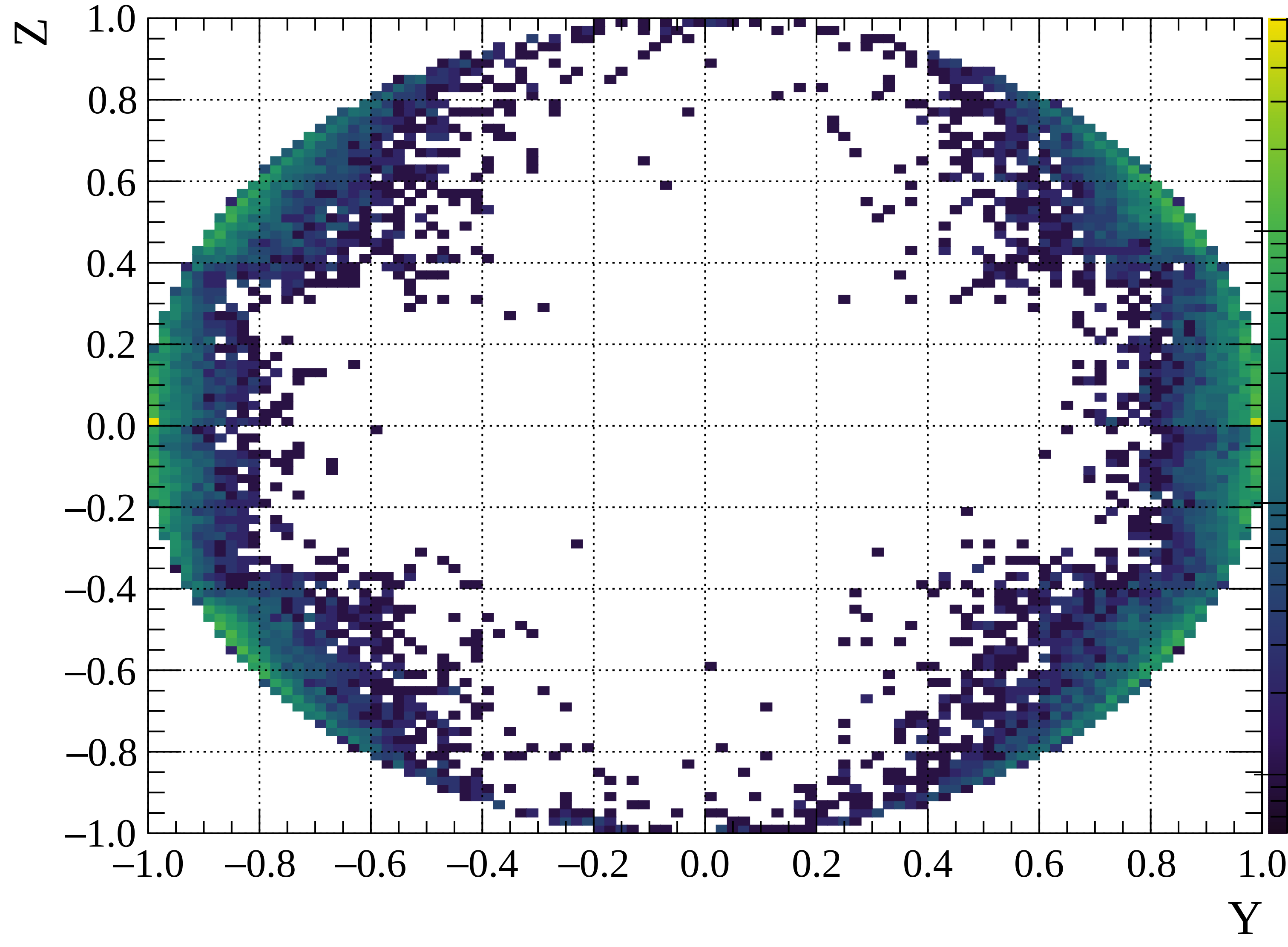


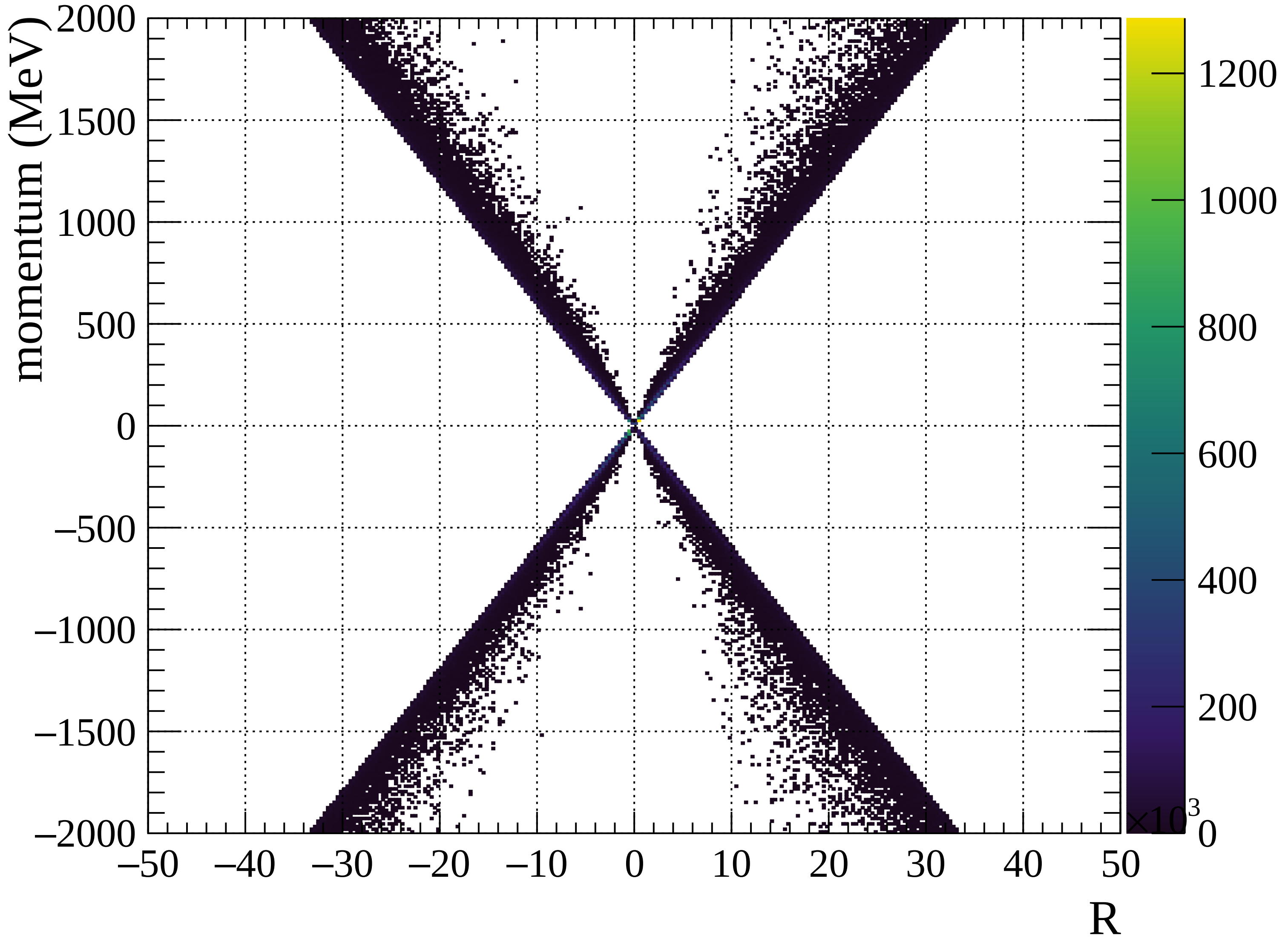
Counts
1000
800
600
400
200
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
X direction


Counts
1000
800
600
400
200
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Y direction

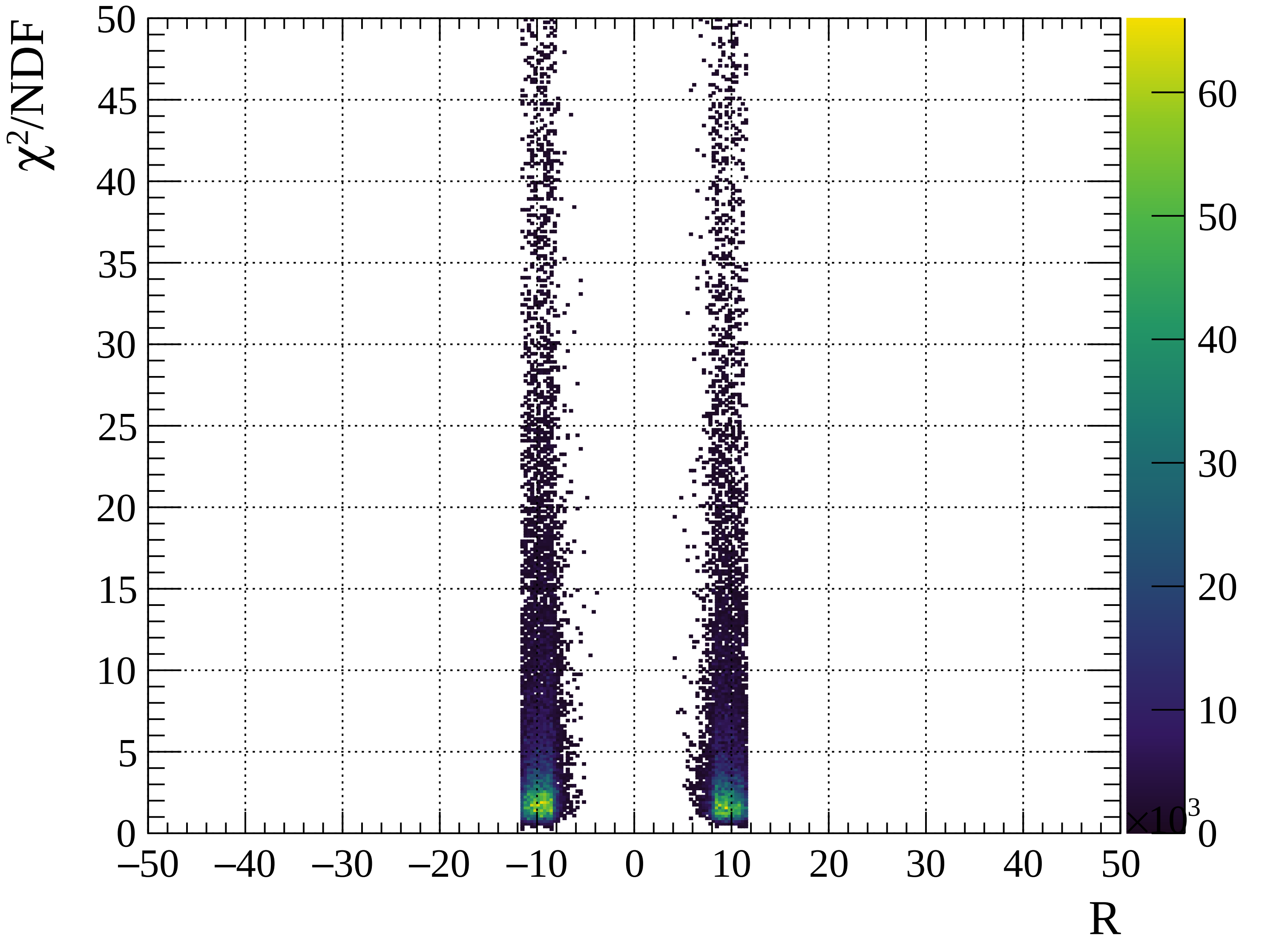

Counts
1000
800
600
400
200
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Z direction


Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X

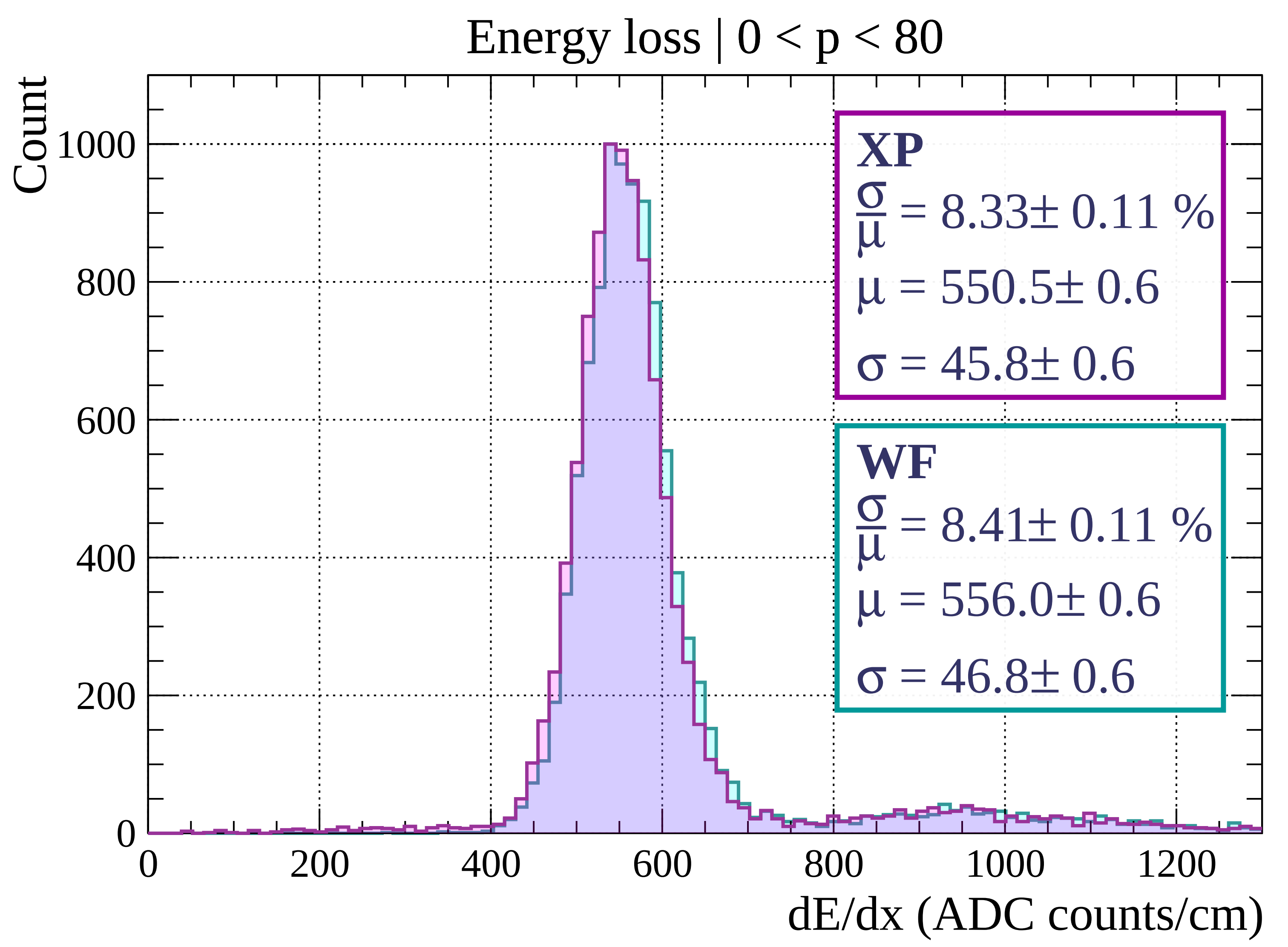

Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y

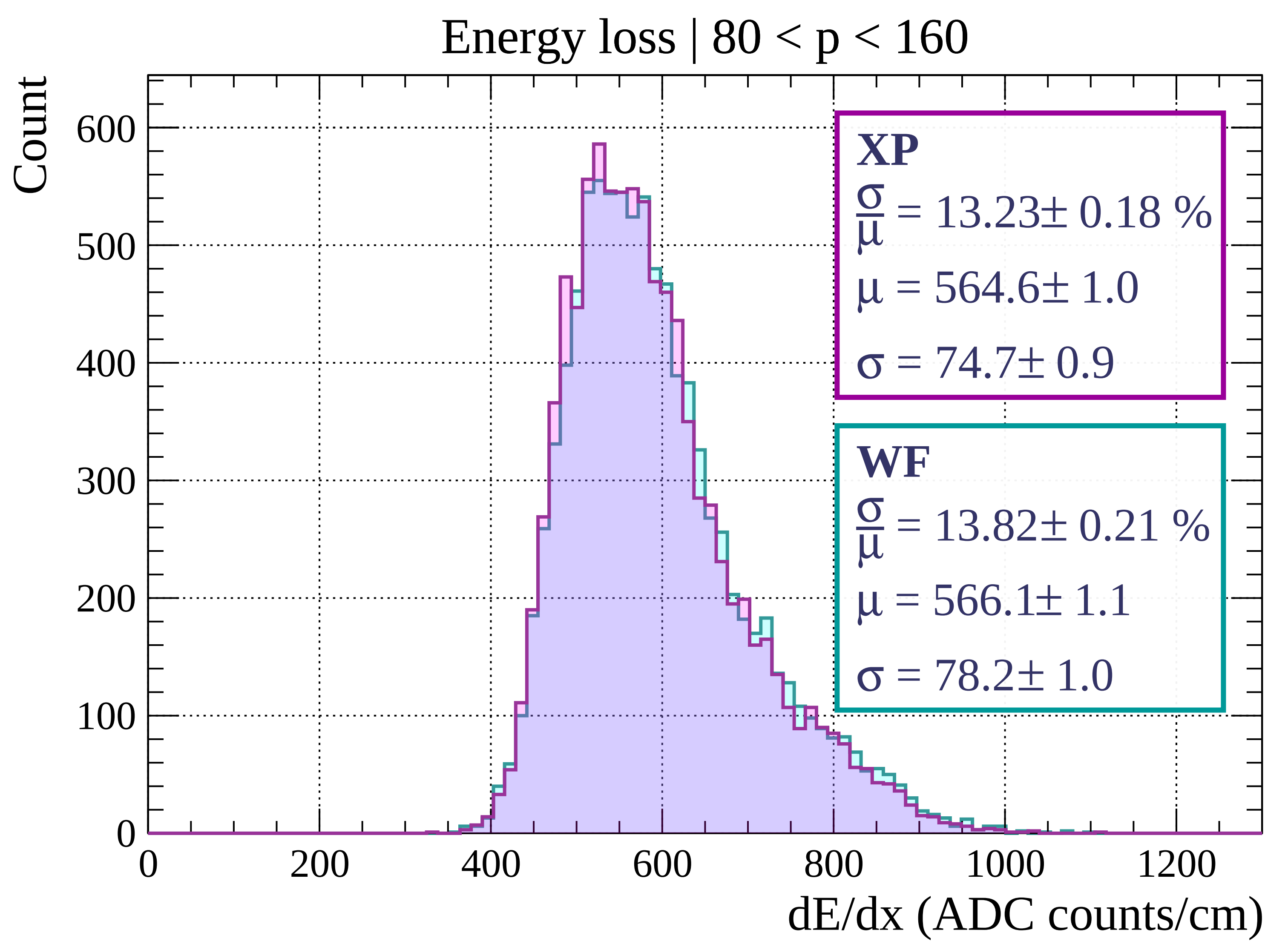


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
1200
1000
800
600
400
200
0
$\times 10^3$

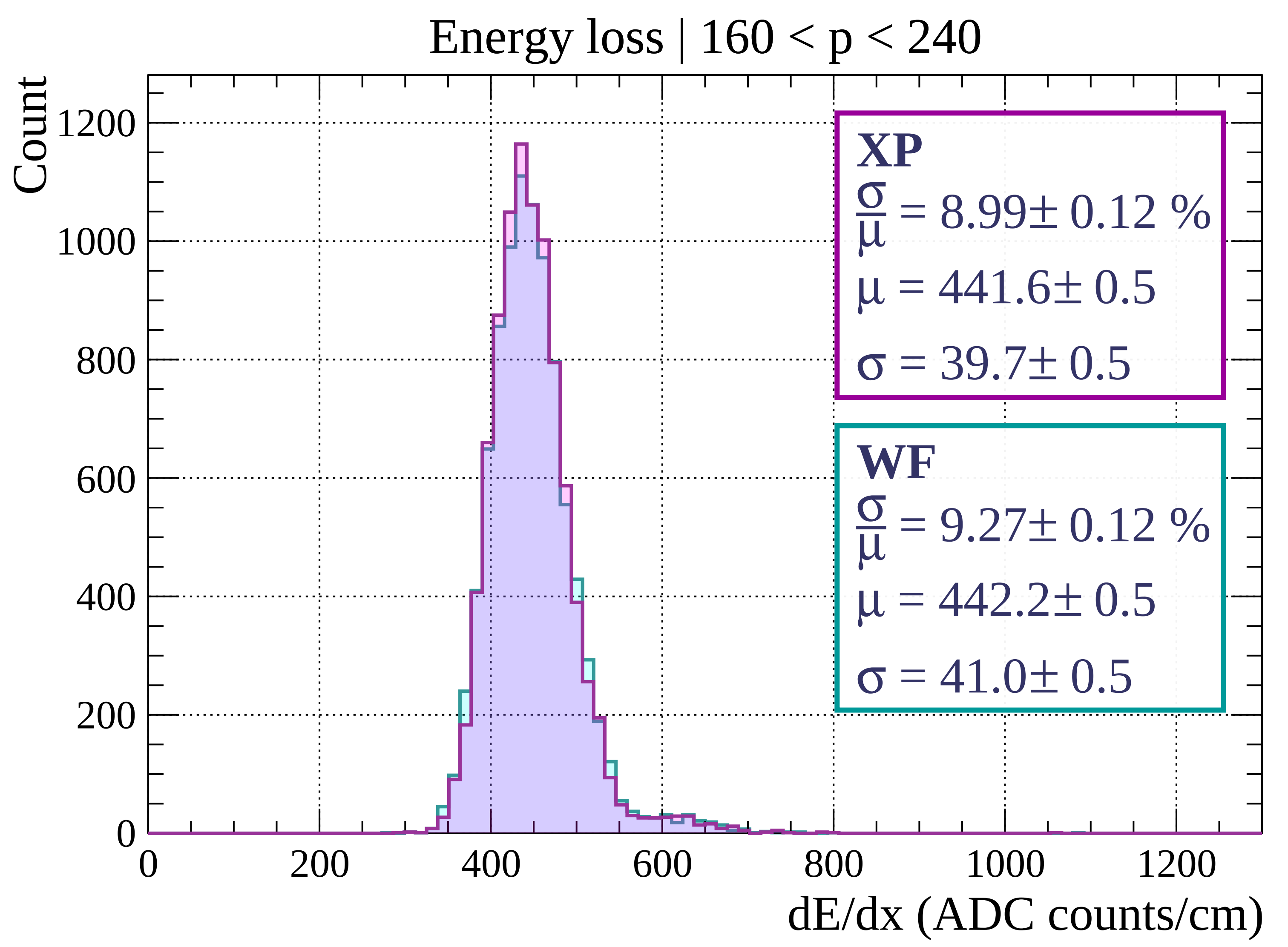


χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
60
50
40
30
20
10
0
$\times 10^3$

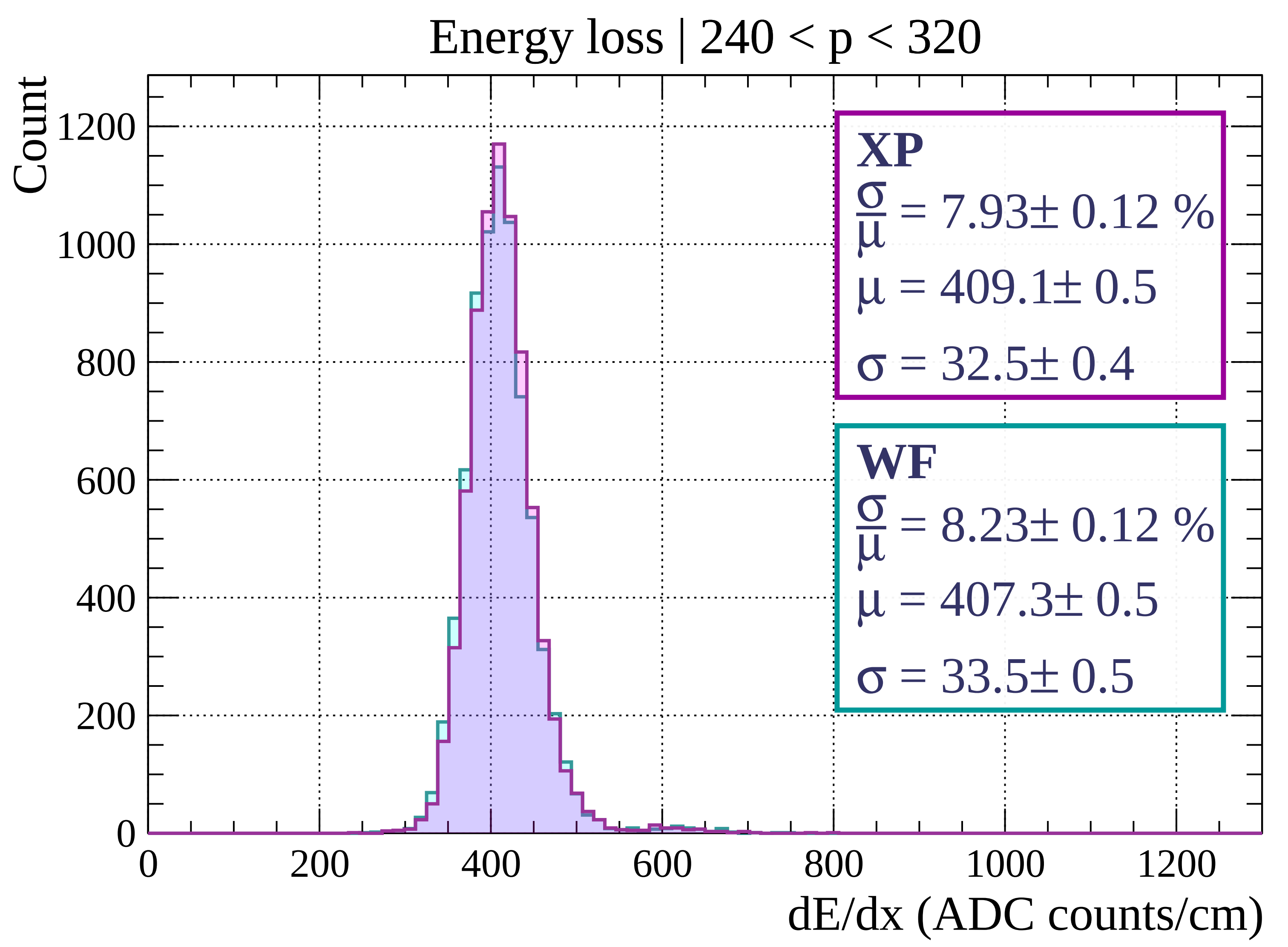

Energy loss | 0 < p < 80
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.33± 0.11 %
μ = 550.5± 0.6
σ = 45.8± 0.6
WF
$\frac{\sigma}{\mu}$ = 8.41± 0.11 %
μ = 556.0± 0.6
σ = 46.8± 0.6
0
200
400
600
800
1000
1200



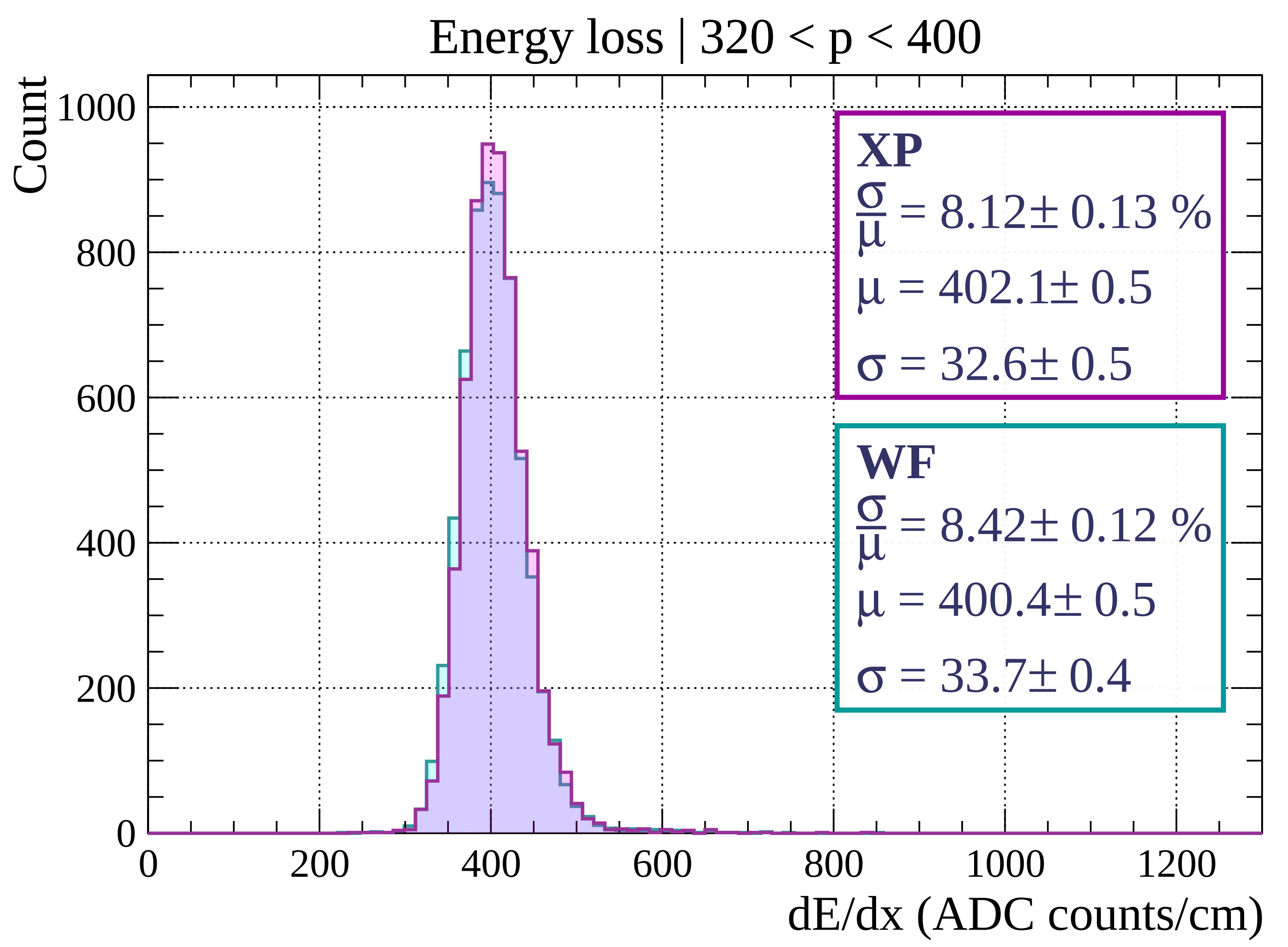
Energy loss | 80 < p < 160
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 13.23± 0.18 %
μ = 564.6± 1.0
σ = 74.7± 0.9
WF
$\frac{\sigma}{\mu}$ = 13.82± 0.21 %
μ = 566.1± 1.1
σ = 78.2± 1.0



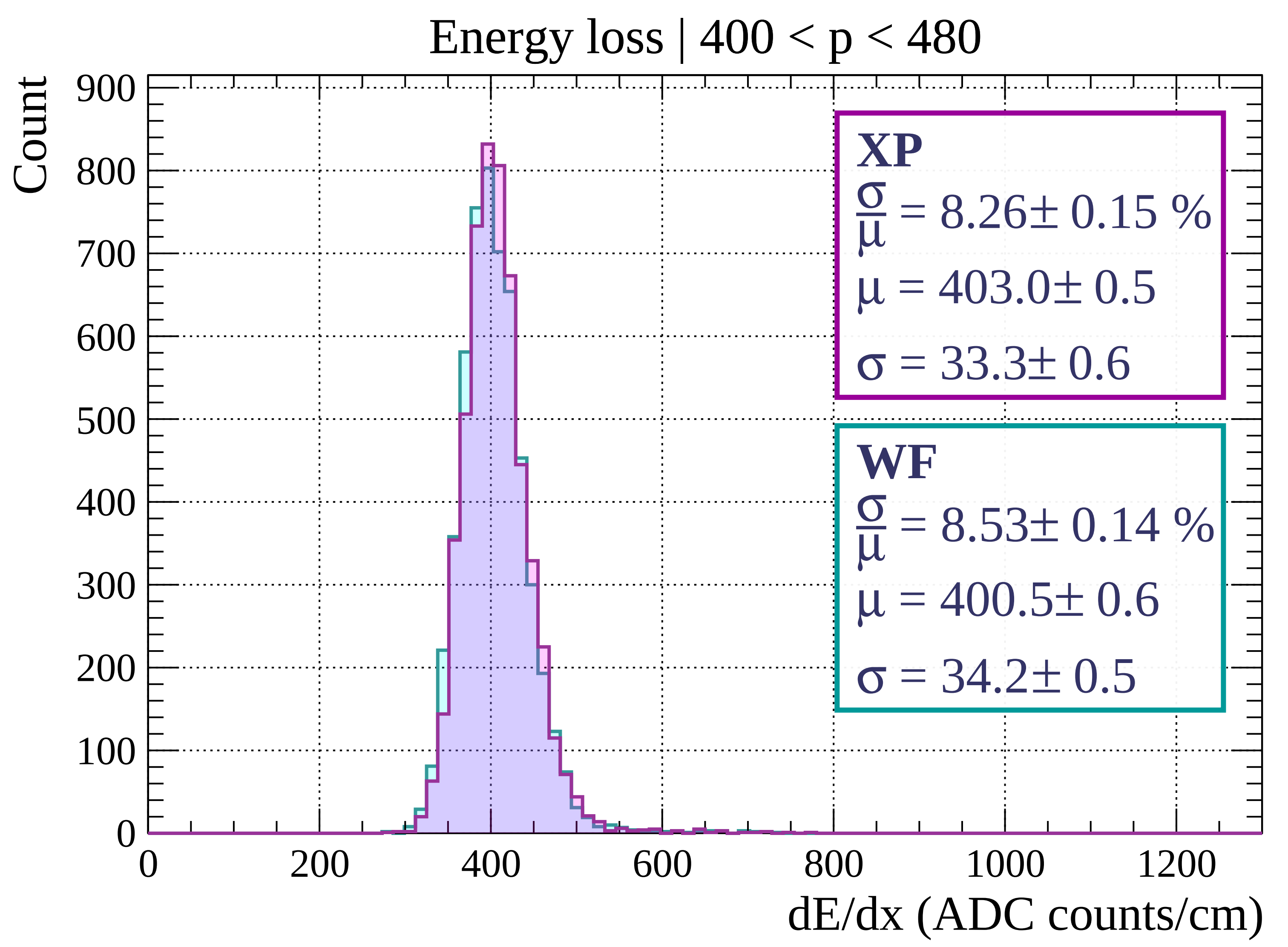
Energy loss | 160 < p < 240
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.99± 0.12 %
μ = 441.6± 0.5
σ = 39.7± 0.5
WF
σ/μ = 9.27± 0.12 %
μ = 442.2± 0.5
σ = 41.0± 0.5



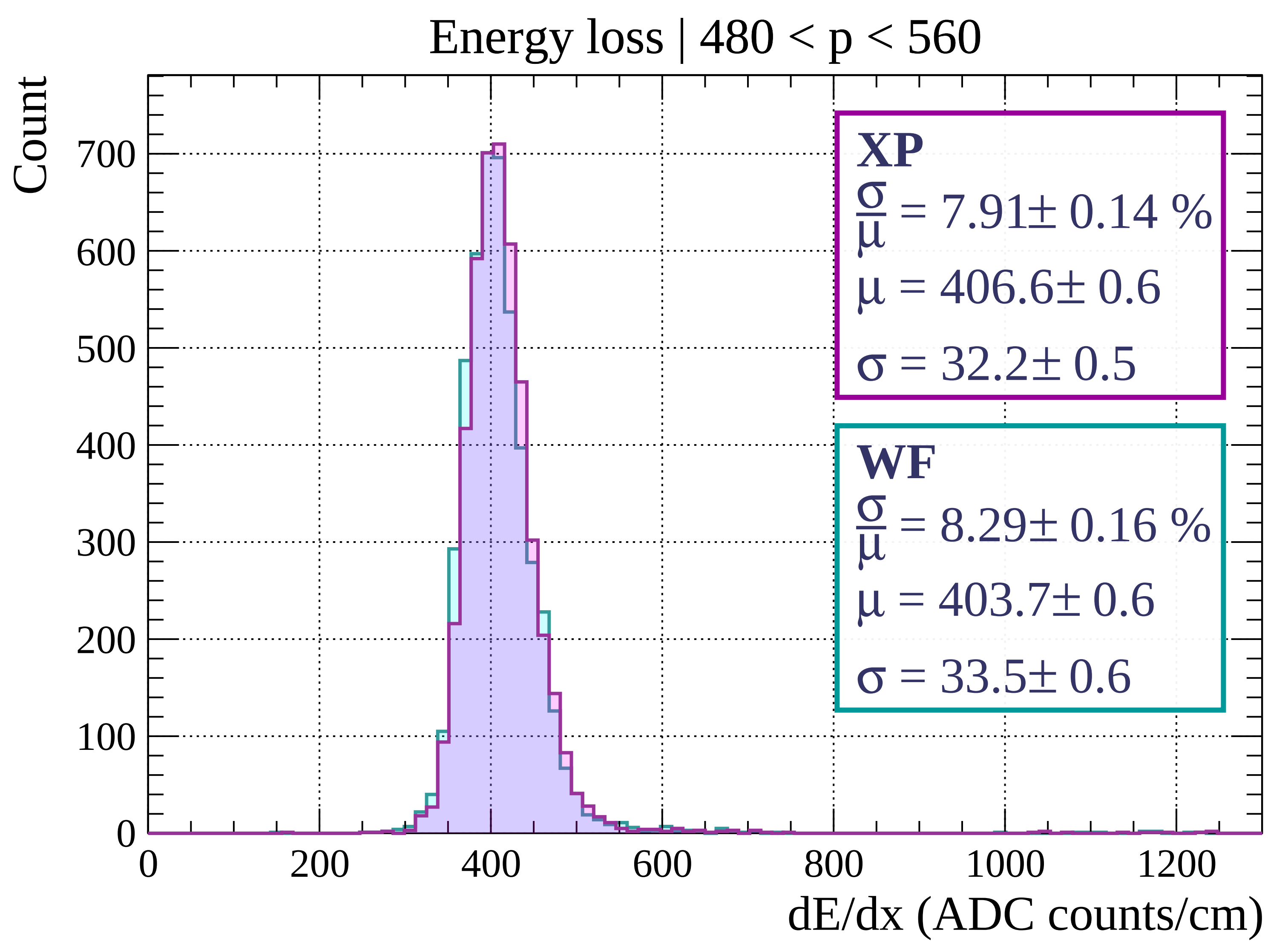
Energy loss | 240 < p < 320
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 7.93± 0.12 %
μ = 409.1± 0.5
σ = 32.5± 0.4
WF
σ/μ = 8.23± 0.12 %
μ = 407.3± 0.5
σ = 33.5± 0.5

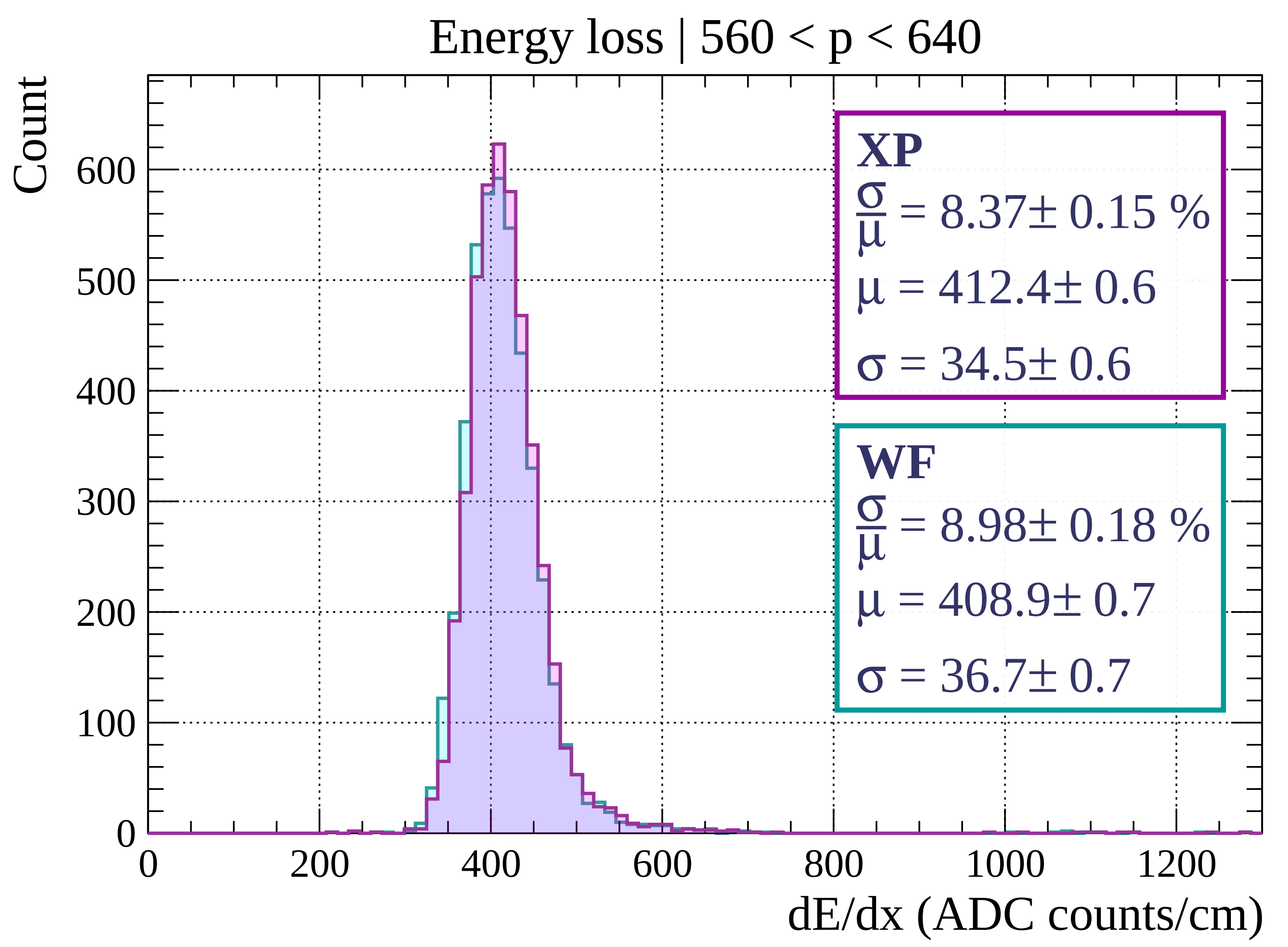


Energy loss | 320 < p < 400
Count
0
200
400
600
800
1000
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.12± 0.13 %
μ = 402.1± 0.5
σ = 32.6± 0.5
WF
$\frac{\sigma}{\mu}$ = 8.42± 0.12 %
μ = 400.4± 0.5
σ = 33.7± 0.4

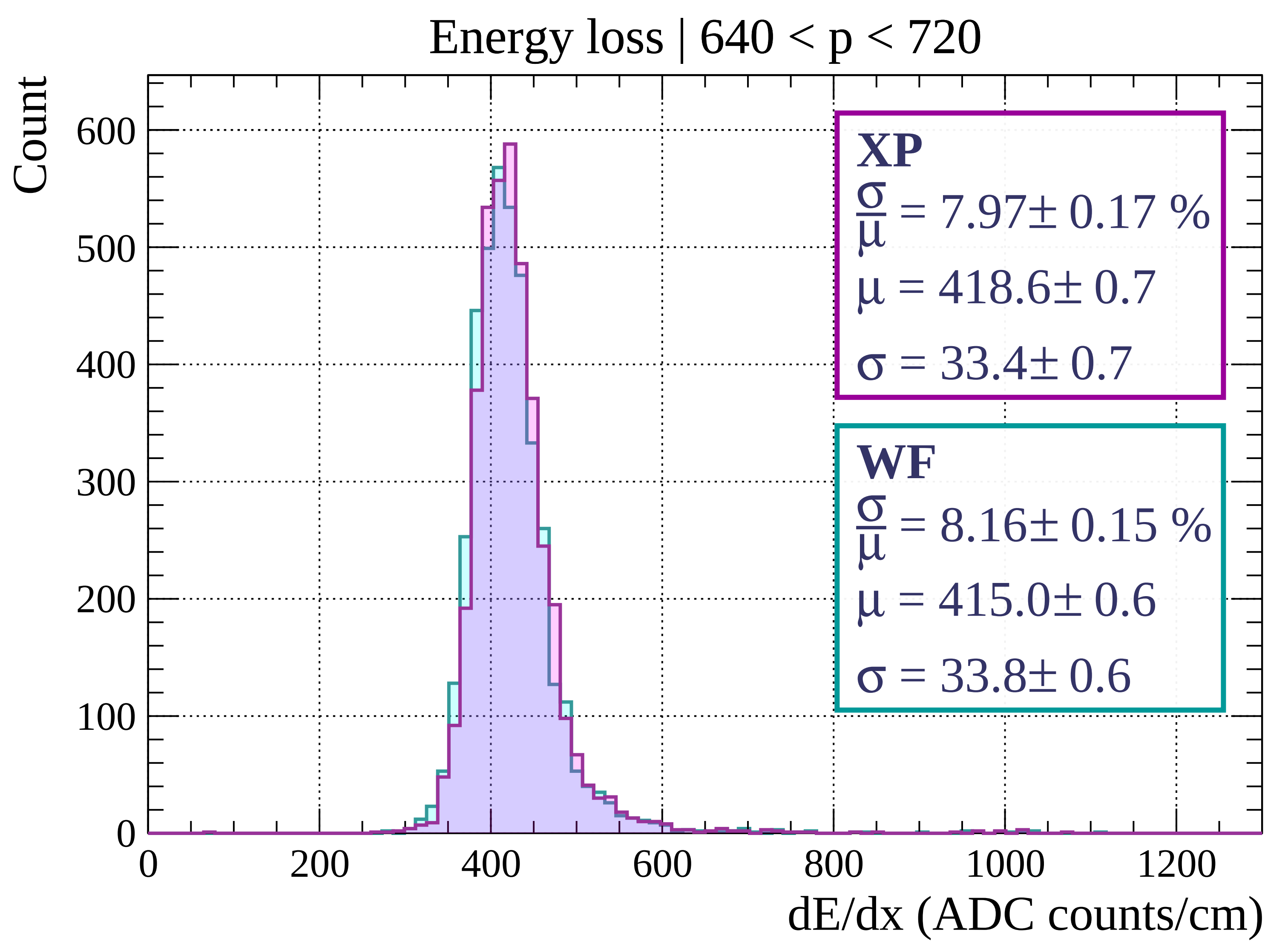


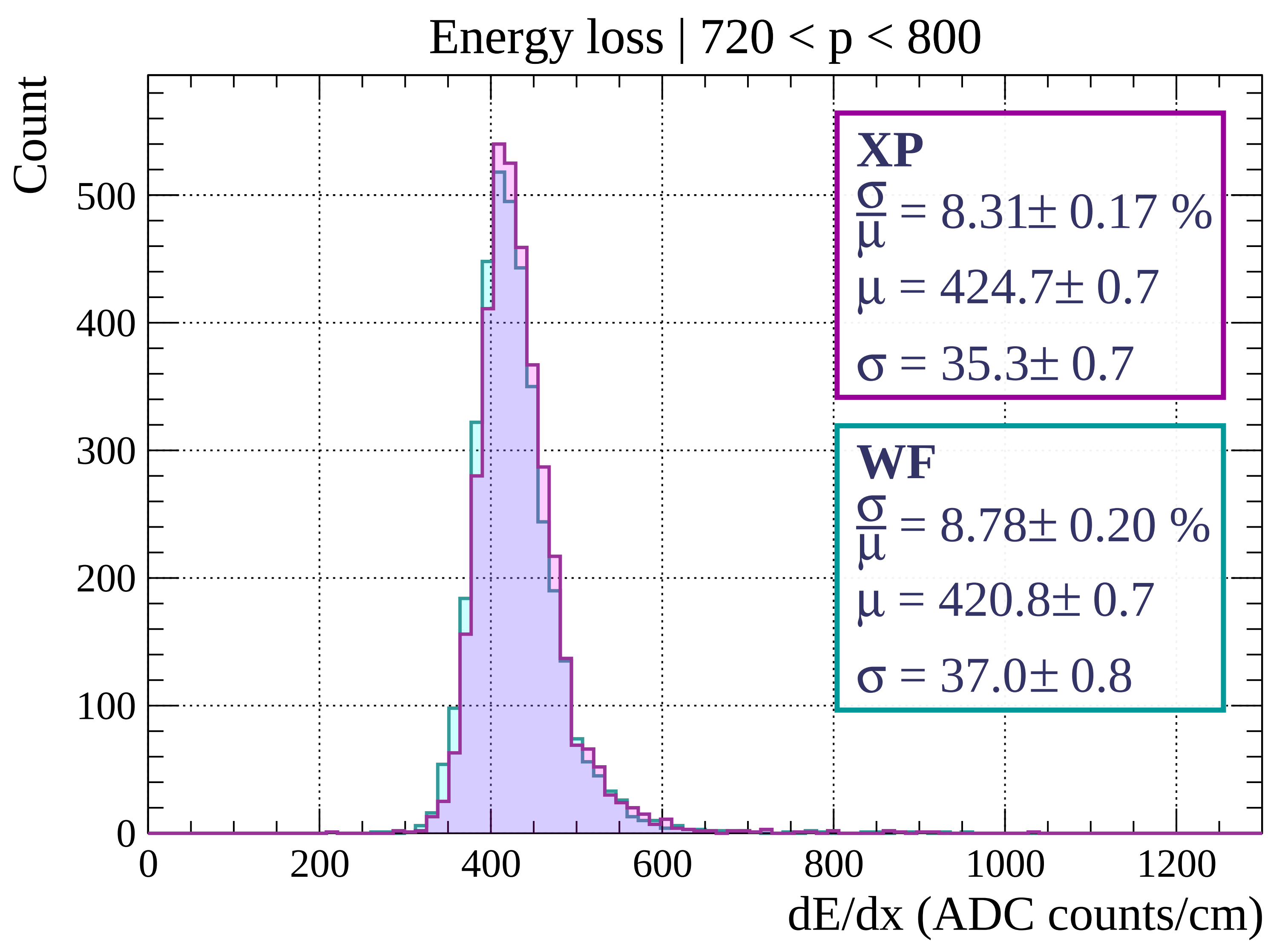
Energy loss | 400 < p < 480
Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.26± 0.15 %
μ = 403.0± 0.5
σ = 33.3± 0.6
WF
$\frac{\sigma}{\mu}$ = 8.53± 0.14 %
μ = 400.5± 0.6
σ = 34.2± 0.5

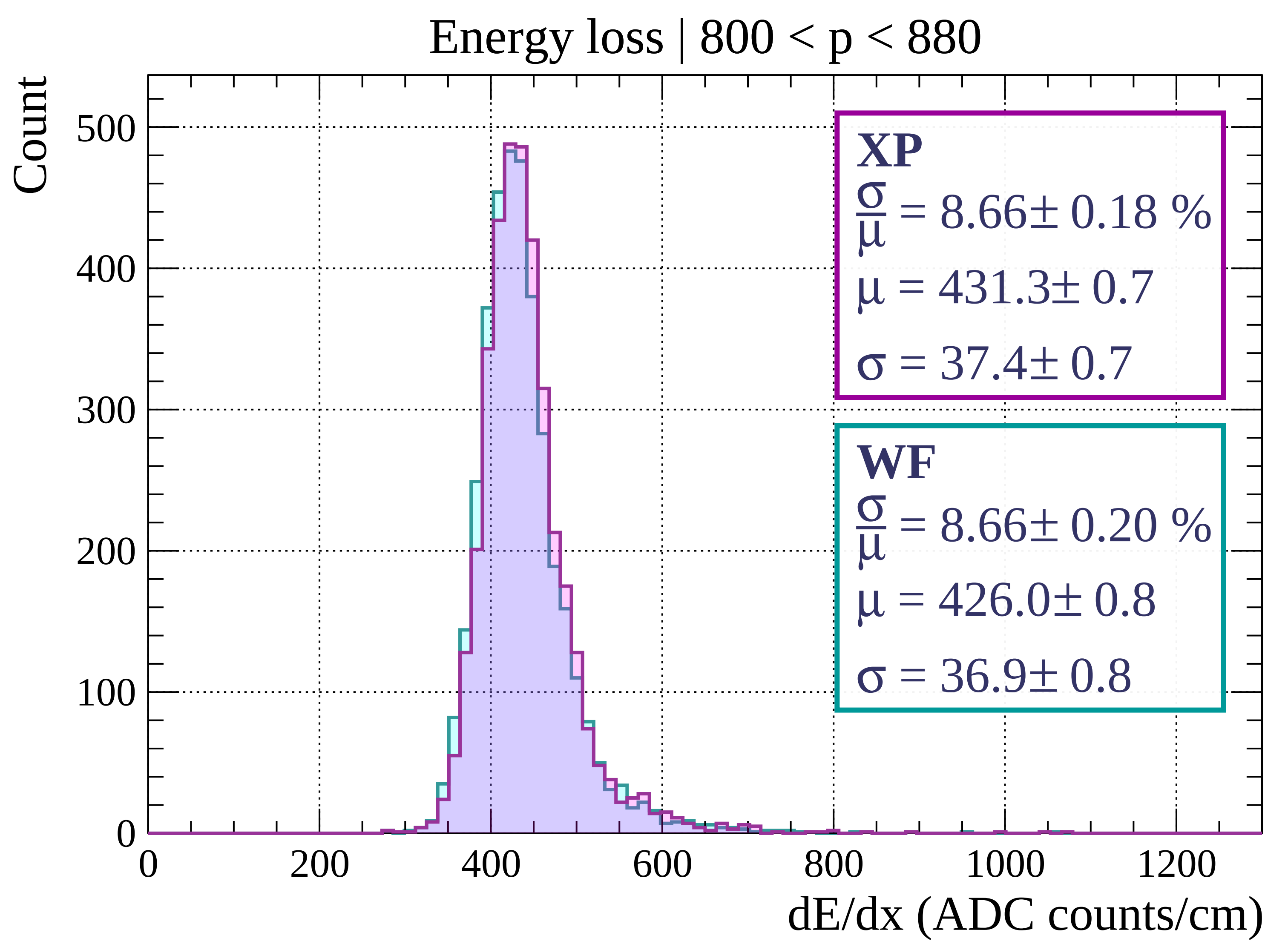

Energy loss | 480 < p < 560
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 7.91± 0.14 %
μ = 406.6± 0.6
σ = 32.2± 0.5
WF
σ/μ = 8.29± 0.16 %
μ = 403.7± 0.6
σ = 33.5± 0.6

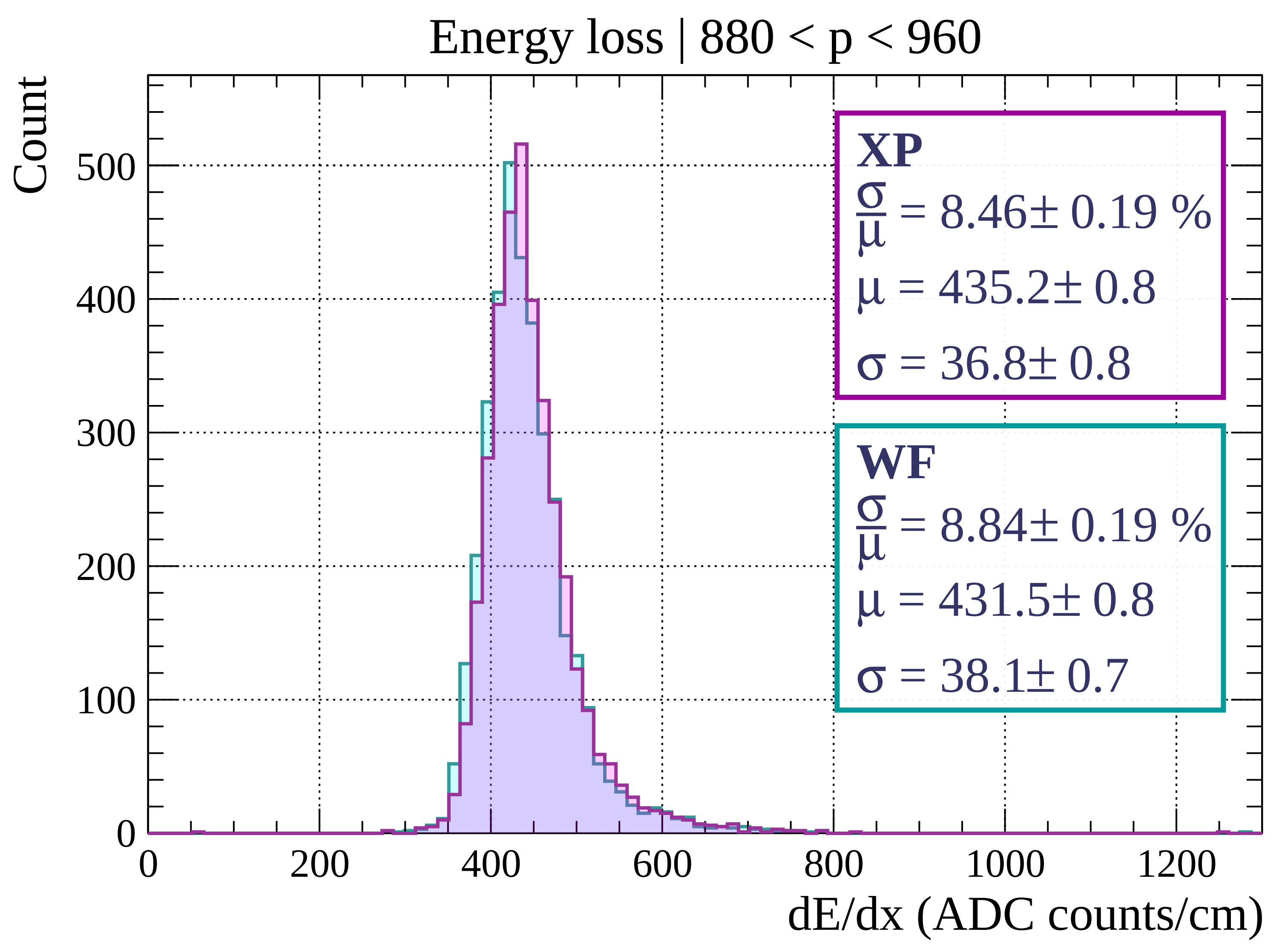


Energy loss | 560 < p < 640
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 8.37± 0.15 %
μ = 412.4± 0.6
σ = 34.5± 0.6
WF
σ/μ = 8.98± 0.18 %
μ = 408.9± 0.7
σ = 36.7± 0.7

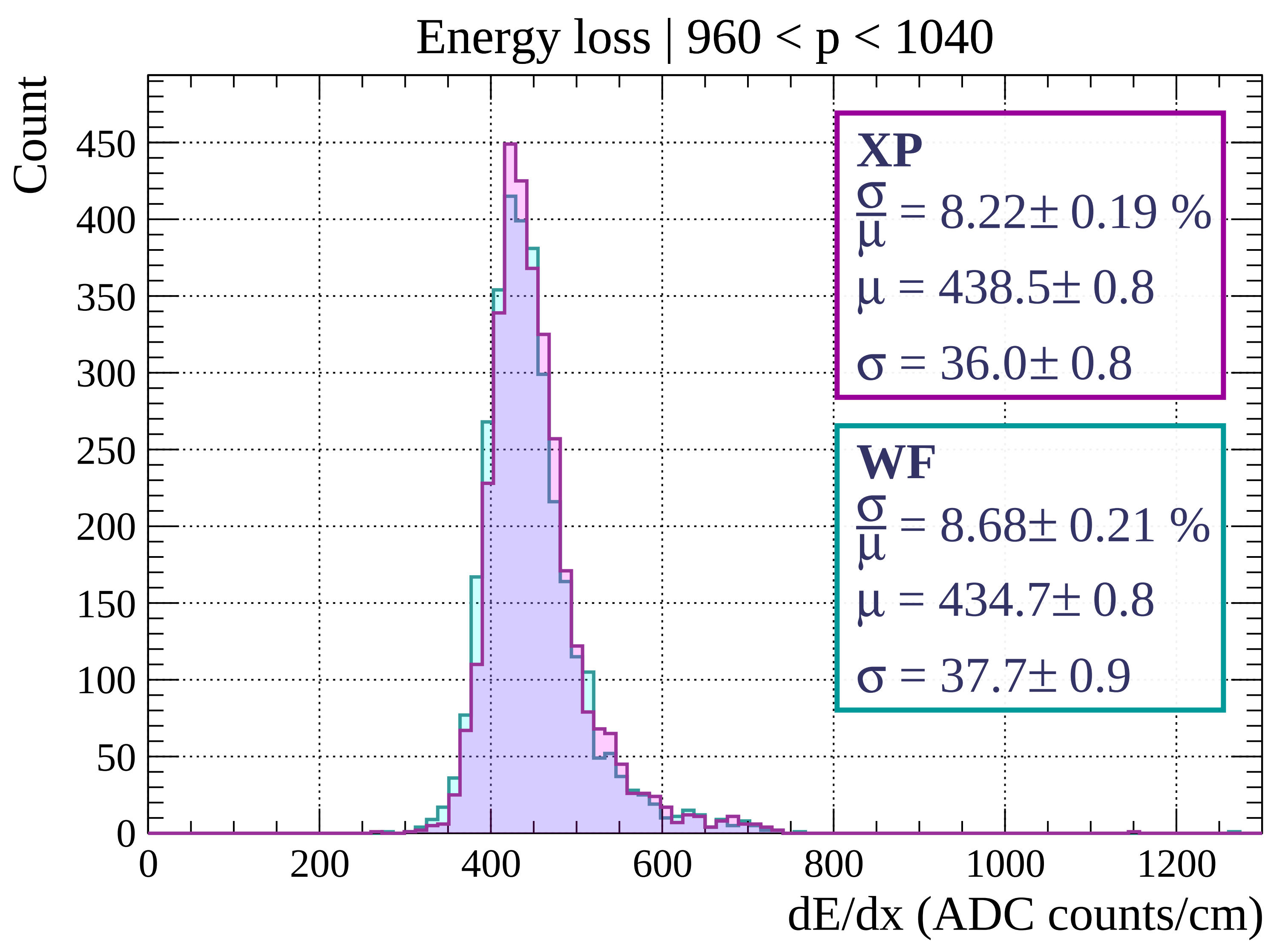

Energy loss | 640 < p < 720
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.97± 0.17 %
μ = 418.6± 0.7
σ = 33.4± 0.7
WF
$\frac{\sigma}{\mu}$ = 8.16± 0.15 %
μ = 415.0± 0.6
σ = 33.8± 0.6


Energy loss | 720 < p < 800
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.31± 0.17 %
μ = 424.7± 0.7
σ = 35.3± 0.7
WF
$\frac{\sigma}{\mu}$ = 8.78± 0.20 %
μ = 420.8± 0.7
σ = 37.0± 0.8

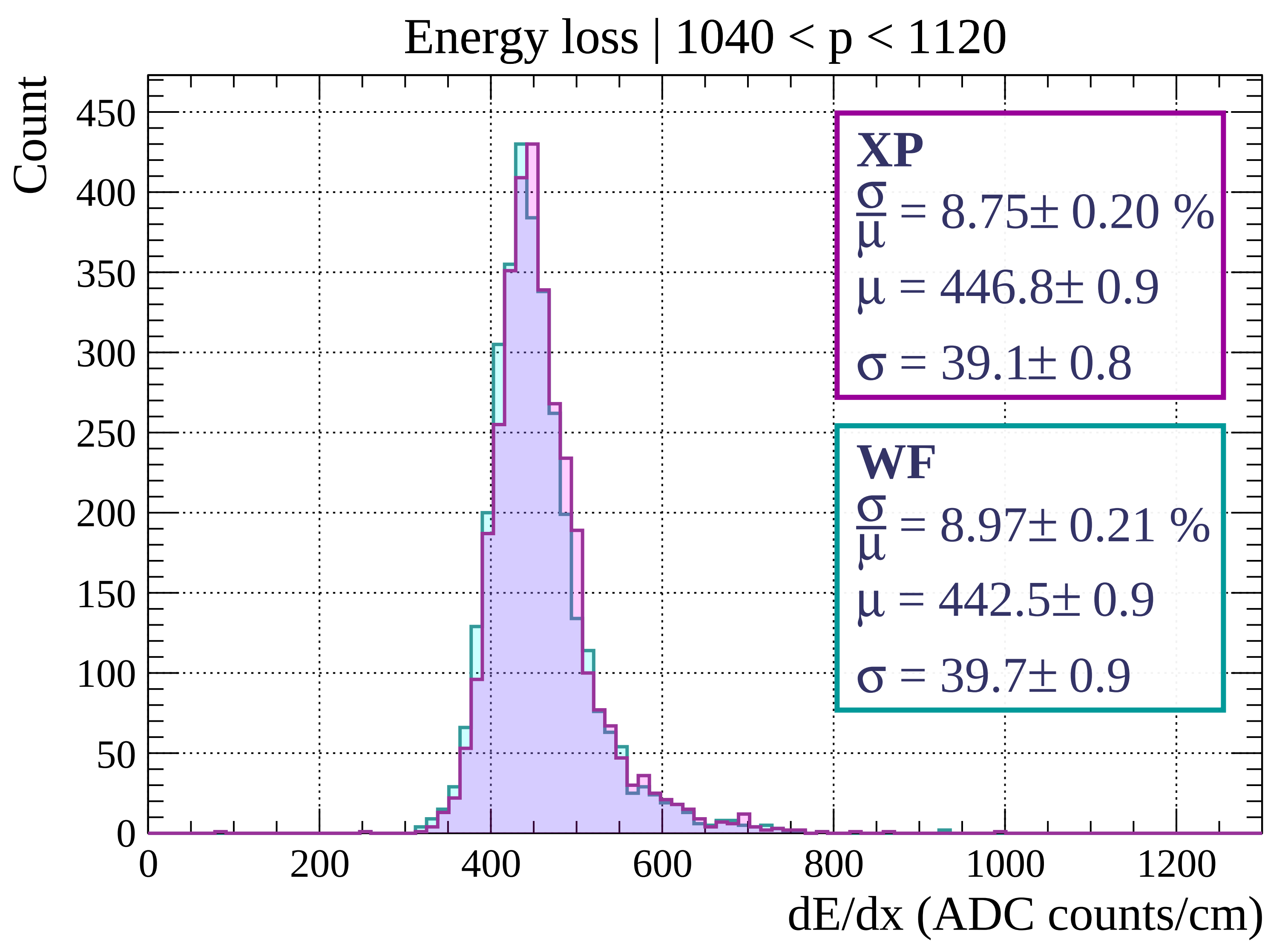

Energy loss | 800 < p < 880
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 8.66± 0.18 %
μ = 431.3± 0.7
σ = 37.4± 0.7
WF
σ/μ = 8.66± 0.20 %
μ = 426.0± 0.8
σ = 36.9± 0.8

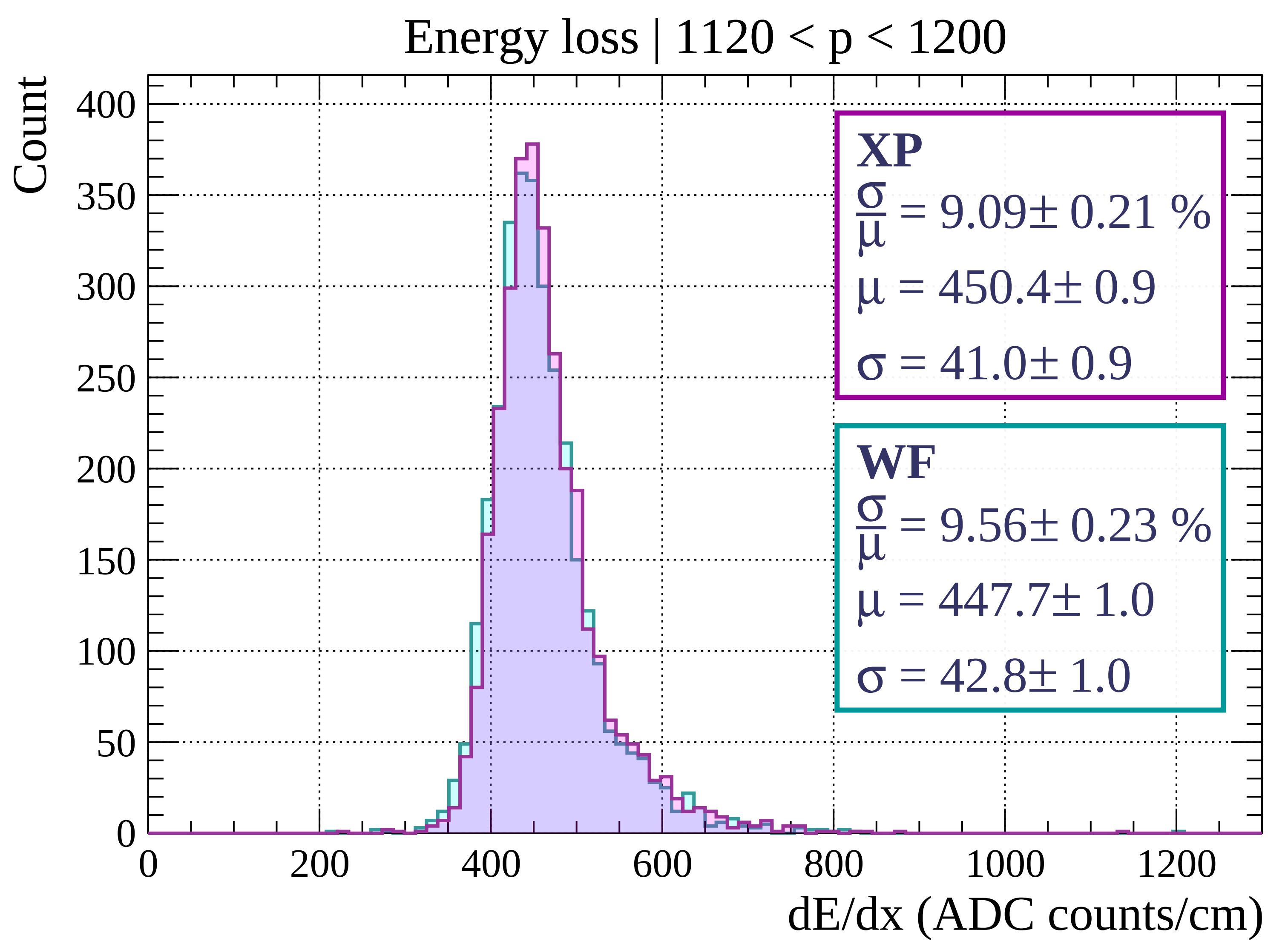


Energy loss | 880 < p < 960
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.46± 0.19 %
μ = 435.2± 0.8
σ = 36.8± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.84± 0.19 %
μ = 431.5± 0.8
σ = 38.1± 0.7

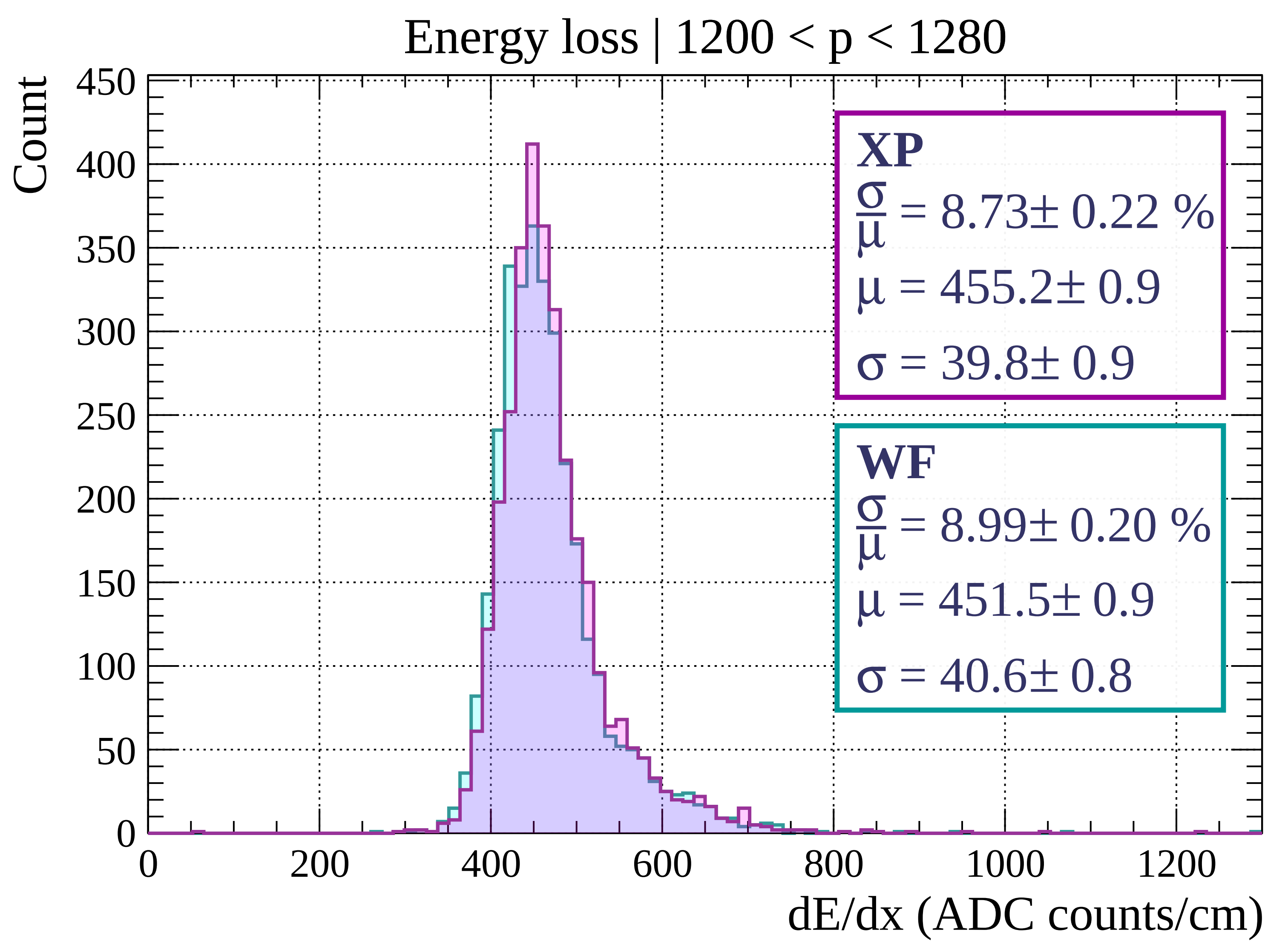


Energy loss | 960 < p < 1040
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.22± 0.19 %
μ = 438.5± 0.8
σ = 36.0± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.68± 0.21 %
μ = 434.7± 0.8
σ = 37.7± 0.9

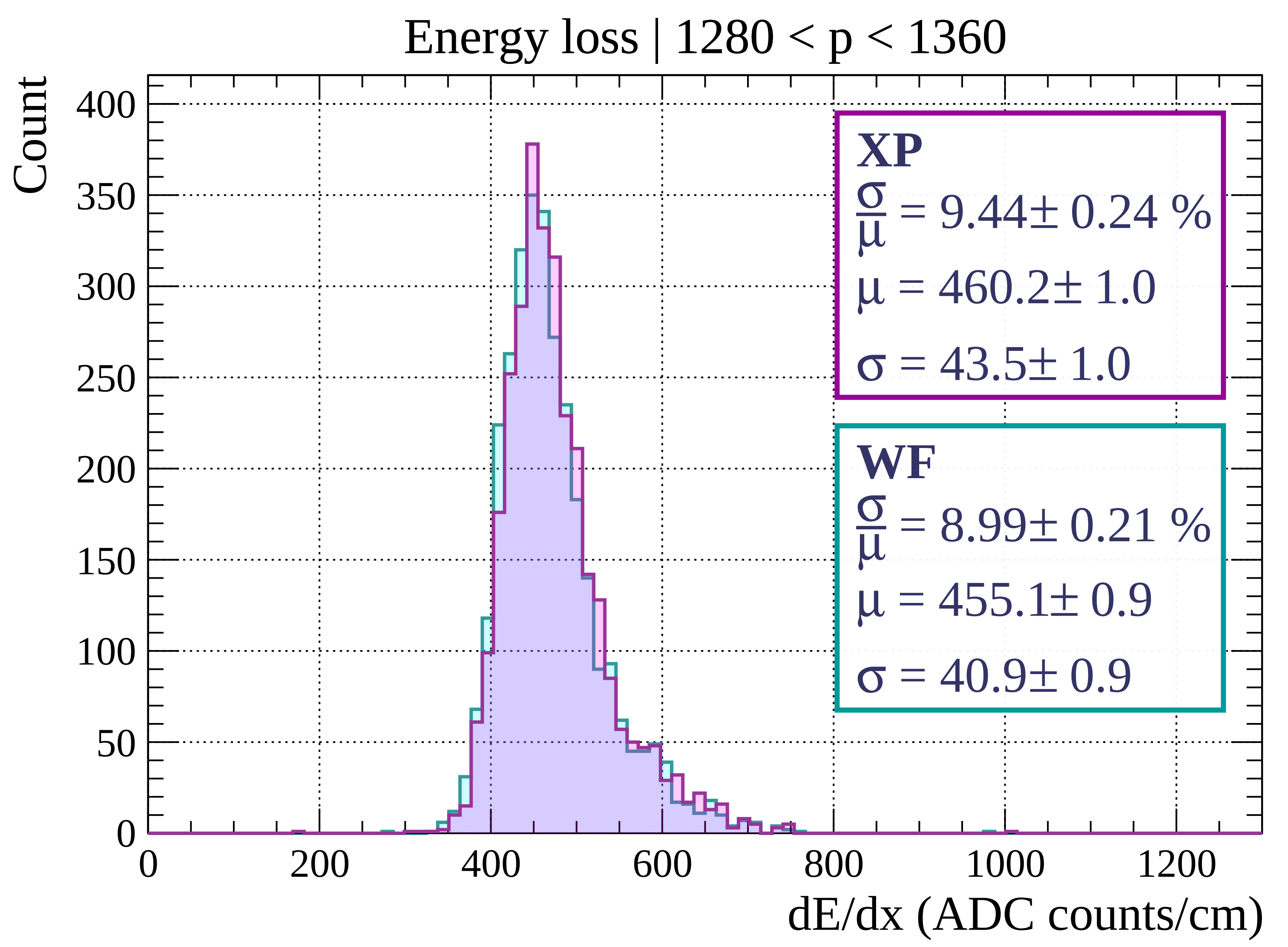

Energy loss | 1040 < p < 1120
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400
450
XP
$\frac{\sigma}{\mu}$ = 8.75± 0.20 %
μ = 446.8± 0.9
σ = 39.1± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.97± 0.21 %
μ = 442.5± 0.9
σ = 39.7± 0.9



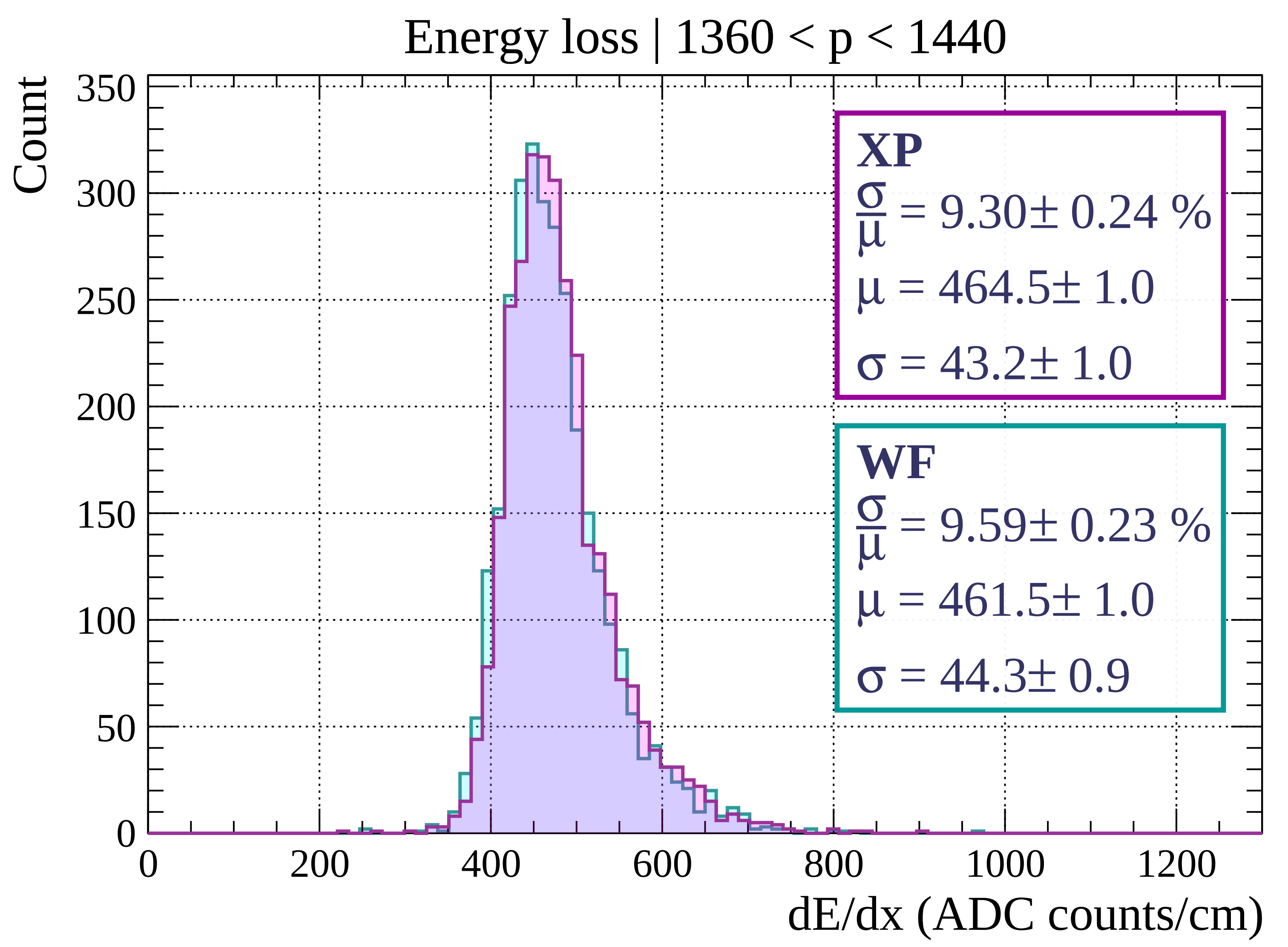
Energy loss | 1120 < p < 1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
XP
$\frac{\sigma}{\mu}$ = 9.09 ± 0.21 %
μ = 450.4 ± 0.9
σ = 41.0 ± 0.9
WF
$\frac{\sigma}{\mu}$ = 9.56 ± 0.23 %
μ = 447.7 ± 1.0
σ = 42.8 ± 1.0



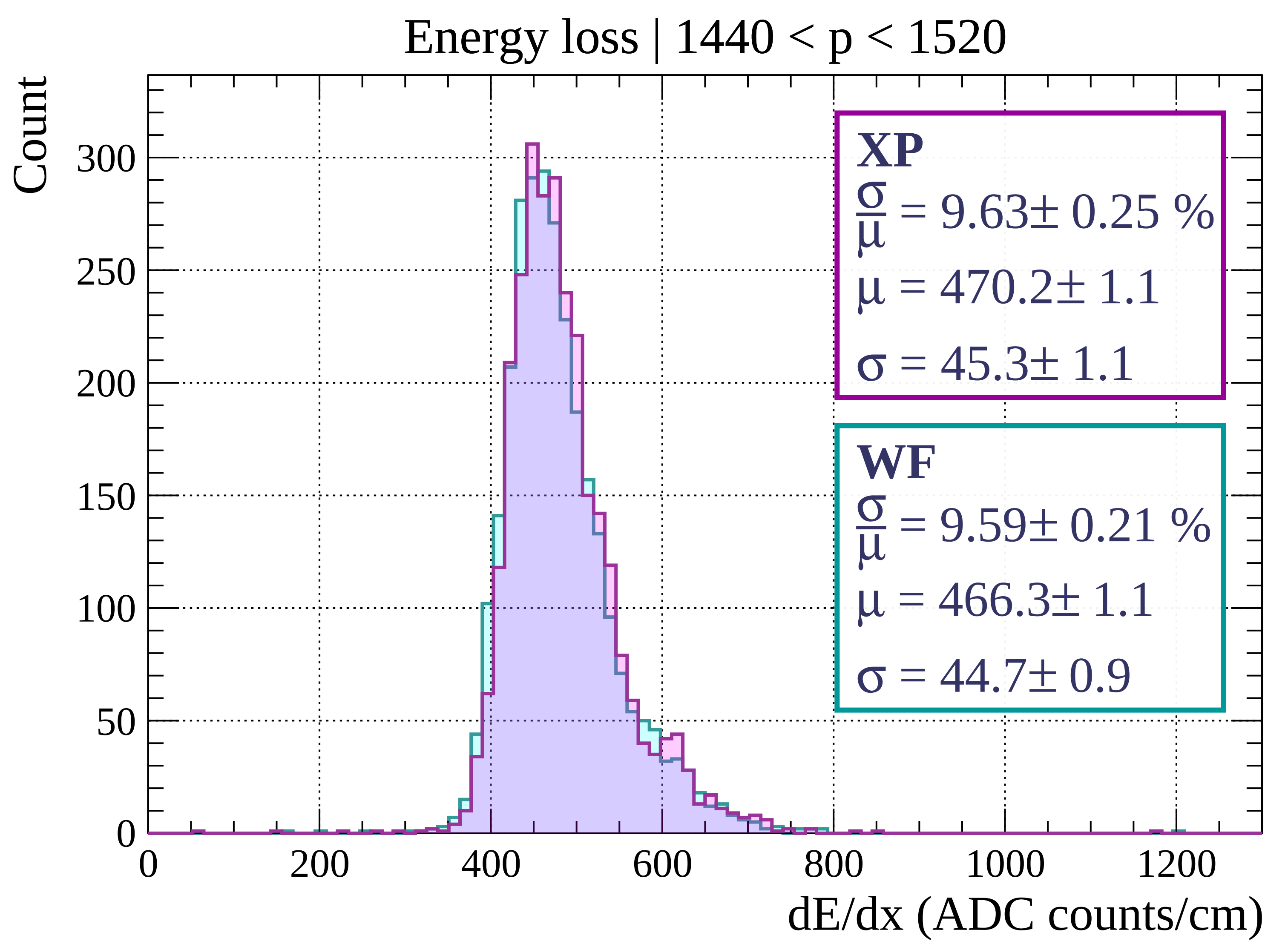
Energy loss | 1200 < p < 1280
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
450
XP
σ/μ = 8.73± 0.22 %
μ = 455.2± 0.9
σ = 39.8± 0.9
WF
σ/μ = 8.99± 0.20 %
μ = 451.5± 0.9
σ = 40.6± 0.8

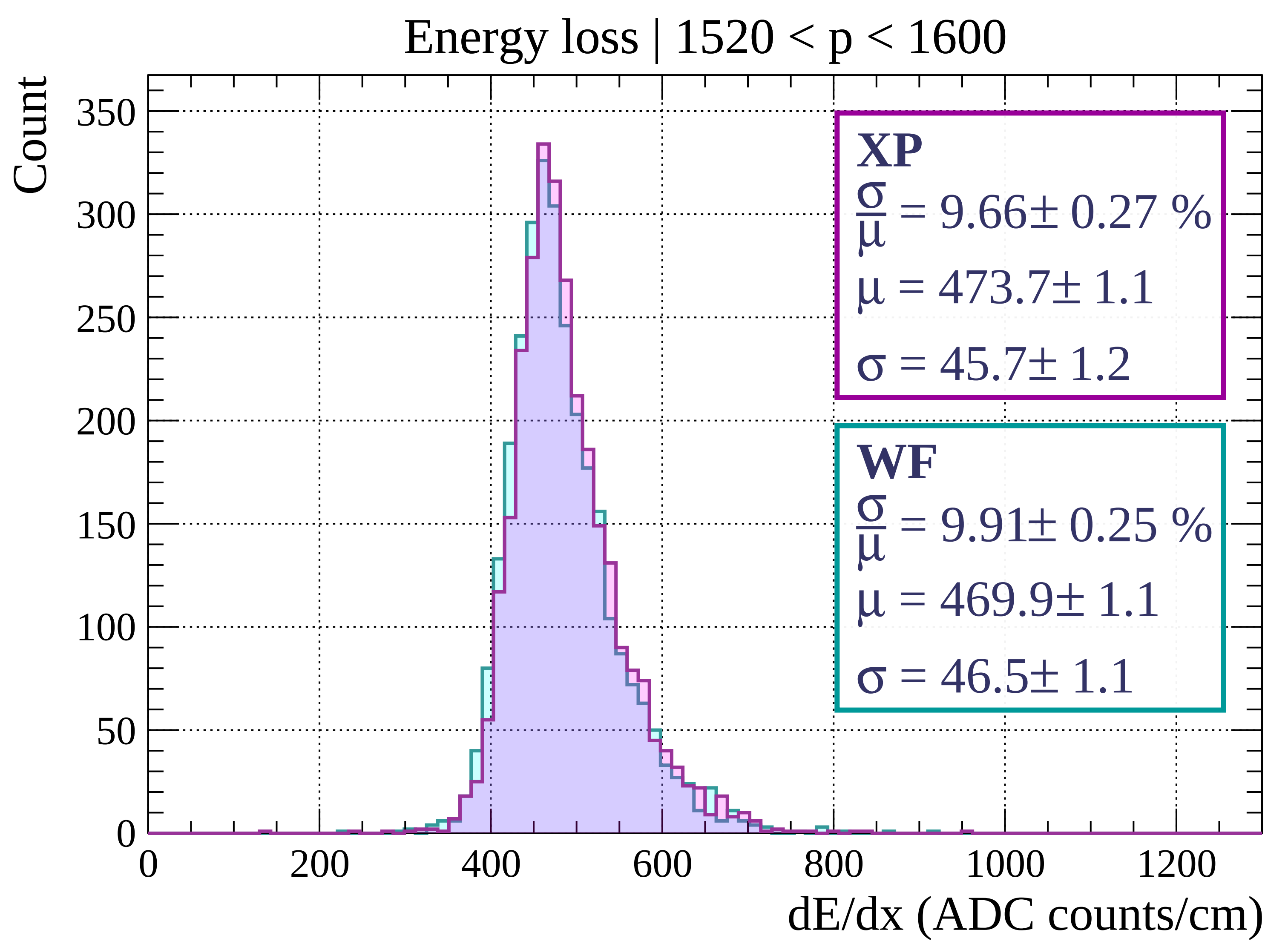


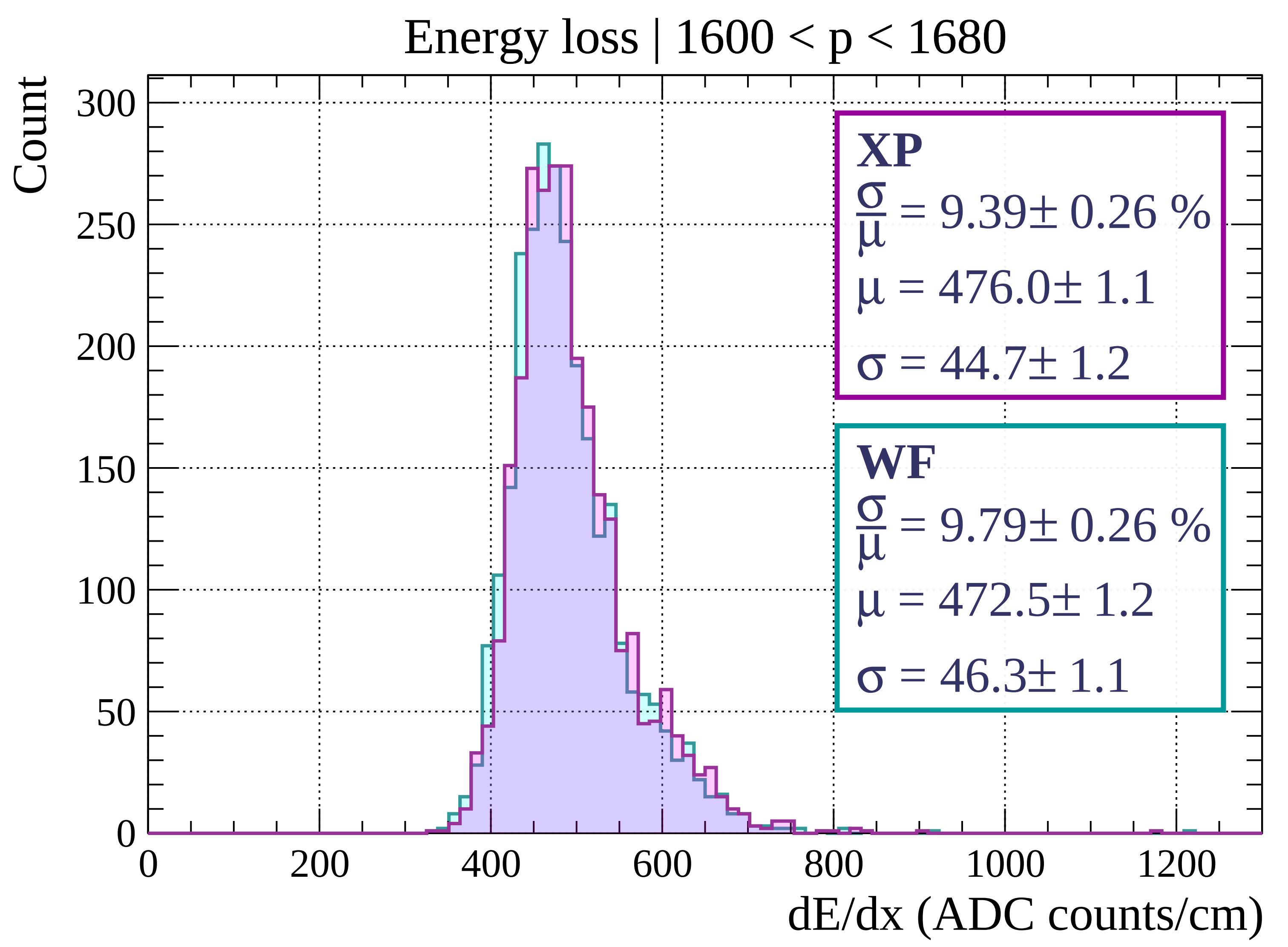
Energy loss | 1280 < p < 1360
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.44± 0.24 %
μ = 460.2± 1.0
σ = 43.5± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.99± 0.21 %
μ = 455.1± 0.9
σ = 40.9± 0.9



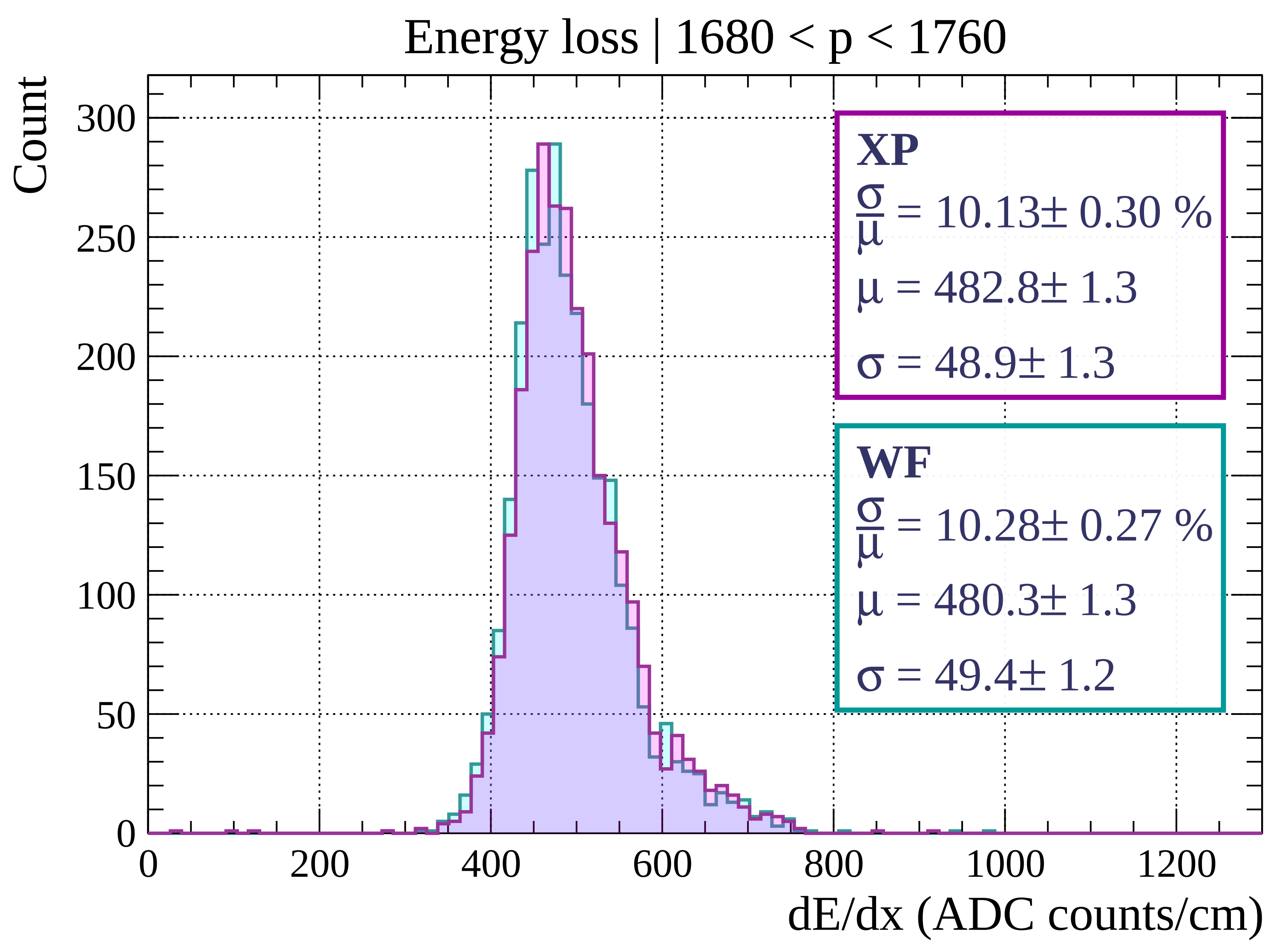
Energy loss | 1360 < p < 1440
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.30± 0.24 %
μ = 464.5± 1.0
σ = 43.2± 1.0
WF
$\frac{\sigma}{\mu}$ = 9.59± 0.23 %
μ = 461.5± 1.0
σ = 44.3± 0.9


Energy loss | 1440 < p < 1520
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
XP
$\frac{\sigma}{\mu}$ = 9.63± 0.25 %
μ = 470.2± 1.1
σ = 45.3± 1.1
WF
$\frac{\sigma}{\mu}$ = 9.59± 0.21 %
μ = 466.3± 1.1
σ = 44.7± 0.9

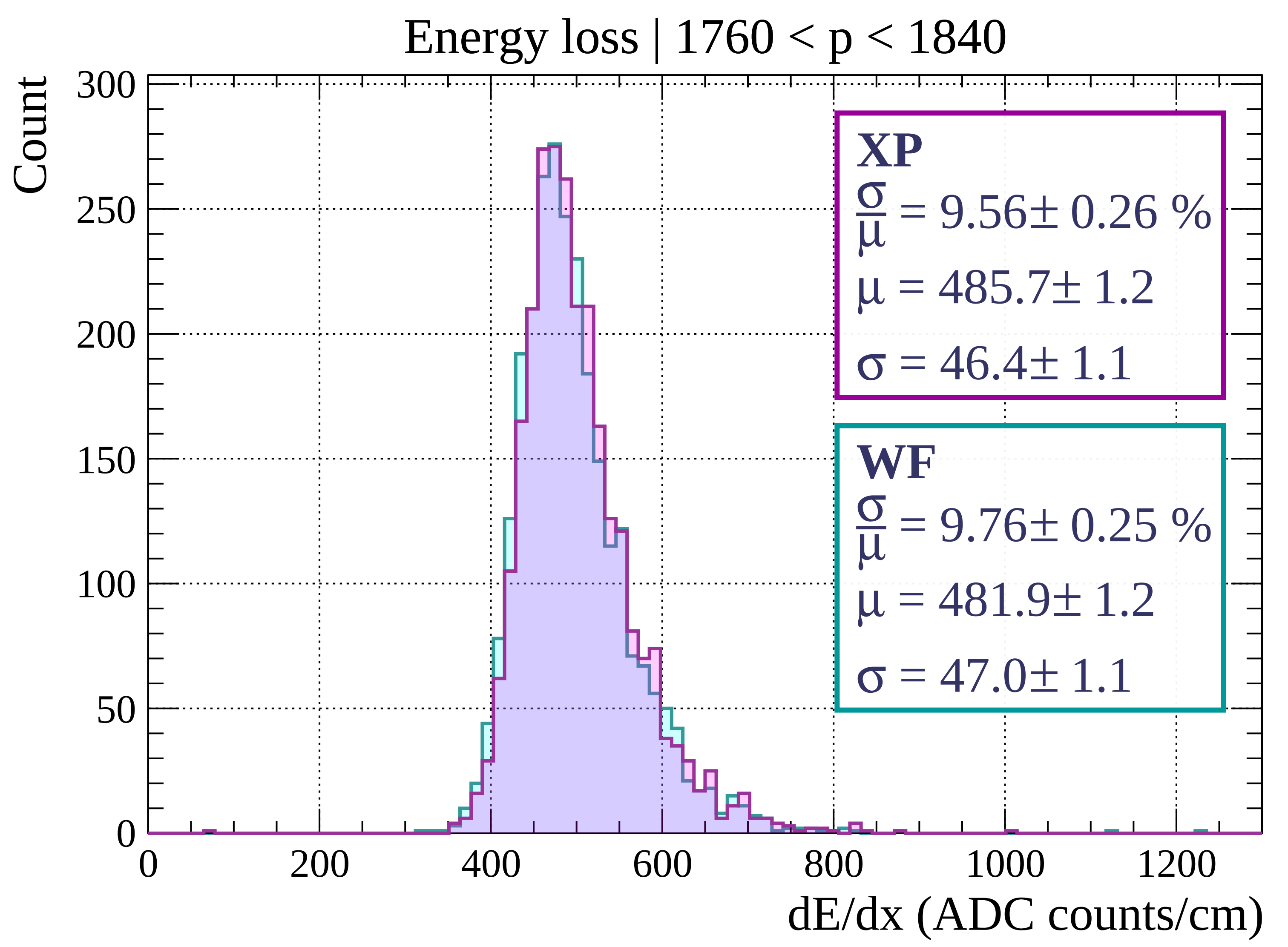

Energy loss | 1520 < p < 1600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
XP
$\frac{\sigma}{\mu}$ = 9.66± 0.27 %
μ = 473.7± 1.1
σ = 45.7± 1.2
WF
$\frac{\sigma}{\mu}$ = 9.91± 0.25 %
μ = 469.9± 1.1
σ = 46.5± 1.1



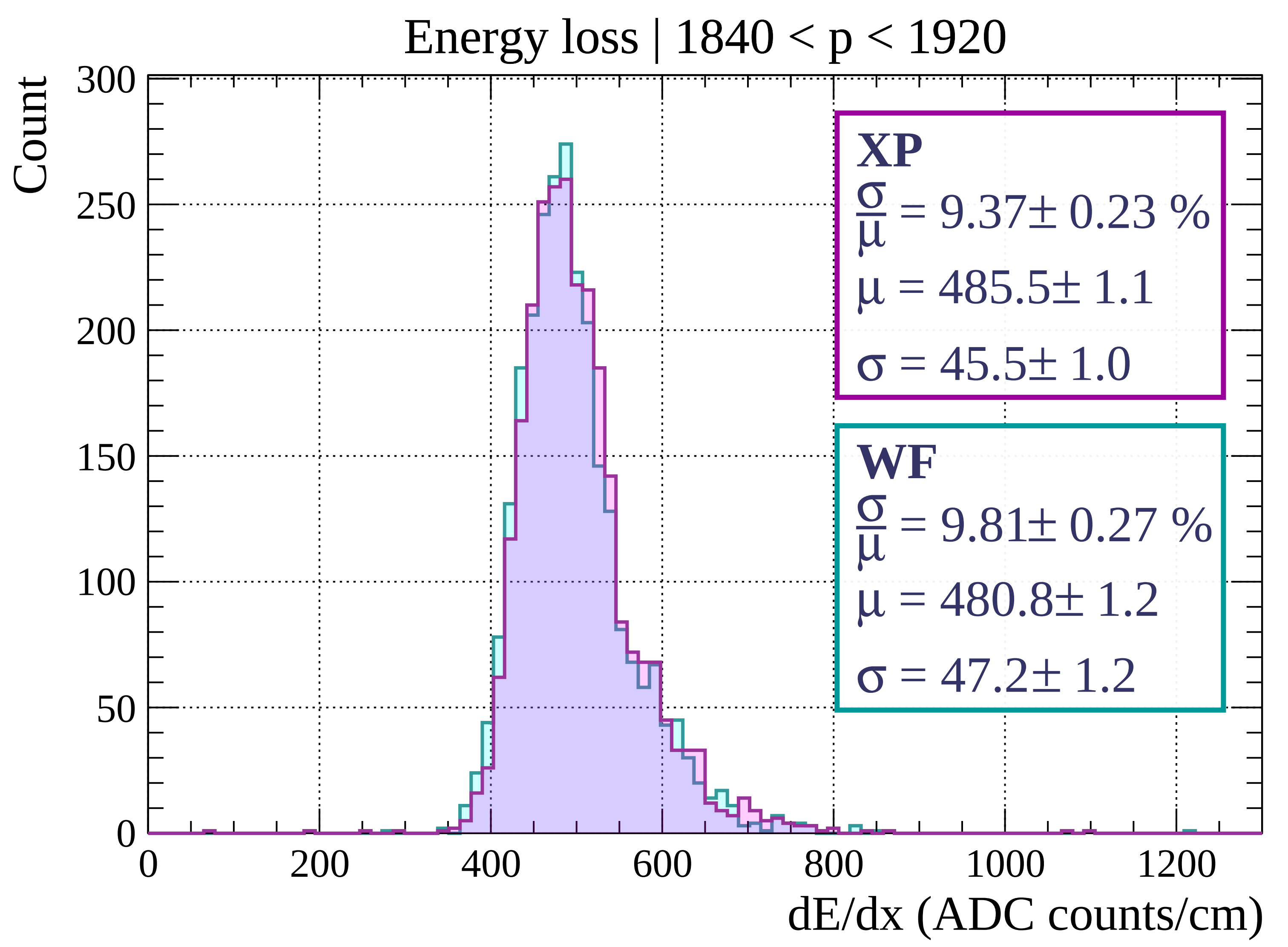
Energy loss | 1600 < p < 1680
Count
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.39± 0.26 %
μ = 476.0± 1.1
σ = 44.7± 1.2
WF
σ/μ = 9.79± 0.26 %
μ = 472.5± 1.2
σ = 46.3± 1.1



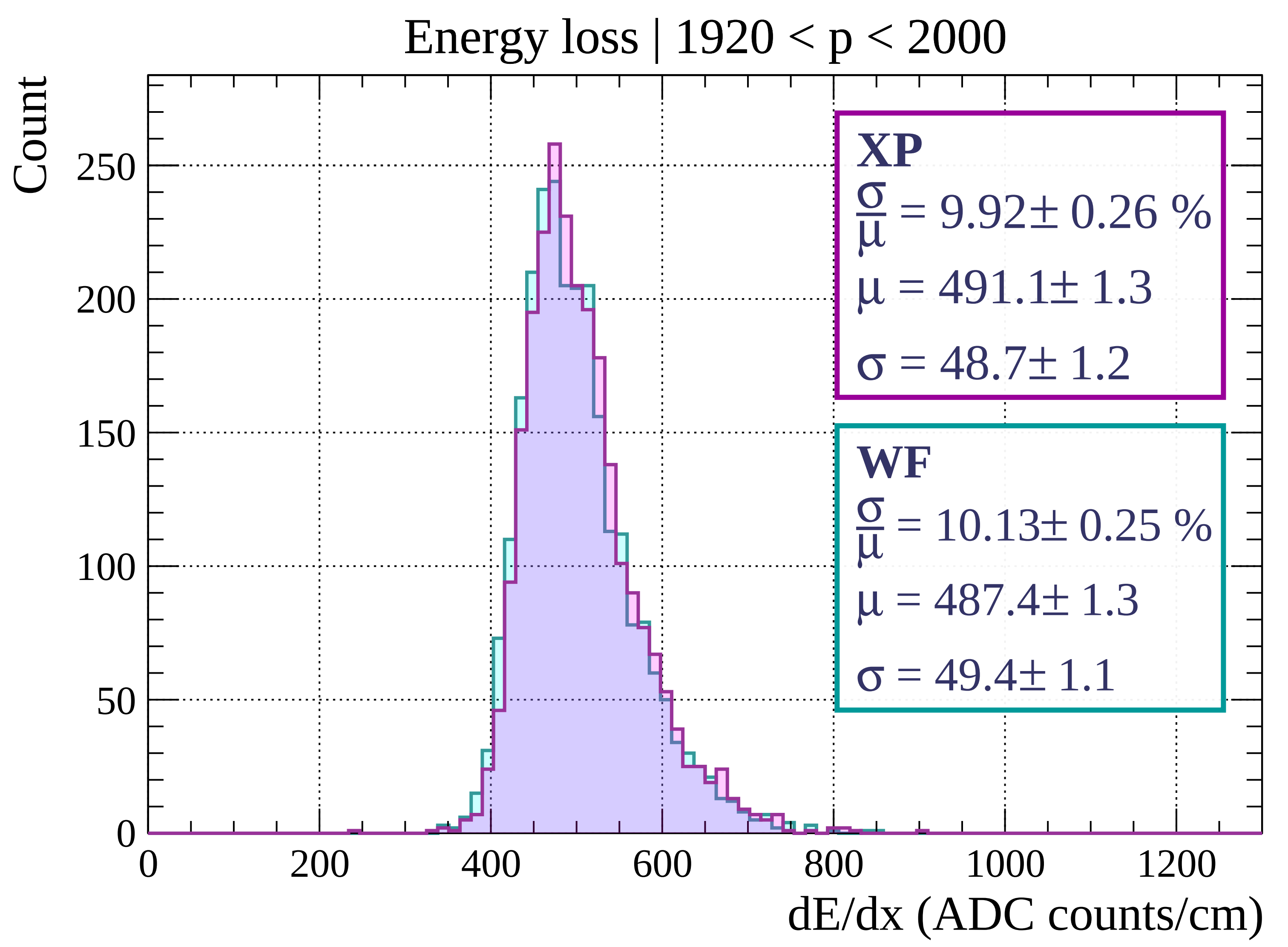
Energy loss | 1680 < p < 1760
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.13± 0.30 %
μ = 482.8± 1.3
σ = 48.9± 1.3
WF
σ/μ = 10.28± 0.27 %
μ = 480.3± 1.3
σ = 49.4± 1.2



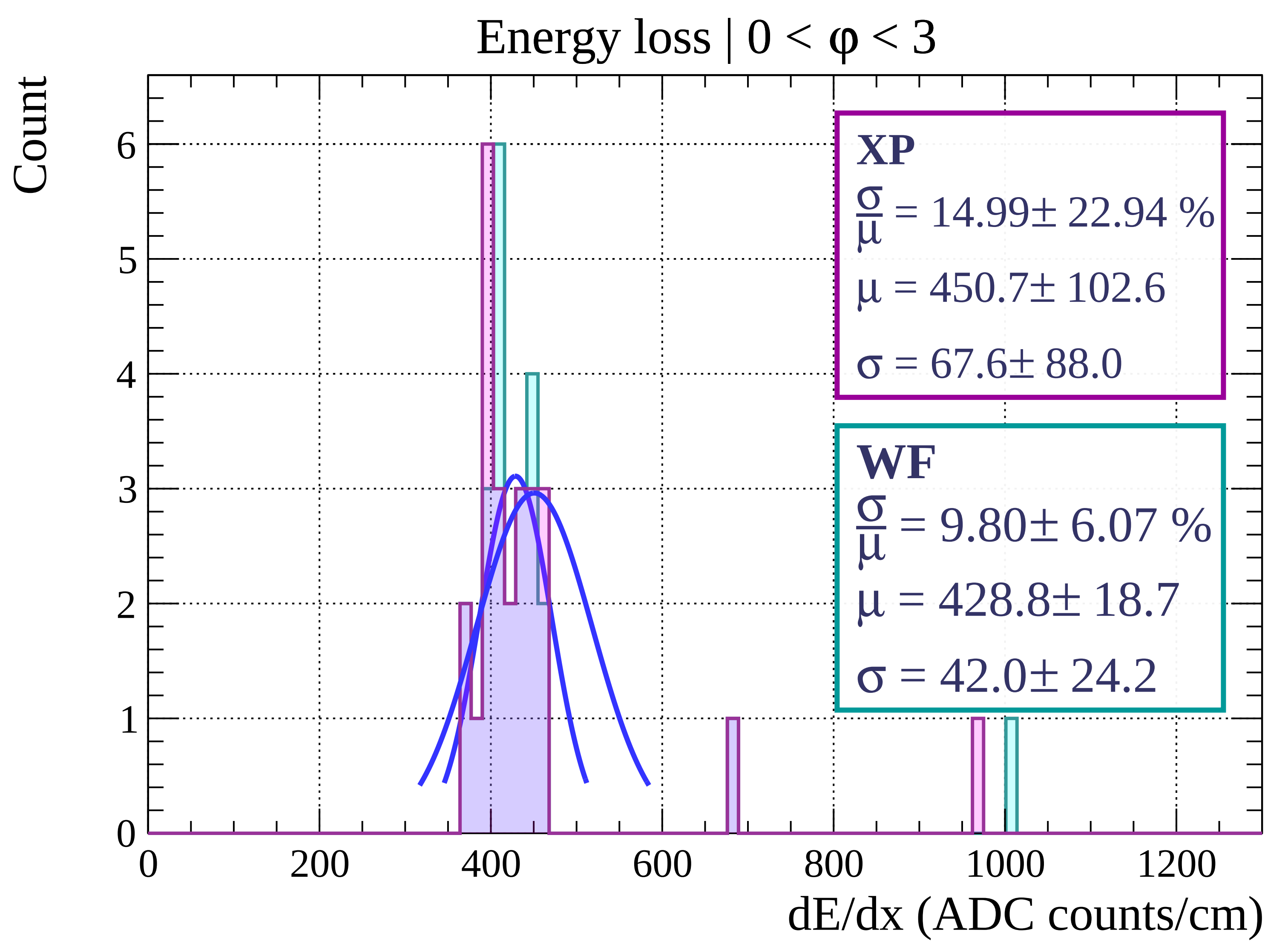
Energy loss | 1760 < p < 1840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
XP
σ/μ = 9.56± 0.26 %
μ = 485.7± 1.2
σ = 46.4± 1.1
WF
σ/μ = 9.76± 0.25 %
μ = 481.9± 1.2
σ = 47.0± 1.1



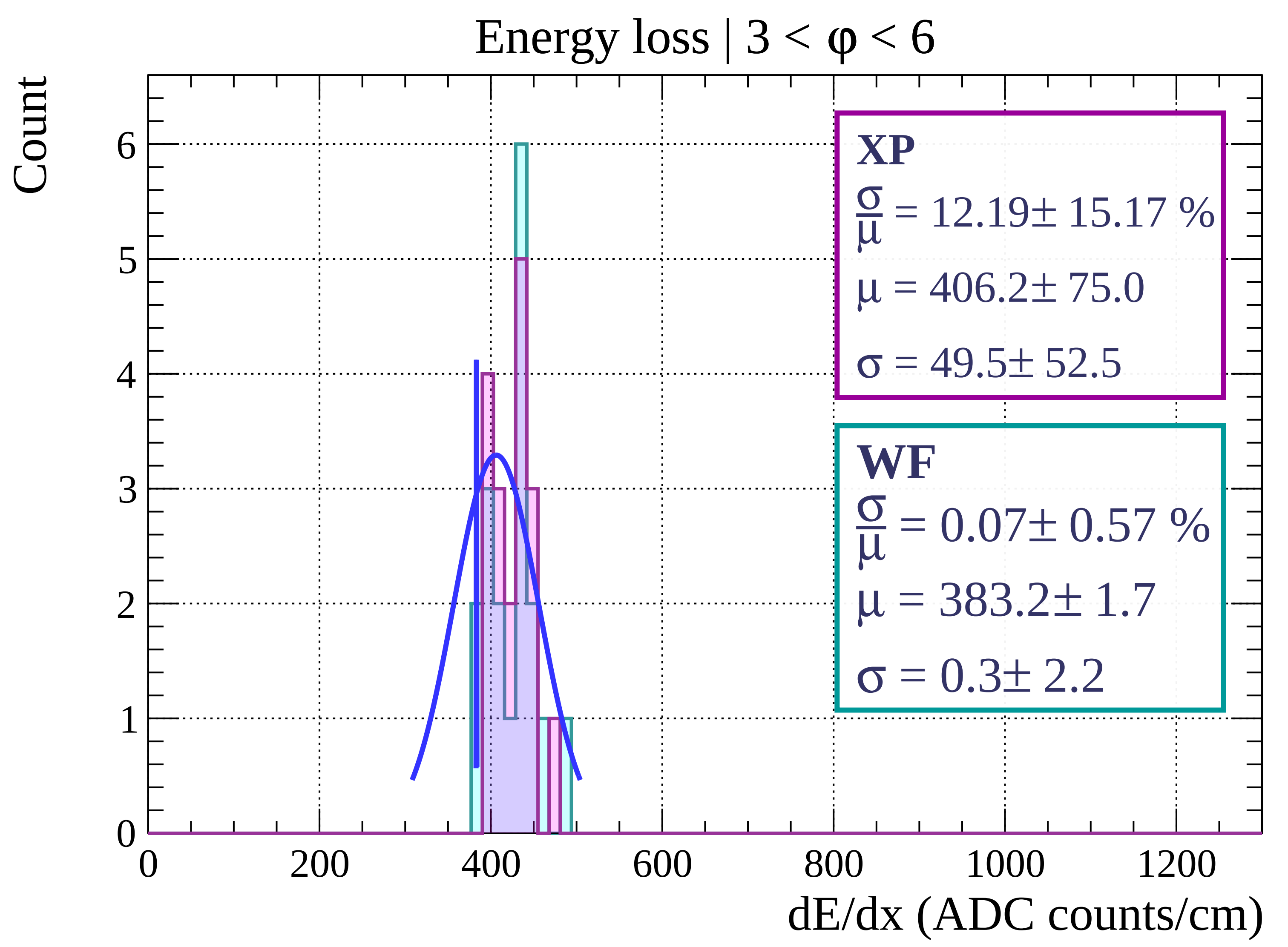
Energy loss | 1840 < p < 1920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
XP
σ/μ = 9.37± 0.23 %
μ = 485.5± 1.1
σ = 45.5± 1.0
WF
σ/μ = 9.81± 0.27 %
μ = 480.8± 1.2
σ = 47.2± 1.2



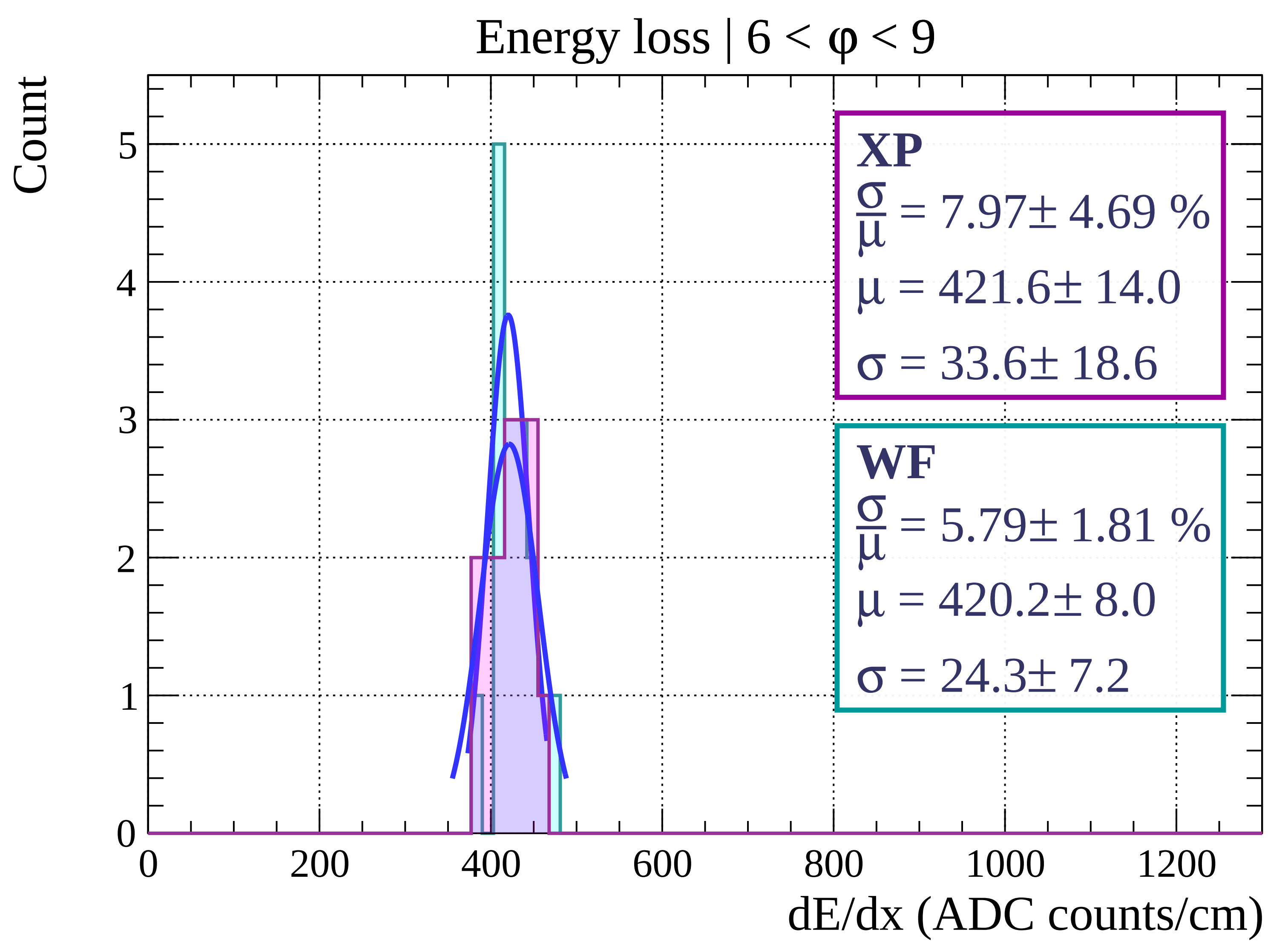
Energy loss | 1920 < p < 2000
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.92± 0.26 %
μ = 491.1± 1.3
σ = 48.7± 1.2
WF
$\frac{\sigma}{\mu}$ = 10.13± 0.25 %
μ = 487.4± 1.3
σ = 49.4± 1.1

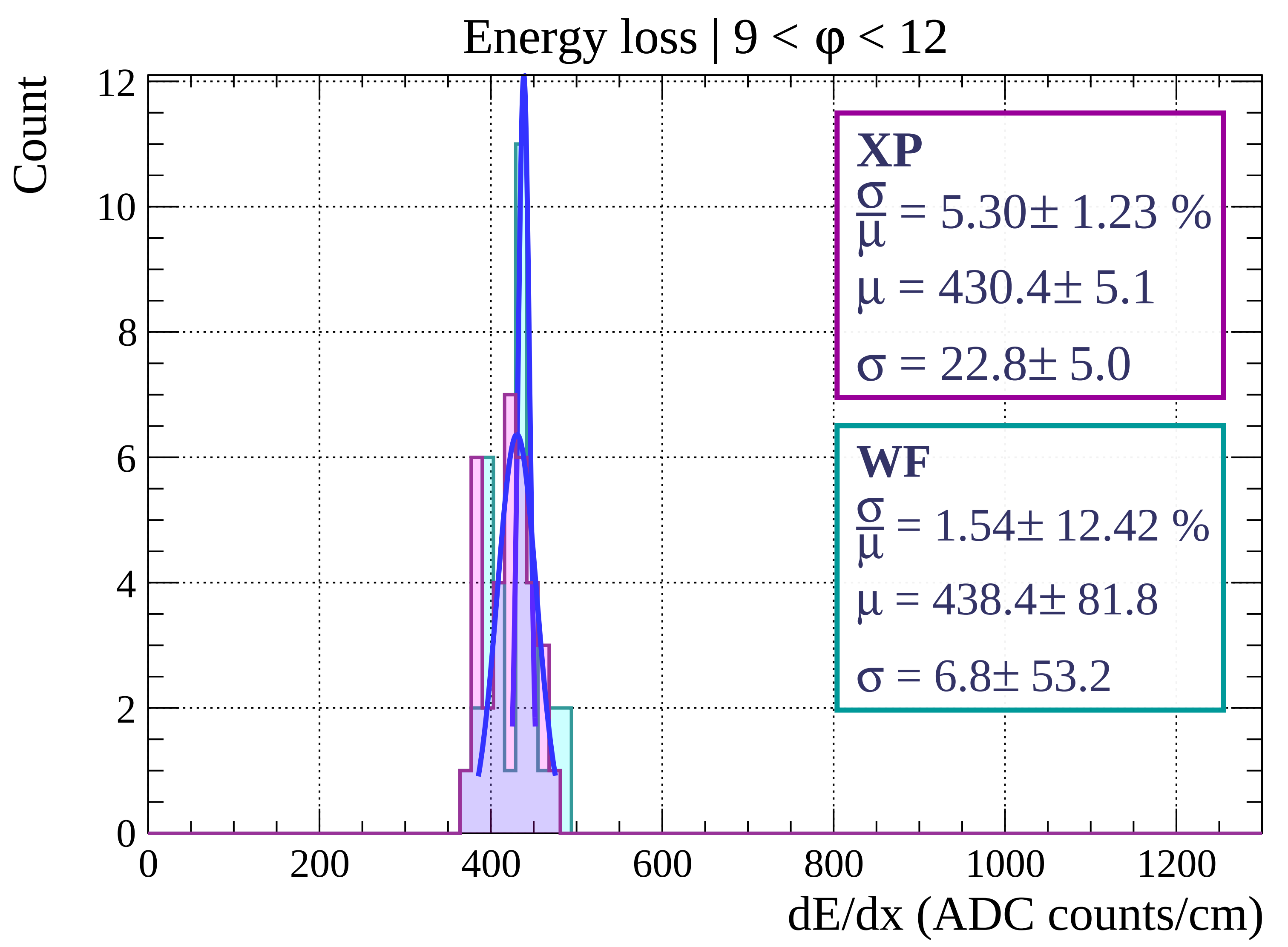


Energy loss | 0 < φ < 3
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
XP
σ/μ = 14.99± 22.94 %
μ = 450.7± 102.6
σ = 67.6± 88.0
WF
σ/μ = 9.80± 6.07 %
μ = 428.8± 18.7
σ = 42.0± 24.2

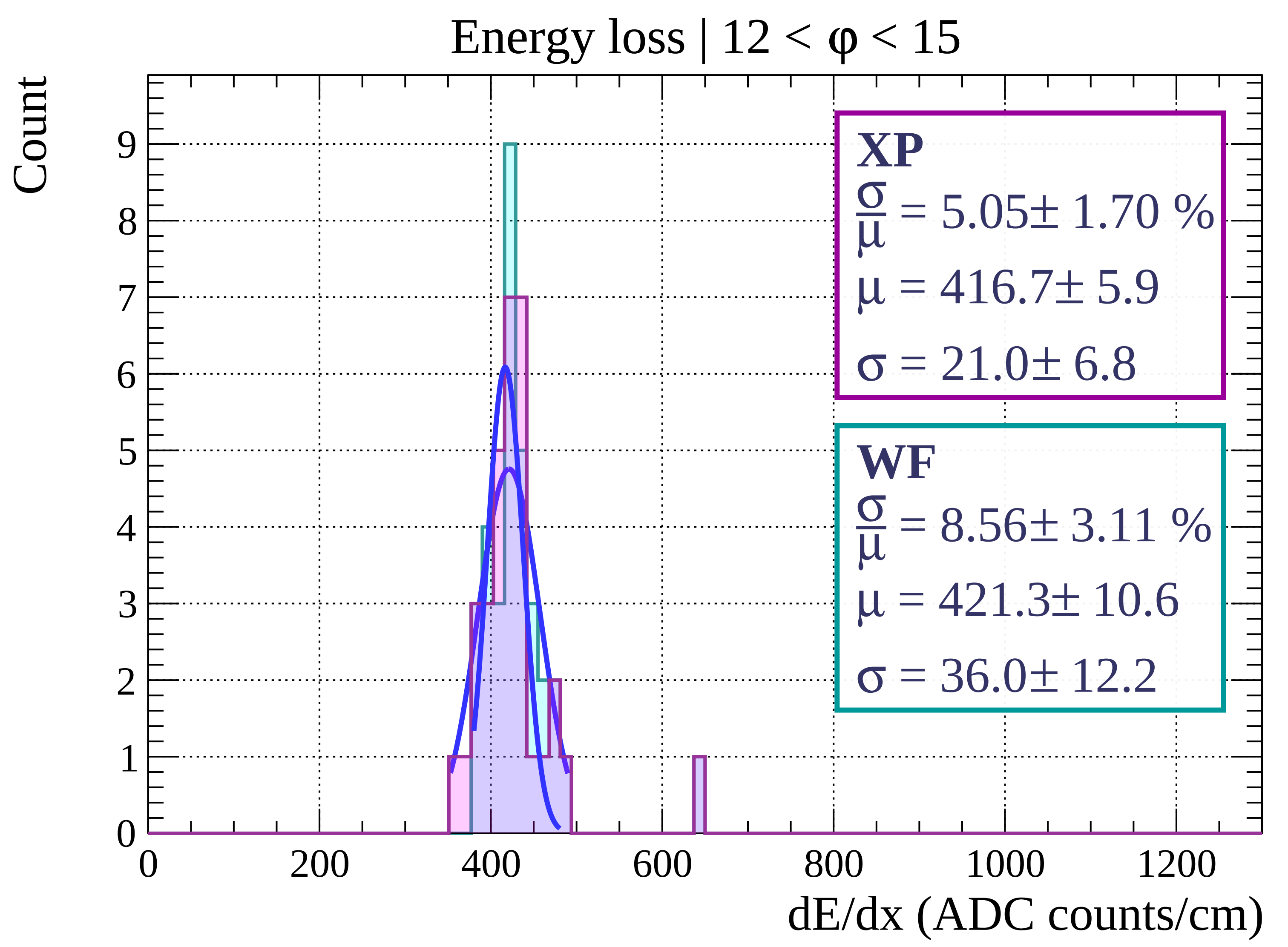


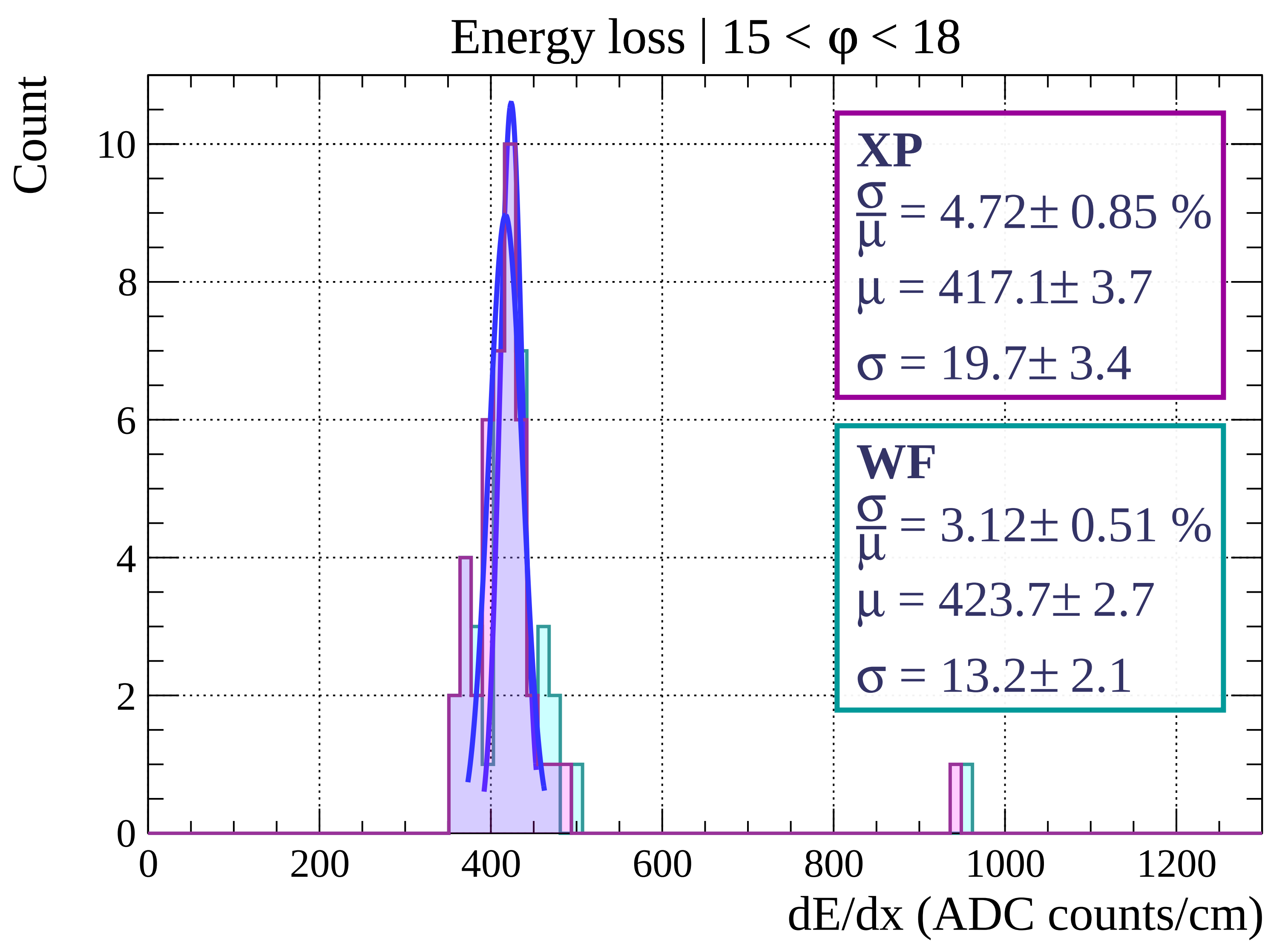
Energy loss | 3 < φ < 6
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
XP
$\frac{\sigma}{\mu}$ = 12.19± 15.17 %
μ = 406.2± 75.0
σ = 49.5± 52.5
WF
$\frac{\sigma}{\mu}$ = 0.07± 0.57 %
μ = 383.2± 1.7
σ = 0.3± 2.2

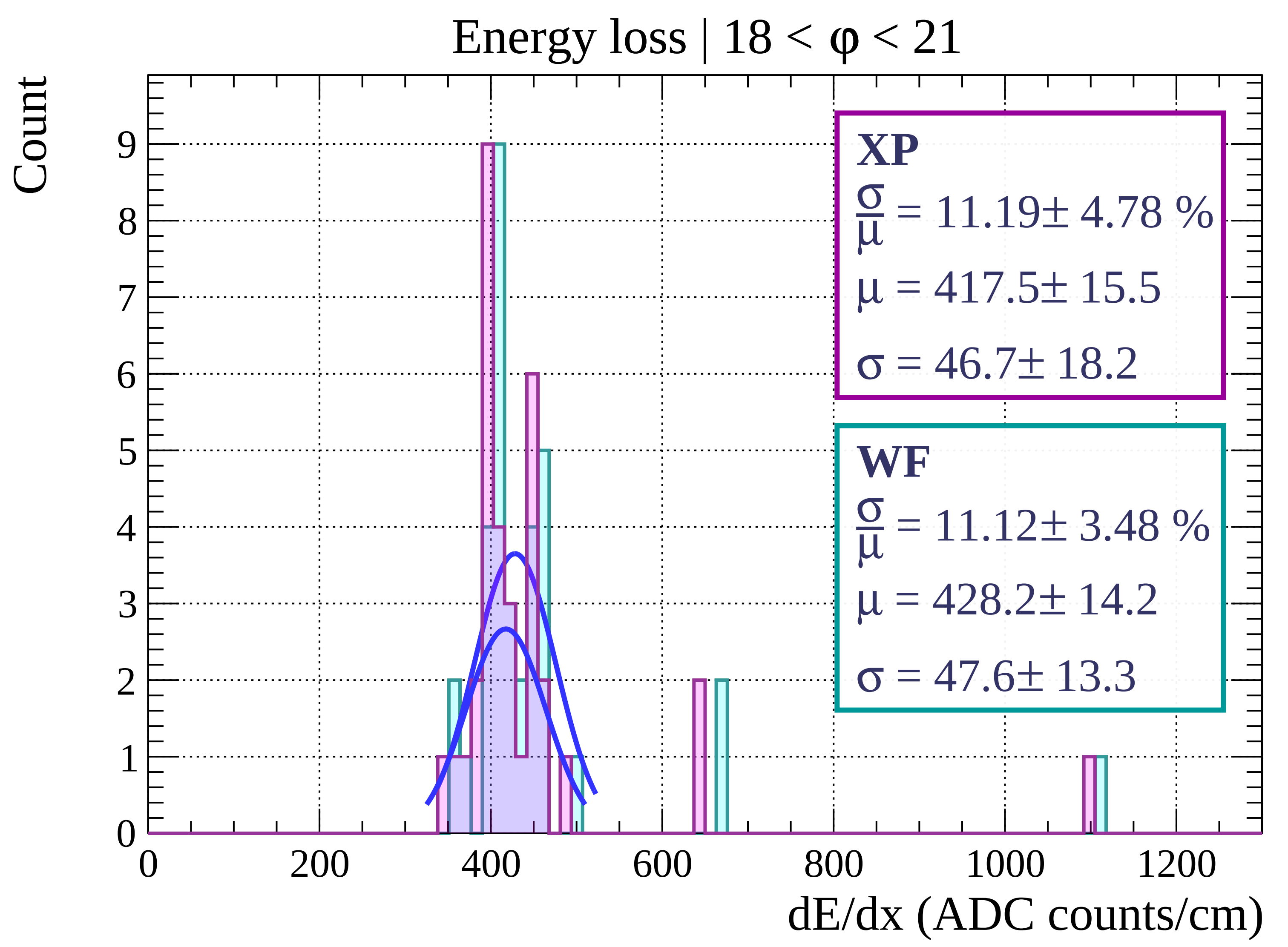

Energy loss | 6 < φ < 9
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.97± 4.69 %
μ = 421.6± 14.0
σ = 33.6± 18.6
WF
$\frac{\sigma}{\mu}$ = 5.79± 1.81 %
μ = 420.2± 8.0
σ = 24.3± 7.2



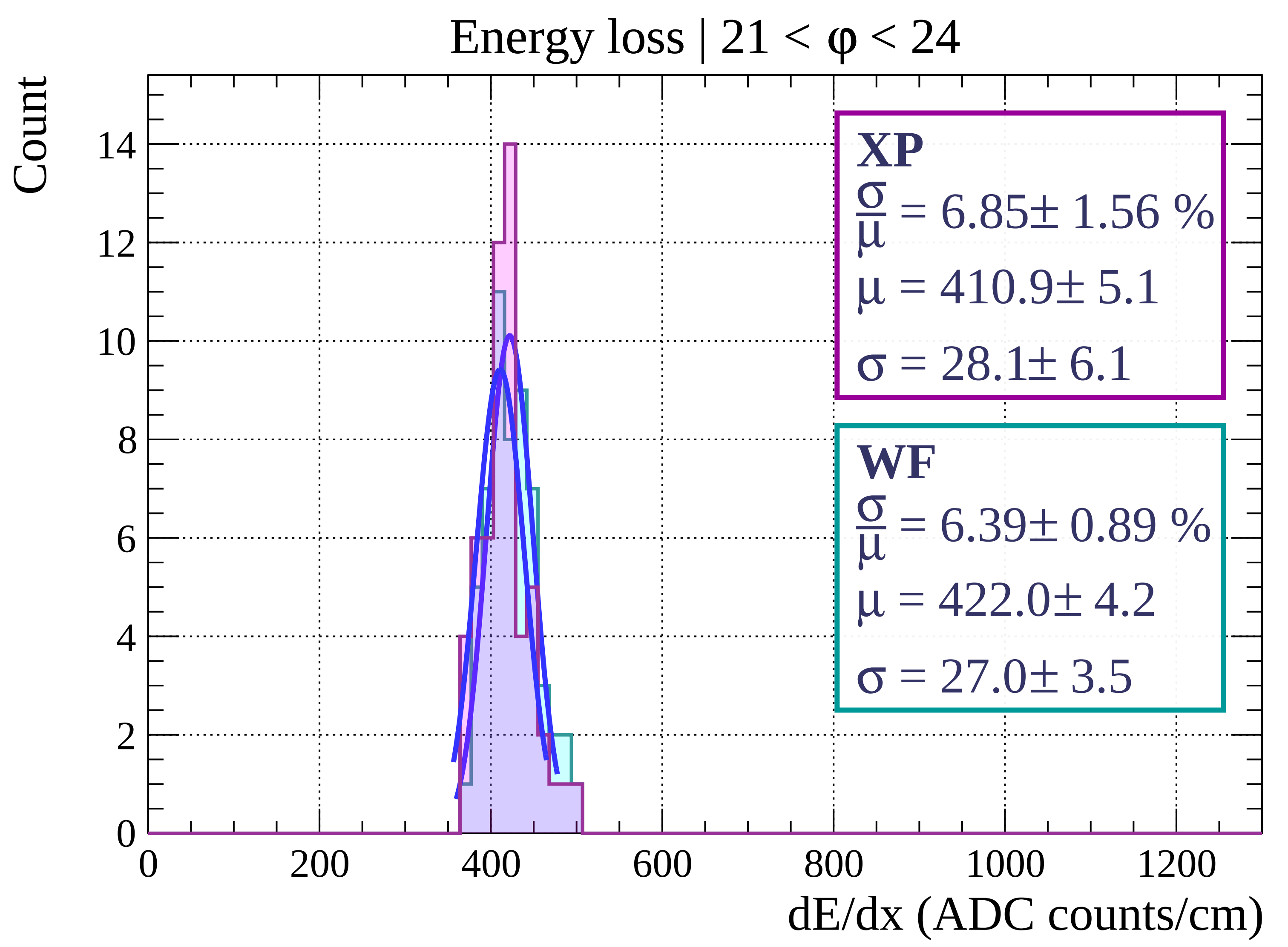
Energy loss | 9 < φ < 12
Count
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 5.30± 1.23 %
μ = 430.4± 5.1
σ = 22.8± 5.0
WF
$\frac{\sigma}{\mu}$ = 1.54± 12.42 %
μ = 438.4± 81.8
σ = 6.8± 53.2


Energy loss | 12 < φ < 15
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
9
XP
σ/μ = 5.05± 1.70 %
μ = 416.7± 5.9
σ = 21.0± 6.8
WF
σ/μ = 8.56± 3.11 %
μ = 421.3± 10.6
σ = 36.0± 12.2

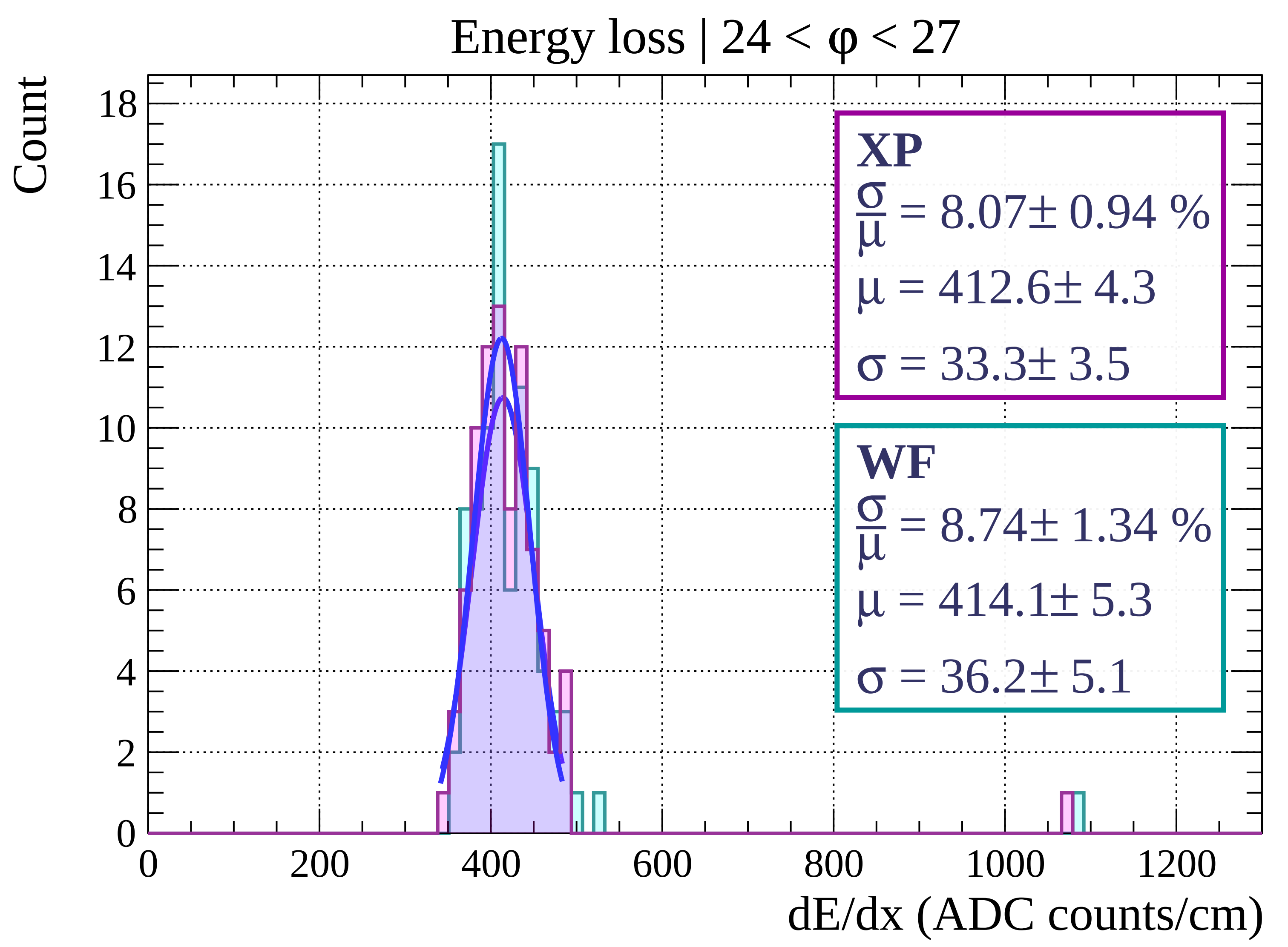

Energy loss | 15 < φ < 18
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
XP
σ/μ = 4.72± 0.85 %
μ = 417.1± 3.7
σ = 19.7± 3.4
WF
σ/μ = 3.12± 0.51 %
μ = 423.7± 2.7
σ = 13.2± 2.1

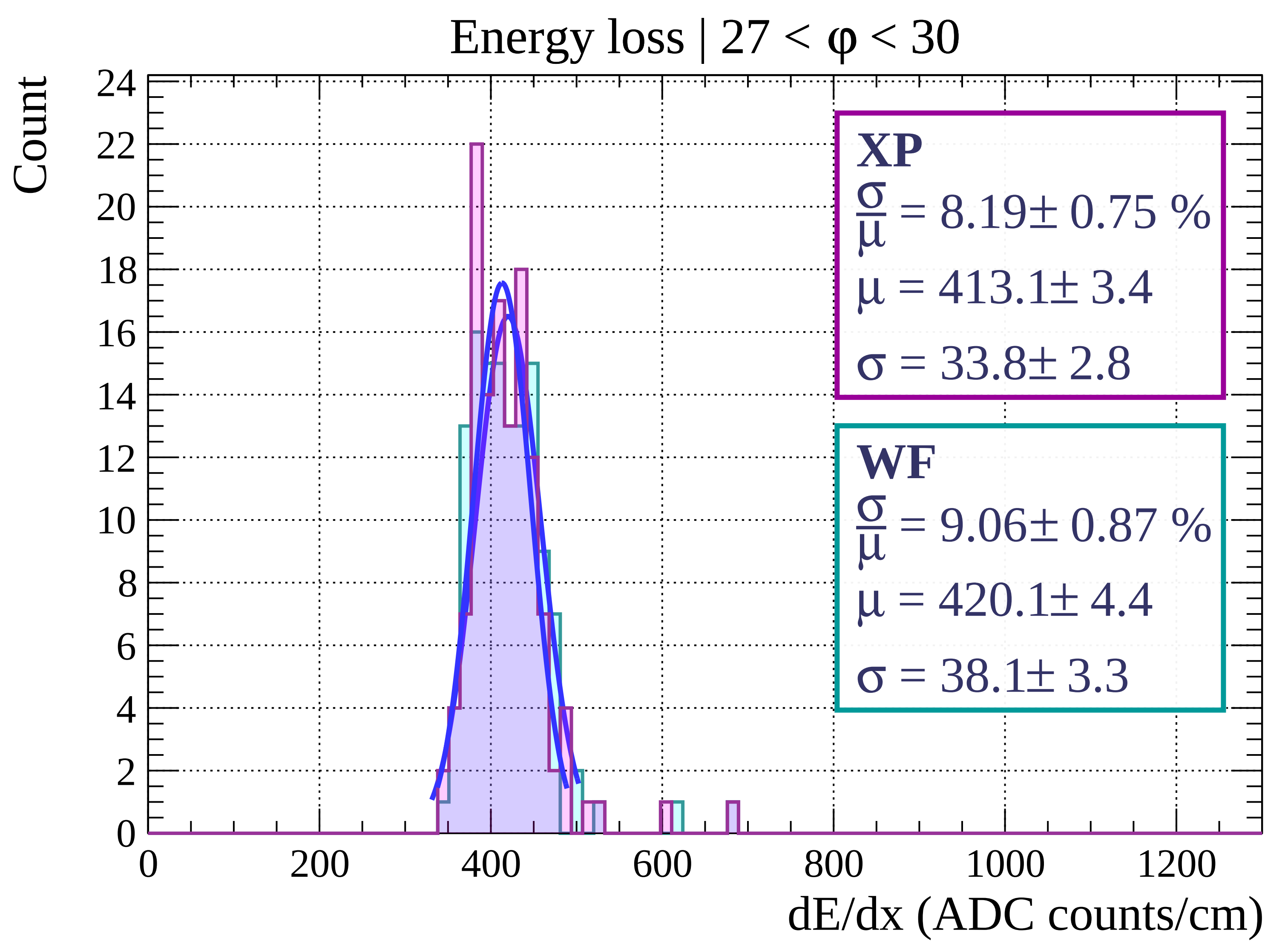


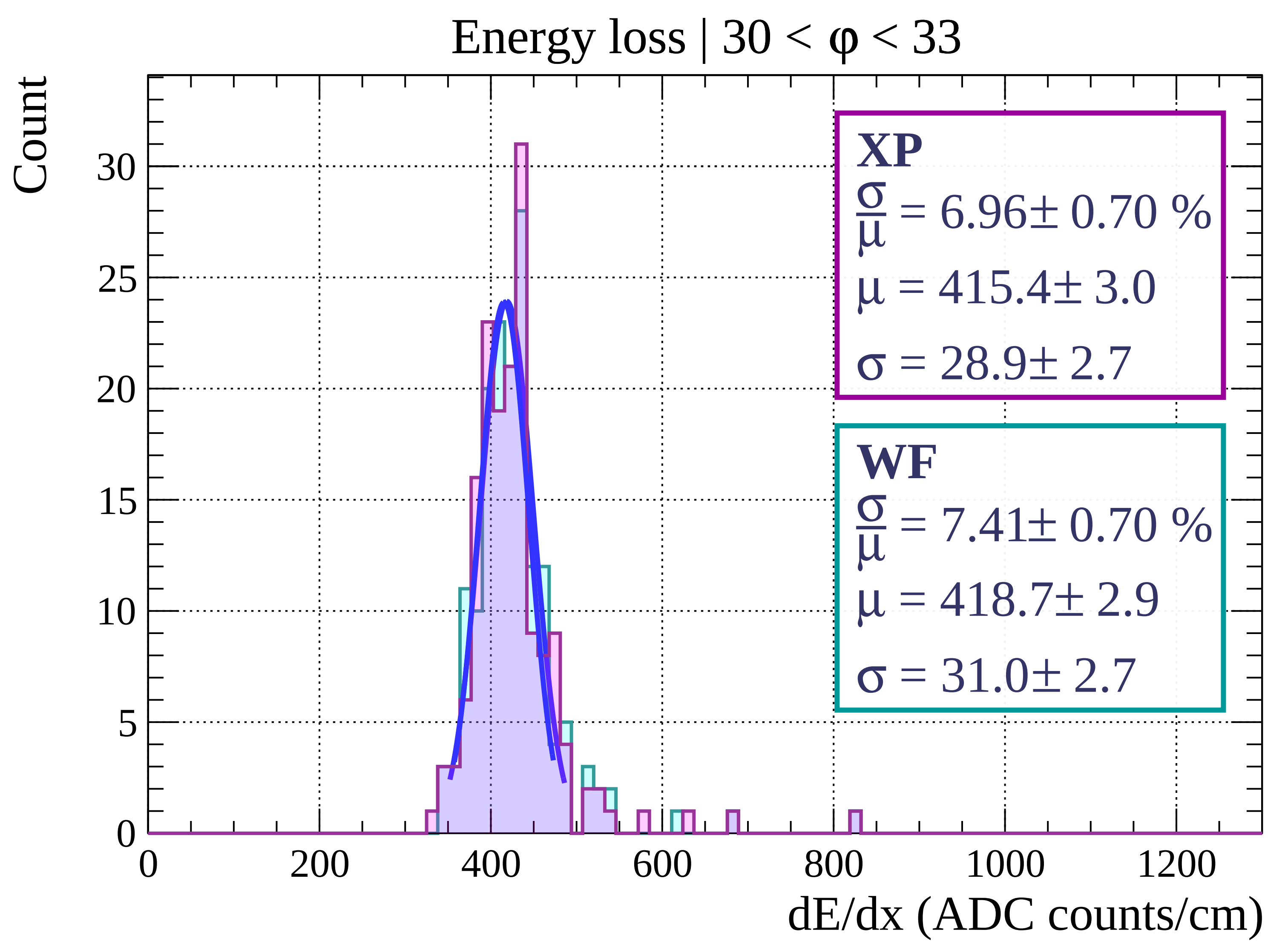
Energy loss | 18 < φ < 21
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.19± 4.78 %
μ = 417.5± 15.5
σ = 46.7± 18.2
WF
σ/μ = 11.12± 3.48 %
μ = 428.2± 14.2
σ = 47.6± 13.3

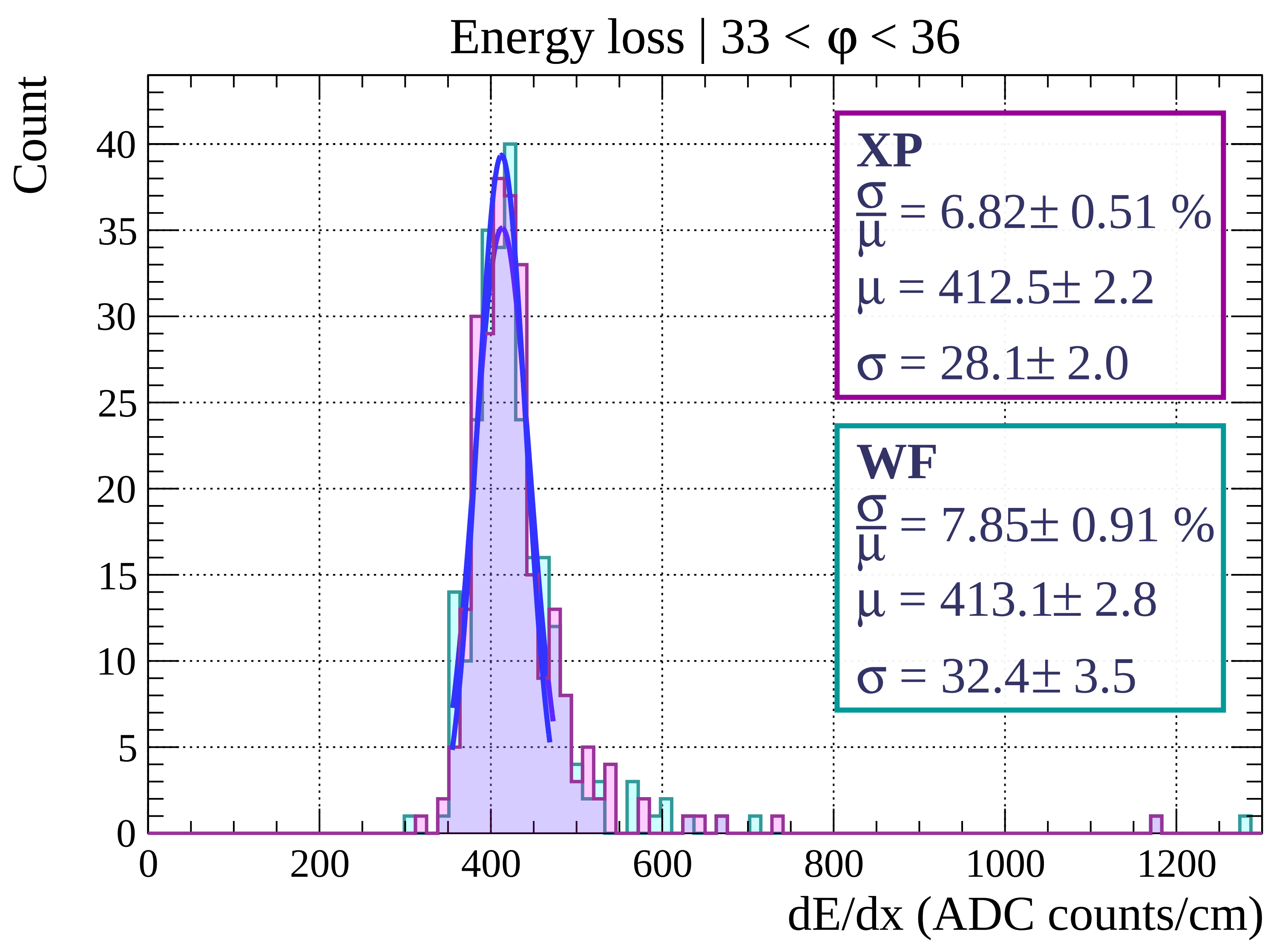


Energy loss | 21 < φ < 24
Count
14
12
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.85± 1.56 %
μ = 410.9± 5.1
σ = 28.1± 6.1
WF
$\frac{\sigma}{\mu}$ = 6.39± 0.89 %
μ = 422.0± 4.2
σ = 27.0± 3.5

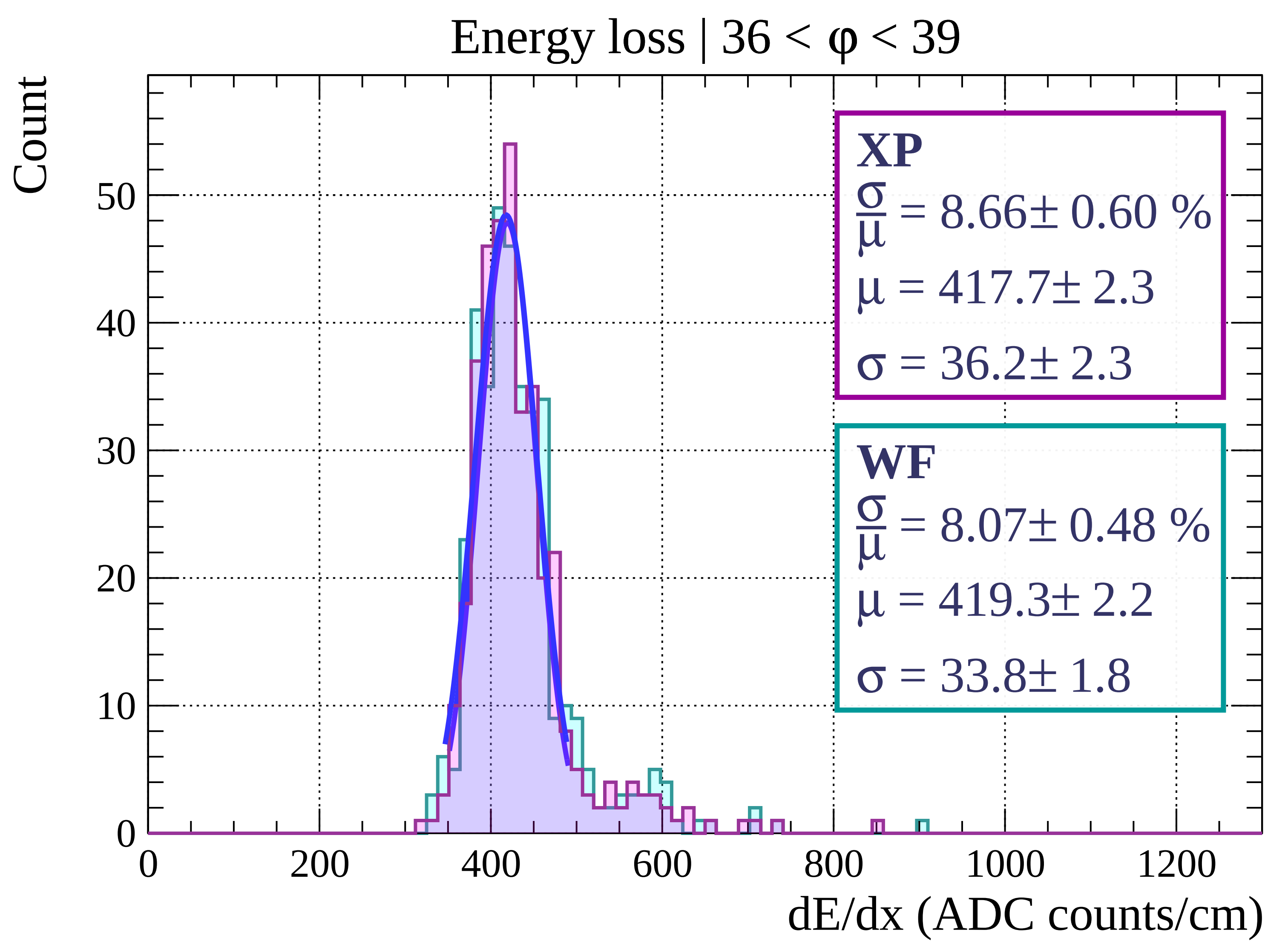

Energy loss | 24 < φ < 27
Count
18
16
14
12
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.07± 0.94 %
μ = 412.6± 4.3
σ = 33.3± 3.5
WF
σ/μ = 8.74± 1.34 %
μ = 414.1± 5.3
σ = 36.2± 5.1


Energy loss | 27 < φ < 30
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.19± 0.75 %
μ = 413.1± 3.4
σ = 33.8± 2.8
WF
$\frac{\sigma}{\mu}$ = 9.06± 0.87 %
μ = 420.1± 4.4
σ = 38.1± 3.3

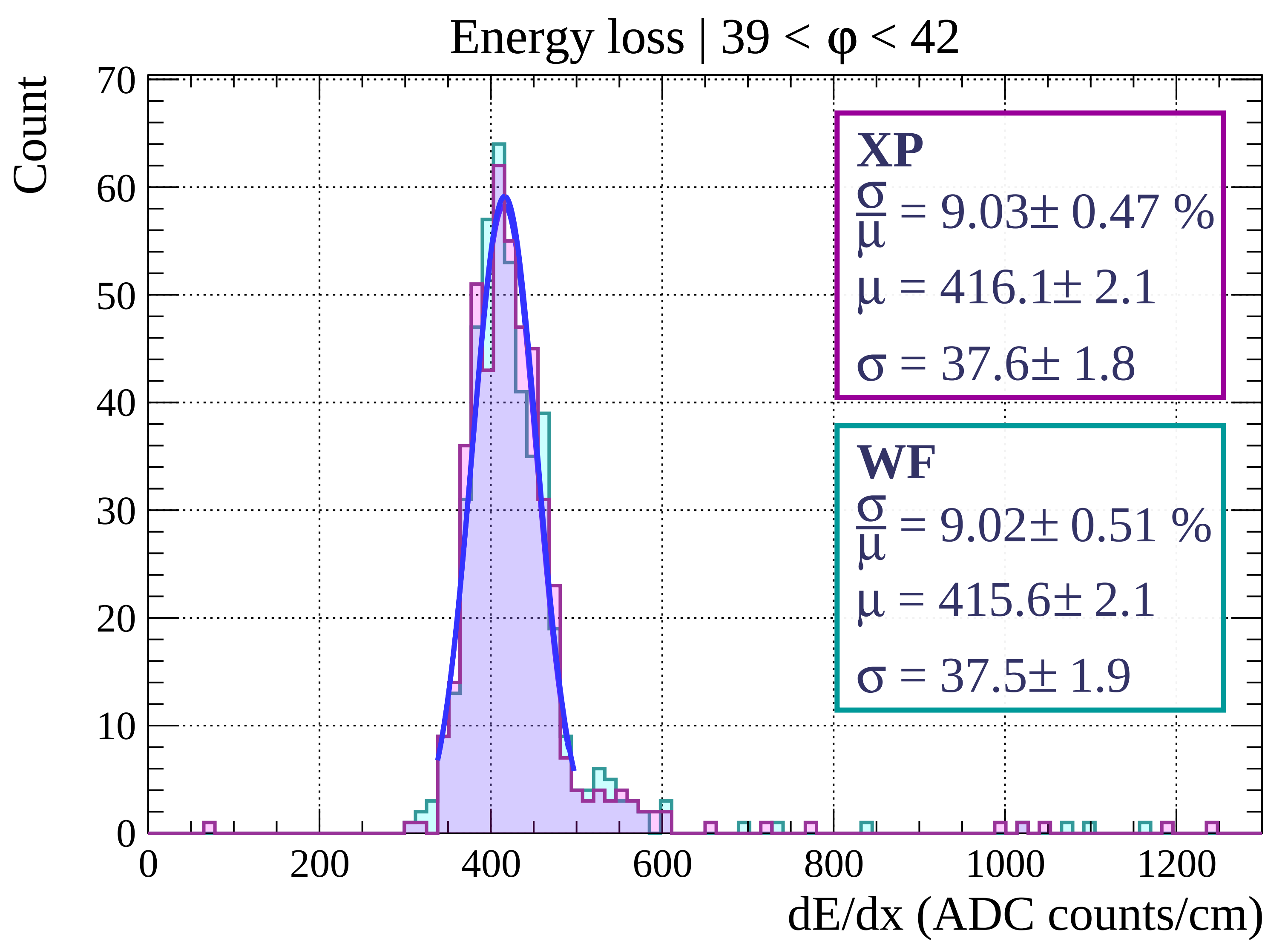

Energy loss | 30 < φ < 33
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
XP
σ/μ = 6.96± 0.70 %
μ = 415.4± 3.0
σ = 28.9± 2.7
WF
σ/μ = 7.41± 0.70 %
μ = 418.7± 2.9
σ = 31.0± 2.7



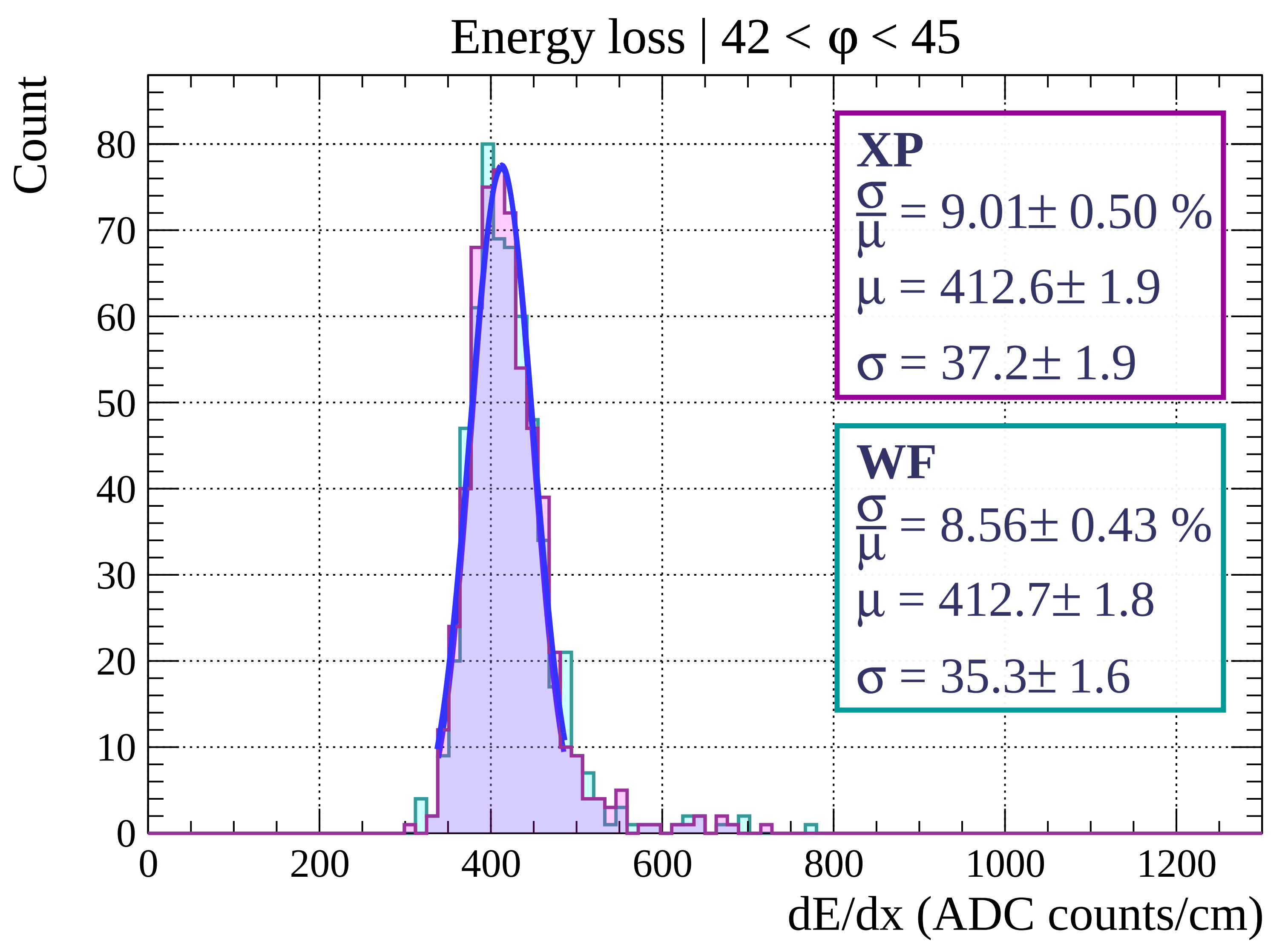
Energy loss | 33 < φ < 36
Count
40
35
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 6.82± 0.51 %
μ = 412.5± 2.2
σ = 28.1± 2.0
WF
σ/μ = 7.85± 0.91 %
μ = 413.1± 2.8
σ = 32.4± 3.5

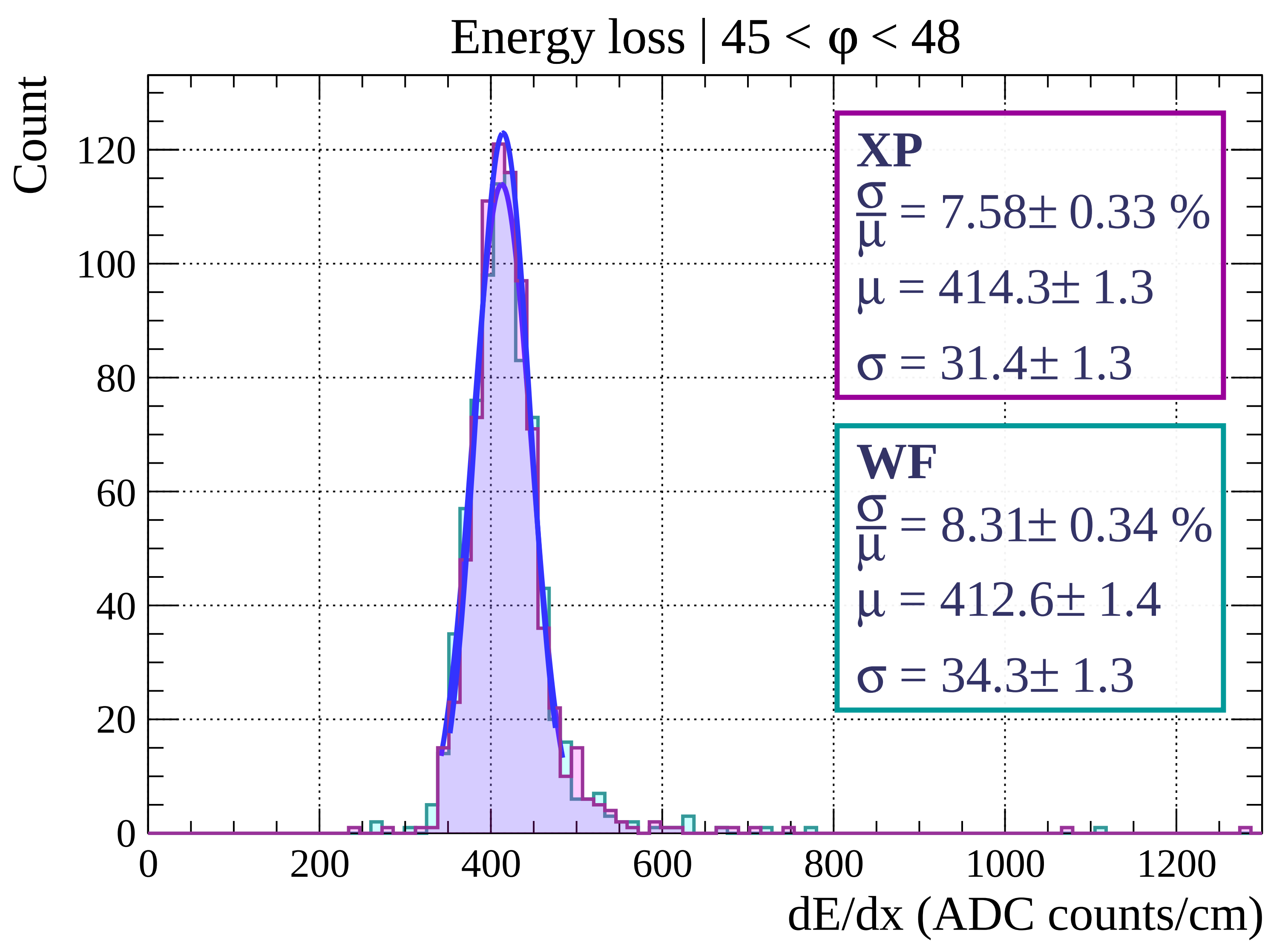


Energy loss | 36 < φ < 39
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.66± 0.60 %
μ = 417.7± 2.3
σ = 36.2± 2.3
WF
$\frac{\sigma}{\mu}$ = 8.07± 0.48 %
μ = 419.3± 2.2
σ = 33.8± 1.8

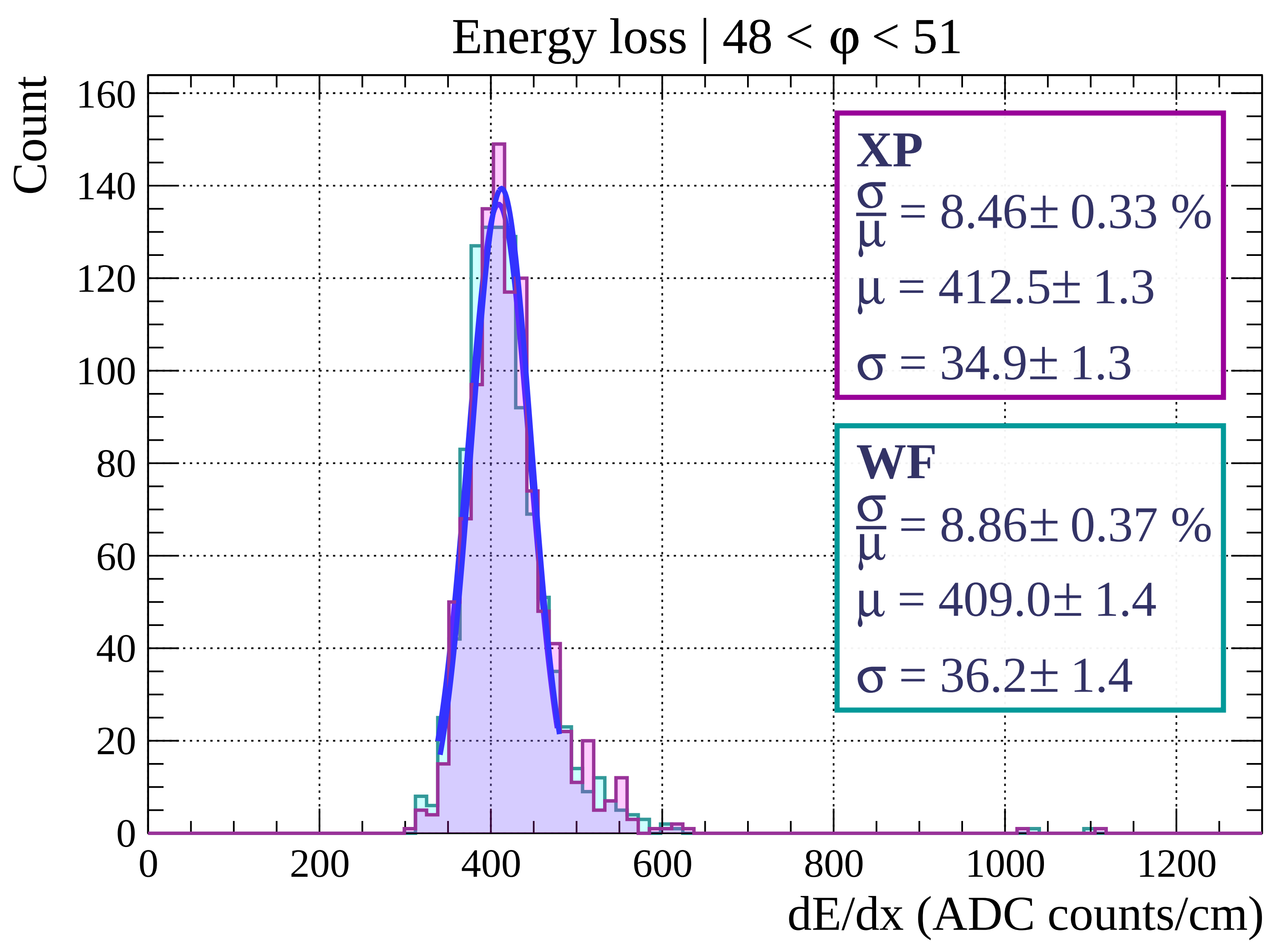


Energy loss | 39 < φ < 42
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
XP
$\frac{\sigma}{\mu}$ = 9.03± 0.47 %
μ = 416.1± 2.1
σ = 37.6± 1.8
WF
$\frac{\sigma}{\mu}$ = 9.02± 0.51 %
μ = 415.6± 2.1
σ = 37.5± 1.9

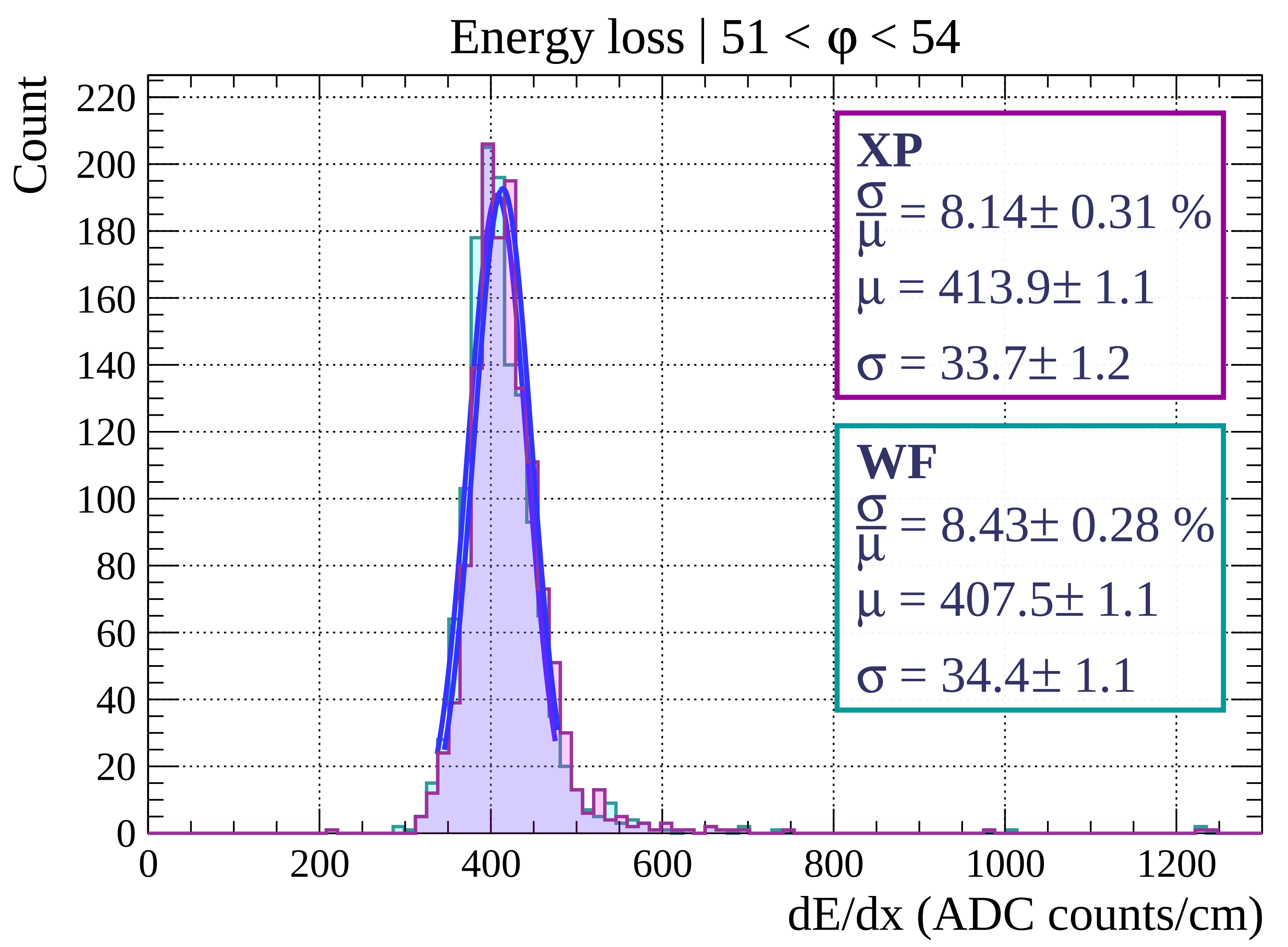

Energy loss | 42 < φ < 45
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.01± 0.50 %
μ = 412.6± 1.9
σ = 37.2± 1.9
WF
σ/μ = 8.56± 0.43 %
μ = 412.7± 1.8
σ = 35.3± 1.6

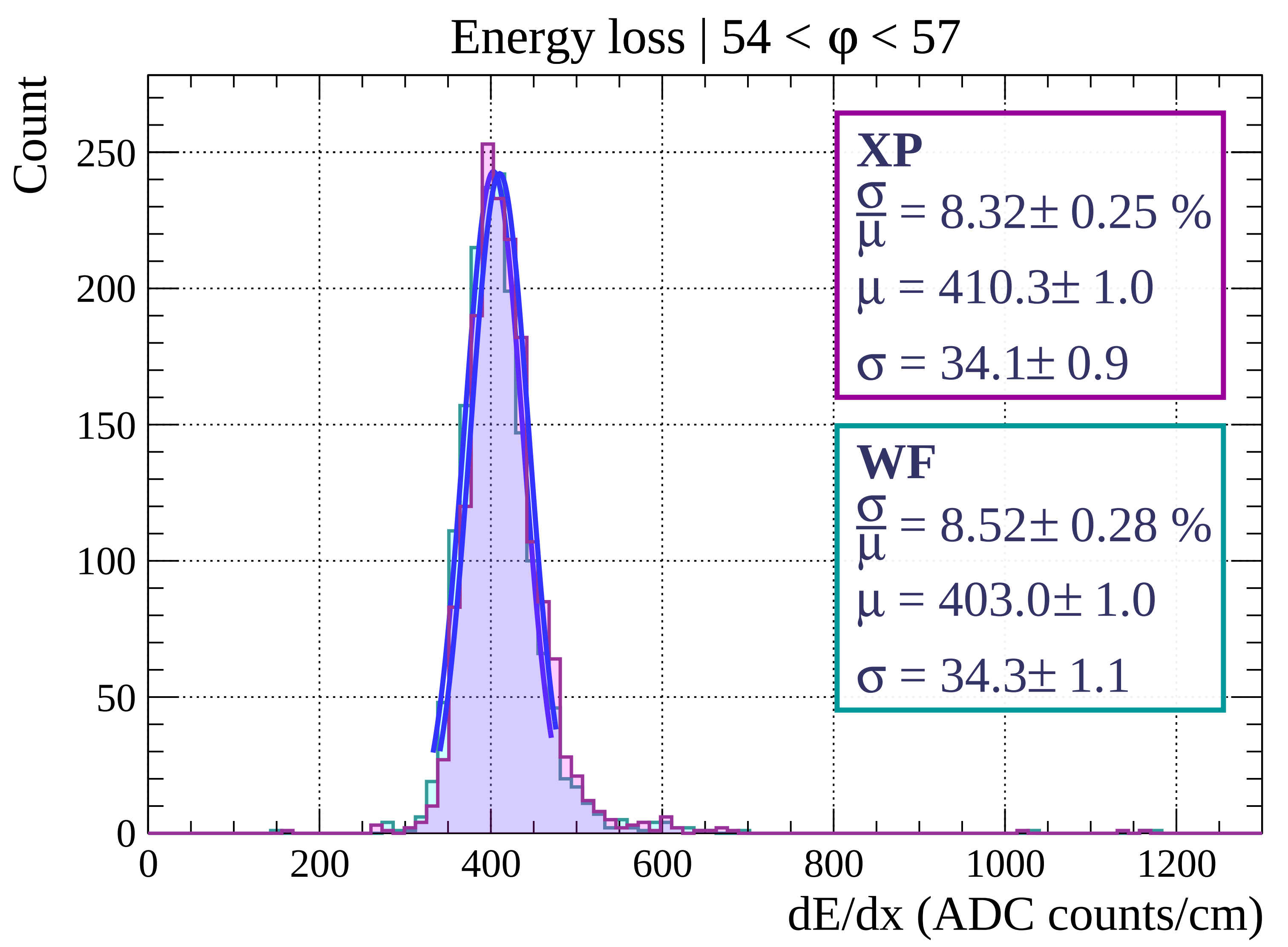


Energy loss | 45 < φ < 48
Count
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.58± 0.33 %
μ = 414.3± 1.3
σ = 31.4± 1.3
WF
$\frac{\sigma}{\mu}$ = 8.31± 0.34 %
μ = 412.6± 1.4
σ = 34.3± 1.3

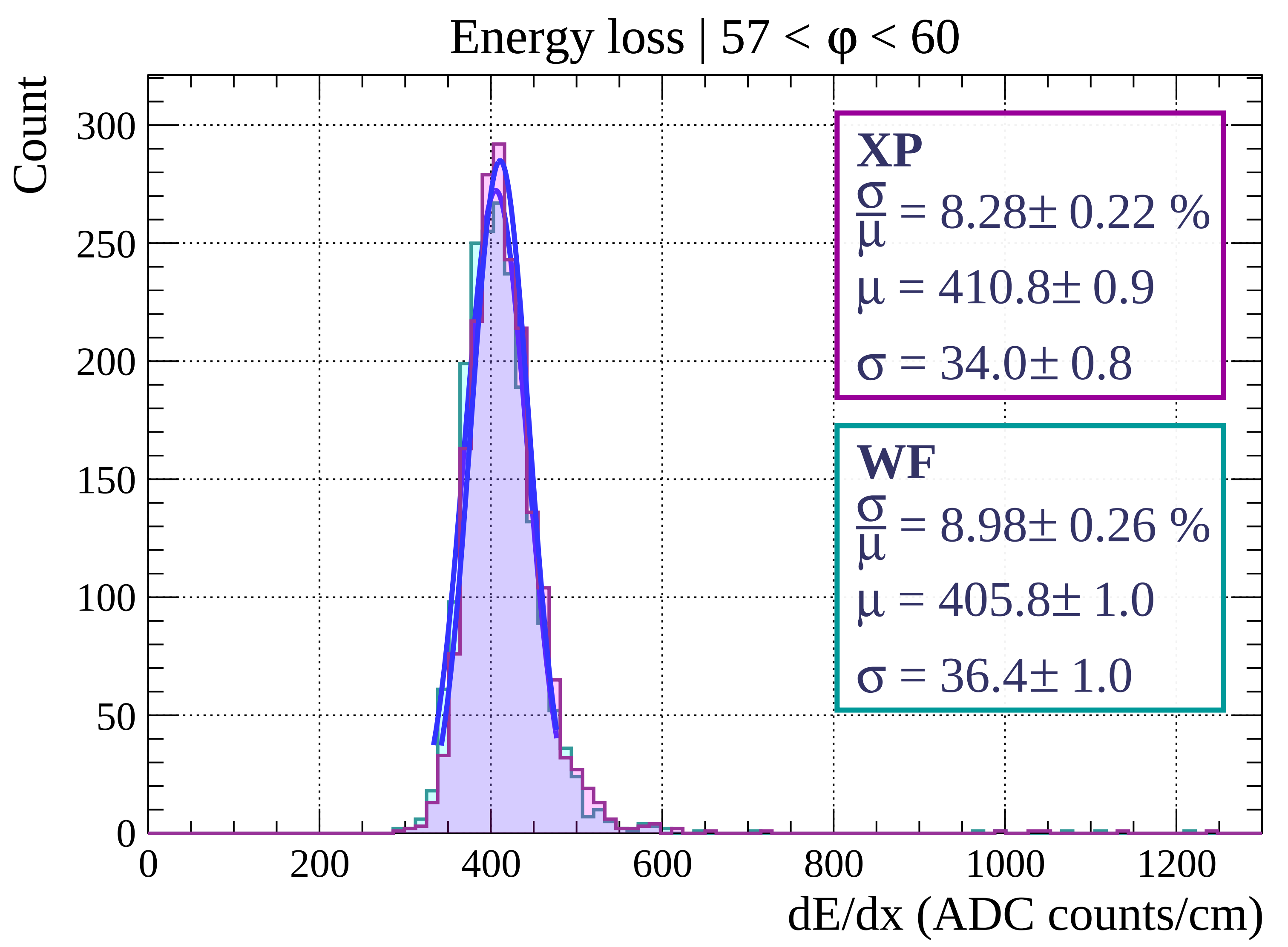


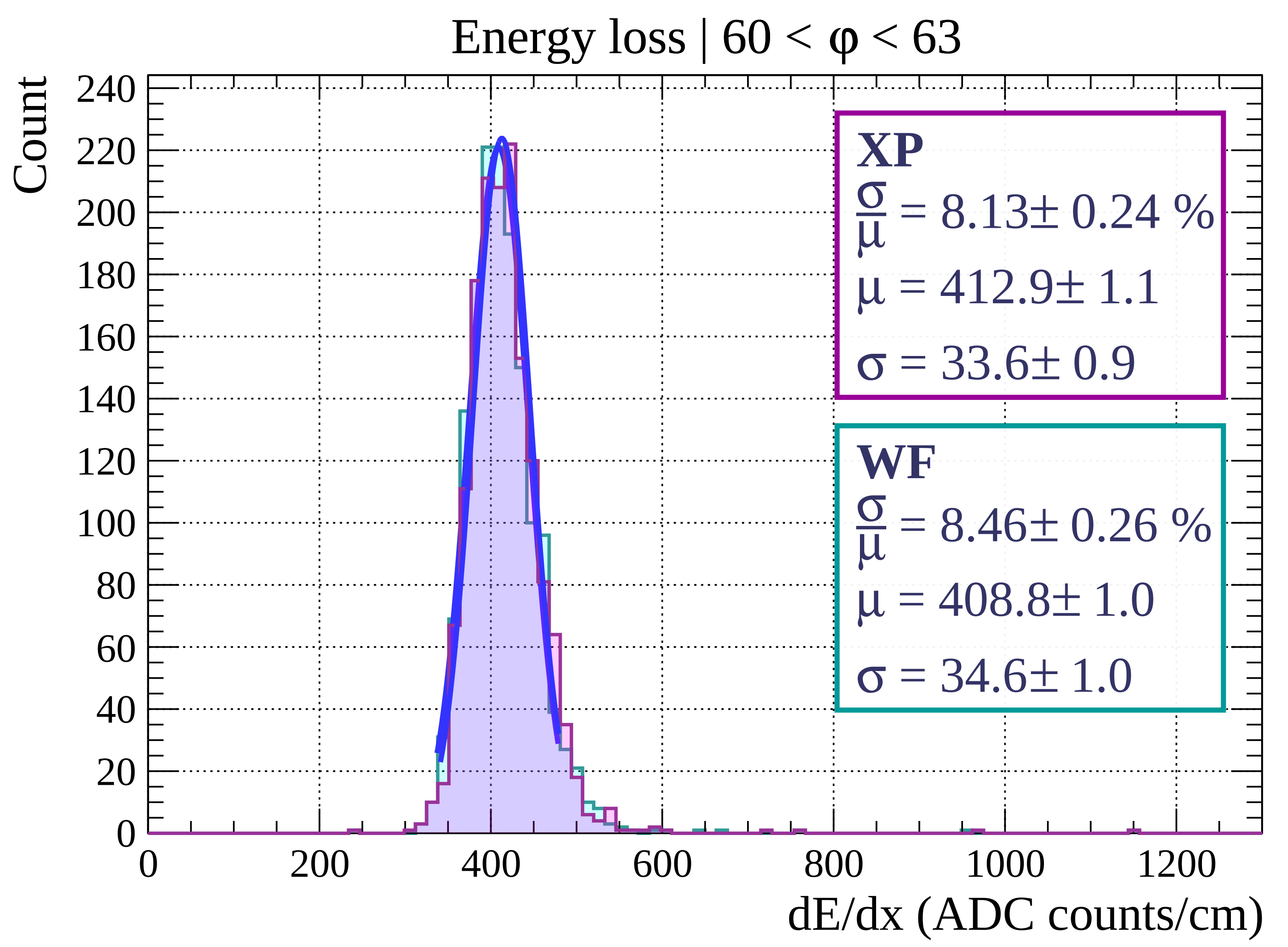
Energy loss | 48 < φ < 51
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.46± 0.33 %
μ = 412.5± 1.3
σ = 34.9± 1.3
WF
σ/μ = 8.86± 0.37 %
μ = 409.0± 1.4
σ = 36.2± 1.4

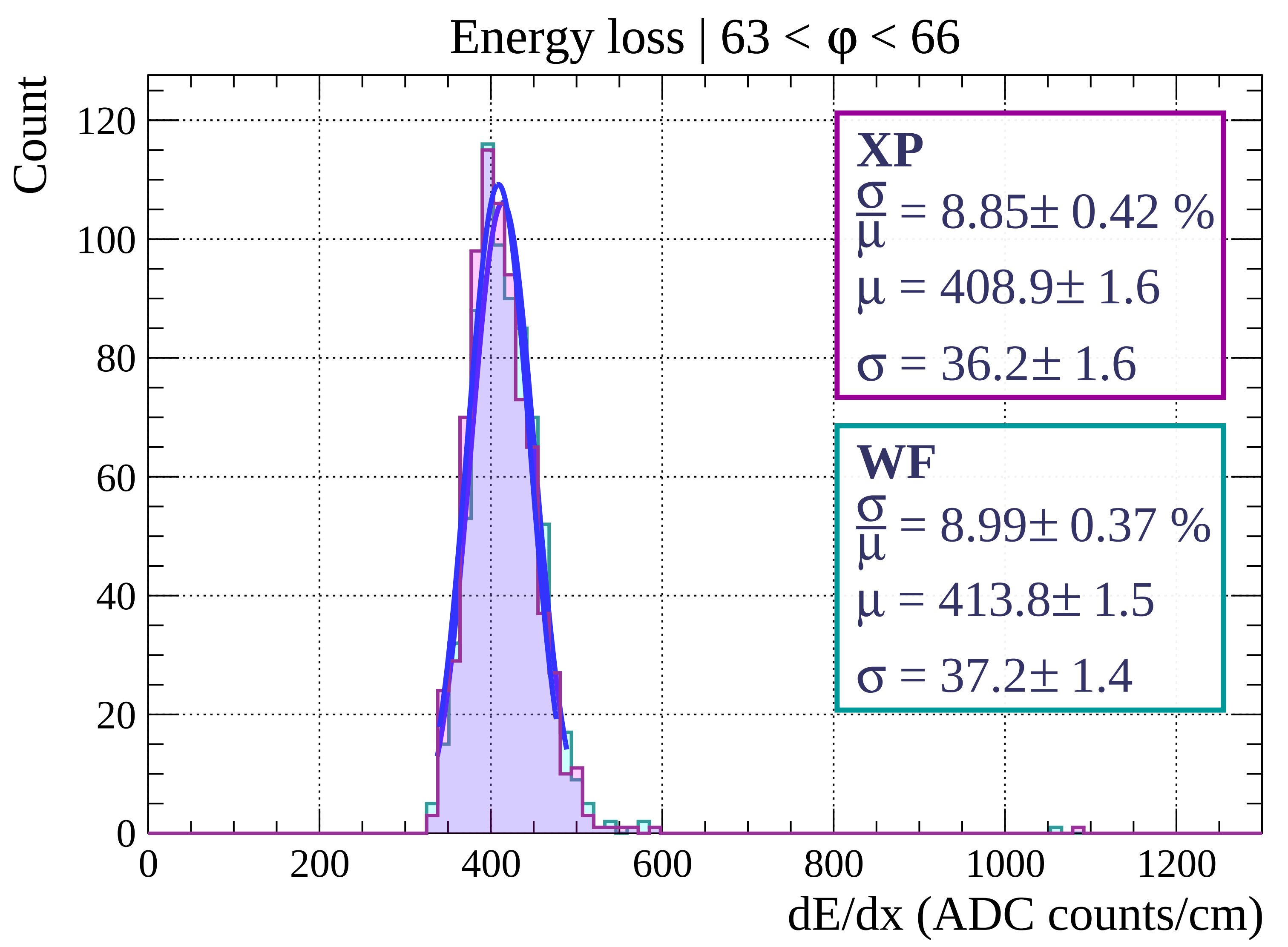

Energy loss | 51 < φ < 54
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
σ/μ = 8.14± 0.31 %
μ = 413.9± 1.1
σ = 33.7± 1.2
WF
σ/μ = 8.43± 0.28 %
μ = 407.5± 1.1
σ = 34.4± 1.1



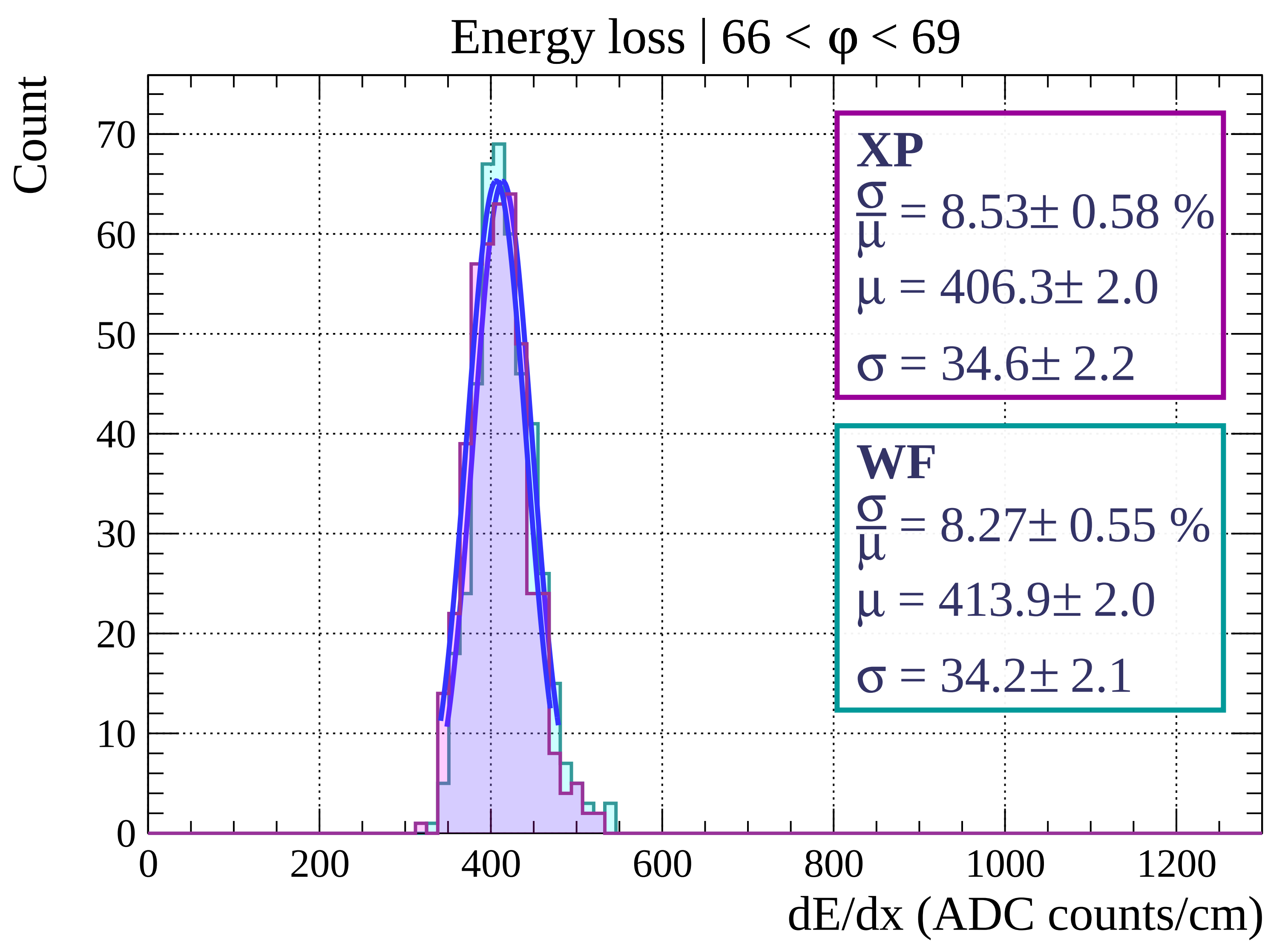
Energy loss | 54 < φ < 57
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 8.32± 0.25 %
μ = 410.3± 1.0
σ = 34.1± 0.9
WF
σ/μ = 8.52± 0.28 %
μ = 403.0± 1.0
σ = 34.3± 1.1

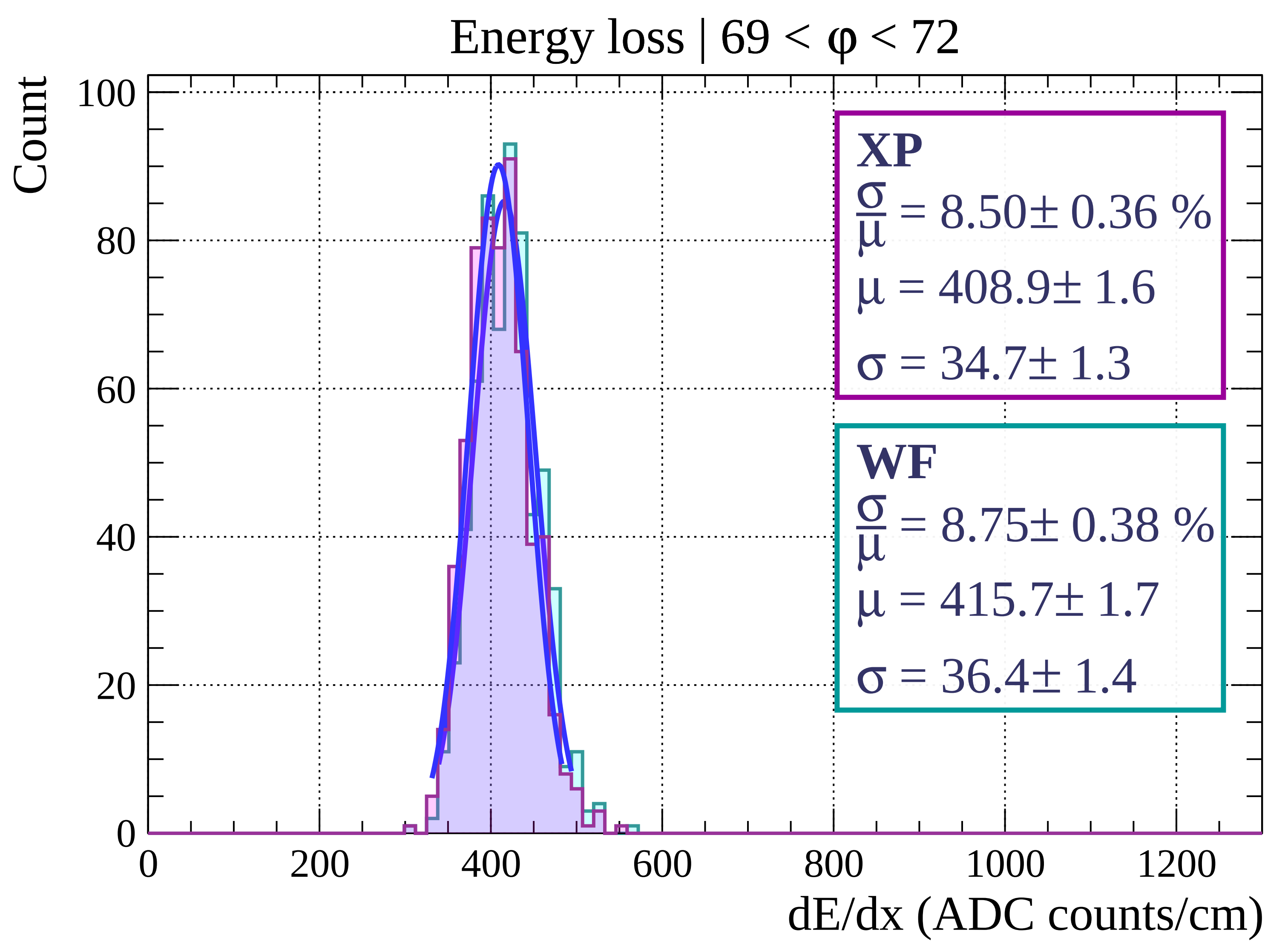

Energy loss | 57 < φ < 60
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.28± 0.22 %
μ = 410.8± 0.9
σ = 34.0± 0.8
WF
σ/μ = 8.98± 0.26 %
μ = 405.8± 1.0
σ = 36.4± 1.0


Energy loss | 60 < φ < 63
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
240
XP
$\frac{\sigma}{\mu}$ = 8.13± 0.24 %
μ = 412.9± 1.1
σ = 33.6± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.46± 0.26 %
μ = 408.8± 1.0
σ = 34.6± 1.0

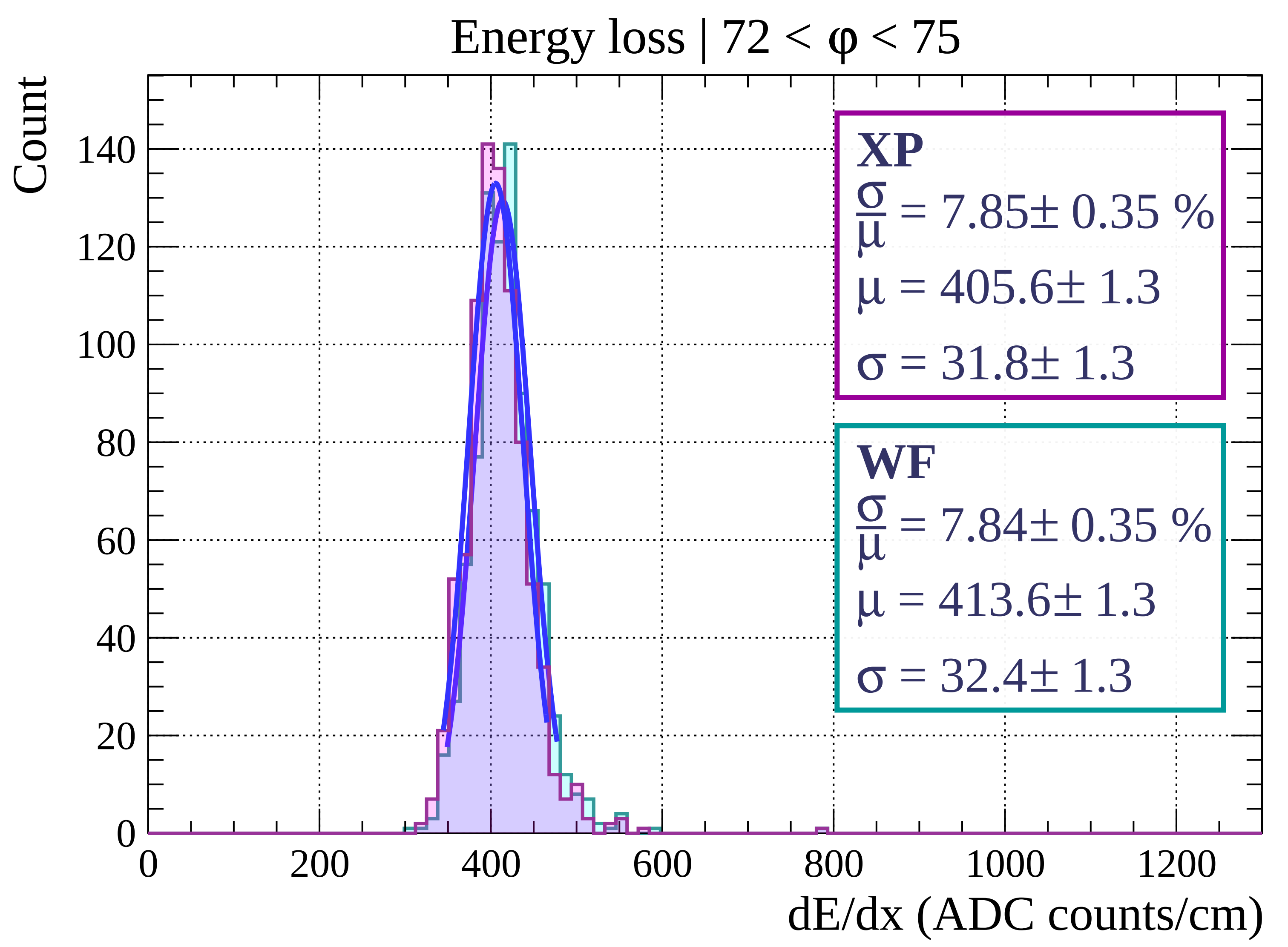


Energy loss | 63 < φ < 66
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 8.85± 0.42 %
μ = 408.9± 1.6
σ = 36.2± 1.6
WF
σ/μ = 8.99± 0.37 %
μ = 413.8± 1.5
σ = 37.2± 1.4

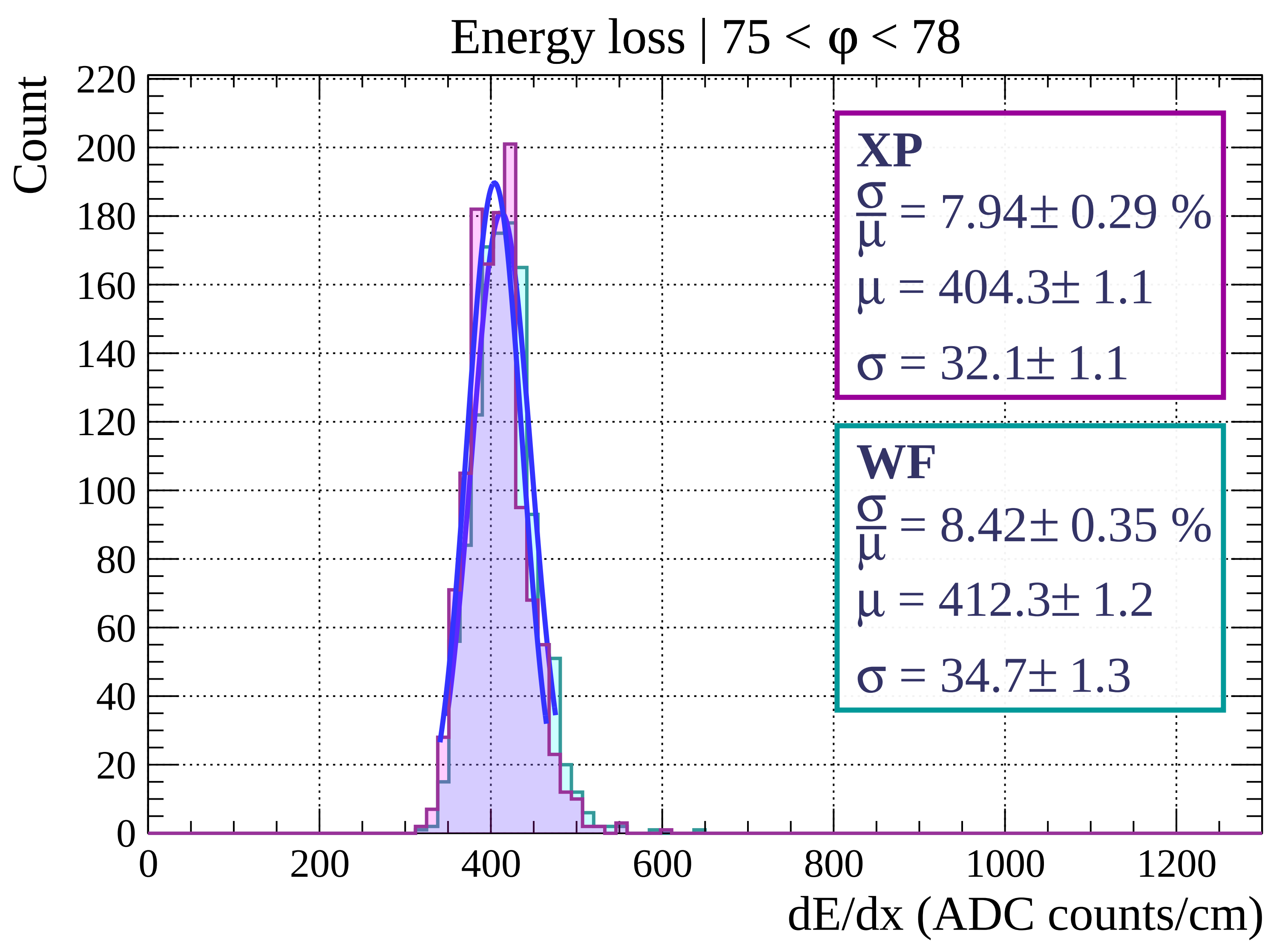

Energy loss | 66 < φ < 69
Count
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.53± 0.58 %
μ = 406.3± 2.0
σ = 34.6± 2.2
WF
σ/μ = 8.27± 0.55 %
μ = 413.9± 2.0
σ = 34.2± 2.1



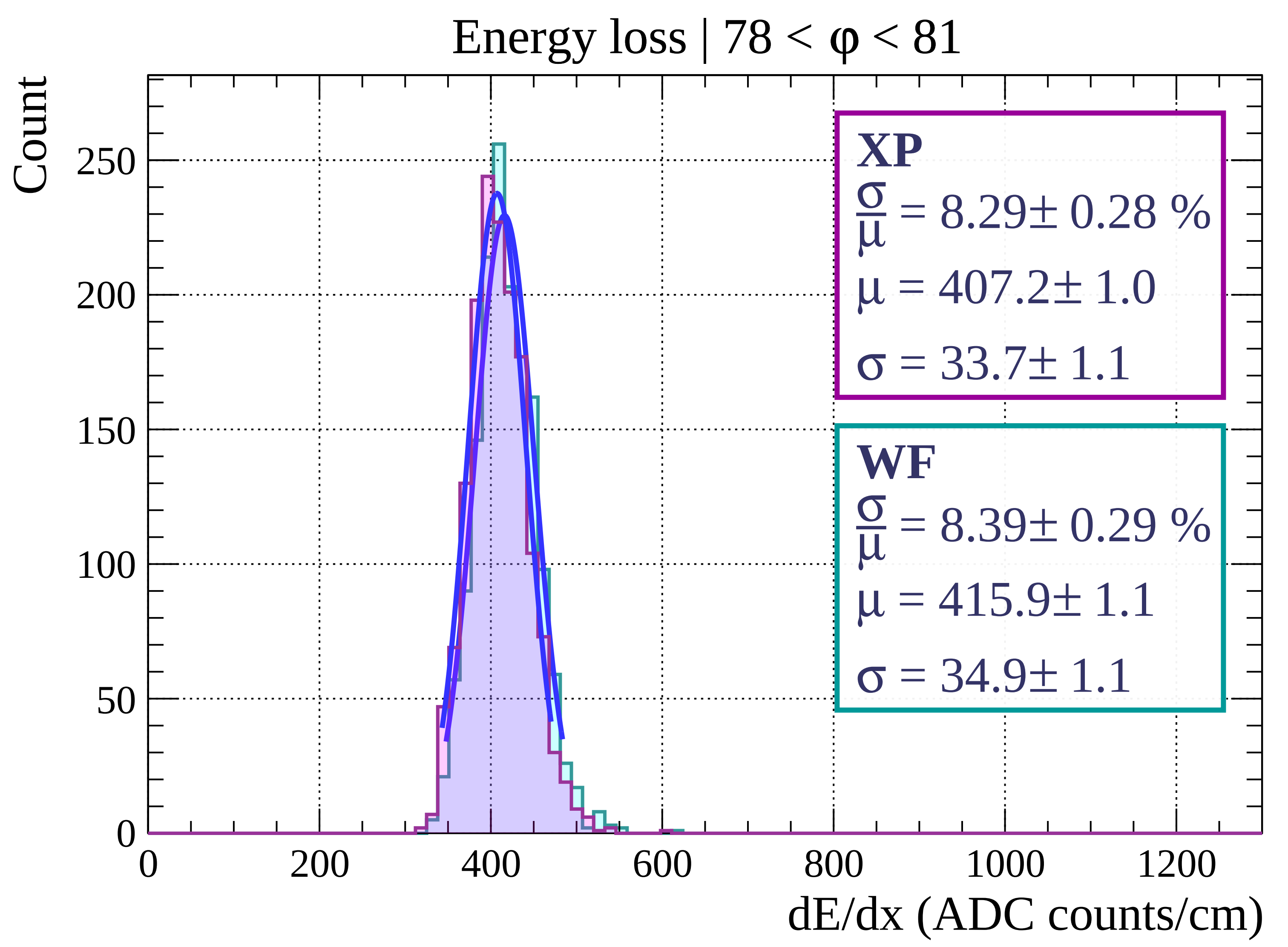
Energy loss | 69 < φ < 72
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.50± 0.36 %
μ = 408.9± 1.6
σ = 34.7± 1.3
WF
$\frac{\sigma}{\mu}$ = 8.75± 0.38 %
μ = 415.7± 1.7
σ = 36.4± 1.4

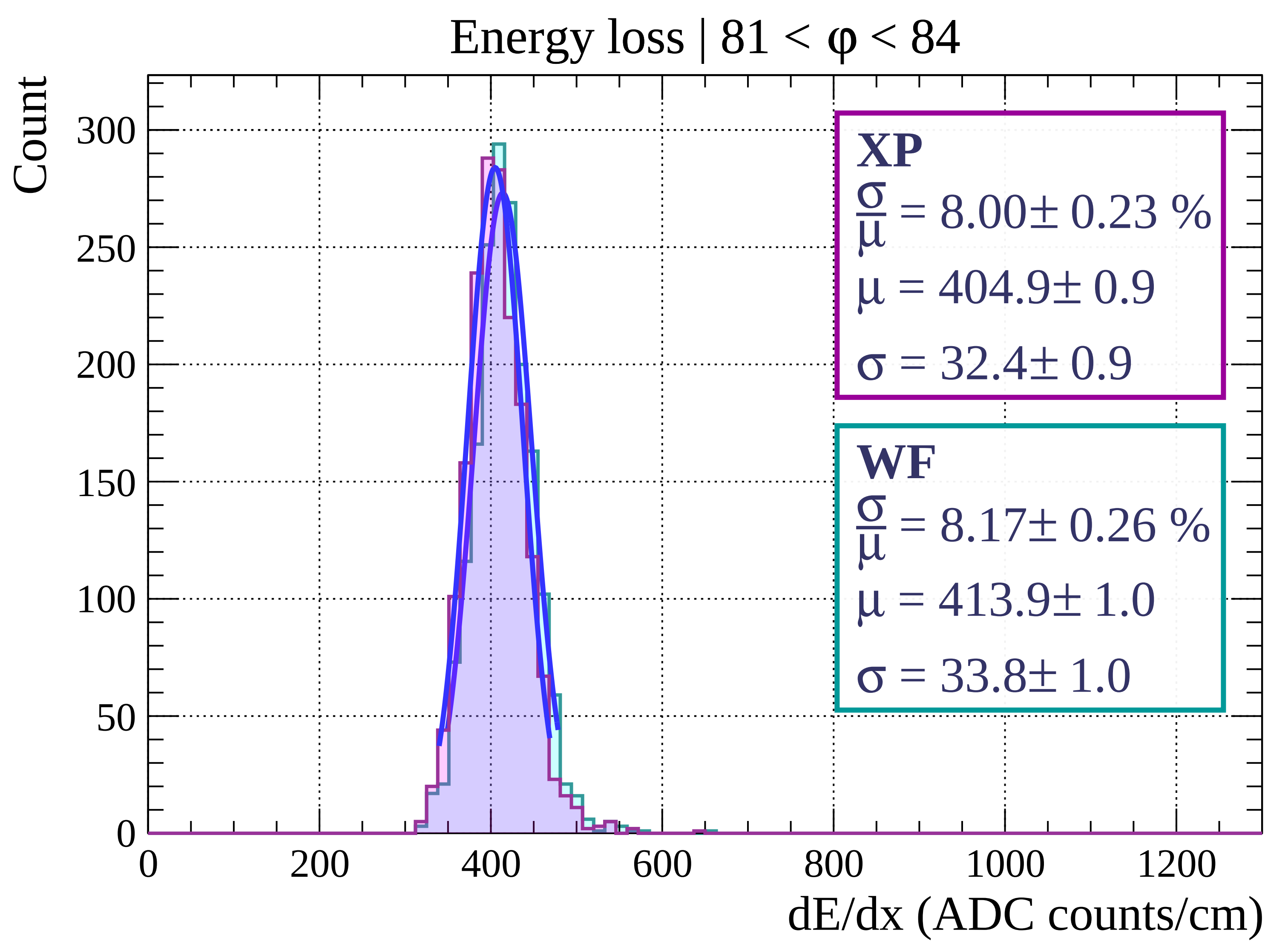


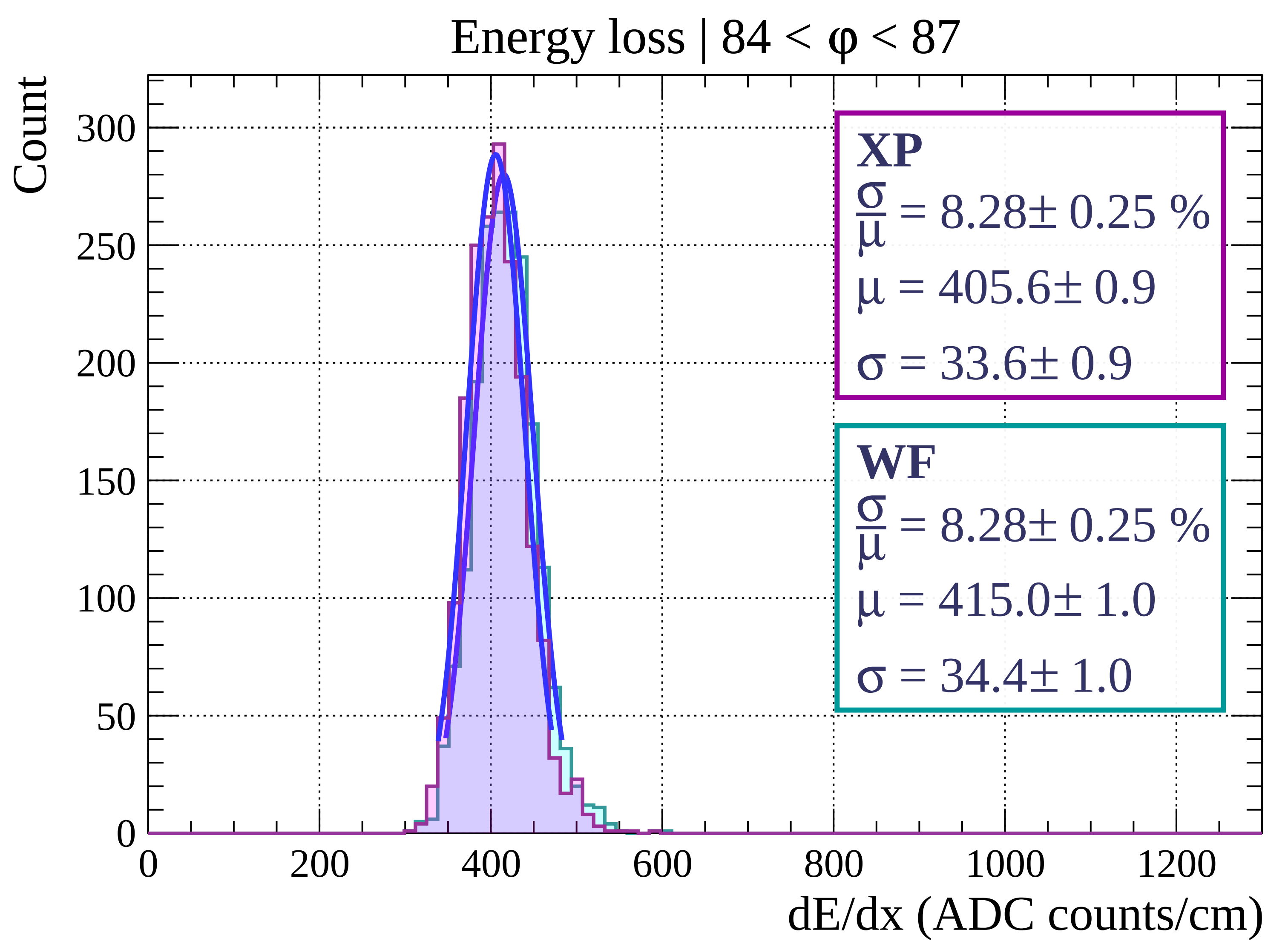
Energy loss | 72 < φ < 75
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 7.85± 0.35 %
μ = 405.6± 1.3
σ = 31.8± 1.3
WF
σ/μ = 7.84± 0.35 %
μ = 413.6± 1.3
σ = 32.4± 1.3



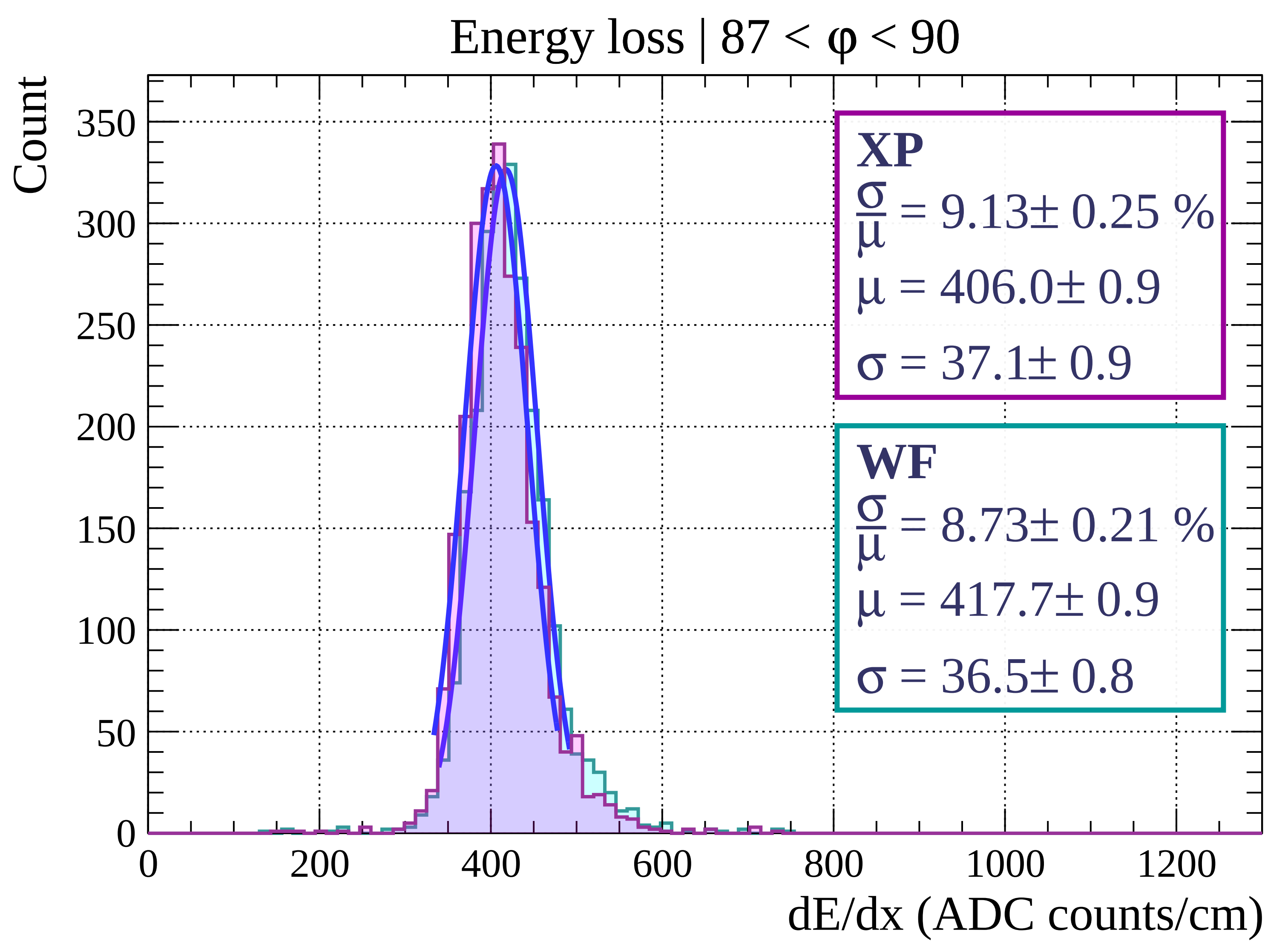
Energy loss | 75 < φ < 78
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
XP
σ/μ = 7.94± 0.29 %
μ = 404.3± 1.1
σ = 32.1± 1.1
WF
σ/μ = 8.42± 0.35 %
μ = 412.3± 1.2
σ = 34.7± 1.3


Energy loss | 78 < φ < 81
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.29± 0.28 %
μ = 407.2± 1.0
σ = 33.7± 1.1
WF
σ/μ = 8.39± 0.29 %
μ = 415.9± 1.1
σ = 34.9± 1.1

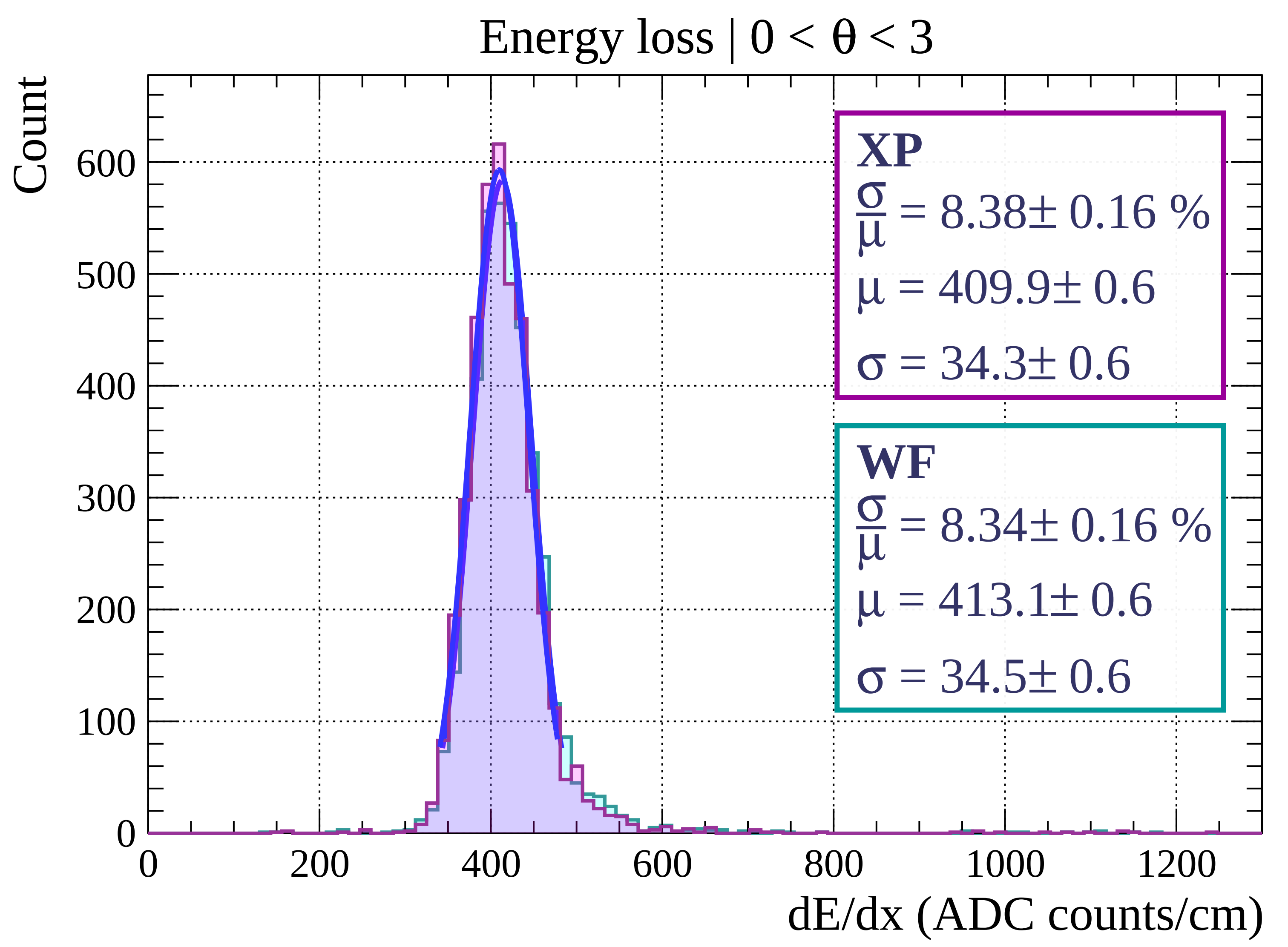

Energy loss | 81 < φ < 84
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 8.00± 0.23 %
μ = 404.9± 0.9
σ = 32.4± 0.9
WF
σ/μ = 8.17± 0.26 %
μ = 413.9± 1.0
σ = 33.8± 1.0



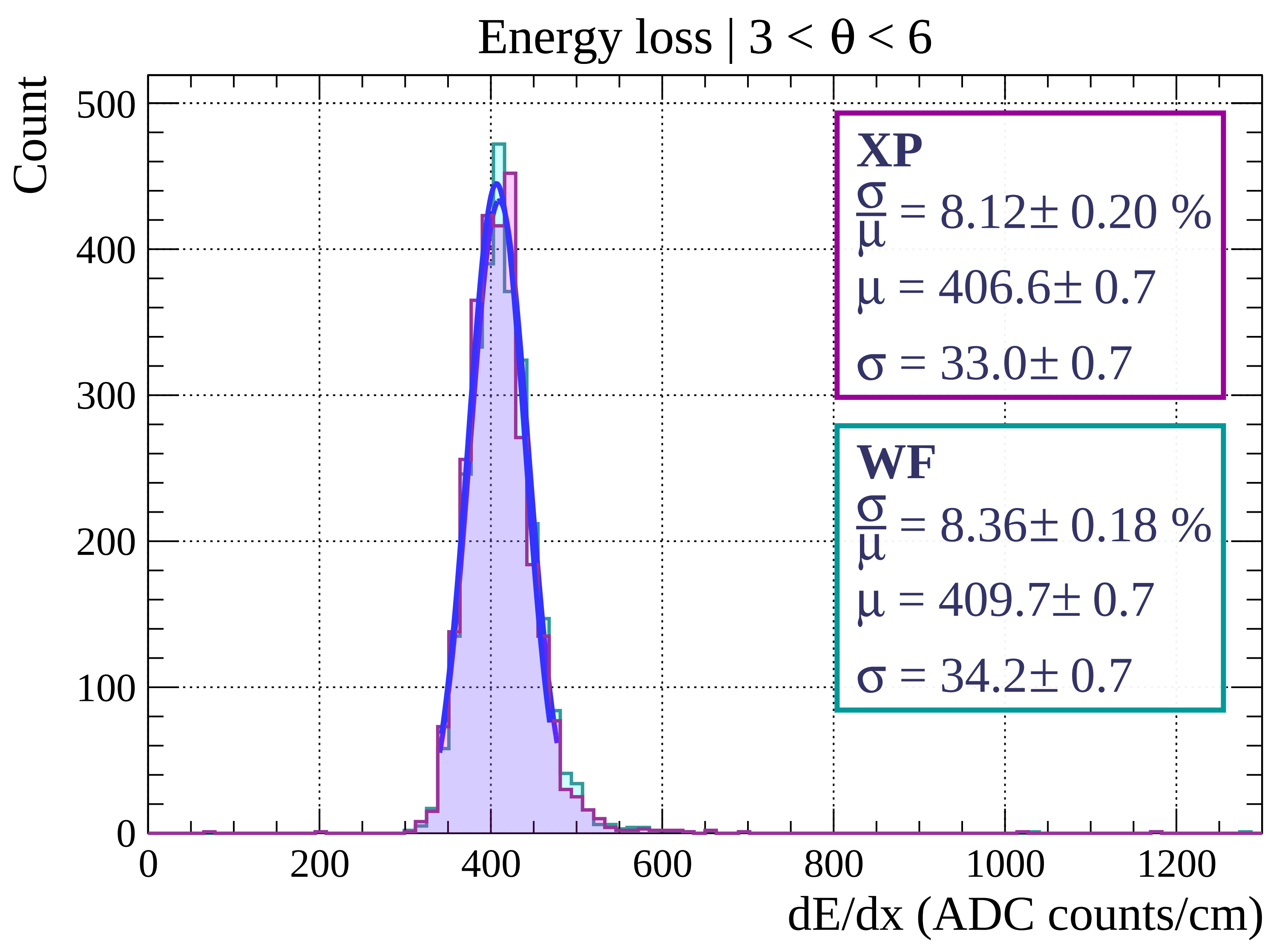
Energy loss | 84 < φ < 87
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.28± 0.25 %
μ = 405.6± 0.9
σ = 33.6± 0.9
WF
$\frac{\sigma}{\mu}$ = 8.28± 0.25 %
μ = 415.0± 1.0
σ = 34.4± 1.0



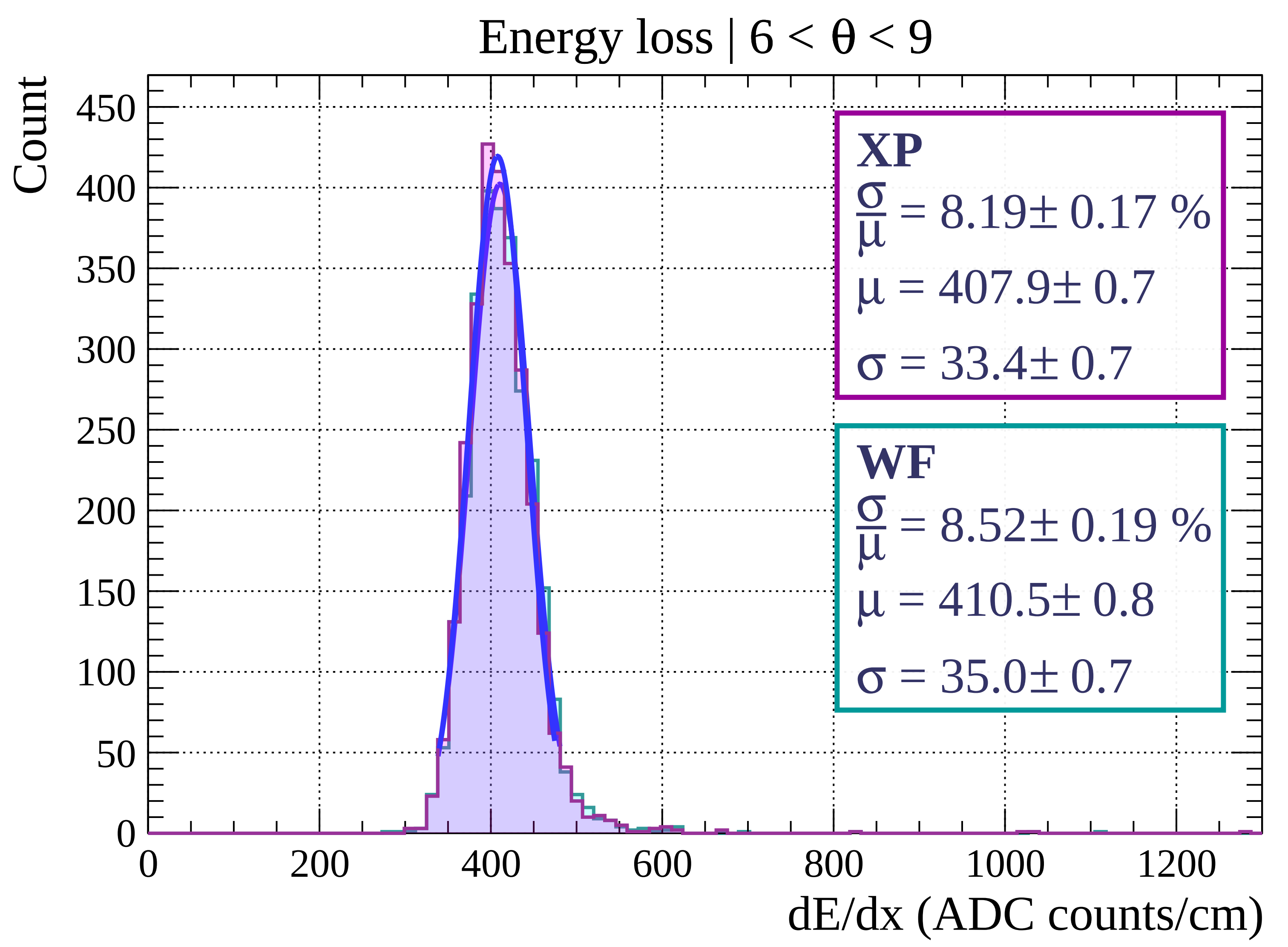
Energy loss | 87 < φ < 90
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 9.13± 0.25 %
μ = 406.0± 0.9
σ = 37.1± 0.9
WF
σ/μ = 8.73± 0.21 %
μ = 417.7± 0.9
σ = 36.5± 0.8



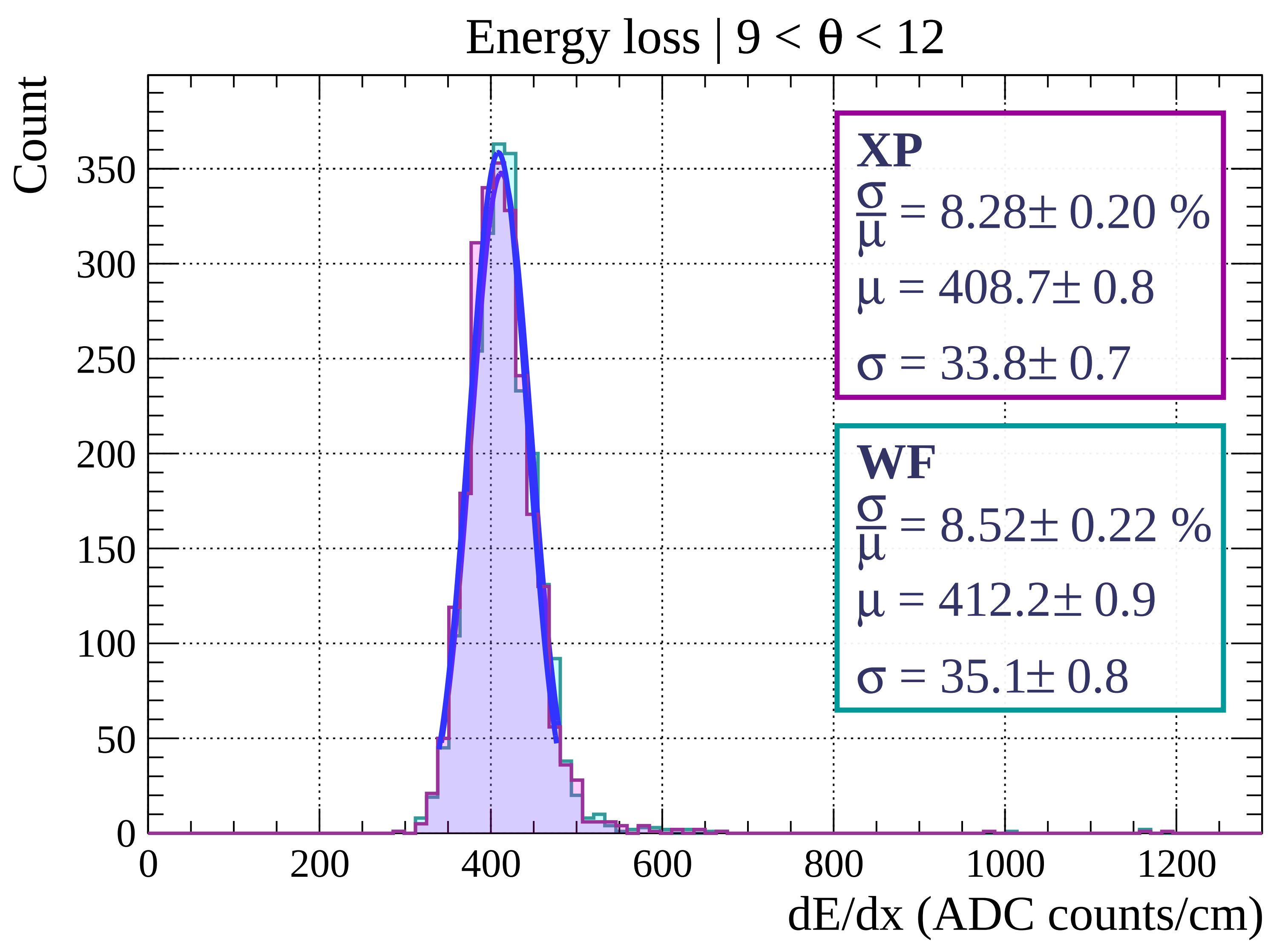
Energy loss | 0 < θ < 3
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.38± 0.16 %
μ = 409.9± 0.6
σ = 34.3± 0.6
WF
$\frac{\sigma}{\mu}$ = 8.34± 0.16 %
μ = 413.1± 0.6
σ = 34.5± 0.6



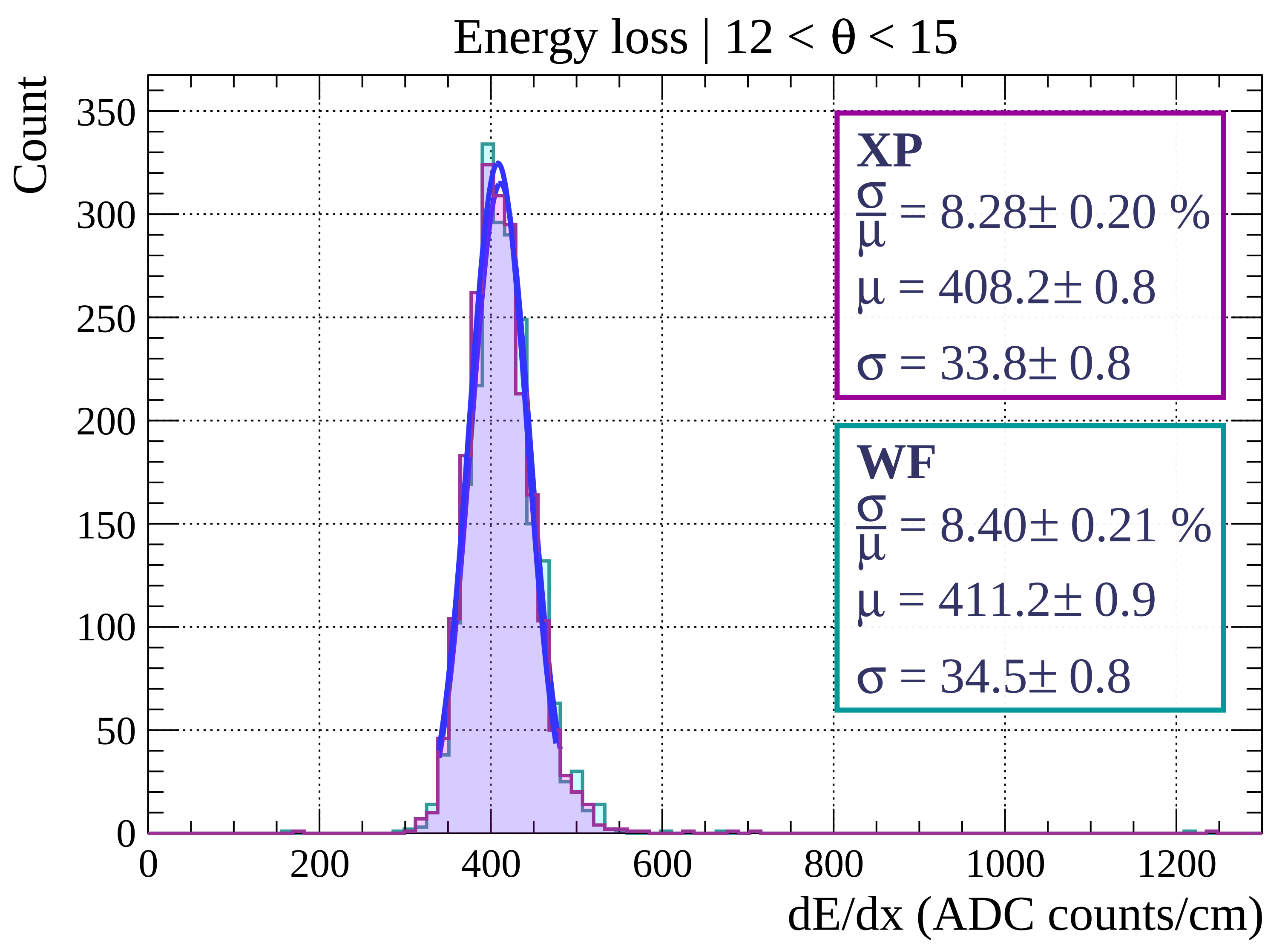
Energy loss | 3 < θ < 6
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 8.12± 0.20 %
μ = 406.6± 0.7
σ = 33.0± 0.7
WF
σ/μ = 8.36± 0.18 %
μ = 409.7± 0.7
σ = 34.2± 0.7

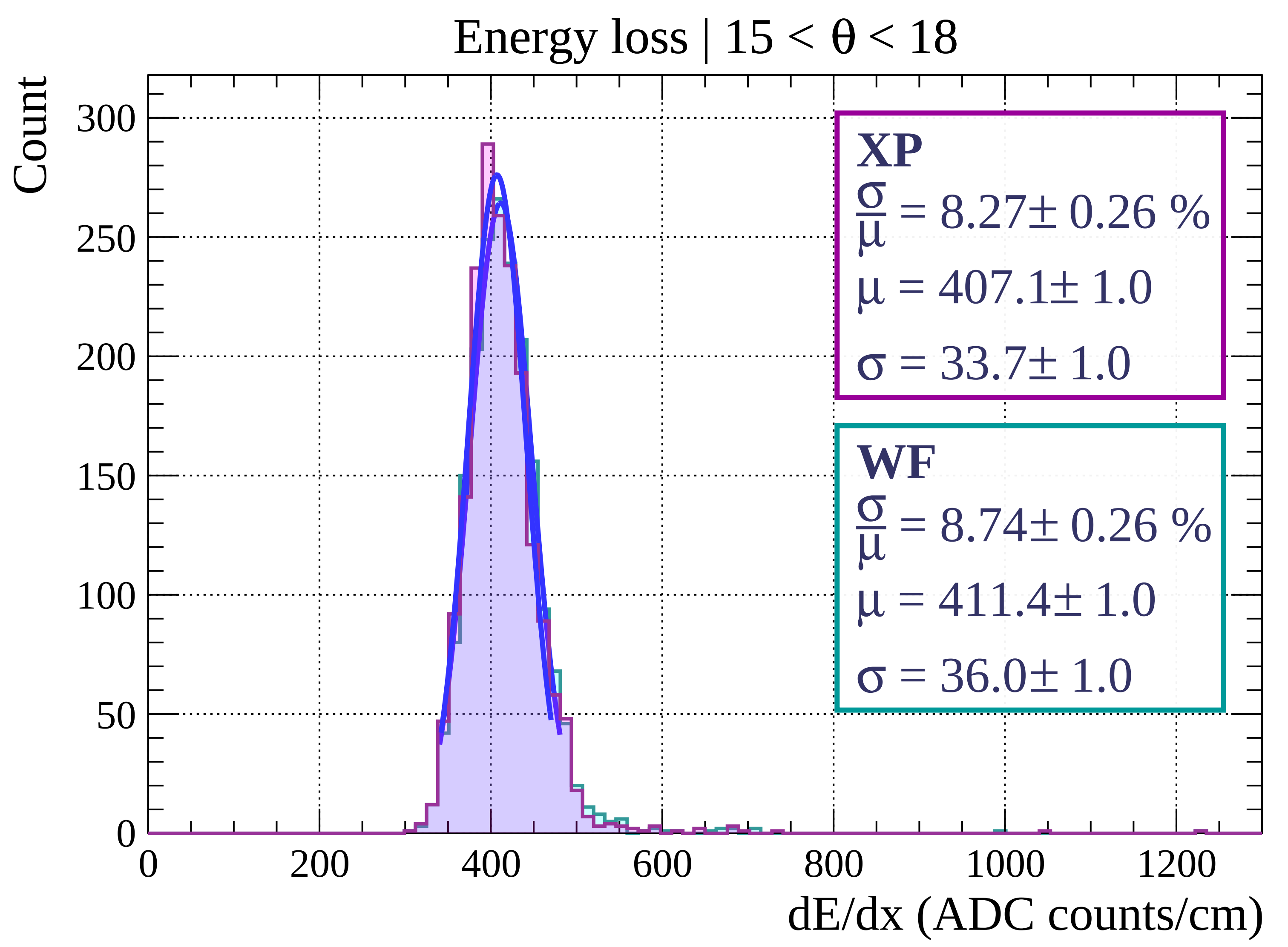


Energy loss | 6 < θ < 9
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
450
XP
$\frac{\sigma}{\mu}$ = 8.19± 0.17 %
μ = 407.9± 0.7
σ = 33.4± 0.7
WF
$\frac{\sigma}{\mu}$ = 8.52± 0.19 %
μ = 410.5± 0.8
σ = 35.0± 0.7

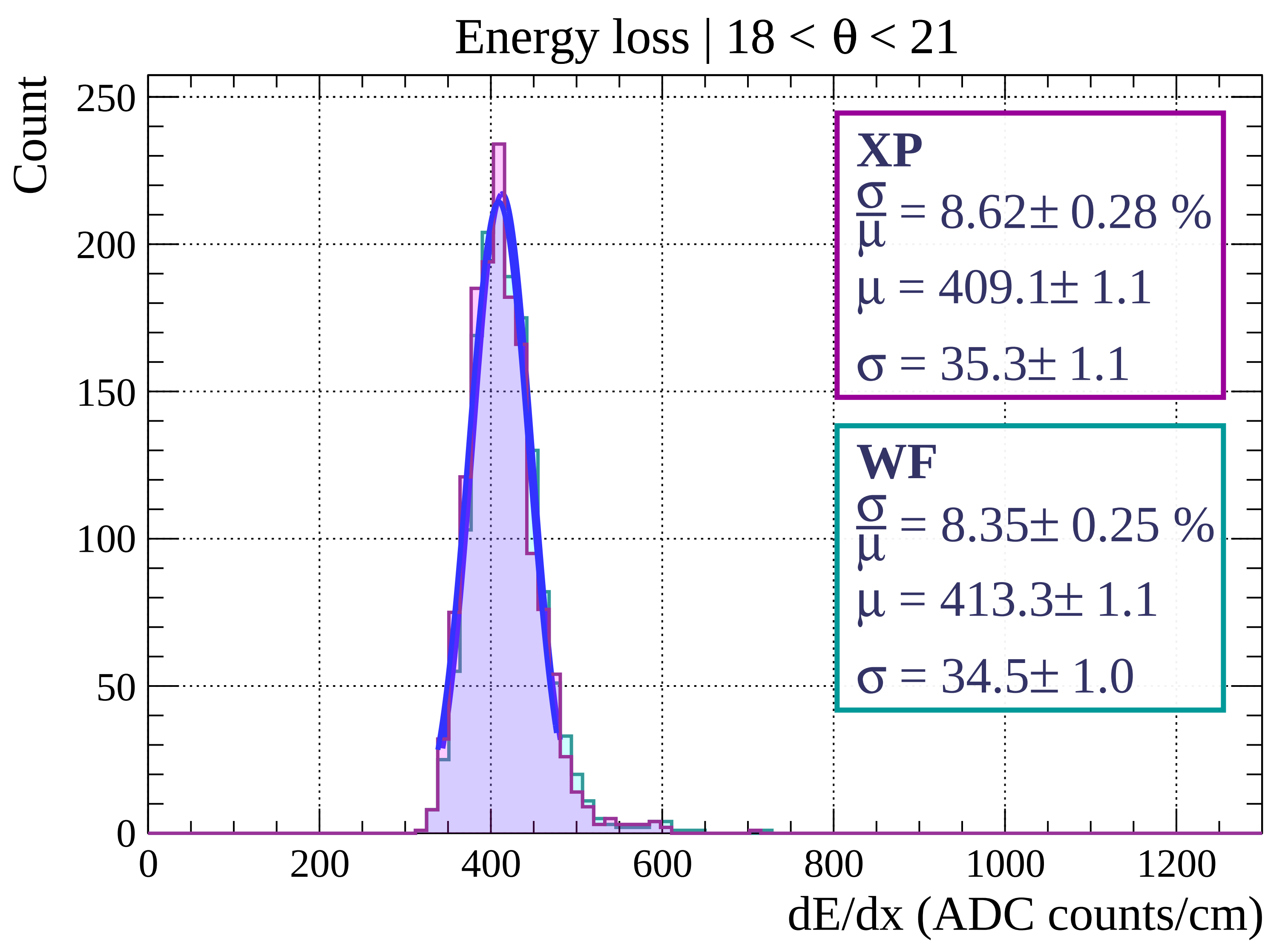


Energy loss | 9 < θ < 12
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 8.28± 0.20 %
μ = 408.7± 0.8
σ = 33.8± 0.7
WF
σ/μ = 8.52± 0.22 %
μ = 412.2± 0.9
σ = 35.1± 0.8

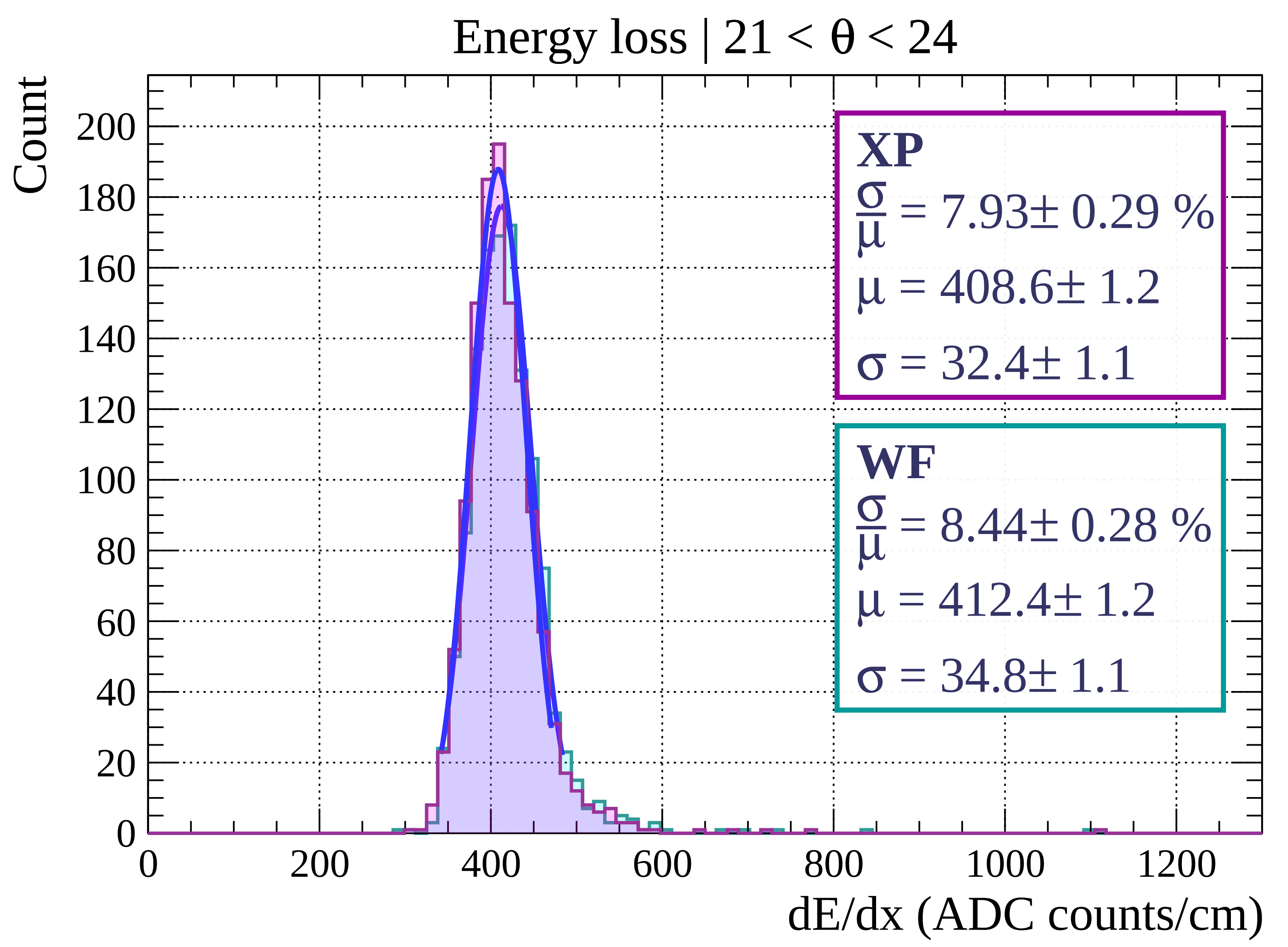

Energy loss | 12 < θ < 15
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 8.28± 0.20 %
μ = 408.2± 0.8
σ = 33.8± 0.8
WF
σ/μ = 8.40± 0.21 %
μ = 411.2± 0.9
σ = 34.5± 0.8



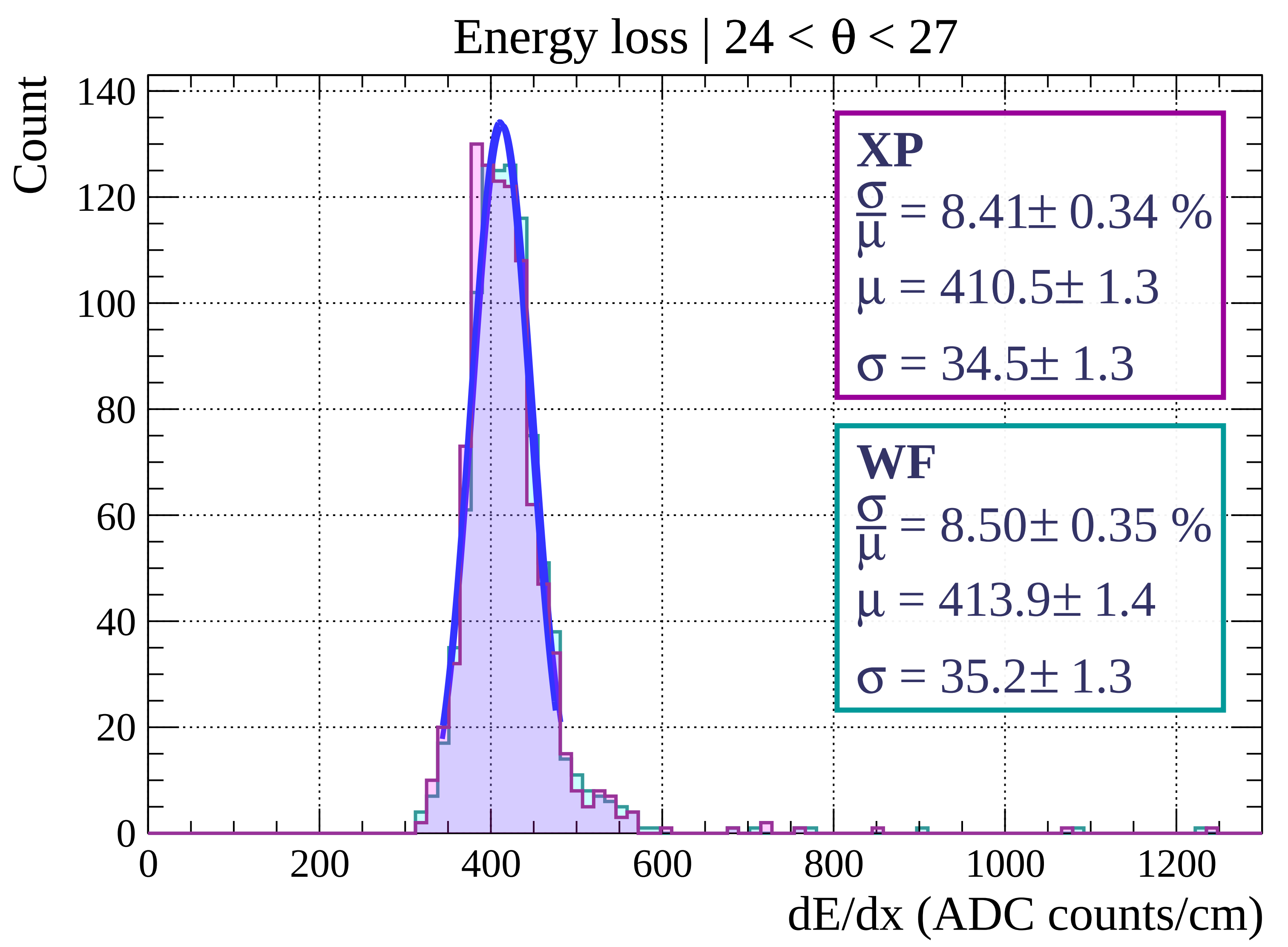
Energy loss | 15 < θ < 18
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.27± 0.26 %
μ = 407.1± 1.0
σ = 33.7± 1.0
WF
σ/μ = 8.74± 0.26 %
μ = 411.4± 1.0
σ = 36.0± 1.0



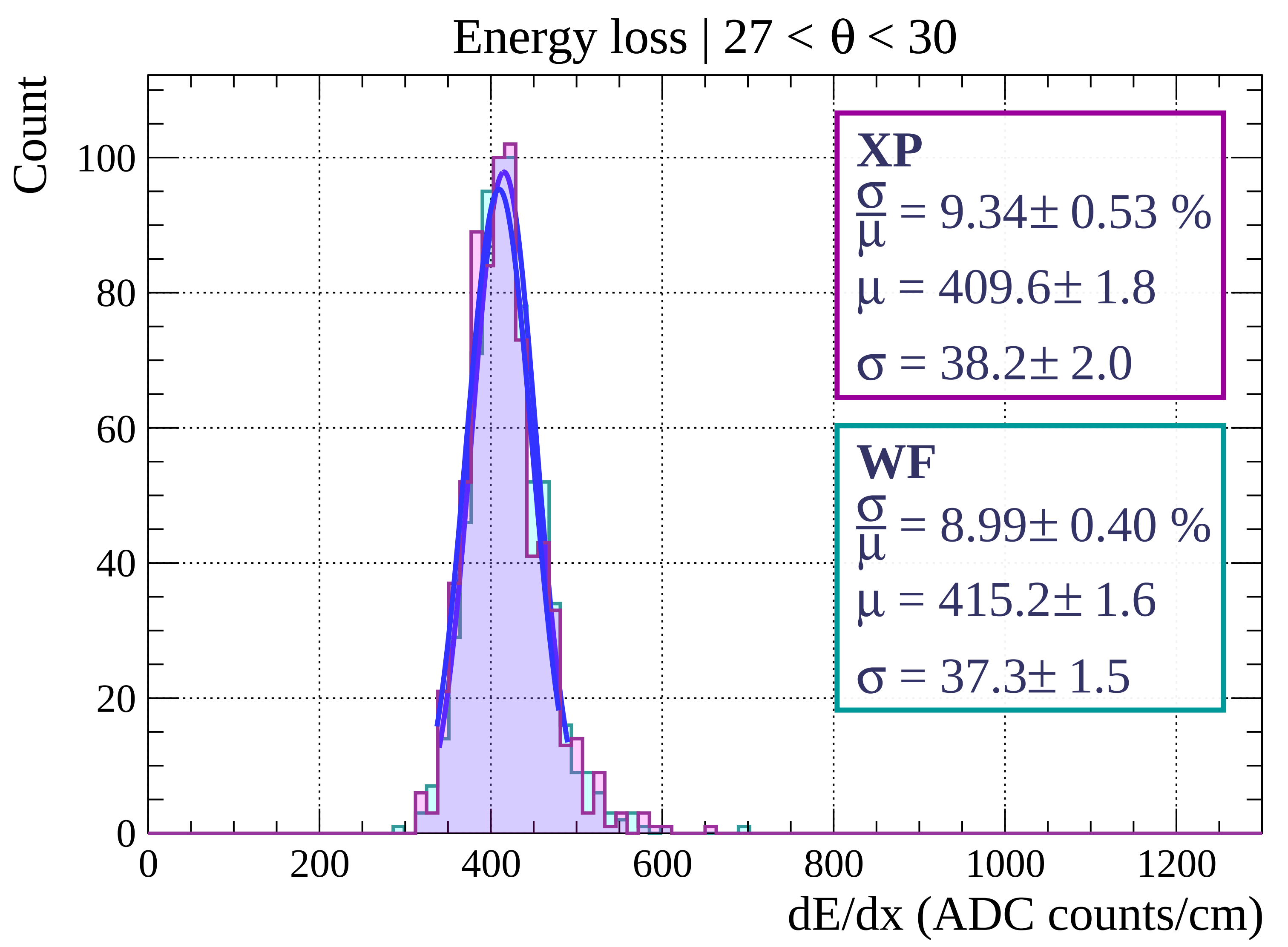
Energy loss | 18 < θ < 21
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.62± 0.28 %
μ = 409.1± 1.1
σ = 35.3± 1.1
WF
σ/μ = 8.35± 0.25 %
μ = 413.3± 1.1
σ = 34.5± 1.0
0
50
100
150
200
250
0
200
400
600
800
1000
1200



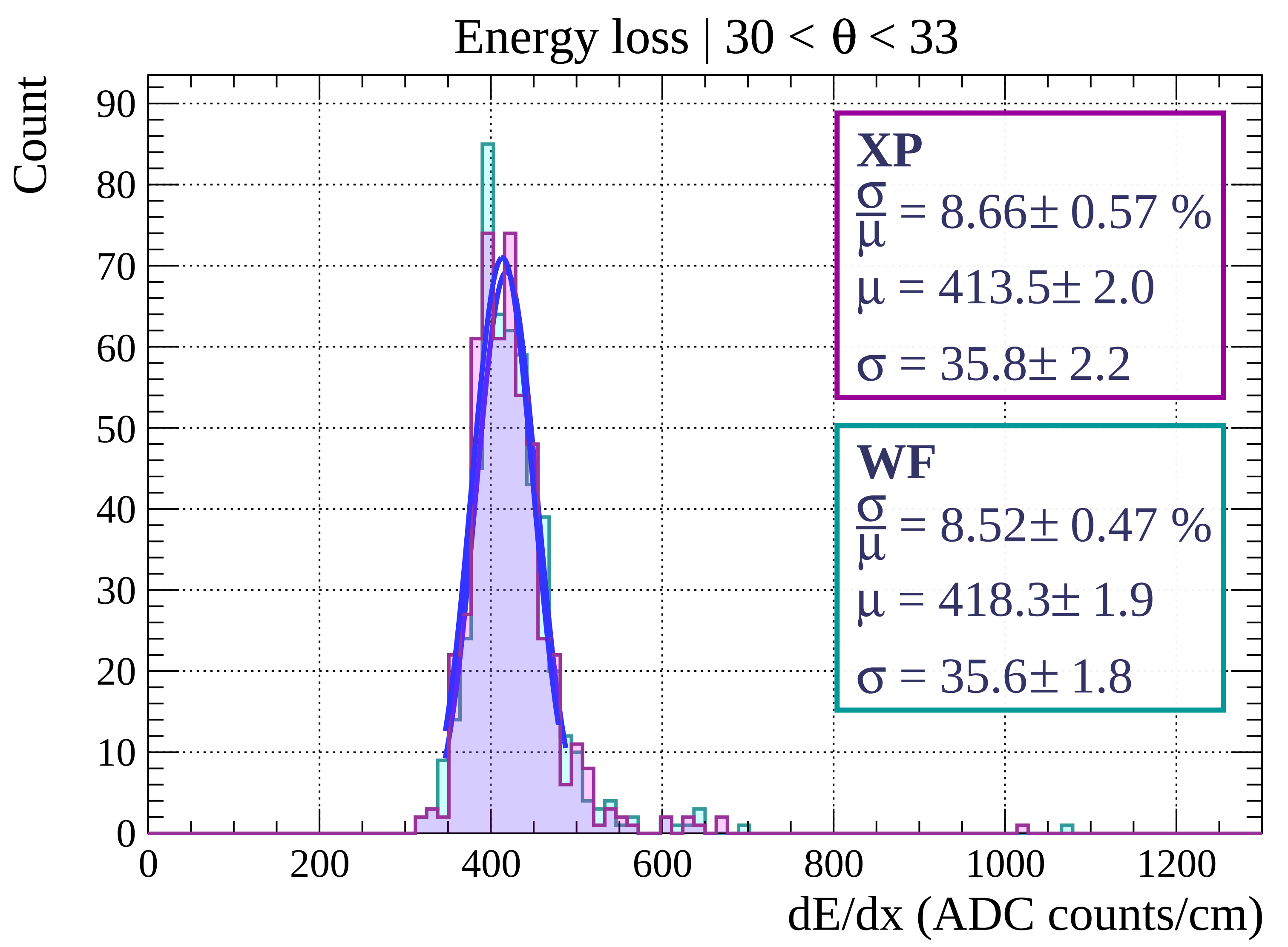
Energy loss | 21 < θ < 24
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
σ/μ = 7.93± 0.29 %
μ = 408.6± 1.2
σ = 32.4± 1.1
WF
σ/μ = 8.44± 0.28 %
μ = 412.4± 1.2
σ = 34.8± 1.1



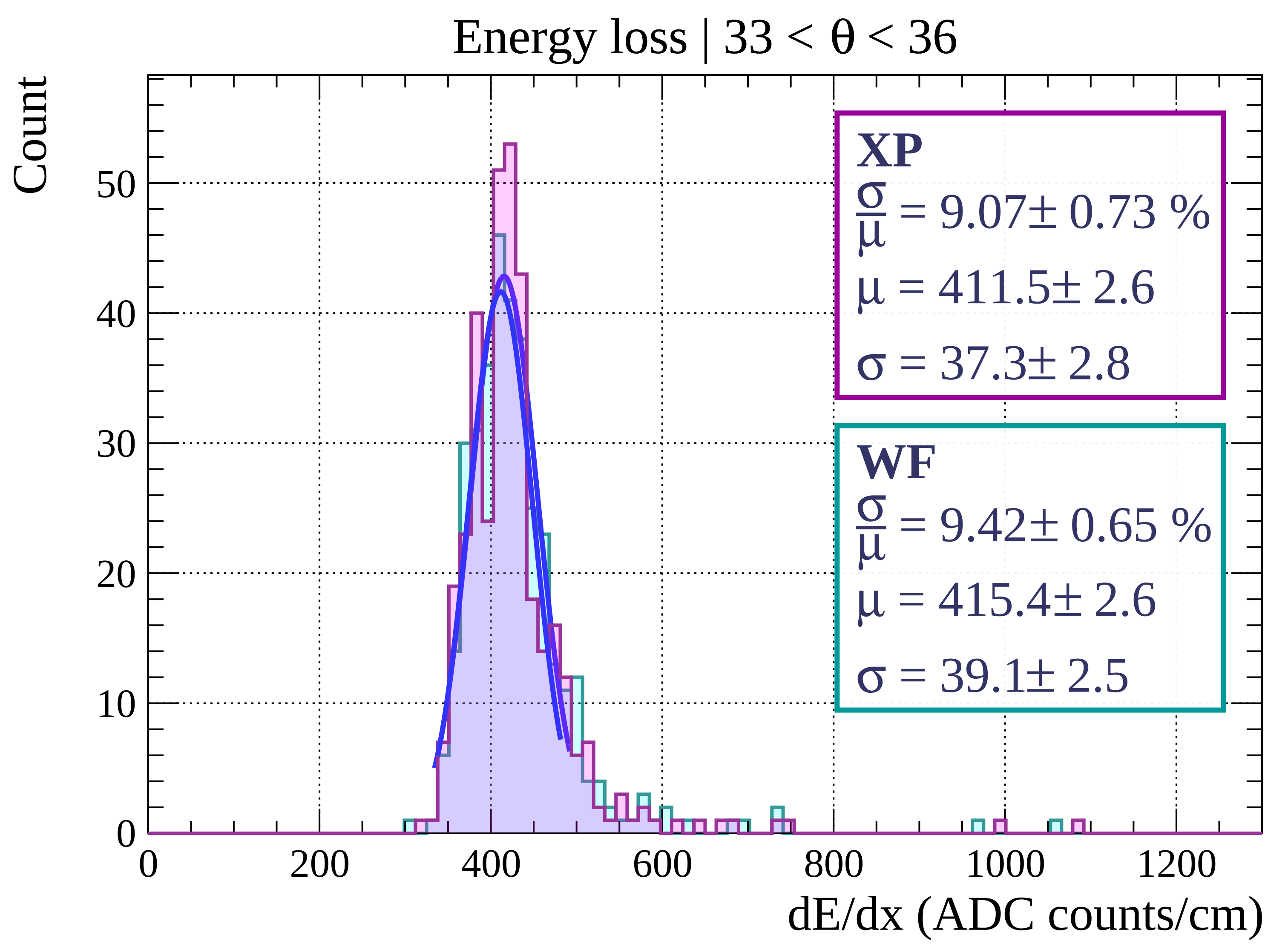
Energy loss | 24 < θ < 27
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP
σ/μ = 8.41± 0.34 %
μ = 410.5± 1.3
σ = 34.5± 1.3
WF
σ/μ = 8.50± 0.35 %
μ = 413.9± 1.4
σ = 35.2± 1.3



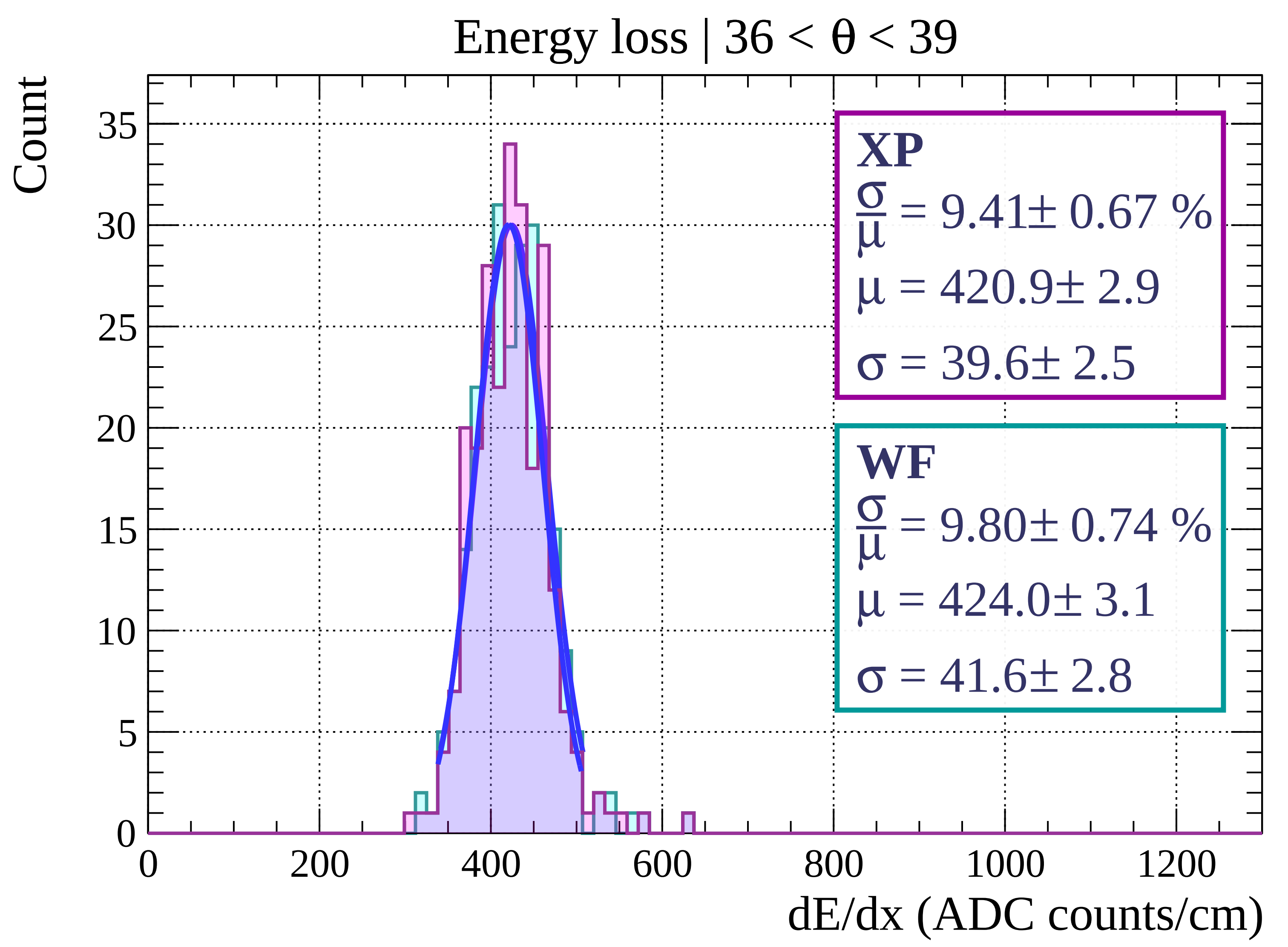
Energy loss | 27 < θ < 30
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.34± 0.53 %
μ = 409.6± 1.8
σ = 38.2± 2.0
WF
$\frac{\sigma}{\mu}$ = 8.99± 0.40 %
μ = 415.2± 1.6
σ = 37.3± 1.5



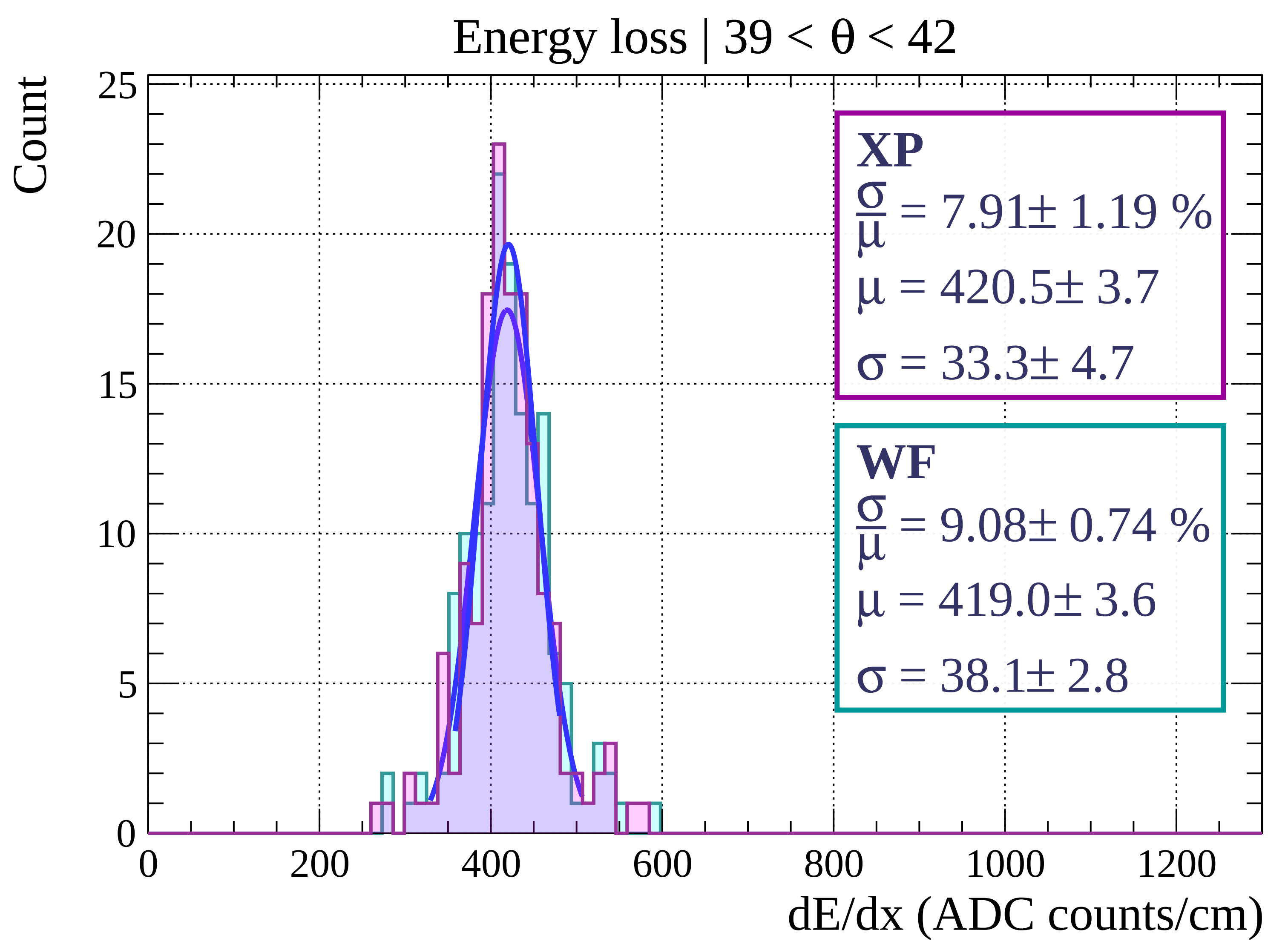
Energy loss | 30 < θ < 33
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90
XP
σ/μ = 8.66± 0.57 %
μ = 413.5± 2.0
σ = 35.8± 2.2
WF
σ/μ = 8.52± 0.47 %
μ = 418.3± 1.9
σ = 35.6± 1.8



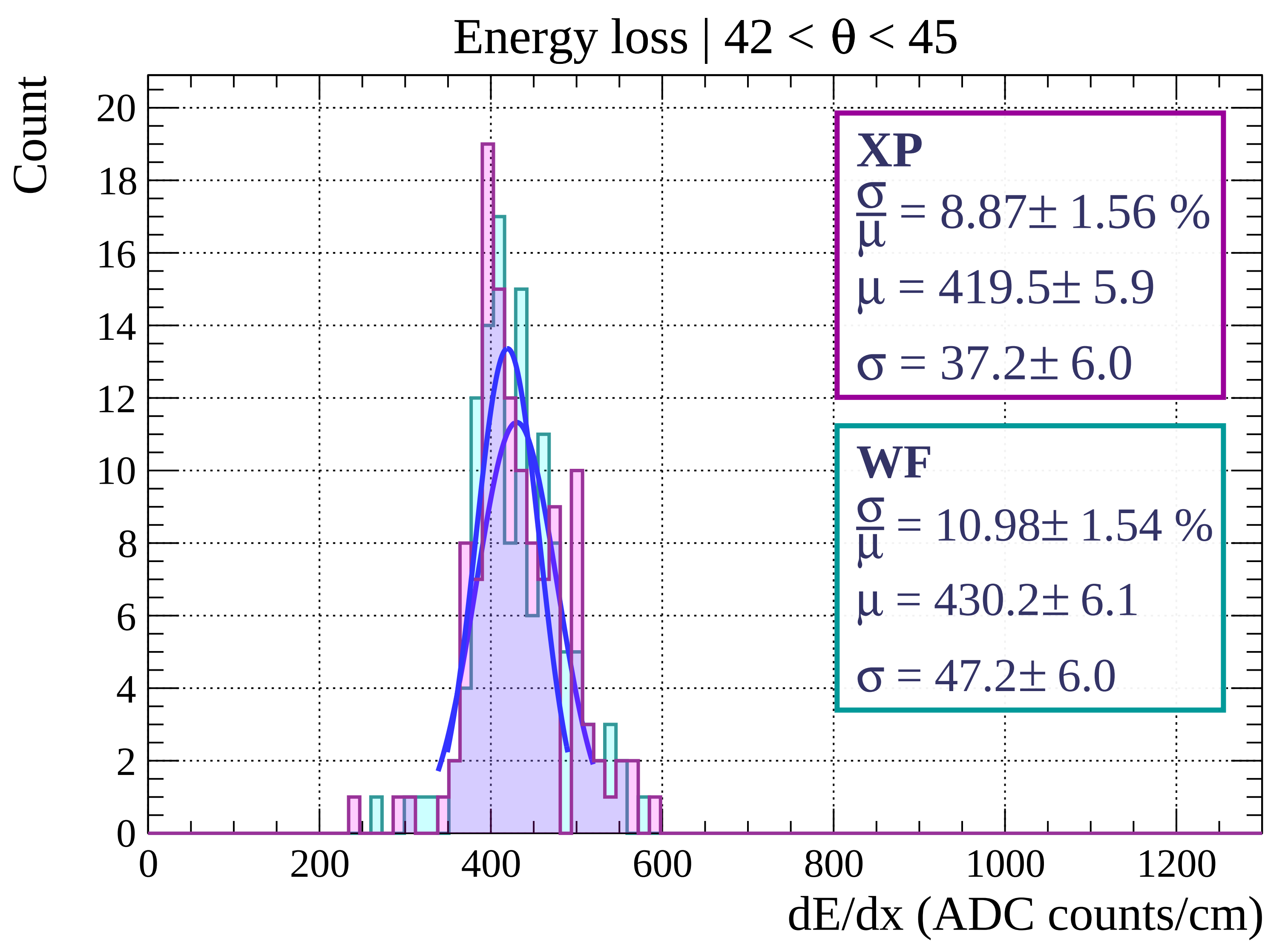
Energy loss | 33 < θ < 36
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.07± 0.73 %
μ = 411.5± 2.6
σ = 37.3± 2.8
WF
$\frac{\sigma}{\mu}$ = 9.42± 0.65 %
μ = 415.4± 2.6
σ = 39.1± 2.5

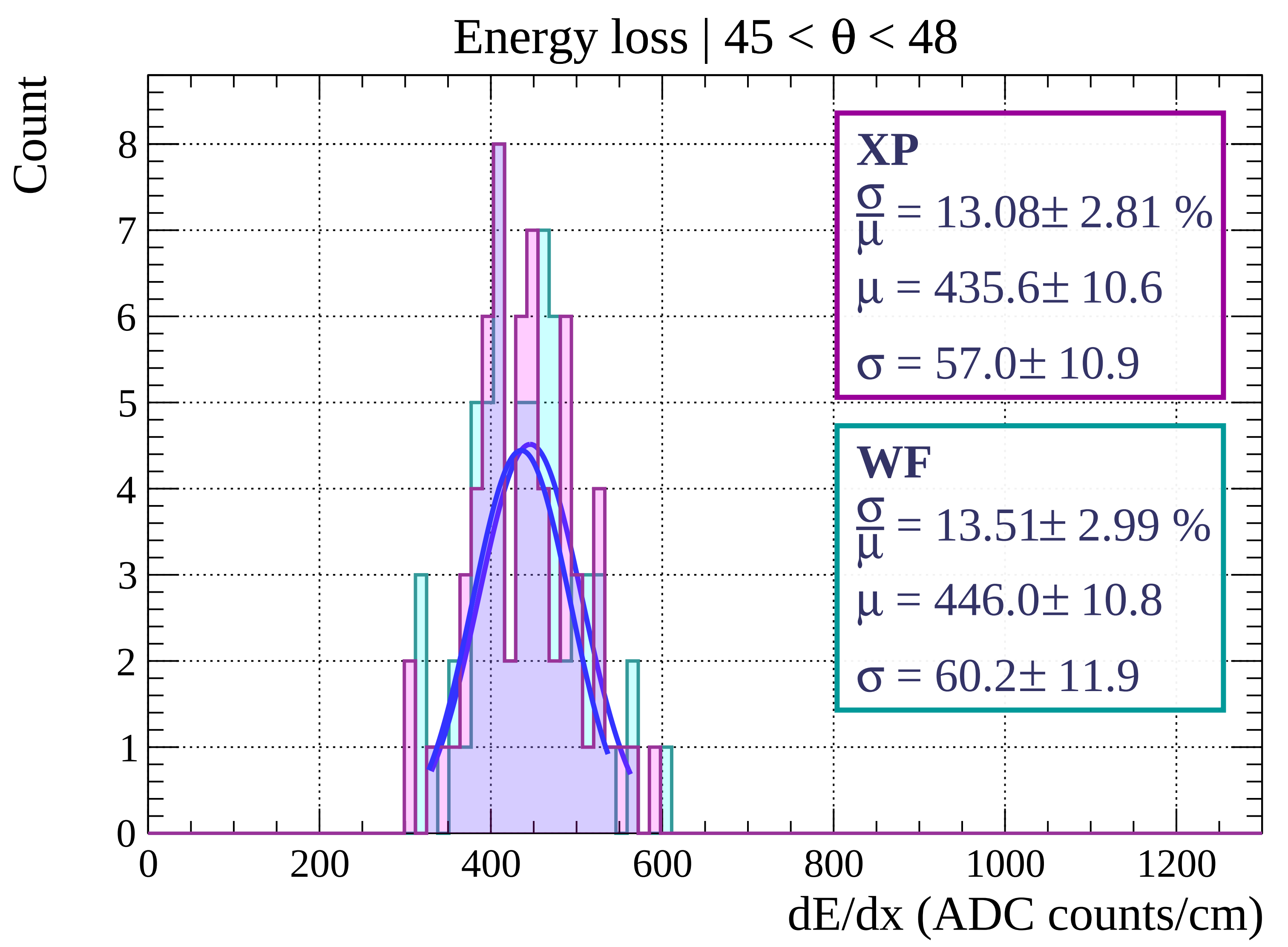


Energy loss | 36 < θ < 39
Count
35
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.41± 0.67 %
μ = 420.9± 2.9
σ = 39.6± 2.5
WF
σ/μ = 9.80± 0.74 %
μ = 424.0± 3.1
σ = 41.6± 2.8

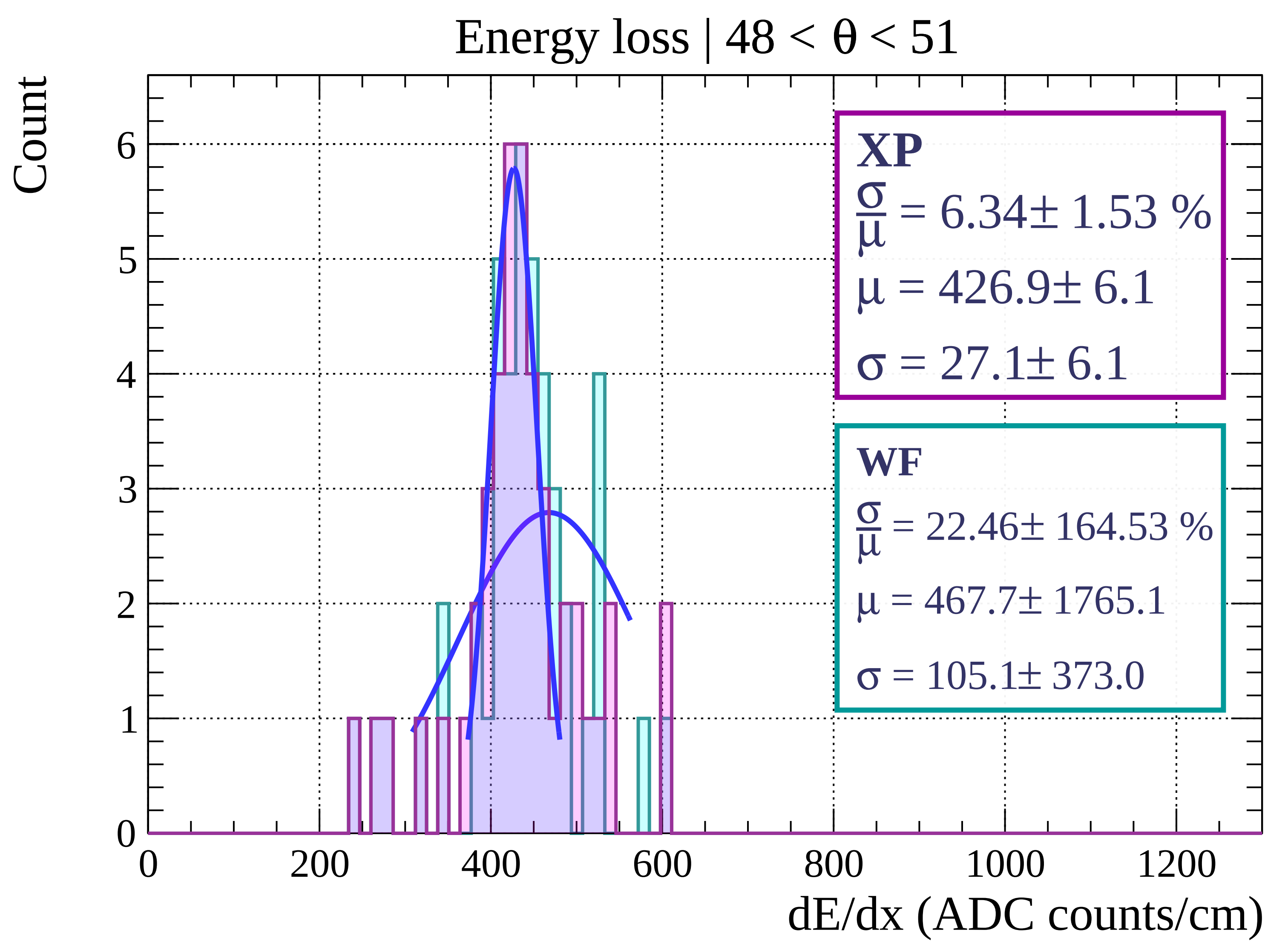


Energy loss | 39 < θ < 42
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 7.91± 1.19 %
μ = 420.5± 3.7
σ = 33.3± 4.7
WF
σ/μ = 9.08± 0.74 %
μ = 419.0± 3.6
σ = 38.1± 2.8

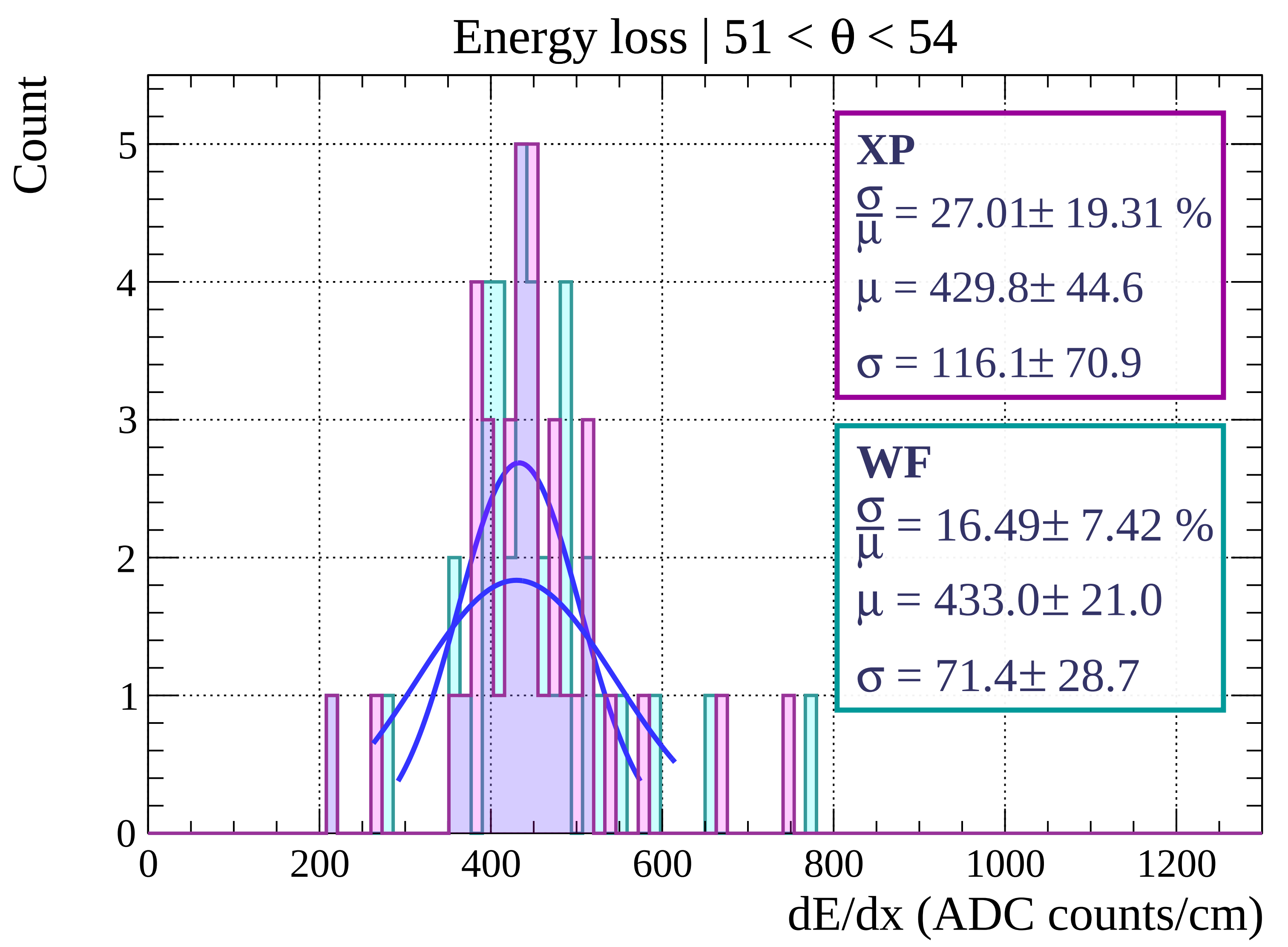

Energy loss | 42 < θ < 45
Count
XP
σ/μ = 8.87± 1.56 %
μ = 419.5± 5.9
σ = 37.2± 6.0
WF
σ/μ = 10.98± 1.54 %
μ = 430.2± 6.1
σ = 47.2± 6.0
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18
20



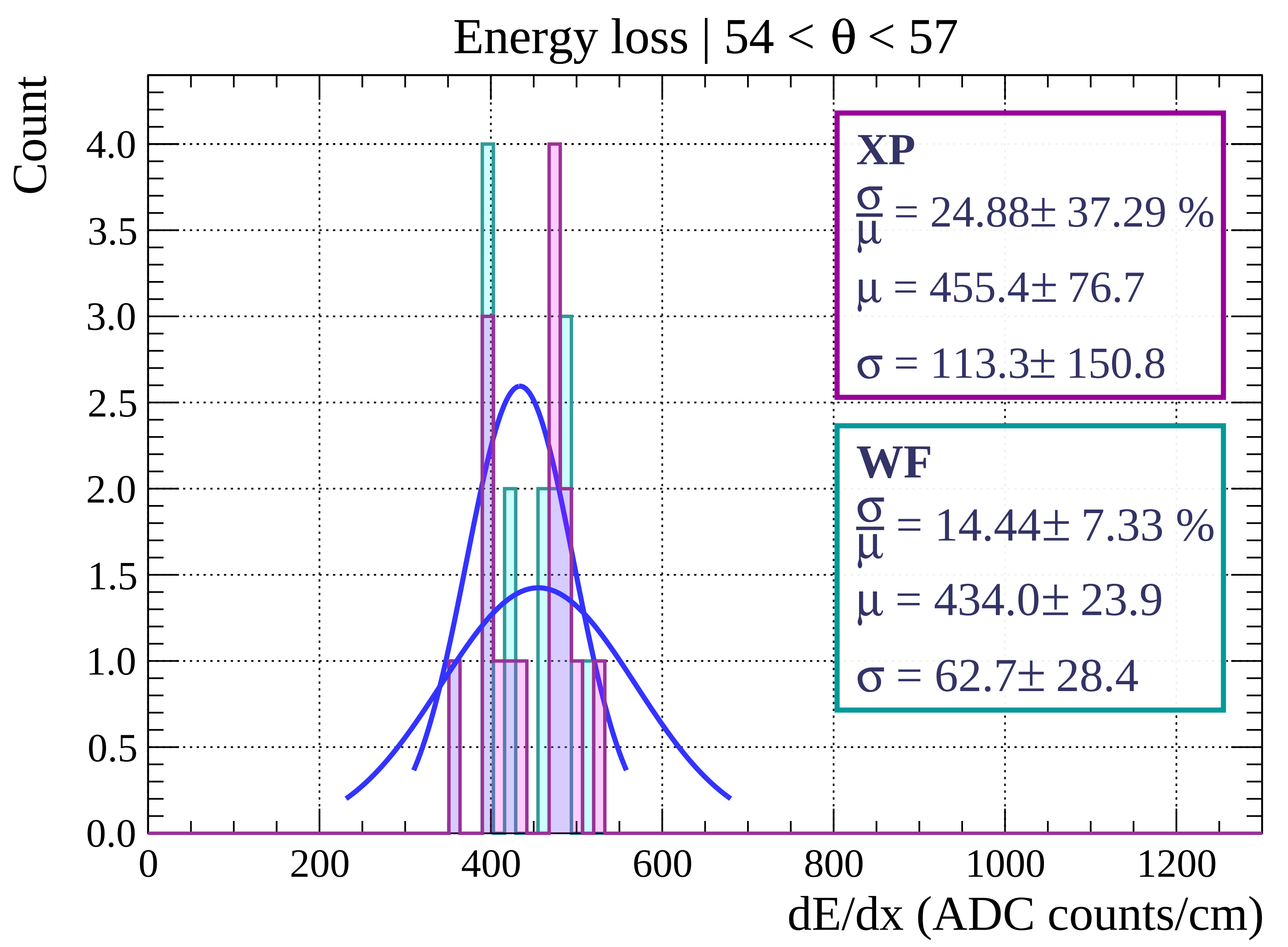
Energy loss | 45 < θ < 48
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
XP
$\frac{\sigma}{\mu}$ = 13.08± 2.81 %
μ = 435.6± 10.6
σ = 57.0± 10.9
WF
$\frac{\sigma}{\mu}$ = 13.51± 2.99 %
μ = 446.0± 10.8
σ = 60.2± 11.9



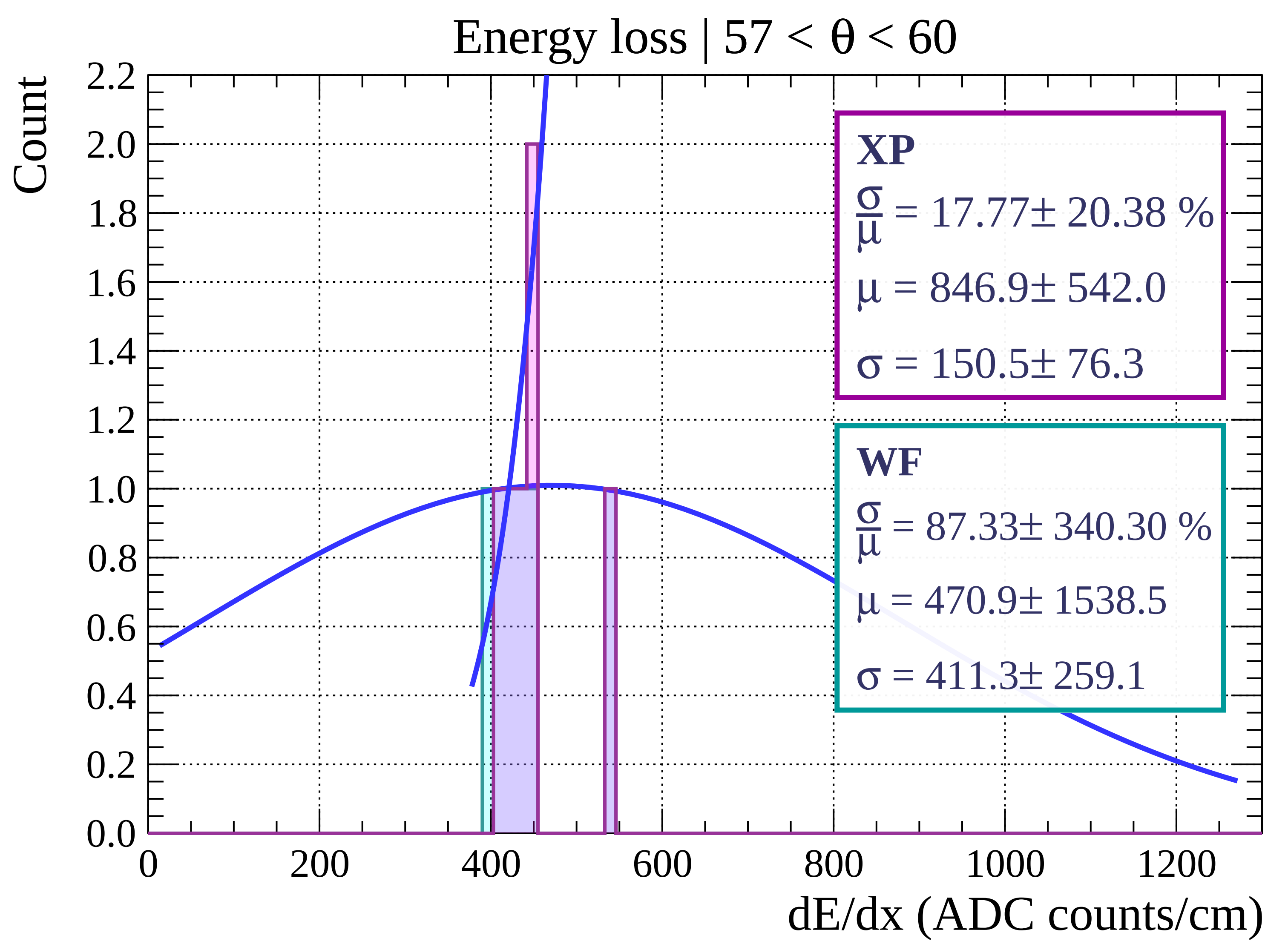
Energy loss | 48 < θ < 51
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200
XP
σ/μ = 6.34± 1.53 %
μ = 426.9± 6.1
σ = 27.1± 6.1
WF
σ/μ = 22.46± 164.53 %
μ = 467.7± 1765.1
σ = 105.1± 373.0



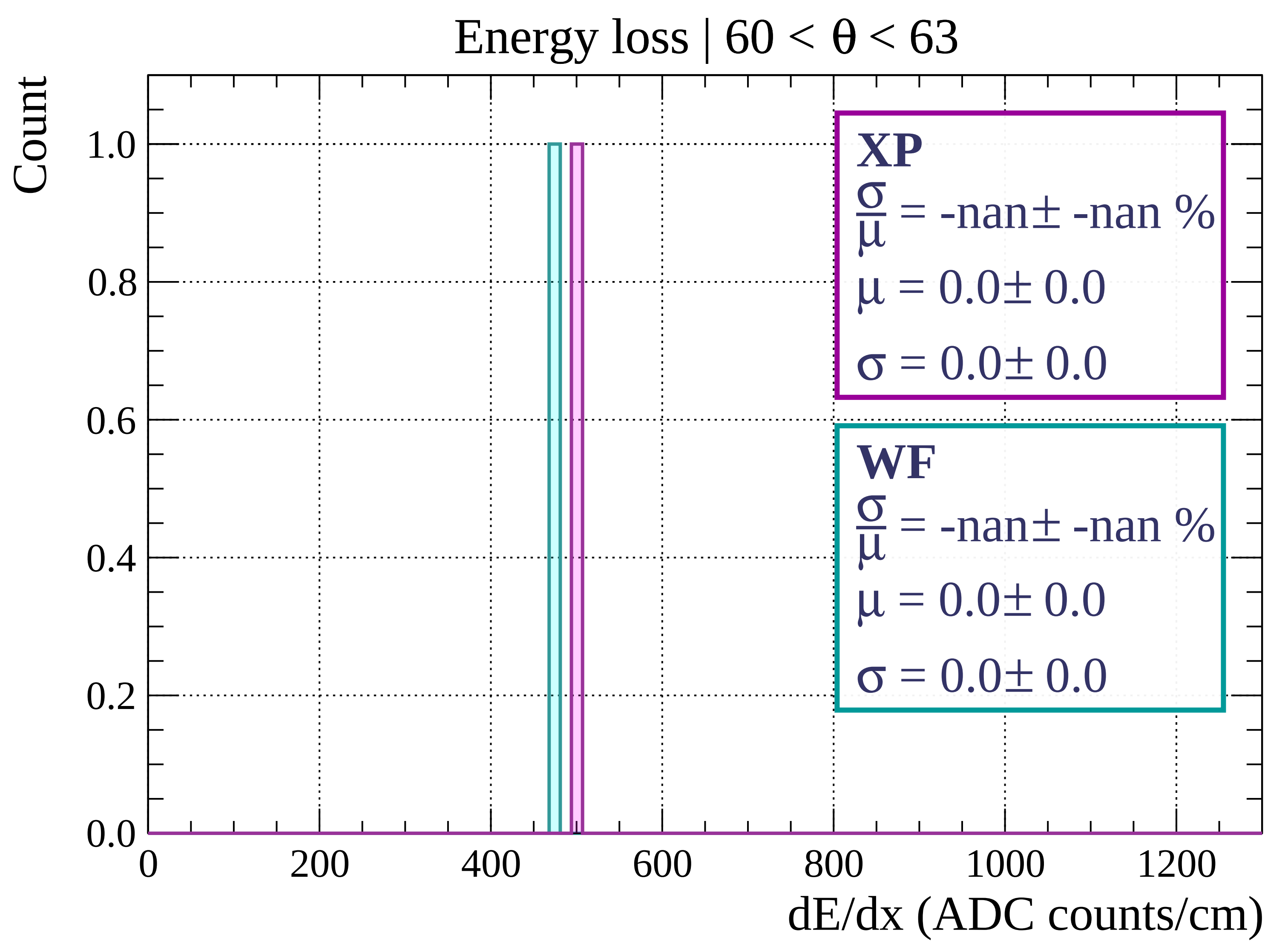
Energy loss | 51 < θ < 54
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
XP
$\frac{\sigma}{\mu}$ = 27.01± 19.31 %
μ = 429.8± 44.6
σ = 116.1± 70.9
WF
$\frac{\sigma}{\mu}$ = 16.49± 7.42 %
μ = 433.0± 21.0
σ = 71.4± 28.7



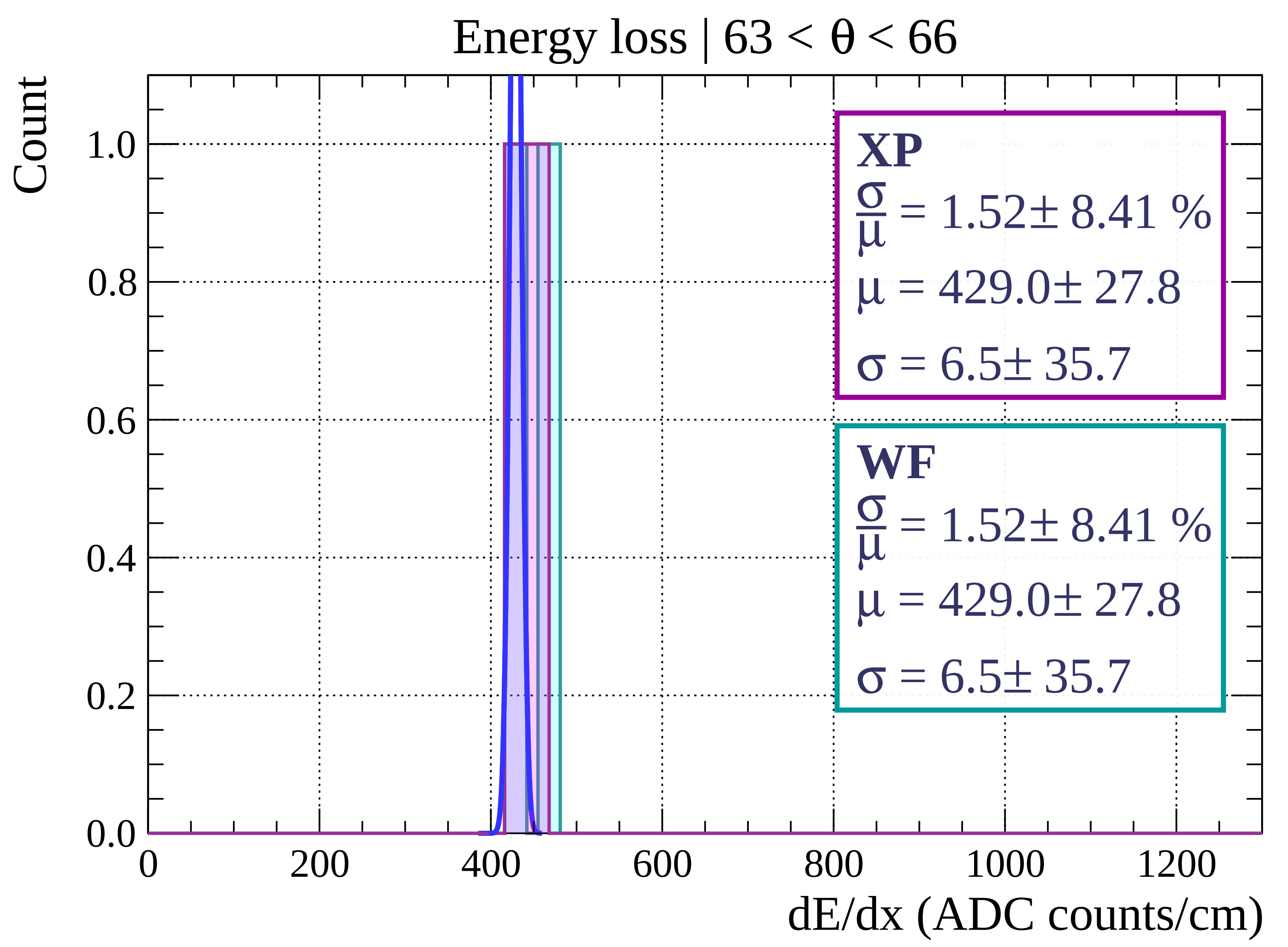
Energy loss | 54 < θ < 57
Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 24.88± 37.29 %
μ = 455.4± 76.7
σ = 113.3± 150.8
WF
$\frac{\sigma}{\mu}$ = 14.44± 7.33 %
μ = 434.0± 23.9
σ = 62.7± 28.4



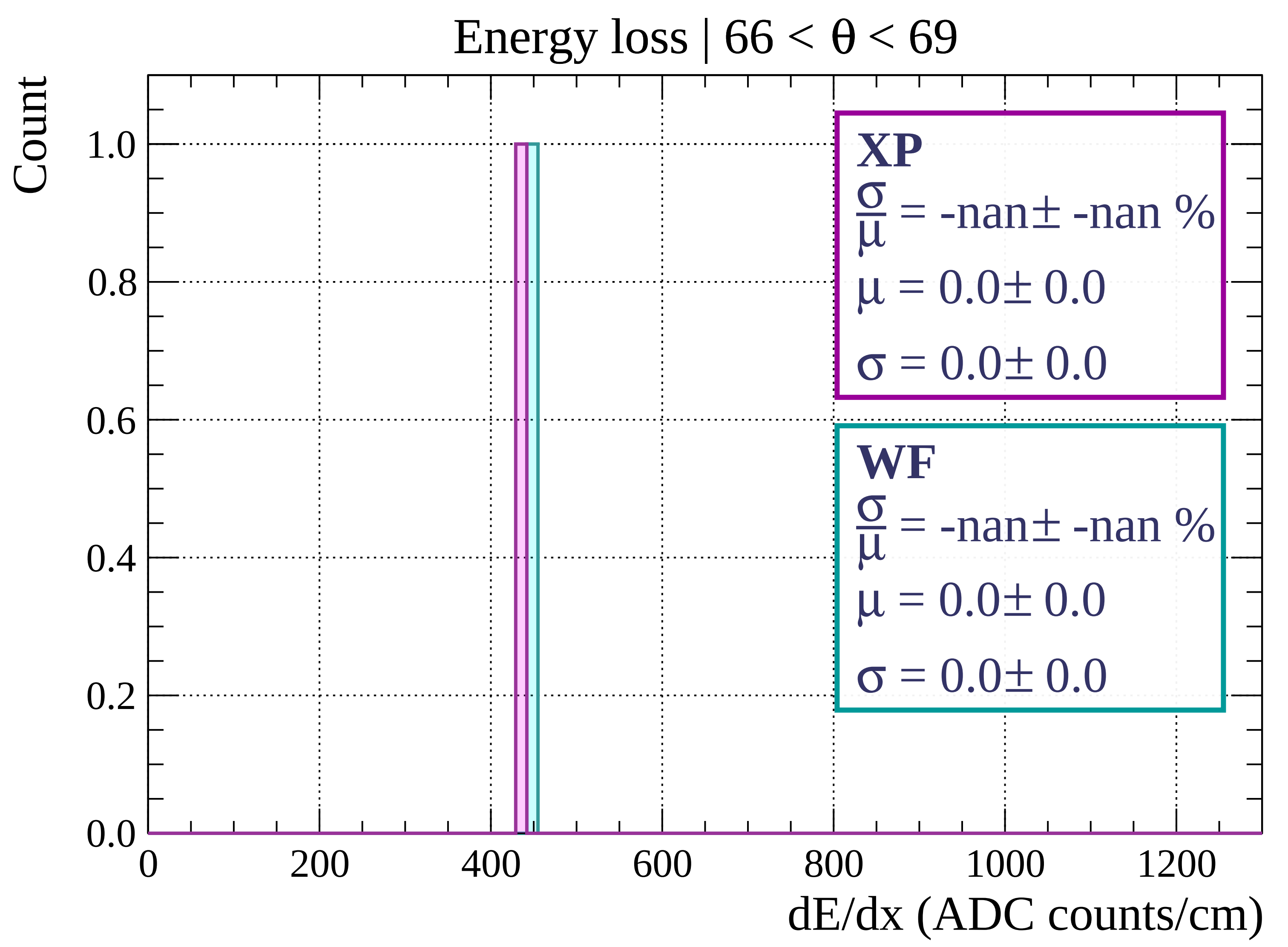
Energy loss | 57 < θ < 60
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 17.77± 20.38 %
μ = 846.9± 542.0
σ = 150.5± 76.3
WF
$\frac{\sigma}{\mu}$ = 87.33± 340.30 %
μ = 470.9± 1538.5
σ = 411.3± 259.1



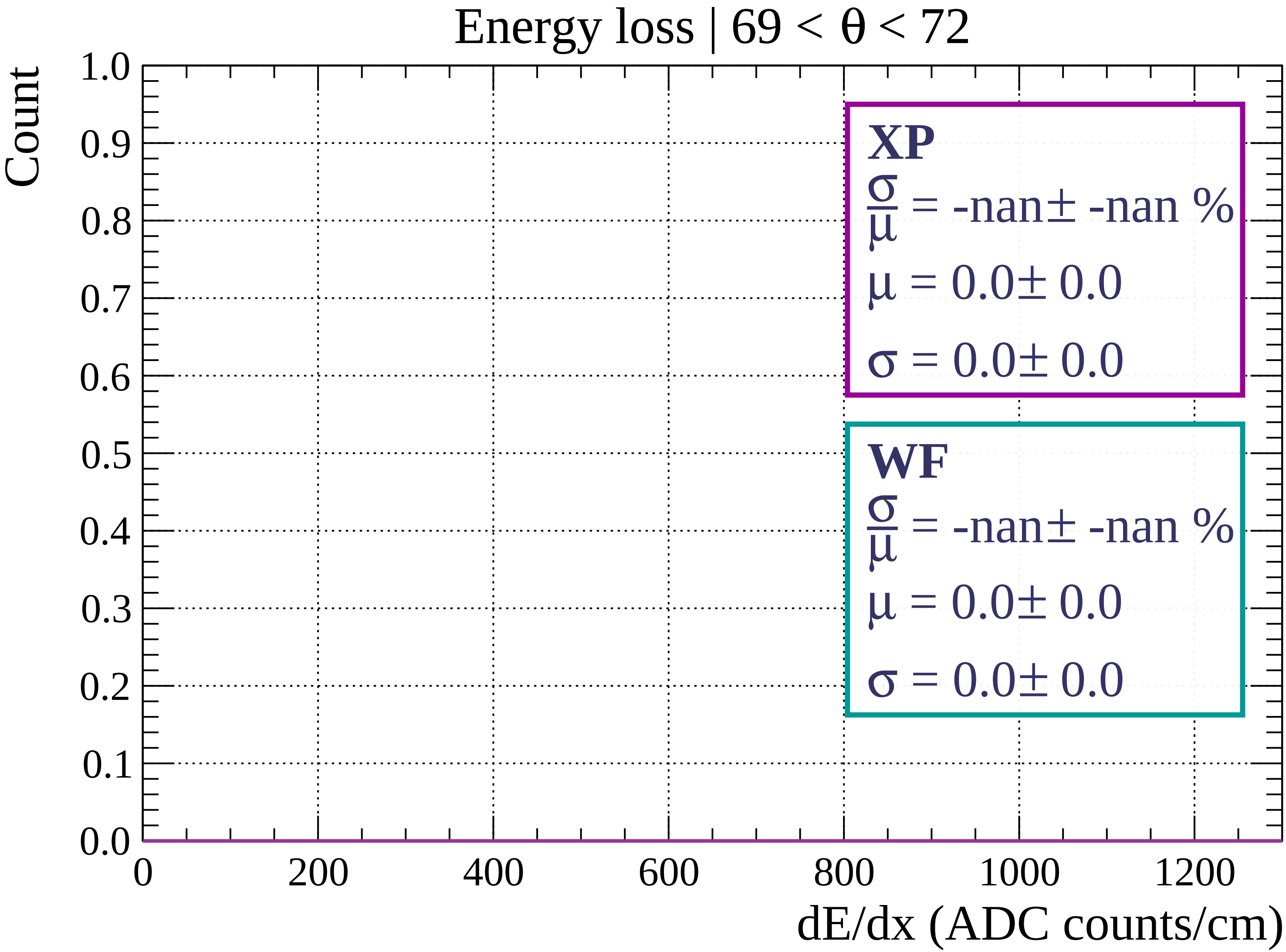
Energy loss | 60 < θ < 63
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



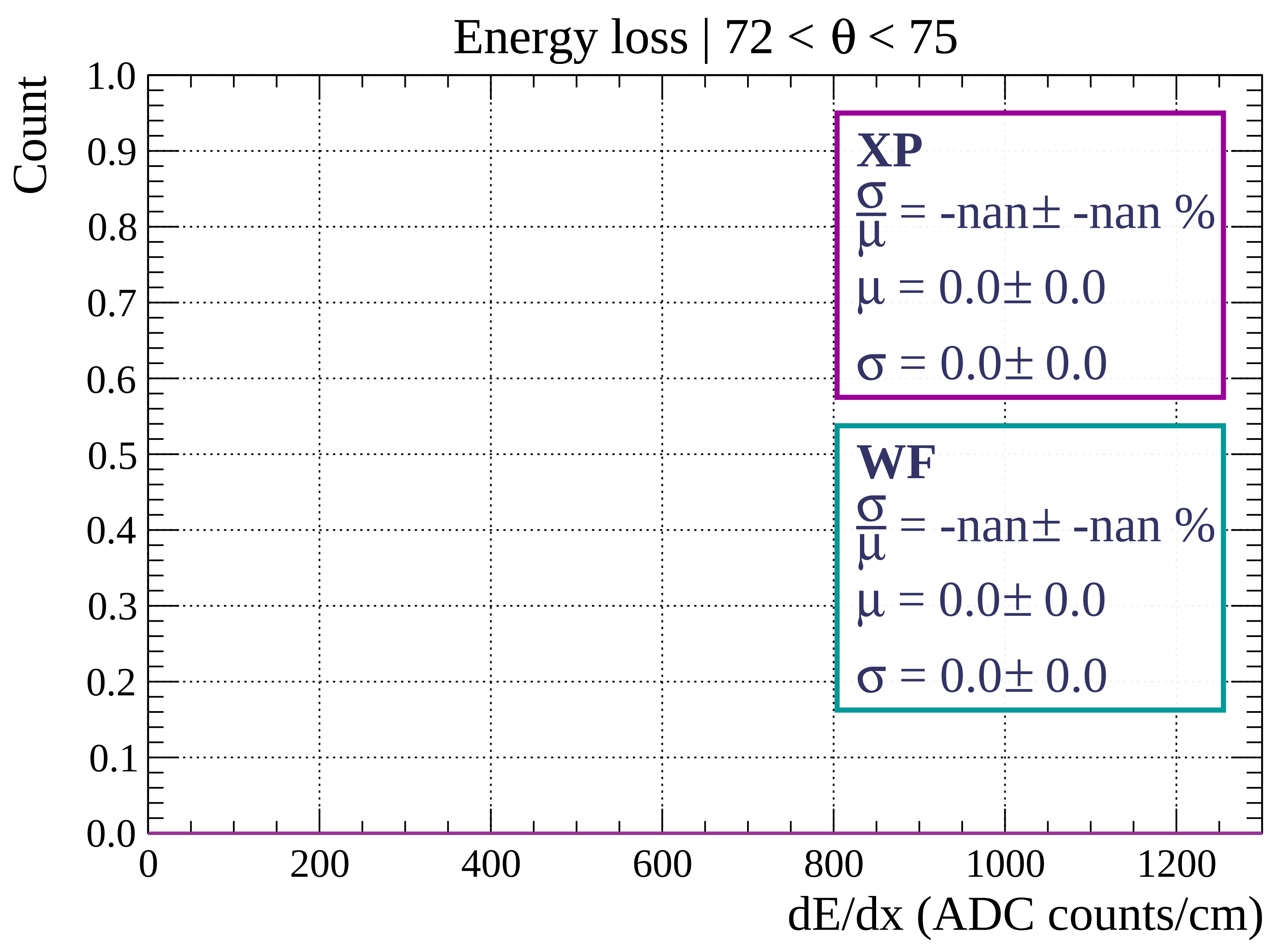
Energy loss | 63 < θ < 66
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 1.52± 8.41 %
μ = 429.0± 27.8
σ = 6.5± 35.7
WF
σ/μ = 1.52± 8.41 %
μ = 429.0± 27.8
σ = 6.5± 35.7

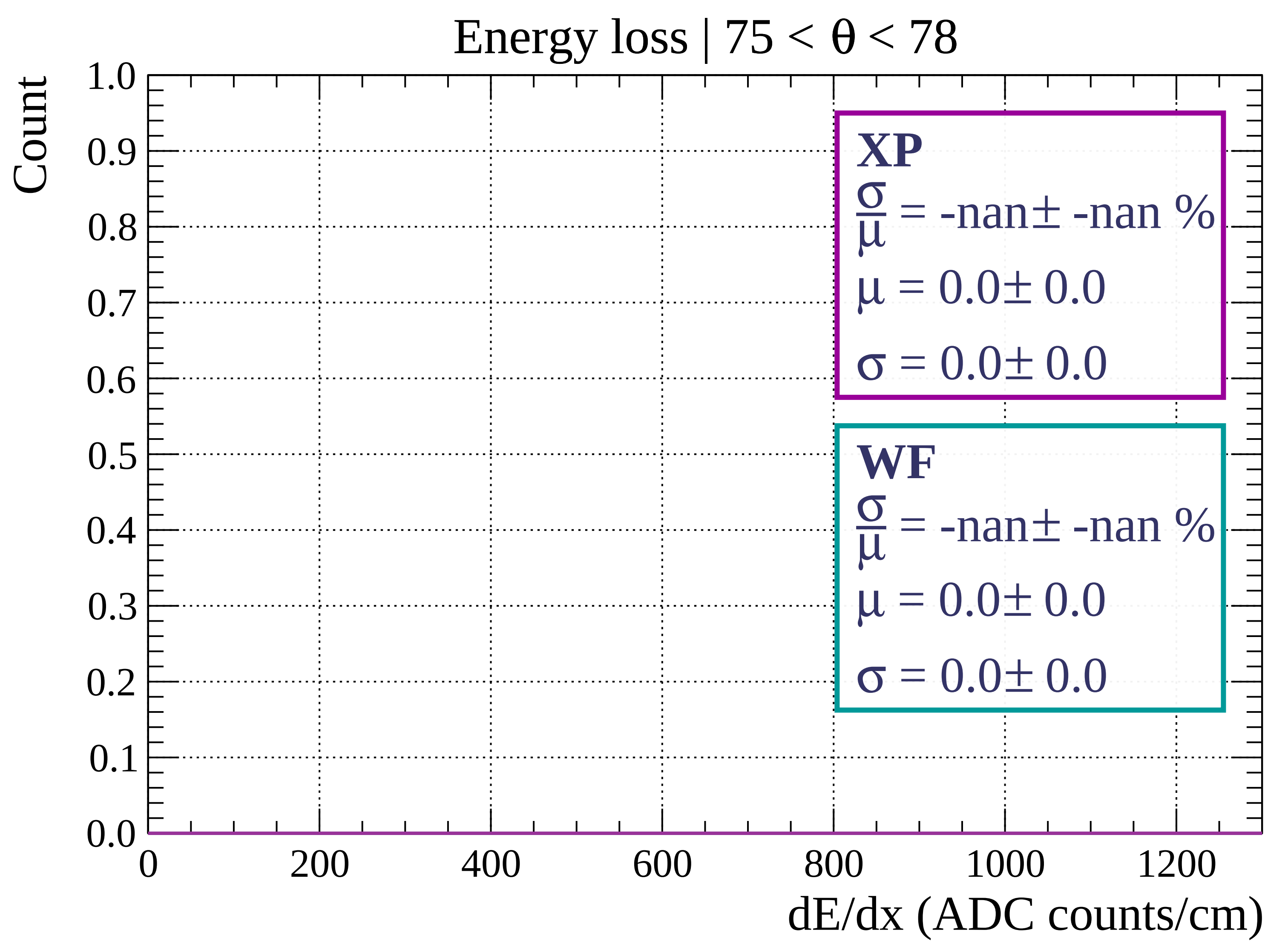


Energy loss | 66 < θ < 69
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

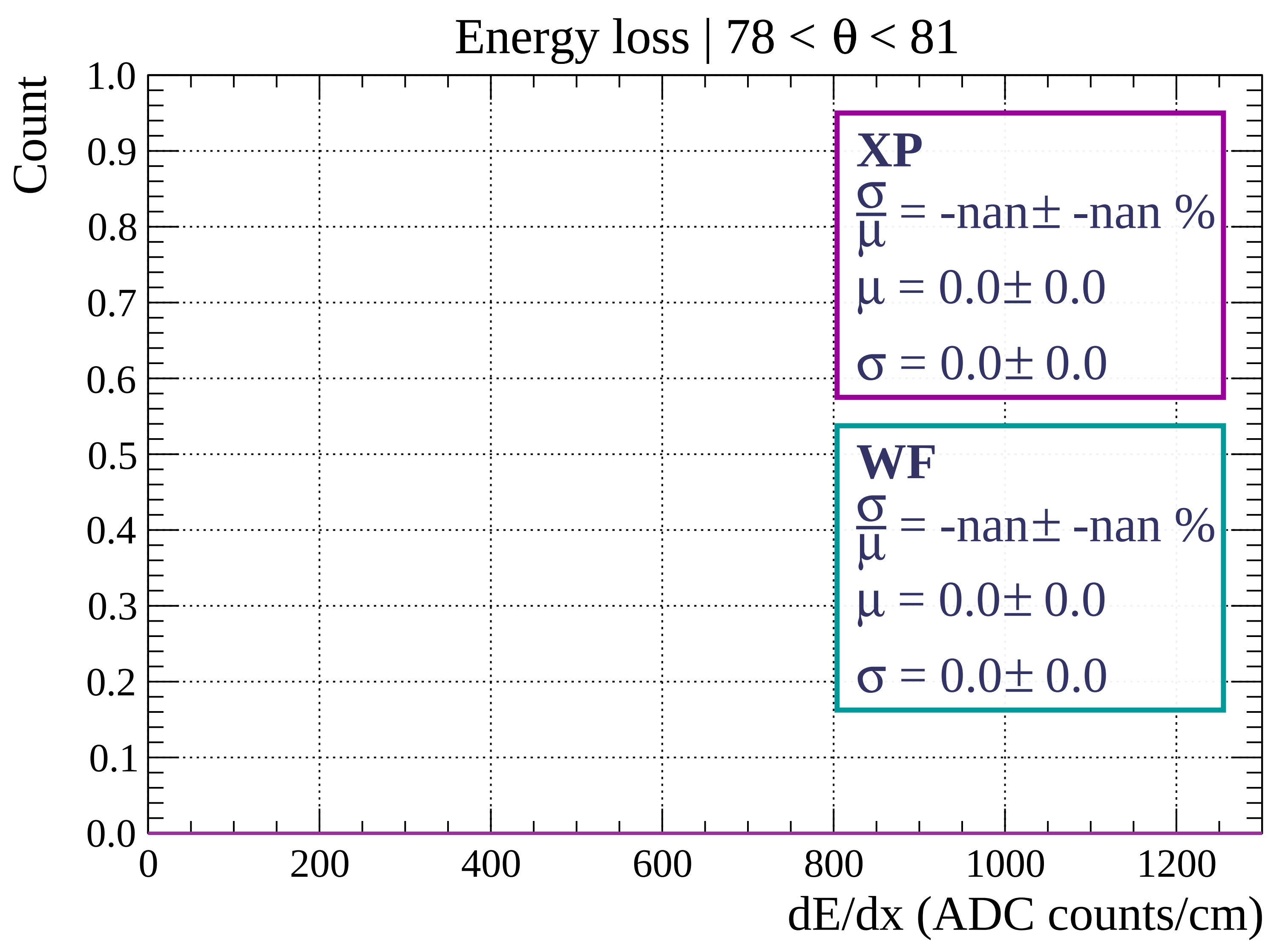


Energy loss | 69 < θ < 72
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

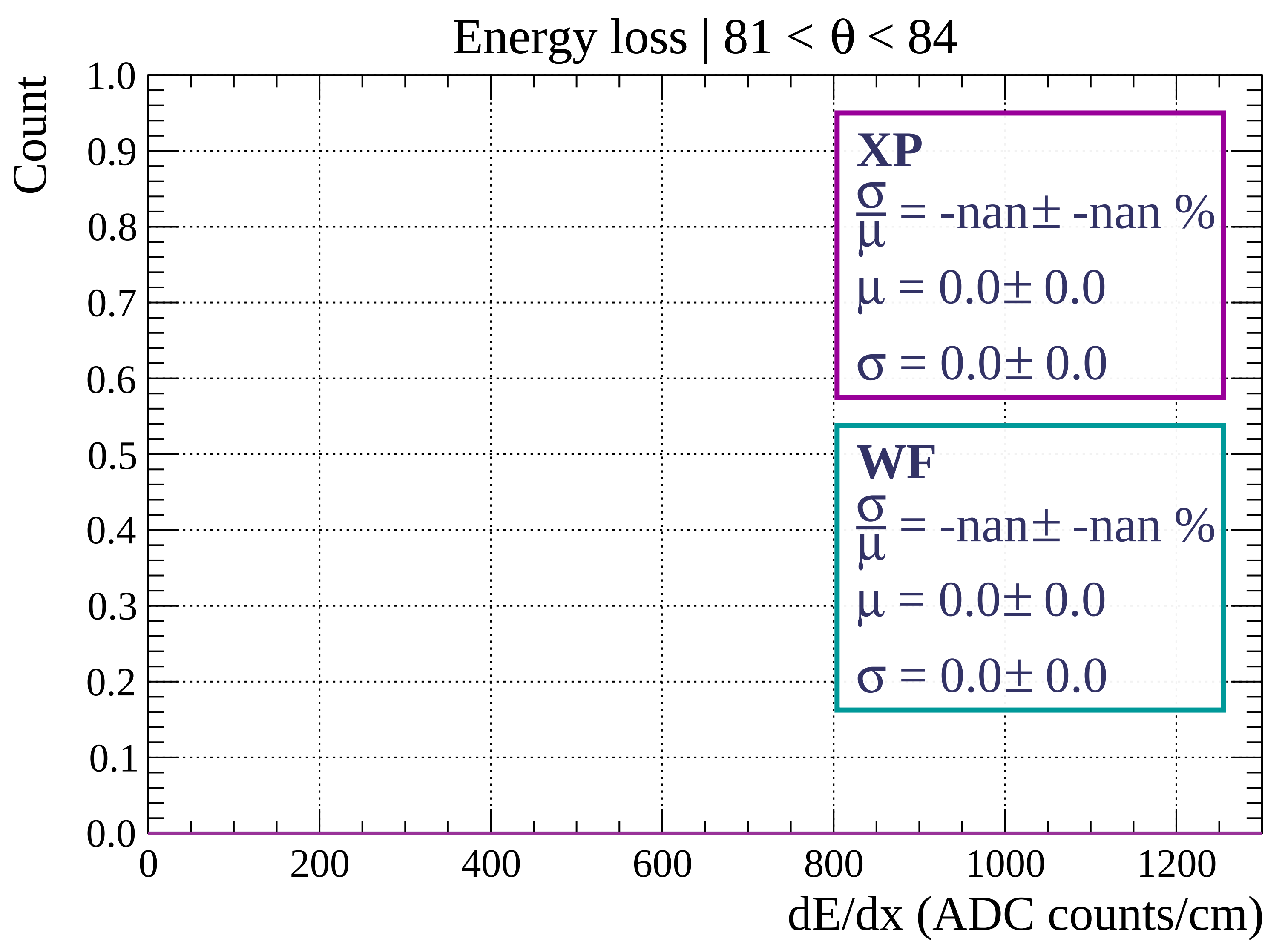

Energy loss | 72 < θ < 75
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

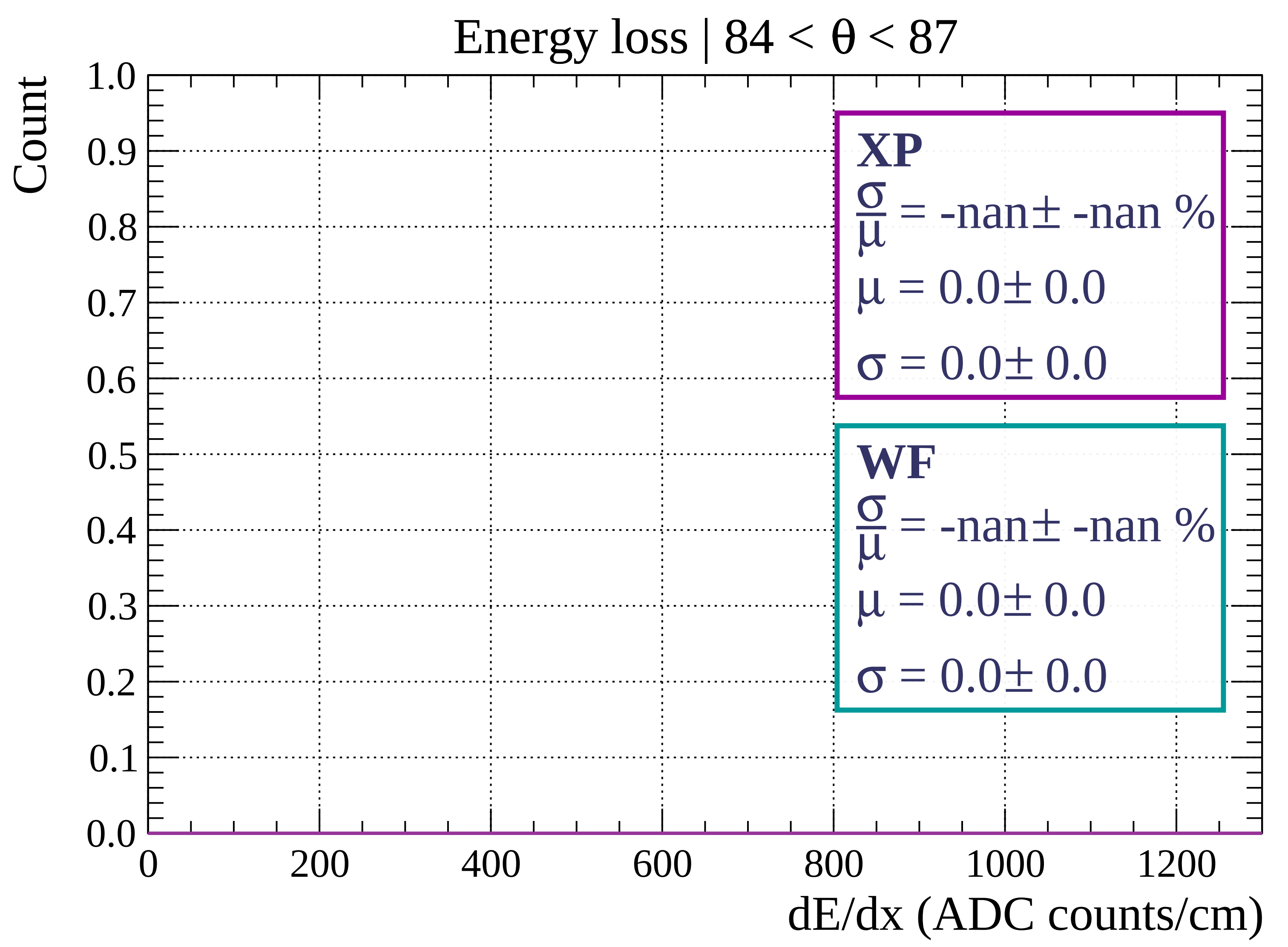


Energy loss | 75 < θ < 78
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

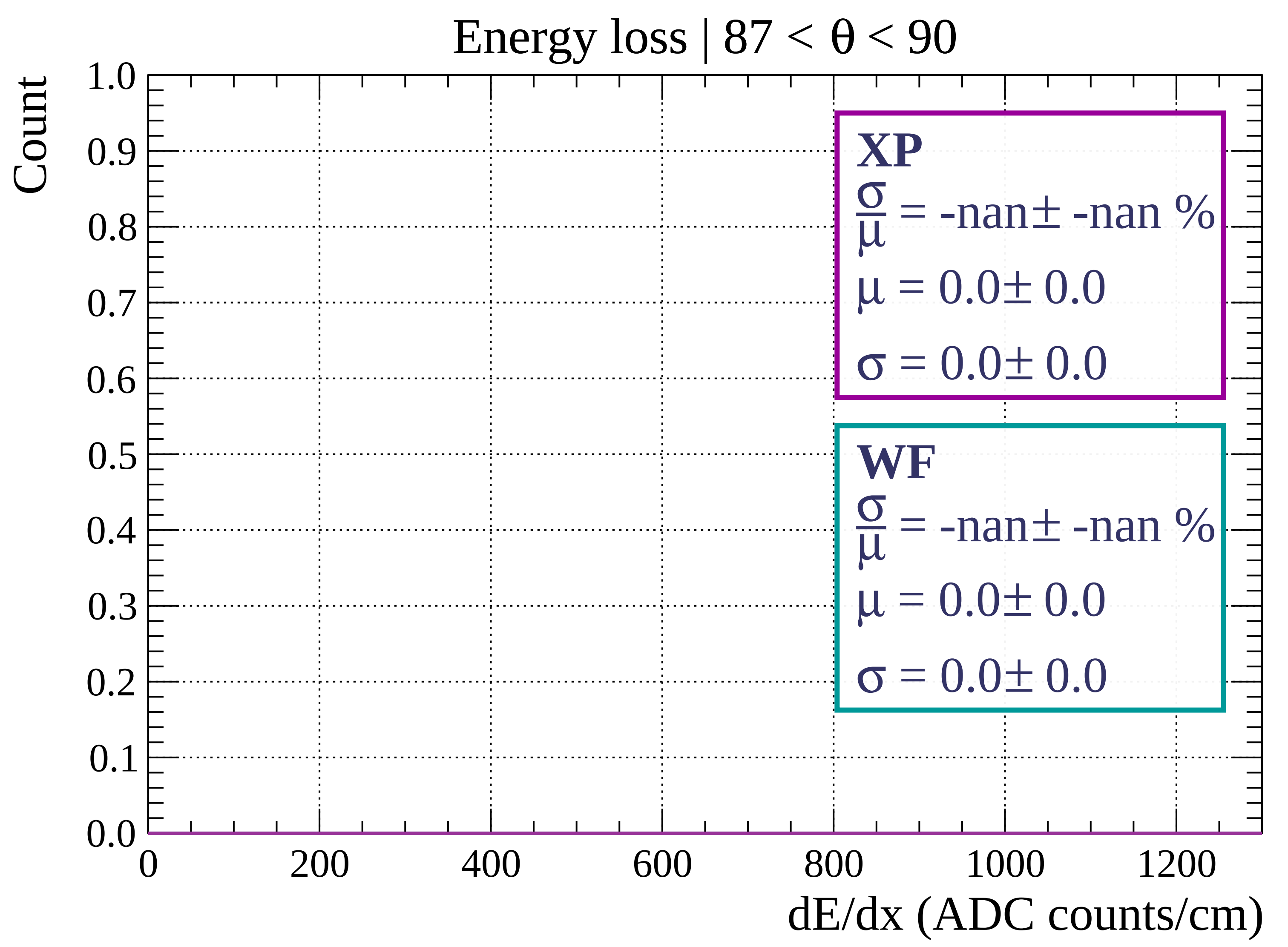


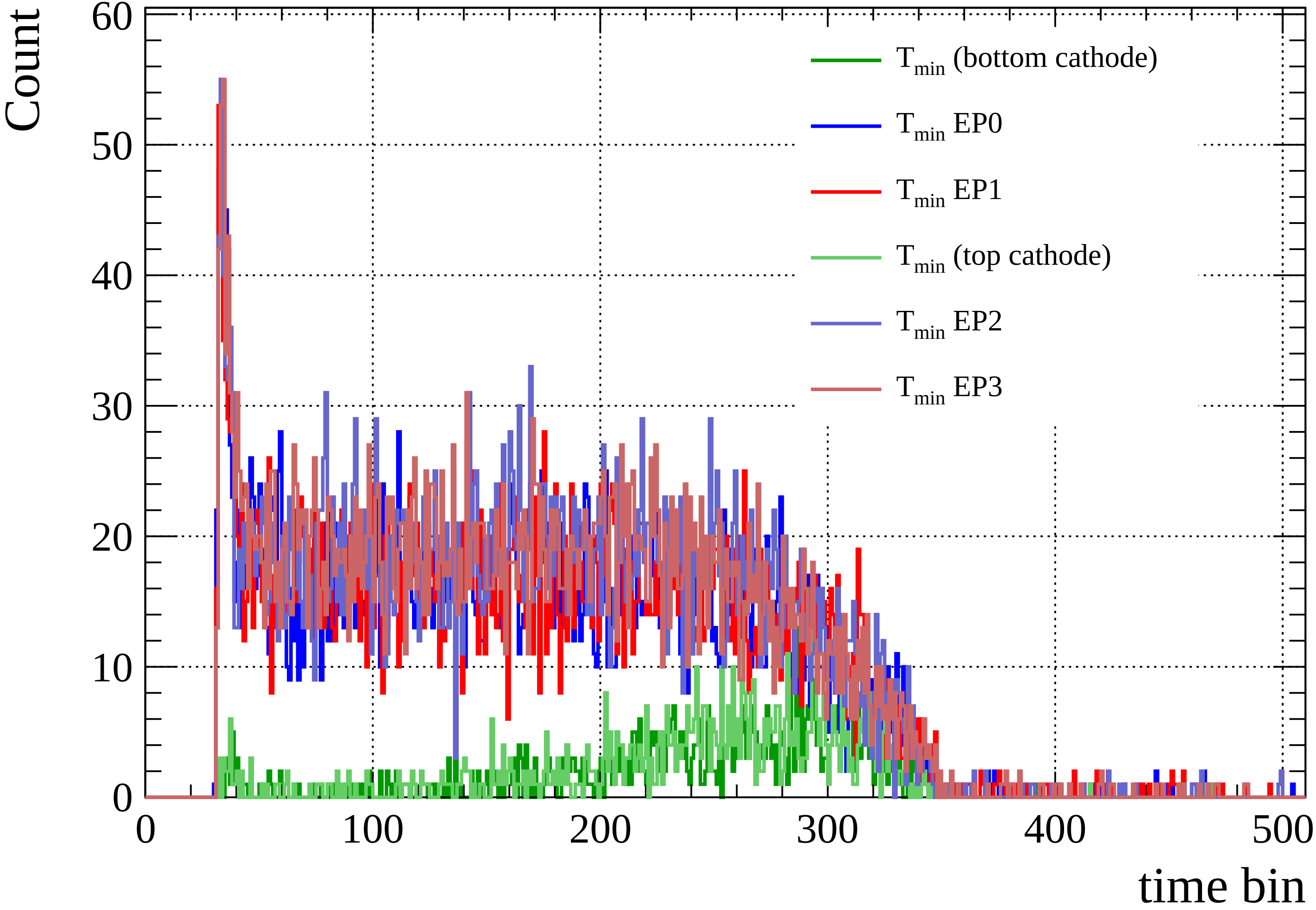
Energy loss | 78 < θ < 81
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

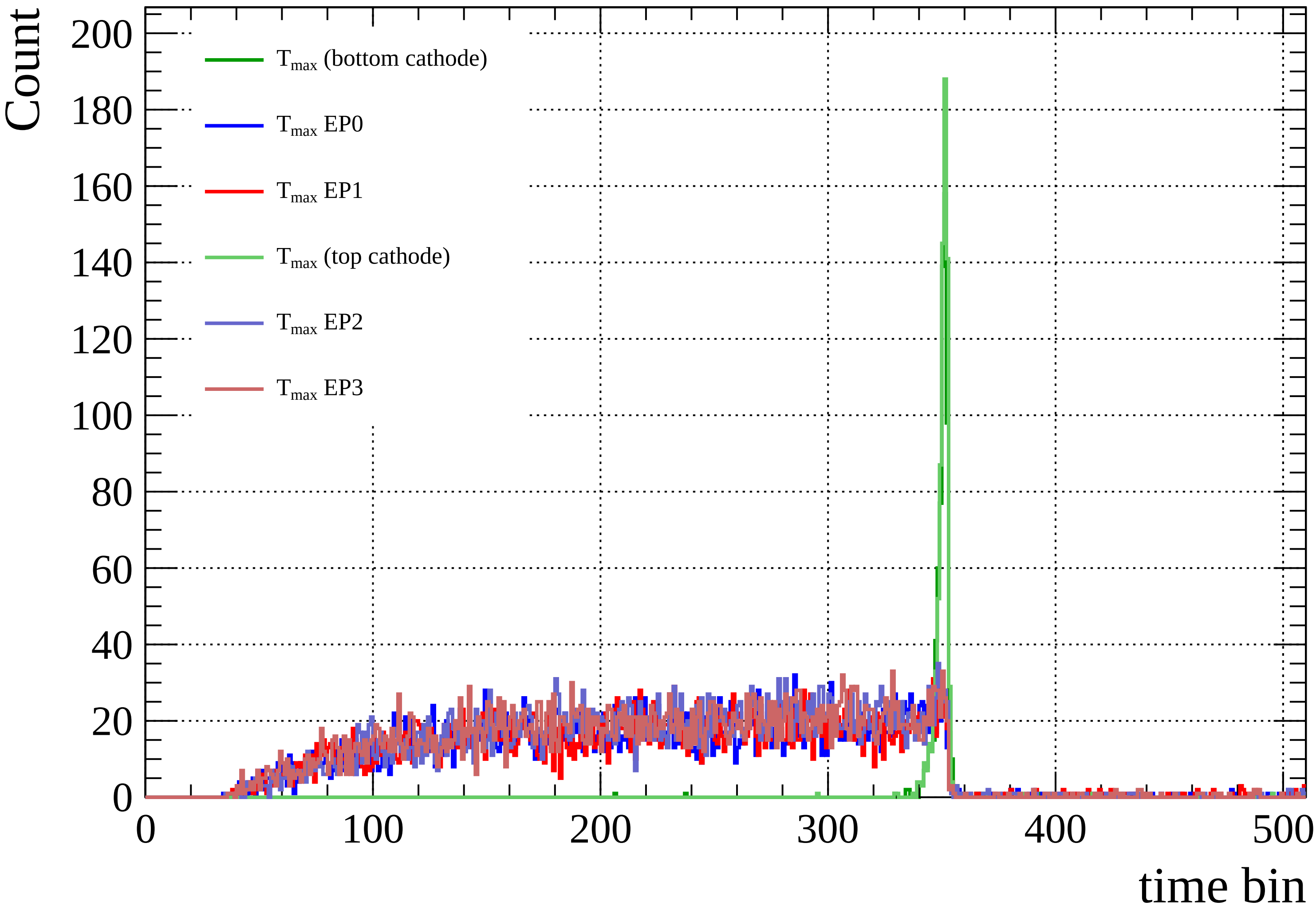

Energy loss | 81 < θ < 84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Energy loss | 84 < θ < 87
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 87 < θ < 90
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Count
time bin
0
10
20
30
40
50
60
0
100
200
300
400
500
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3


Count
time bin
0
100
200
300
400
500
0
20
40
60
80
100
120
140
160
180
200
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3